



BASE24-pos[®]

Merchant Host Interface Manual

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Preface

The ***BASE24-pos Merchant Host Interface Manual*** is designed to give readers a thorough understanding of the BASE24-pos Merchant Host Interface (MHI) module. This manual is designed to provide a general overview of the versatility of the MHI module and terminals that meet its specifications. It provides information on topics important in gaining a basic understanding of the MHI module, such as supported features. However, this manual is also helpful to readers who require a detailed explanation of the MHI module and its functionality. It provides detailed information including how to configure the MHI module, the files used with the MHI module, and the transaction flows used with the MHI module.

Audience

There are three types of readers for this manual. One reader reads this manual to get a general overview of the versatility of the MHI module. Another reader is a merchant host who wants to process using terminals that are compatible with the MHI module. The third reader of this manual is a customer who is purchasing the MHI module and is contracting with a merchant host.

Prerequisites

Although a knowledge of BASE24-pos is encouraged for the customer who is reading this manual, the manual is designed to be a stand-alone manual. Those who seek an overview of BASE24-pos processing should be able to find as much information about BASE24-pos in this manual as they will need to perform their functions. However, if more information about BASE24-pos is desired, readers should refer to the ***BASE24-pos Transaction Processing Manual***.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic

is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers that do not need the additional detail and at the same time provide the source for readers that require the additional information. This manual contains references to the following BASE24 publications:

- The ***BASE24 Base Files Maintenance Manual*** discusses the base files used with the MHI module. It also provides additional information about BASE24-pos files maintenance.
- The ***BASE24 Device Control Manual*** discusses in detail the Device Control Terminal (DCT) screens used with the MHI module.
- The ***BASE24 Logical Network Configuration File (LCONF) Manual*** discusses in detail the assigns and params used in configuring the MHI module.
- The ***BASE24 Log Message Reference Manual*** discusses in detail how to access log messages generated by the MHI module.
- The ***BASE24 Text Command Reference Manual*** discusses in detail text commands initiated from a network control facility.
- The ***BASE24-pos Settlement and Reporting Manual*** discusses in detail BASE24-pos settlement, cutover, balancing, and reporting.
- The ***BASE24-pos Transaction Processing Manual*** discusses in detail transaction processing performed by the MHI module and other Device Handler processes.
- The ***BASE24-pos Transaction Security Manual*** discusses in detail PINs, message authentication codes (MACs), and dynamic key management used by the MHI module and other Device Handler processes.
- The ***BASE24-pos Files Maintenance Manual*** discusses the following BASE24-pos files used with the MHI module: POS Terminal Data File (PTDF), POS Transaction Log File (PTLF), and POS Retailer Definition File (PRDF). It also provides additional information about BASE24-pos files maintenance.
- The ***BASE24-pos Router Authorization Manual*** describes Router and Authorization module initialization.
- The ***BASE24 CRT Access Manual*** contains additional information about BASE24 files maintenance.
- The ***BASE24 Tokens Manual*** provides information about tokens used with the BASE24-pos Standard Internal Message.

- The *NET24-XPNET Network Control Operators Guide* and the *NET24-XPNET Network Control Command Reference Manual* are companion manuals that network operators can use together to perform network control functions.
- The *XPNET BASE24 Configuration Guidelines* contains information on the Log Message Configuration File (LMCF).

In addition, the ANSI X9.19-1986 standard, *Financial Institution Retail Message Authentication*, describes the standard used for the support of message authentication codes (MACs).

Software Release

This manual documents standard processing for BASE24-pos release 5.3.

Customer-specific code modifications (that is, RPQs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

“Conventions Used in This Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, “Introduction,” provides an overview of the MHI module and its functionality. It also defines the responsibilities that ACI, the customer, and the vendor have when implementing the MHI module. This section includes information explaining how the MHI module can be integrated with BASE24-pos. This section also discusses the role of the MHI module during transaction processing.

Section 2, “Terminal Configuration,” explains the options available to customers configuring their POS terminals. This section explains the files that must be set up in order to tailor MHI module configuration options to meet the needs of a particular customer. This section includes descriptions of the Logical Network Configuration File (LCONF) assigns and params used by the MHI module.

Section 3, “BASE24-pos Merchant Host Interface Module,” contains information about the MHI module. It includes an overview of the module, a description of each file used by the MHI module during processing, a description of the steps the MHI module performs during initialization and warmbooting, and a list of the network control facility text commands recognized by the MHI module.

Section 4, “The BASE24 Standard Message,” contains information about the MHI message, its optional data fields, and how the MHI module formats responses to send to terminals.

Section 5, “Download Data,” contains information about the role of the MHI module when downloading configuration data to terminals for use during processing. It includes information about the records that can be fully or partially downloaded to terminals from the Merchant Terminal Configuration File (MCNF), the POS Terminal Data File (PTDF), and the POS Retailer Definition File (PRDF).

Section 6, “BASE24-pos Merchant Totals Processing,” describes the Merchant Totals process used to accumulate retailer and group totals. The Merchant Totals process is a process separate from the MHI module.

Section 7, “MHI Module Configuration Considerations,” discusses information on topics the customer must consider when deciding how to configure the MHI module. It includes message sequencing options, message authentication code (MAC) information, draft capture options, transaction accumulation totals information, handshaking transaction support, and BASE24-mail support. In addition, section 7 contains information on how customers can configure receipt formats. It includes the standard header fields that are typically printed on receipts, the parameters that need to be set in order for the MHI module to determine the type of response to return to the terminal, and the parameters to determine the exact text of the response.

Section 8, “Configuration Files Maintenance,” contains the files used in configuring the MHI module. BASE24-pos files maintenance manipulates these files through the use of formatted screens. This section shows the formatted screens used by the MHI module and describes each field shown on these screens. It also includes the procedures an operator can take to configure each screen to meet specific requirements.

Section 9, “Transaction Message Flows,” contains descriptions of transaction flows that occur between the terminal and the MHI module. This section includes BASE24-mail transaction flows.

Appendix A, “American Express Standard Industry Formats,” defines the American Express standard industry formats used with the AMEX Data Collection option. The AMEX Data Collection option allows customers to perform draft capture on transactions originating from American Express cardholders.

Appendix B, “Category Code Examples,” provides examples of how the Category Code field in the Download/Mail Key optional data field (field identifier (FID) V) is used during downloads. The category code indicates to the MHI module whether a download is in order, and to the terminal whether download information is available from the MHI module.

Appendix C, “Track Data Processing,” describes how the MHI module formats Track 1 and Track 2 information in the BASE24-pos Standard Internal Message (PSTM).

Readers can use the index by field name to locate information about a particular screen field and the index by data name to locate information about a particular field from a file or record structure.

Publication Identification

Two entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, PS-DH020-01 for the ***BASE24-pos Merchant Host Interface Manual***) appears on every page to assist readers in applying updates to the appropriate manual. The publication date (for example, 11/1998 for November, 1998) identifies the last time material on a particular page has been revised. This date can be used to verify that the reader has the most current information and is particularly helpful when dealing with HELP24 to answer operating questions. The print record, located on the page immediately following the title page, is also helpful when verifying that a manual contains the most current information. An entry is added to the print record each time a BASE24 publication is modified in any way. Each print record entry includes the date of the change, a unique version number, and the type of change (new, update, or reissue). Version 1 is assigned to a *new* manual. Later versions can be updates or reissues, depending on the changes being made. An *update* typically involves changes on a small number of pages throughout the manual. A *reissue* occurs when all pages of the manual are replaced because changes affect a large number of pages.

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Conventions Used in this Manual

This section explains how fields and unlabeled fields on screens are documented in this manual. It also explains a change in Nucleus process terminology.

Field Descriptions

Each field appearing on a files maintenance screen is listed by name and then described. Field descriptions briefly summarize the contents, purpose, and permissible values, as shown in the following example taken from Merchant Host Configuration File (MHCF) screen 1.

Example

LOGON/LOGOFF MSG — Indicates whether the MHI module should expect the specified DPC to send network management messages to log the DPC on and off. Valid values are as follows:

- 0 = No network management messages are sent.
- 1 = Network management messages are required.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	MHCF.PROCESSING-FLG.LOGON-FLG

Explanation

Each field description is completed by one or more of the following items of information:

Item	Description
Field Length	Specifies the size of the field and the type of characters that can be entered. This length refers to the field and valid values for the files maintenance screen, not the field in the DDLs. Possible values are alphabetic, alphanumeric, hexadecimal, and numeric. The term <i>alphanumeric</i> includes all alphabetic, numeric, and special characters that can be entered from a keyboard without using a control sequence. The term <i>hexadecimal</i> includes all numbers and the letters A through F. When a field value cannot be modified by the operator, the field length is System-protected.
Occurs	Indicates the number of times the field can be displayed on the screen. This information is given only if the field can be displayed multiple times.
Required Field	Specifies whether you have to enter a value in the field. Possible values are Yes and No. Some fields are required only under certain conditions. In this case, the entry is Yes, followed by the conditions that determine whether the field is required.
Default Value	Specifies the value that is automatically placed in the field when the screen is first displayed or when the F8 key is pressed to clear the screen.
Data Name	Provides the DDL name associated with the field appearing on the screen. Data names are included in the documentation to assist in communicating screen and field issues to your technical staff. Note that screen data is not always stored in the BASE24 database as it appears on the screen or as it is described in the field description. If you need information on how screen data is actually stored, consult the DDLs.

Unlabeled Fields on Screens

Angle brackets (< >) indicate an unlabeled field on a screen. An unlabeled field is a field that is present on a screen but is not preceded by an identifying literal label. For the purposes of documenting the field, a label has been assigned and appears inside the angle brackets. A multiple line unlabeled field is displayed as a shaded area and also has a label in angle brackets.

In the field descriptions for the screen, the unlabeled field appears according to its place on the screen and is identified by the same label.

Unlabeled fields are not included in the index by field name; they appear by subject in the main index.

Nucleus Process Terminology Change

There is one process that is the hub of an XPNET system. It manages messages queues; controls all lines, stations, and processes in the system; and serves as an interprocess interface. Throughout this manual, this process is referred to as the XPNET process.

In other manuals, you may see references to the Nucleus process as performing these same functions. These two terms (Nucleus process and XPNET process) should be viewed as synonymous. The Nucleus process and the XPNET process perform the same functions and processing. As manuals are reissued, references to the Nucleus process will be replaced with references to the XPNET process.

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Section 1

Introduction

This section provides an overview of the BASE24-pos Merchant Host Interface (MHI) module. It provides an overview of the MHI module and its functionality. This section also contains the following information about the MHI module:

- Supported features
- Supported transactions
- File usage
- Integration with BASE24-pos
- MHI version information

Supported Features

The BASE24-pos Merchant Host Interface (MHI) allows institutions to select and configure the features necessary to meet their processing requirements. This subsection describes the following features:

- Expanded transaction set
- Host control and terminal control modes
- Configurable messages
- Multiple language support
- Configurable receipts
- Download options
- Draft capture options
- American Express data collection support
- BASE24-pos settlement and cutover support
- Terminal balancing
- BASE24 transaction security options
- Message sequencing support
- PIN encryption
- Message authentication
- Retailer or group totals processing
- Log message generation
- BASE24-mail support
- PS2000 support
- Track 1 and 2 support
- Address verification
- Port usage tracking
- Maximum returns and adjustments

Expanded Transaction Set

The MHI module can handle processing for a total of 25 different POS transactions sent from the terminal, including both financial and administrative transactions. Customer flexibility and functionality are greatly increased because the MHI module supports a wide variety of transaction types. These include purchase-with-cash-back transactions, preauthorization purchase transactions, preauthorization purchase-completion transactions, and request-mail transactions. A complete list of the transaction set available with the MHI module is shown in the topic “MHI Transaction Support” later in this section.

Host Control and Terminal Control Modes

The MHI module can operate in two different modes: host control mode or terminal control mode. These modes allow for different levels of participation by BASE24 in terms of processing.

If the UPDATE PTDF field in the Merchant Host Configuration File (MHCF) is set to zero, the MHI module is running in host control mode. In host control mode, only one POS Terminal Data File (PTDF) record exists for each data processing center (DPC).

When the MHI module is operating in host control mode the institution wants BASE24 to authorize transactions. The merchant host or controller is responsible for much of the transaction processing. Some restrictions apply when the MHI module is running in host control mode. These include no terminal totals, and limited duplicate checking. It is assumed that the merchant host or controller provides some of the required support. The customer defines by DPC whether the MHI module is operating in host control mode.

If the UPDATE PTDF field in the Merchant Host Configuration File (MHCF) is set to one, the MHI module is running in terminal control mode. In terminal control mode, one POS Terminal Data File (PTDF) record exists for each terminal attached to the merchant host or controller.

When the MHI module is operating in terminal control mode, the institution places more processing responsibility on BASE24. Not only is BASE24 responsible for authorizing transactions, it is also responsible for much of the device functionality (provided by the MHI module). For example, if the MHI module is running in terminal control mode, terminal, retailer, and group totals can be accumulated, and downloads can be performed.

Configurable Messages

The MHI message format allows customers to configure message requests and responses. The MHI module can read requests from the terminal that include any number of optional data fields. These optional data fields can be in the request in any order. The data fields included in responses sent to the terminal can also be configured by customers.

Customers can select a different message format for each transaction type, if they want. The request message and the response message for each of these transaction types can also consist of different fields. The fields included in the request message can be in any order the customer chooses.

All MHI messages consist of a standard header and optional data fields. The standard header is mandatory and can stand alone. However, it can be followed by any number of optional data fields. Optional data fields are subject to the requirements of the transaction. This means that certain transactions may require the presence of certain data fields in their messages. The only other restrictions on the number of optional data fields that can be included in the message are the protocol limitations, the size of the terminal read buffer, and the size of the MHI module read buffer. The MHI module read buffer is configurable to a maximum of 4,088 bytes, including all data communication characters associated with the message.

Multiple Language Support

Customers have the option of formatting display responses to a terminal in one of three languages. The MHI module can use information contained in the terminal request to access language tables configured by the customer in the Merchant Terminal Response File (MRSP) to format its response, or it can use information contained in the POS Terminal Data File (PTDF) to determine the language.

If no language is specified in the request, terminal responses are displayed in the default language set up in the MHI module device-dependent area of the PTDF. If a language table is specified in the request, the information in the MRSP tables override the default value set up in the PTDF. In short, the language table information must be in the request from the terminal in order to be used. Otherwise, the default value in the PTDF is used. Setting up the multiple language support option is explained in section 8. For more information on the PTDF, refer to the ***BASE24-pos Files Maintenance Manual***.

Configurable Receipts

The MHI module does not format receipts. Customers must determine if receipts are necessary and then supply the optional data for the receipts using the Merchant Terminal Configuration File (MCNF). It is the responsibility of the cardholder to configure each response in the MCNF to include sufficient data for the terminal to format and print a receipt.

The MHI module has the capability of returning a response, determined by the customer, in up to three languages as well as returning a free format response. These options are set up in the Merchant Terminal Response File (MRSP).

Download Options

The MHI module supports both full and partial downloads. The terminal can receive full configuration file information or single records from the MCNF. The download is requested at the terminal level. The download options available with the MHI module are flexible, allowing the customer to send more than 1,000 bytes of discretionary download configuration data to the terminal. In addition, download data records can pertain to as many terminals as the customer wants. A separate download record is not required for each terminal.

Draft Capture Options

The MHI module provides an efficient method of draft capture with third-generation terminals. The draft capture option common for first-generation terminals consists of authorizing the transaction online and sending a paper follow-up. This method requires merchants to send the paper follow-up before receiving payment.

The draft capture option common for second-generation terminals consists of authorizing the transaction online and sending a separate transaction at the end of a batch to verify that the transaction was completed correctly. In effect, this consists of two transactions for each purchase. Since most of the transactions that are authorized are completed, this requires almost twice the amount of disk space and communication costs. Second-generation terminals also can use the draft capture option offered with first-generation terminals.

The draft capture option offered by the MHI module overcomes the problems encountered with first- and second-generation terminals. With the MHI module, a purchase requires only one transaction. In the one-step process, the MHI module

oversees authorization and captures the data necessary to pay the merchant. If a transaction is not completed correctly, the clerk who entered the transaction sends an adjustment transaction. This procedure compensates for the second-generation terminals' system of ensuring that transactions are completed correctly. The MHI module also supports transactions that require paper follow-up.

American Express Data Collection Support

The MHI module provides customers with the option of performing data capture on transactions originating from American Express cardholders. American Express has defined standard industry formats for various types of services (for example, auto rental, retail, and restaurant). Each of these formats requires different data to be present in the request message from the terminal. The MHI module is able to recognize the format by the information contained in the request message and subsequently the MHI module can capture and process transactions using these formats.

BASE24-pos Settlement and Cutover Support

BASE24-pos provides an efficient method of settlement and cutover. Settlement, in terms of BASE24-pos, is actually a reconciliation effort whereby the transaction activity of the current day is closed out and reported, and the system is shown to be in balance. The term cutover simply refers to the point in time where transactions stop being logged to the business of the current day and start being logged to the business of the next day. It refers to the closing of the current posting day and the opening of the next.

BASE24-pos settlement differs from the settlement carried out by participating BASE24-pos financial entities (for example, retailers, retailer sponsors, cardholders, card issuers, interchanges) in that BASE24-pos does not move funds to settle accounts. It does, however, provide the support that allows these financial entities to initiate the movement of funds. An extensive set of reports provides financial entities with the information they require to settle their accounts. BASE24-pos also provides the POS Transaction Log File (PTLF), which keeps a record of each transaction processed by BASE24-pos during its defined reporting day. In addition, BASE24-pos provides an automated method to cut over terminals, retailers, institutions, interchanges, and the BASE24-pos network to a new posting day. For more information on BASE24-pos settlement, cutover, and reporting, refer to the ***BASE24-pos Settlement and Reporting Manual***.

Terminal Balancing

The MHI module provides several methods for merchants to balance terminals with BASE24-pos. Merchants have the option of performing a series of subtotal request transactions to determine if BASE24-pos and the terminal have accumulated the same totals. Merchants then have the option of entering adjustment transactions, if necessary, before closing the batch.

Merchants also can receive a report a day later, or communicate with the service provider to receive terminal balancing information. The BASE24-pos report that contains terminal balancing information is the Retailer Reconciliation Report (B24POS35). This report is used to balance transaction activity at the terminal batch level. Additional totals are provided by shift and day, if the terminal supports these balancing levels. Cumulative totals are provided by retailer, group, region, institution, and network. The B24POS35 is divided into the following totals sections:

- Batch totals, shift totals, day totals, and network day totals
- Retailer totals—all terminals
- Region totals—all retailers
- Group totals—all regions
- Institution totals—all groups

Refer to the ***BASE24-pos Settlement and Reporting Manual*** for more information about the Retailer Reconciliation Report.

BASE24 Transaction Security

The MHI module offers various ways of ensuring that transactions are transmitted and secured correctly. These include options that ensure every transaction is being processed only once and options that allow customers to select from various PIN encryption methods.

In addition, the MHI module supports the use of message authentication codes (MACs). MACs are used to verify whether data has been accidentally or fraudulently altered. Any data sent between the terminal and the MHI module can be verified through the use of the MACs. The details necessary to use this standard have been developed by ACI to use with BASE24-pos and the MHI module. The customer can decide whether to use MAC verification with a particular transaction by configuring MAC parameters in the MCNF.

Setting up MAC options is explained in section 8. MACs are explained in section 7. PIN encryption is also explained in section 7. For more information on BASE24 transaction security, refer to the ***BASE24 Transaction Security Manual***.

Message Sequencing Support

Message sequencing is a method for ensuring that every message being sent from the terminal is being received only once and in the correct order. There are two methods available for message sequencing. One method calls for the MHI module to check the transmission number, which is a field in the standard header, in order to detect and drop duplicate requests. The other method consists of comparing message sequence numbers to positively identify the sequencing of a request. If message sequence numbers are included in the message, the MHI module can ensure that no messages have been lost or transmitted out of sequence. The message sequence number method is more reliable, but requires an additional field be included in the message. For more information on the sequence number field, refer to section 4.

PIN Encryption

PIN encryption can be accomplished using hardware or software. The PIN encryption method used is set in the PIN ENCRYPTION TYPE field in the POS Terminal Data File (PTDF). These are standards that are used throughout the financial industry. The customer may choose to not encrypt PINs. For information on the PIN encryption methods available, refer to the ***BASE24-pos Files Maintenance Manual***.

Message Authentication

Any data sent between the terminal and the MHI module can be verified with the use of the message authentication technique known as ANSI X9.19 Financial Institution Retail Message Authentication. ACI has developed the details necessary to use this standard with BASE24-pos and the MHI module. This is referred to as message authentication code (MAC) verification. MAC verification can have performance impacts depending on the customer's configuration.

Retailer or Group Totals Processing

The Merchant Totals process provides retailer or group totals to terminals connected to the merchant host. The Merchant Totals process is separate from the MHI module. The Merchant Totals process calculates retailer or group totals when it receives a subtotals request from the MHI module. For more information on the Merchant Totals process, refer to section 6.

Log Message Generation

The MHI module generates log messages to inform operators of the status of the MHI module. Log messages can be used to assist operators with their daily tasks or to notify them of problems occurring in the system.

In addition, ACI has developed a method that allows customers to view log message documentation electronically. This online log message documentation provides customers with greater flexibility by allowing them to print log message documentation as needed at their location or to electronically manipulate the data to suit their needs. The online log message documentation provides the message number, severity, and text for each log message generated by the MHI module. In addition, it also provides the cause of each message and the recommended action to take.

Accessing the documentation for log message generated by the MHI module is discussed later in this manual. For a complete explanation of online log message documentation, refer to the ***BASE24 Log Message Reference Manual***.

BASE24-mail Support

The MHI module is fully compatible with release 5.3 of BASE24-mail. BASE24-mail allows host processors to send informational messages to POS terminals throughout a network. Mail messages can also be transmitted from POS terminals to the host processor. BASE24-mail uses the technology of the Tandem computer to provide reliable, fail-safe service 24 hours a day, seven days a week.

PS2000 Support

The MHI module is compatible with Visa Payment Service 2000 (PS2000). PS2000 is a Visa program that ensures that the portion of the transaction that is authorized is the same as the portion that is cleared and settled. All phases of the

transaction are linked, allowing the institution to reduce back office expenses. The MHI module supports the PS2000 program for the passenger transport, hotel, automobile rental, direct marketing, and fuel market segments.

Track 1 and Track 2 Support

The MHI module formats both Track 1 and Track 2 information from the native message into the BASE24-pos Standard Internal Message (PSTM). The MHI module fills the Track 2 Data field of the PSTM regardless of the type of track data available in the native message. When Track 1 data is available, the MHI module adds the Track 1 token to the PSTM. For more information on how the MHI module formats track data, see appendix C.

Address Verification

The MHI module is compatible with the BASE24-pos Address Verification add-on module. Address verification assists merchants in controlling losses that can result from credit card fraud during transactions where the cardholder cannot be readily identified. The MHI module places address verification information in the PSTM. See section 4 for information about the data fields used to format address verification data in the PSTM. For more information on the BASE24-pos add-on Address Verification module, refer to the ***BASE24-pos Address Verification Manual***.

Port Usage Tracking

The MHI module can track station or port usage. Port usage tracking gives an institution the ability to analyze, control, and balance network service usage. The MHI module retrieves the station ID from the network message header and places it in the Station ID token. The token is appended to the PSTM and can be configured to be logged to the POS Transaction Log File (PTLF). Refer to the ***BASE24 Tokens Manual*** for more information on the Station ID token.

Maximum Returns and Adjustments

The MHI module can be configured to prevent a clerk from accidentally exceeding the return and adjustment limits defined in the PTDF. When a return or adjustment limit is exceeded, the MHI module can decline the transaction, refer the transaction, or continue processing based on an LCONF param. For more information about checking return and adjustment limits, refer to section 3.

MHI Transaction Support

All of the transactions supported by the MHI module must originate with a request from the terminal. A response is sent from the MHI module for each request received unless the MHI module is unable to read the POS Terminal Data File (PTDF) record that corresponds to the terminal sending the request. In this case, no response is sent and the terminal times out. Message requests and responses are explained in more detail later.

A complete list of transactions supported by the MHI module is shown in the following table. Each transaction contained in the table is followed by a short description.

Transaction	Description
Normal Purchase	Purchase of goods or services at the point of sale using a credit or debit card.
Preauthorization Purchase	Online check of a cardholder's account to determine if funds are available for an intended purchase. The amount preauthorized is placed on hold in the cardholder's account until the purchase is completed, a timer expires, or the hold is removed.
Preauthorization Purchase Completion	Completion of a purchase that was preauthorized. The actual amount of the purchase is entered using this transaction and the hold for the amount entered in the preauthorization purchase transaction is released. The transaction is then completed using the correct amount.
Mail or Telephone Order	Purchase of goods or services through the mail or a telephone call.
Merchandise Return	Return of merchandise to a retailer for refund. The refund is electronically credited to the cardholder account.
Cash Advance	Cash disbursement from a cardholder's credit card account at the point of sale.
Card Verification	Online check to see if a cardholder's card is valid.
Balance Inquiry	Online inquiry into the balance of a cardholder's account.

Transaction	Description
Purchase with Cash Back	Combination purchase and cash withdrawal at the point of sale. The requested transaction amount is higher than the price of the purchase. The cardholder receives the difference in cash. This transaction is used only with debit cards.
Check Verification	Online check to see if a cardholder's check is tied to a valid checking account. BASE24-pos supports this transaction in pass-through mode. This means BASE24-pos provides a link between the transaction and destination by simply passing the transaction to the destination without performing authorization.
Check Guarantee	Electronically reserves funds in a cardholder's checking account to cover a check written by the cardholder. BASE24-pos supports this transaction in pass-through mode. This means BASE24-pos provides a link between the transaction and destination by simply passing the transaction to the destination without performing authorization.
Purchase Adjustment	Corrects an error in a previously completed purchase transaction.
Merchandise Return Adjustment	Corrects an error in a previously completed merchandise return transaction.
Cash Advance Adjustment	Corrects an error in a previously completed cash advance transaction.
Cash Back Adjustment	Corrects an error in a previously completed purchase with cash back transaction. Adjustments can be made only to the cash back amount.
Logon Request	Changes the clerk ID in the POS Terminal Data File (PTDF) and accumulates totals for the new employee. It also automatically performs an implied logoff of the previous employee, if required.
Logoff Request	Clears the clerk ID in the PTDF.

Transaction	Description
Close Batch Request	Ends the current batch and opens a new batch to which data is logged and accumulated. When a close batch transaction is received, the system creates a batch totals record from the terminal totals in the PTDF and writes it to the PTLF, if batch totals are configured to be logged. It then clears the batch totals in the PTDF. If the terminal can send its own batch totals, the system also compares the terminal totals to the PTDF totals, and, if the debits, credits, and adjustments do not match exactly, writes the terminal totals to the PTLF immediately following the PTDF totals, if the totals are configured to be logged. Note: When a batch is closed the system automatically writes a service totals record to the PTLF (prior to the batch totals record), if service totals are configured to be logged, and clears the service totals in the PTDF. The system also automatically writes a clerk totals record to the PTLF (prior to the batch totals record), if clerk totals are configured to be logged, and clears the clerk totals in the PTDF.

Transaction	Description
Close Shift Request	Ends the current shift and opens a new shift to which data is logged and accumulated. When a close shift transaction is received, the system creates a shift totals record from the terminal totals in the PTDF and writes it to the PTLF. It then clears the shift totals in the PTDF. If the terminal can send its own shift totals, the system also compares the terminal totals to the PTDF totals, and, if the debits, credits, and adjustments do not match exactly, writes the terminal totals to the PTLF immediately following the PTDF totals. Note: Close shift transactions, when entered, must be immediately preceded by a close batch transaction. If not, the system will deny the close shift transaction.
Close Day Request	Ends the current terminal day and begins a new day. When a close day transaction is received, the system creates a daily totals record from the terminal totals in the PTDF and writes it to the PTLF. It then clears the terminal daily totals in the PTDF that have accumulated since the last close day transaction or since the totals were last cleared. If the terminal can send its own daily totals, the system also compares the terminal totals to the PTDF totals, and, if the debits, credits, and adjustments do not match exactly, writes the terminal totals to the PTLF immediately following the PTDF totals. Note: Close day transactions, when entered, must be immediately preceded by a close shift transaction. If not, the system denies the close day transaction. Close day transactions should only be entered once during a reporting day in order to maintain reporting integrity.

Transaction	Description
Employee Subtotals Request	Obtains the count and amount of transactions that have been entered at a point-of-sale terminal by a specific employee since the totals were last cleared from the PTDF. This transaction requests PTDF clerk totals. Clerk totals are retrieved from the clerk totals records that have already been logged to the PTLF. The Employee Subtotals Request can be used when balancing a terminal to determine whether the system totals are in agreement with the actual totals at the terminal. This transaction can be entered at any time and is not logged. It also does not affect PTDF totals.
Batch Subtotals Request	Obtains the count and amount of draft capture transactions that have been entered at a point-of-sale terminal since the last close batch request transaction or since the totals were last cleared from the PTDF. This transaction can be used when balancing a terminal to determine whether the PTDF totals are in agreement with the actual totals at the terminal. This transaction can be entered at any time and is not logged. It also does not affect PTDF totals.
Shift Subtotals Request	Obtains the count and amount of draft capture transactions that have been entered at a point-of-sale terminal since the last close shift request transaction or since the totals were last cleared. This transaction can be used when balancing a terminal to determine whether the PTDF totals are in agreement with the actual totals at the terminal. This transaction can be entered at any time and is not logged. It also does not affect PTDF totals.

Transaction	Description
Day Subtotals Request	Obtains the count and amount of draft capture transactions that have been entered at a point-of-sale terminal since the last close day request transaction or since the totals were cleared. This transaction can be used when balancing a terminal to determine whether the PTDF totals are in agreement with the actual totals at the terminal. This transaction can be entered at any time and is not logged. It also does not affect PTDF totals.
Retailer/Group Subtotals Request	Obtains the count and amount of draft capture transactions that have been entered at point-of-sale terminals for a given retailer group. This transaction can be entered at any time and is not logged. It also does not affect the PTDF totals.
Read Mail Request	Obtains a mail message intended for the point-of-sale terminal.
Mail Delivered Request	Notifies the Mail process that mail in a read mail request was delivered to the terminal.
Send Mail Request	Sends a mail message from the point-of-sale terminal to another destination.
Download Request	Causes a full or partial download to be sent to the terminal.
Handshake Request	Verifies the status of the communications link and, optionally, the terminal communication and authentication keys.

MHI File Usage

The following table provides a comprehensive list of all files used by the MHI module. The MHI module uses both base files and BASE24-pos files and, therefore, the table indicates whether each file is a base file or a BASE24-pos file.

File Name	File Type
Logical Network Configuration File (LCONF)	Base
Log Message Configuration File (LMCF)	Base
Merchant Terminal Configuration File (MCNF)	BASE24-pos
Merchant Terminal Response File (MRSP)	BASE24-pos
Merchant Host Configuration File (MHCF)	BASE24-pos
Merchant Regulation E File (MREG)	BASE24-pos
Merchant Totals Configuration File (MTCF)	BASE24-pos
POS Retailer Definition File (PRDF)	BASE24-pos
POS Terminal Data File (PTDF)	BASE24-pos
POS Transaction Log File (PTLF)	BASE24-pos

Refer to the ***XPNET BASE24 Configuration Guidelines*** for more information about the LMCF. Information about configuring the other files listed above can be found in section 2 of this manual.

Integration with BASE24-pos

The MHI module can be integrated with BASE24-pos. BASE24-pos is a complete point-of-sale authorization and switching system that uses the technology of the Tandem computer to provide reliable, fault-tolerant service 24 hours a day, seven days a week.

BASE24-pos offers routing of transactions based on card type and retailer floor and ceiling limits. If the primary destination for authorization is not available, BASE24-pos allows the transaction to be sent to up to two additional authorization centers before a decision is made about approving, denying, or referring the transaction. Destinations to which transactions can be routed include host computers and interchange networks.

In addition, BASE24-pos offers a comprehensive set of authorization options on the Tandem. These include local and national negative identification, positive identification, and positive identification with account balances. Another authorization option for credit card accounts, parametric authorization, enables institutions to define additional criteria on which to base the authorization of transaction requests that exceed standard cardholder limits.

BASE24-pos also includes online control of all network communications elements and POS terminals, as well as a comprehensive set of daily and periodic reports for service providers and individual retailers.

Transaction Flow

When a merchant host sends a message to BASE24, it is first received by the XPNET process, the transaction manager, which then sends the message to the bound Device Handler/Router/Authorization process.

The MHI module is bound into a Router/Authorization process. Any messages coming into the bound process from a POS device or merchant host are first received by the Router module which then translates the message to the POS Standard Internal Message (PSTM) and routes the message to the Device Handler module. Messages coming into the bound process from any other destination are also received first by the Router module, but are routed instead to the Authorization module to be logged to the POS Transaction Log File (PTLF).

For more information about the Router module, refer to the ***BASE24-pos Transaction Processing Manual***.

MHI Versions

The MHI module is available in two versions: release 3.4 and release 5.3. These releases of the MHI module correspond to release 3.4 and release 5.3 of BASE24-pos. Customers should note that release 5.3 of the MHI module is designed to accommodate terminals that have firmware designed for release 5.3 of the MHI module. The release 5.3 MHI module also supports the release 3.4 native message format.

Section 2

Terminal Configuration

The BASE24-pos Merchant Host Interface (MHI) module allows customers to configure their POS terminals to meet their exact specifications. For example, customers determine the information to be included in requests and responses for every type of transaction they want to use. In addition, the MHI module allows customers to determine the responses sent to the terminal. These responses, based on the results of authorization processing, can be sent in up to three different languages.

Options supported by the MHI module are based on the way the base, MHI module, and associated BASE24-pos files are set up. These files consist of the following:

- Merchant Terminal Configuration File (MCNF)
- Merchant Terminal Response File (MRSP)
- Merchant Host Configuration File (MHCF)
- Merchant Regulation E File (MREG)
- Merchant Totals Configuration File (MTCF)
- POS Retailer Definition File (PRDF)
- POS Terminal Data File (PTDF)
- POS Transaction Log File (PTLF)
- Logical Network Configuration File (LCONF)

MHI Configuration Files

This section describes the files used to configure the MHI module. The descriptions explain the role of each file in determining the options supported by the MHI module.

The MCNF, MHCF, MREG, MRSP, and MTCF are used exclusively with the MHI module. The LCONF is a BASE24 file vital in determining MHI module configuration issues. BASE24-pos files PTDF and PRDF also contain information set up by the customer in order to allow the MHI module to support the necessary options. Each of these files is explained below and on the following pages. How these files are used during processing is discussed in section 3.

Merchant Terminal Configuration File

The Merchant Terminal Configuration File (MCNF) is central to the configuration of the MHI module. It specifies processing parameters, the fields to be included in requests and responses for every transaction type, the fields to be verified through message authentication codes (MACs) within the messages, and all the information downloaded to the terminal.

This file is accessed through BASE24 files maintenance. Instructions on setting up and accessing the MCNF are found in section 8. The Data Definition Language (DDL) uses the acronym MCNFG for the Merchant Terminal Configuration File.

The MCNF contains three types of records. The first record type contains processing data. The second record type contains field map data. The third record type contains download data.

The MCNF is a key-sequenced file, with the primary key consisting of three fields: the TERMINAL GROUP, RECORD TYPE, and RECORD ID.

The value in the TERMINAL GROUP field should correspond to a value in the TERM GROUP field in the POS Terminal Data File (PTDF). The MHI module uses the TERM GROUP field in the PTDF to determine the MCNF records associated with the terminal. Therefore, MCNF records are applicable to an entire group of terminals. The number of terminals within one group is determined by the customer when they set up PTDF records.

The RECORD TYPE field indicates whether the record contains processing data, field map data, or download data. The RECORD ID field identifies the record number within the record type. These record types are discussed on the following pages.

Processing Data

One processing data record exists for each terminal group defined in the MCNF. This record consists of 20 general purpose data fields and 20 flag fields. The first nine general purpose data fields and the first four flag fields are currently used. The remaining data fields are for future use. These fields and flags allow customers to set up general processing parameters used by the MHI module when handling messages.

From this record, customers can configure several timeout options including the amount of time the MHI module waits for responses from the Authorization and Mail processes. Customers can also determine the maximum length of responses from the MHI module to the terminal and whether the MHI module is to append user data to the POS Standard Internal Message (PSTM) with every financial transaction. In addition, customers can determine if totals returned to the terminal in responses are draft capture totals or if the totals include all totals.

Several BASE24-mail options can also be set using this record. Customers can determine whether a terminal is allowed to perform implicit closes and the data processing centers to which mail should be sent.

The MHI module reads this data for a terminal into memory upon receiving the first message from the terminal. When the MHI module receives a warmboot (9501****) message, the data in memory is deleted and this record is reread when the MHI module receives the first message from the terminal.

Field Map Data

Two field map data records exist for each terminal group included in the MCNF. The first record defines the fields to be included in requests for each transaction type and the fields within these request messages to be used in the verification of the MAC. The second record defines the fields to be included in responses for each transaction type and the fields within these response messages to be used in the generation of the MAC.

Although changes to this data can be made online through BASE24 files maintenance, the customer and vendor must coordinate this data and not rely on the file to determine the fields to be included in messages or the fields to be included in the MAC. The field maps do not determine the data sent in requests. Terminals do not receive the field map data. It is the responsibility of the terminal software to determine the data in requests. The purpose of the field map data in the MCNF is to allow the MHI module to know what to expect from the terminal and how to format the responses to the terminal.

The MHI module reads this data for a terminal into memory upon receiving the first message from the terminal. When the MHI module receives a warmboot (9501****) message, the data in memory is deleted and this record is reread when the MHI module receives the first message from the terminal.

Although the fields within the requests and responses can be determined by the customer, some are required for certain transactions. Information concerning the fields required for specific transactions is included in section 4. Using MACs is described later in this section.

Download Data

There are 12 download records for each terminal group included in the MCNF.

Records 00, 01, and 02 contain a total of 26 40-byte fields that can contain any data the customer wants to download to the terminal or any information the terminal requires.

Records 03, 04, 05, 07, 08, 09, 10, and 11 contain card prefix range processing parameters. Examples of the processing parameters include whether cards within a given card prefix range use draft capture processing, whether cards within a given card prefix range require PINs, and the authorization telephone numbers for each card prefix range. Up to 30 card prefix ranges can be defined in this set of records.

Record 06 contains flags that indicate the fields from the POS Terminal Data File (PTDF) that can be downloaded to the terminal.

Downloads can be initiated by terminals, using a network control facility, or by the Device Control Terminal (DCT). Additional information about downloads is included in section 5. For more information on network control facilities, refer to the *NET24-XPNET Network Control Operations Guide*.

Merchant Terminal Response File

The Merchant Terminal Response File (MRSP) contains a table of terminal response displays, in up to three different languages. The MHI module uses this file to develop a terminal response display that describes the results of transaction processing to the terminal. Responses can contain a maximum of 48 bytes.

The MHI module reads this data for a terminal into memory upon receiving the first message from the terminal. When the MHI module receives a WARMBOOT message, the data in memory is deleted and this record is re-read when the MHI module receives the next message from the terminal.

The MRSP is a key-sequenced file, with the primary key consisting of two fields: the TERMINAL GROUP and RECORD NUMBER. The value in the TERMINAL GROUP field should correspond to a value in the TERM GROUP field in the POS Terminal Data File (PTDF). If no terminal group is specified in the PTDF, a terminal group default value of all asterisks (****) can be used. The MHI module uses the TERM GROUP field in the PTDF to determine the MRSP records associated with the terminal. Therefore, MRSP records are applicable to an entire group of terminals. The number of terminals within one group is determined by the customer when PTDF records are set up.

The MRSP contains a maximum of five records for each terminal group. These records consist of a response code map record, three display text records, and a transaction description table record. The response code map record allows for up to 1000 response codes to be mapped to descriptions defined in the display text records. Since there are three display text records, responses for the terminal can be in one of three languages. The language to be used is determined in the request from the terminal. Language parameters are set up in the LANGUAGE ID field in the PTDF. The transaction description table record allows customers to determine the wording to be used to describe each transaction in the transaction set.

This file is accessed through BASE24 files maintenance. Instructions on setting up and accessing the MRSP are found in section 8. The Data Definition Language (DDL) uses the acronym MRESP for the Merchant Terminal Response File.

Note: The PTDF TERM GROUP field cannot be left blank. If customers want to use the default MRSP, they need to ensure that no MRSP records exist with the specified TERM GROUP.

Merchant Host Configuration File

The Merchant Host Configuration File (MHCF) contains one record for each data processing center (DPC) that communicates with an MHI module. Each record contains processing information specific to a certain DPC including processing flags and station tables. The processing flags configured in the MHCF are used to determine the processing parameters supported by the MHI module such as the DPC type, MAC type, and whether the MHI module is running in terminal or host control mode.

The MHCF is a key-sequenced file with the primary key consisting of two fields: the DPC number and the MHI module name. The MHCF records are used to format a DPC table at MHI module initialization and when a warmboot (9501****) command is performed. When a transaction is received from the merchant host, the DPC number in the message is used to access the appropriate DPC table entry.

This file is accessed through BASE24 files maintenance. Instructions on setting up and accessing the MHCF are discussed in section 8.

Merchant Regulation E File

The Merchant Regulation E File (MREG) is a key-sequenced file with the terminal ID as the primary key. The MREG contains Regulation E data for the specified terminal. Data in the MREG is used only when the MHI module is running in host control mode and when parameters in the Merchant Host Configuration File (MHCF) are configured to allow the MHI module to use the MREG data. The Regulation E data in the MREG overrides the corresponding data in the POS Terminal Data File (PTDF).

This file is accessed through BASE24 files maintenance. Instructions for setting up and accessing the MREG are discussed in section 8.

Merchant Totals Configuration File

The Merchant Totals Configuration File (MTCF) contains records defining totaling groups for the purpose of obtaining group totals on a retailer or group totals request transaction. One record exists for each terminal within a totaling group. This allows an unlimited number of terminals to be included in one totaling group.

The MTCF is a relative file with an alternate key consisting of three fields: the RETAILER ID, the TERMINAL IDS, and the TOTALLING GROUP. It is used by the Merchant Totals process to determine which terminal ID records to include in the group totals in response to a retailer or group totals request transaction. This data is read into extended memory.

This file is accessed through BASE24 files maintenance. Instructions for setting up and accessing the MTCF are discussed in section 8.

POS Terminal Data File

The POS Terminal Data File (PTDF) contains one record for each POS terminal in the network, including terminals associated with the MHI module. When the MHI module is running in host control mode, the PTDF contains one terminal record per DPC. Additional fields used by terminals are contained in the PTDF device-dependent area.

When the MHI module is running in terminal control module, it uses the TERMINAL ID field (contained with the standard header of all request messages) to access the appropriate record in the PTDF. When the MHI module is running in host control mode, it uses the PTDF TERMINAL ID field on MHCF screen 1 to access the appropriate PTDF record. The USE MESSAGE TERM ID field on MHCF screen 1 is used when the MHI module is running in host control mode to determine whether Regulation E data is taken from the message header or from the PTDF. Regulation E data taken from the message header is set up in the MHI Regulation E File (MREG).

For more information on the PTDF, refer to the ***BASE24-pos Files Maintenance Manual***. For more information on the MREG and MHCF, refer to section 8.

POS Transaction Log File

The POS Transaction Log File (PTLF) contains information about the transactions processed by BASE24-pos. Both approved and declined financial transactions are logged to the PTLF. When a close batch request, a close shift request, or a close day request is received from the merchant host, the totals are logged to the PTLF only if they are configured to do so on PTDF screen 3.

For more information on the PTLF, refer to the ***BASE24-pos Files Maintenance Manual***.

POS Retailer Definition File

The POS Retailer Definition File (PRDF) contains one record for each retailer in the network. The MHI module uses the PRDF to retrieve the name of the merchant when this information is to be downloaded to the terminal. The customer determines if this information is to be downloaded to the terminal by setting a flag in the MCNF. For more information on the PRDF, refer to the ***BASE24-pos Files Maintenance Manual***.

Formatting PSTM.USER-DATA

Information from a financial request message is stored in the user data field in the PSTM if the FORMAT PSTM field in the MCNF is set to Y (Yes). The PSTM.USER-DATA field is written to the PTLF and can be extracted for host usage. This information is always formatted if the MHI module is running in host control mode.

The PSTM.USER-DATA field is formatted from the following request fields.

FID	Field	PIC
K	Business Date	9(06)
N	Customer ID	X(24)
O	Customer ID Type	X(02)
Q	Echo Data	X(16)
U	Language Code	X(01)
a	Optional Data	X(80)
d	Retailer ID	X(12)
i	Original Sequence Number	X(09)
Header Field	Terminal ID	X(16)

Logical Network Configuration File

One Logical Network Configuration File (LCONF) exists for each logical network. It contains the assigns and params associated with the network.

Assigns associate the physical names of objects in the logical network to names used in BASE24 code. Assign information must be present; otherwise BASE24 is unable to process. Most assigns link the actual physical file names to logical file names as they are referenced in BASE24 program code. For example, in order to access the Merchant Terminal Configuration File (MCNF), a process or server must be able to identify the fully qualified name of the file as it exists in a logical network. The LCONF must contain an assign linking the MCNF name to the actual physical file. Other assigns link the physical names for different hardware to the names as they are referenced in the BASE24 program code. For example, the types of hardware referenced in BASE24 program code includes spooler locations, terminal locations, and security devices.

Params set processing values at the logical network level. In some instances when param information is not present, a process abends during initialization; however, some params are not critical for processing.

For more information about these assigns and params, refer to the ***BASE24 Logical Network Configuration File Manual***.

Assigns

The MHI module requires several assigns in the LCONF in order to perform processing. These assigns are as follows and are in addition to any assigns required by the Router/Authorization module bound to the MHI module.

POS-MHI-MCNFG — Contains the name of the Merchant Terminal Configuration File (MCNF).

POS-MHI-MRESP — Contains the name of the Merchant Terminal Response File (MRSP).

POS-MHI-MHCF — Contains the name of the Merchant Host Configuration File (MHCF).

POS-MHI-MREG — Contains the name of the MHI Regulation E File (MREG).

POS-MHI-MTCF — Contains the name of the Merchant Totals Configuration File (MTCF). The MTCF is used exclusively by the Merchant Totals process.

POS-PTDF — Contains the name of the POS Terminal Data File (PTDF).

Params

Several params are used by the MHI module during processing. These params are as follows and are in addition to any params required by the Router/Authorization module bound to the MHI module.

POS-DH-CONS-MAC-ERR-LMT — The number of consecutive security MAC errors that can occur before the MHI module takes corrective actions. After the number of consecutive security MAC errors in this param have occurred, the MHI module generates a new MAC key and sends it in the next message to the device. The default for this param is 3.

The POS-DH-CONS-MAC-ERR-LMT param can contain any numeric value from zero through 32000. A value of zero indicates that the MHI module does not take any corrective actions regardless of the number of consecutive security MAC errors.

POS-DH-LMT-EXCEED-DISP — Specifies the action the MHI module takes when the return or adjustment limits are exceeded. Valid values are as follows:

- 0 = Continue processing
- 1 = Deny the transaction
- 2 = Refer the transaction

The default is 0.

POS-DH-MAX-CNFGS — Contains the maximum number of configuration records to be stored in memory at one time by an MHI module. The default value for this param is 10. To increase performance, this param should be set to at least three times the number of terminal groups to be processed by one MHI module (for example, allow one processing record and two format records for each terminal group).

POS-DH-MAX-IN-CORE — Contains the maximum number of PTDF records that can be held in memory at one time. The value in this param should be an estimate of the number of devices that will have transactions in progress at one time. The default for this param is 25.

Since the value in this parameter represents the largest number of transactions that can be processed at one time, the value cannot be greater than that of the POS-DH-MAX-TDFS param.

POS-DH-MAX-TDFS — Contains the maximum number of PTDF records to be handled by one MHI module. The default value for this param is 25. For example, if 100 terminals are routed to an MHI module, this param should be 100. On the other hand, if service-based routing is used and 1000 terminals can be processed, this param should be 1000.

When the MHI module is initialized or warmbooted, space is allocated in memory to build a table. This table stores the record positions of all the PTDF records that the MHI module can access. The POS-DH-MAX-TDFS param determines the table size and, therefore, the maximum number of terminals that each MHI module can service.

POS-MHI-MAX-REQUESTS — The maximum number of POS Standard Internal Message (PSTM) records the MHI module can store in memory at one time. The value of this param should be an estimate of the maximum number of host control transactions in progress at one time. This param can be set to any numeric value greater than zero and is used only when the MHI module is running in host control mode. The default for this param is 50.

In host control mode, PSTM records are stored in a table in memory while transactions are in progress. A PSTM record is retained in memory each time a transaction is received from a device until an authorizing response is received.

Note: If the MHI module is operating in terminal control mode, the PSTM table is not required. Therefore, this param should be set to its lowest valid value (which is 1) to minimize the use of extended memory.

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Section 3

BASE24-pos Merchant Host Interface Module

The Merchant Host Interface (MHI) module is used in place of the Device Handler module as far as the BASE24-pos Device Handler/Router/Authorization process is concerned. In BASE24-pos, the Device Handler module is bound together with a Router/Authorization module to create the Device Handler/Router/Authorization process. This process is composed of three modules—Device Handler, Router, and Authorization. Each module has distinct functions, but they operate together and are defined as one bound process. Because the three modules are bound, transactions passed between them do not travel through the XPNET process.

The Router and Authorization modules are the same for each Device Handler/Router/Authorization process. However, the Device Handler module differs between Device Handler/Router/Authorization processes depending on the type of POS device attached to the system. For example, a different Device Handler module is required for VISA II devices than is required for MHI devices.

This section is devoted to describing the processing associated with the MHI module. For more information on the Router and Authorization modules, refer to the ***BASE24-pos Router/Authorization Manual***.

This section describes the processing performed by the MHI module. It includes the following information about the MHI module:

- An overview of the MHI module
- A description of each file used by the MHI module during processing
- A description of the steps the MHI module performs during initialization and warmbooting
- A list of network control facility text commands recognized by the MHI module
- A description of log message generation and documentation

MHI Module Overview

The BASE24-pos Merchant Host Interface (MHI) module is an acquirer of transactions originating from terminals controlled by another host or controller. It is designed to accept transactions to be authorized and to optionally accumulate totals by the terminal.

The MHI module can operate in two different modes concerning the control of the devices on the host: terminal control or host control. The institution defines, by data processing center (DPC), whether host control mode is used by setting the UPDATE PTDF field on Merchant Host Configuration File (MHCF) screen 1 to 0. To use terminal control mode, set the UPDATE PTDF field on MHCF screen 1 to 1.

The MHI module operates in host control mode when the institution wants BASE24 to authorize the transaction. Restrictions to operating in host control mode include no terminal totals and limited duplicate checking. It is assumed in host control mode that the host or controller will provide some of this support. When the MHI module is operating in host control mode, only one POS Terminal Data File (PTDF) record will exist for each DPC.

The MHI module operates in terminal control mode when the institution wants the MHI module to provide all of its device functionality. If the MHI module is operating in terminal control mode, the MHI module provides support for several options including terminal totals and duplicate checking. When the MHI module is operating in terminal control mode, one PTDF record exists for each terminal attached to the host or controller.

Additional functions that the MHI module provides are listed in the following paragraphs.

Enforcing Transaction Sequence

The MHI module can enforce the sequence in which transactions are processed in the system. When the terminal is unable to assign a transaction sequence number, the MHI module assigns one when it receives the transaction.

Updating Accumulator Fields and Status Flags

The MHI module updates the accumulator fields and status flags in the PTDF record for each transaction that is processed. For example, when a transaction passes through the Device Handler/Router/Authorization process, the MHI module updates the accumulation fields BATCH NUMBER, SHIFT NUMBER, NUMBER OF BATCHES, and NUMBER OF SHIFTS in the PTDF by one and updates other fields and flags depending on the type of transaction and the processing required.

Cutting Over Terminals

The MHI module cuts over terminals for daily settlement using a cutover transaction. The institution designates a timeframe in which cutover must take place and designates a transaction to start cutover. When the Device Handler/Router/Authorization process receives that transaction during the designated timeframe, the terminal is cut over. If the MHI module does not perform this cutover, the Settlement Initiator performs a force cutover at the end of the designated window.

Decrypting PINs

The MHI module can decrypt PINs for incoming transactions using software decryption. However, if decryption is done using a hardware security module, the MHI module passes the transaction to the Authorization module. The Authorization module then passes the transaction to the hardware security module. Refer to the ***BASE24 Transaction Security Manual*** for more information on PIN decryption.

Downloading

The MHI module performs both full and partial downloads to the terminals consisting of terminal configuration information. The information is downloaded from load files (primarily from the MCNF).

Controlling Response Timers

The MHI module sets and stops transaction response timers as transactions pass in and out of the MHI module. When the MHI module receives a transaction request from a device, it sets a response timer and then passes the transaction on for processing. When processing completes, the MHI module deletes the timer. If processing does not complete before the timer expires, the transaction is either declined or reversed, depending on what type of transaction timed out.

Validating Return and Adjustment Limits

The MHI module checks terminal return and adjustment limits in the POS Terminal Data File (PTDF) as part of its request validation function. If the limits have been exceeded, the MHI module continues processing the transaction, denies the transaction, or refers the transaction based on the value of the LCONF parameter POS-DH-LMT-EXCEED-DISP.

Refer to the ***BASE24 Logical Network Configuration File Manual*** for more information about the POS-DH-LMT-EXCEED-DISP param. For more information about return and adjustment limit fields in the PTDF, refer to the ***BASE24-pos Files Maintenance Manual***.

Checking Limits

The MHI module checks return and adjustment limits on the following transactions:

- Merchandise return
- Adjustment to merchandise return
- Adjustment to purchase
- Adjustment to cash advance

An adjustment to a merchandise return goes through two sets of checks. After the MHI module performs return limit checks, it then performs adjustment limit checks.

One check is to see if the transaction is under the return or adjustment count limit. The following table describes the return or adjustment count check:

Transaction	Over Count Limit If...
Return	$\text{PTDF.REC.BATCH.CR-CNT} + 1 > \text{PTDF.REC.RETURN-LMT-CNT}$
Adjustment	$\text{PTDF.REC.BATCH.ADJ-CNT} + 1 > \text{PTDF.REC.ADJ-LMT-CNT}$

If the count limit has been exceeded, the MHI module takes the action specified by the POS-DH-LMT-EXCEED-DISP param. This action is described below in the topic “Processing Transactions Exceeding Limits.” If the count limit has not been exceeded, the MHI module checks return or adjustment amount limits.

The following table describes the return or adjustment amount checks:

Transaction	Over Amount Limit If...
Merchandise return	$\text{PTDF.REC.BATCH.CR} + \text{PSTM.TRAN.AMT-1} > \text{PTDF.REC.RETURN-LMT-AMT}$
Adjustment to merchandise return (return amount limit)	$\text{PTDF.REC.BATCH.CR} + \text{PSTM.TRAN.AMT-2} > \text{PTDF.REC.RETURN-LMT-AMT}$
Adjustment to purchase or adjustment to cash advance	$\text{PTDF.REC.BATCH.ADJ} + \text{PSTM.TRAN.AMT-1} > \text{PTDF.REC.ADJ-LMT-AMT}$
Adjustment to merchandise return (adjustments amount limit)	$\text{PTDF.REC.BATCH.ADJ-PSTM.TRAN.AMT-1} > \text{PTDF.REC.ADJ-LMT-AMT}$

If the transaction does not exceed amount limits, the MHI module continues processing as usual. If the transaction exceeds the amount limit, the MHI module takes the action specified by the POS-DH-LMT-EXCEED-DISP param, as discussed below.

Processing Transactions Exceeding Limits

If the MHI module determines that a return or adjustment transaction has exceeded a count or amount limit, it performs the following steps:

1. Logs a message.
2. Sets the error flag in the BASE24-pos Release 5.0 token to E (return limit exceeded) or A (adjustment limit exceeded).
3. Takes the action specified in the POS-DH-LMT-EXCEED-DISP param, as follows:
 - If the POS-DH-LMT-EXCEED-DISP param is set to 0 (continue processing) or if this is a force-post (message type P) or store-and-forward transaction (message subtype S), the MHI module continues processing the transaction. Limits processing is now complete.
 - If the POS-DH-LMT-EXCEED-DISP param is set to 1 (deny transaction), the MHI module denies the transaction with a response code 094 and continues with step 4.
 - If the POS-DH-LMT-EXCEED-DISP param is set to 2 (refer transaction), the MHI module denies the transaction with a response code 102 and continues with step 4.
4. Sends the transaction to the Router/Authorization module to be logged to the PTLF. This is a log only transaction (message type 9900).
5. Responds to the terminal.

File Usage

The MHI module uses the following files during processing. Each file listed is followed by a brief description explaining how it is used by the MHI module.

Logical Network Configuration File (LCONF) — The assigns and params used during processing are read from the LCONF at initialization and reinitialization, and are maintained in memory.

Log Message Configuration File (LMCF) — The LMCF contains log message information such as text, routing code, and severity, that has been modified by the institution. At initialization, the Device Handler/Router/Authorization process builds a table in memory from the LMCF records to store the modifications. When a log message is generated, the Device Handler/Router/Authorization process accesses the table to verify whether that log message has been modified. If the log message has been modified, the Device Handler/Router/Authorization process logs the modified information instead of the standard log message. Refer to the *XPNET BASE24 Configurations Guidelines* for information on modifying the LMCF.

Merchant Terminal Configuration File (MCNF) — The MCNF is used exclusively by the MHI module. This file is central to the configuration of the MHI module. It specifies processing parameters, the fields to be included in requests and responses for each transaction type, and download data.

Merchant Host Configuration File (MHCF) — The MHCF is used exclusively by the MHI module. The MHCF contains one record for each DPC that communicates with the MHI module. The record contains processing information specific to that DPC (for example, processing flags, timers, and station tables).

Merchant Regulation E File (MREG) — The MREG is used exclusively by the MHI module. This file contains Regulation E data for the specified terminal. Data in the MREG must be in compliance with the Regulation E requirements. If the Regulation E data in the MREG is used, it overrides the corresponding data in the POS Terminal Data File (PTDF). The MREG is used only when the MHI module is operating in host control mode.

Merchant Terminal Response File (MRSP) — The MRSP is used exclusively by the MHI module. This file contains a table of terminal response displays in up to three different languages. The MHI module uses the MRSP to

develop a terminal response display that describes the results of transaction processing to the terminal. The MHI module reads this file into memory at initialization and warmbooting.

Merchant Totals Configuration File (MTCF) — The MTCF is used exclusively by the Merchant Totals process. The MTCF contains records defining totaling groups for the purpose of obtaining group totals on a retailer/group subtotals request. One record exists for each terminal within a totaling group. This allows for an unlimited number of terminals to be in one totaling group.

POS Retailer Definition File (PRDF) — The MHI module accesses the PRDF when processing transactions to retrieve various information about the retailer associated with the transaction. This file is also used during terminal cutover. For more information on the PRDF, refer to the ***BASE24-pos Files Maintenance Manual***.

POS Terminal Data File (PTDF) — The MHI module builds a table of PTDF records in order to access and update PTDF records as transactions are processed. For more information on the PTDF, refer to the ***BASE24-pos Files Maintenance Manual***.

POS Transaction Log File (PTLF) — The POS Transaction Log File (PTLF) contains information about the transactions processed by BASE24-pos. Both approved and declined transactions are logged to the PTLF. When a close batch request, a close shift request, or a close day request transaction is performed, totals are logged to the PTLF (provided the totals are configured in the PTDF to be logged). For more information on the PTLF, refer to the ***BASE24-pos Files Maintenance Manual***.

MHI Module Initialization and Warmboot

This section describes the steps taken by the MHI module when it initializes or is warmbooted. Understanding these steps requires knowledge of the background information provided in the following paragraphs.

In BASE24-pos, Device Handler modules are bound together with Router/Authorization modules. Each of these modules has distinct functions, but they operate together and are defined as one bound process. Because the three modules are bound, transactions passed between them do not travel through the XPNET process. In addition, the Router and Authorization modules are the same for each Device Handler/Router/Authorization process. However, the Device Handler module differs between Device Handler/Router/Authorization processes depending on the type of POS device attached to the system. For example, the MHI module can be used with several types of devices, whereas the Hypercom Device Handler modules are used with Hypercom devices.

When the Device Handler/Router/Authorization process is initialized, the Router module begins initialization first. Part of the initialization process required by the Device Handler module is within the Router module. The steps described in this section assume that the Router module has already initialized. For a complete description of the steps taken by the Router and Authorization modules when initializing, refer to the ***BASE24-pos Router/Authorization Manual***.

The MHI module initializes by performing the following steps:

1. Initializes the Tandem Event Management Service (EMS) with the name of the specific MHI module being initialized and reads the Log Message Configuration File (LMCF).
2. Sets up a pointer used to retrieve MHI device-dependent data from the POS Terminal Data File (PTDF).
3. Retrieves the following assigns and params from the Logical Network Configuration File (LCONF):

Assigns

POS-MHI-MCNFG
POS-MHI-MRESP
POS-MHI-MHCF
POS-MHI-MREG
POS-PTDF

Params

POS-DH-CONS-MAC-ERR-LMT

POS-DH-LMT-EXCEED-DISP

POS-DH-MAX-TDFS

POS-DH-MAX-IN-CORE

POS-DH-MAX-CNFGS

POS-MHI-MAX-REQUESTS

4. Opens the PTDF and sets the file mode. The file mode is set so any process attempting to lock or read a PTDF record while the MHI module has the record locked will receive an error 73. This error indicates the file or record is locked. If the MHI module is unable to open the PTDF, the MHI module abends.
5. Opens the MHCF, MCNF, MRSP, and MREG. If the MHI module is unable to open any of these files, the MHI module abends.
6. Allocates extended memory space for the number of PTDF records it needs at one time. The number of records that can be in memory at one time is specified in the POS-DH-MAX-TDFS param. Separate extended memory space is allocated for the host control PTDF memory block and the terminal control PTDF memory block.
7. Allocates extended memory for the MCNF. If the MHI module is unable to allocate memory for the MCNF, the MHI module abends.
8. Allocates extended memory for the MREG if the file exists. If the MHI module is unable to allocate memory for the MREG, the MHI module abends.
9. Allocates extended memory for the MRSP. If the MHI module is unable to allocate memory for the MRSP, the MHI module abends.
10. Allocates extended memory for the DPC table.
11. Reads the MHCF, initializes the DPC table from the MHCF data, and stores the DPC table in extended memory. If the MHI module is unable to allocate memory for the DPC table, the MHI module abends.
12. Starts performance timers for all DPCs that are marked as up. Each DPC and station is initially marked as down. The DPC is marked up when the first request or DPC logon is received. The stations are marked up as requests are received for them.

13. Allocates extended memory for the PSTM suspended requests table. This step is taken only if this is the first time the MHI module has been initialized.

When the MHI is operating in terminal control mode, this step is complete, and processing continues with step 14. When the MHI module is operating in host control mode, fields from the transaction request and the PTDF are sent as part of the PSTM. After the PSTM has been filled with the proper data from the request, the MHI module suspends an image of the PSTM in extended memory. A pointer to this extended memory is entered into an extended memory table, indexed by the unique host trace number and terminal ID included in every PSTM request. PSTM information is used in processing each transaction.

14. Initializes the retailer and group totals response buffer. This buffer is used to store retailer or group totals.
15. Moves the default MHI device data records from the MRSP into extended memory. These records include response codes, language displays, and transaction descriptions.

Warmbooting the MHI Module

When the MHI module receives a WARMBOOT command, it performs the initialization steps described earlier with two exceptions.

- Step 1 in the initialization procedures indicates the MHI module initializes the logging process with the name of the specific MHI module being initialized. This step is not performed by the MHI module when it receives a WARMBOOT command.
- When the MHI module receives a WARMBOOT command, the MHI module deallocates the extended memory that was previously allocated for the PTDF, MCNF, MRSP, and MREG. The MHI module then closes these files and deallocates the extended memory for the DPC table. The MHI module then performs steps 2 through 14 described in the initialization procedures.

Note: The MHI module does not reallocate extended memory for the PSTM request suspense table. This step is only performed the first time the MHI module is initialized.

Text Commands

Network text commands are used to send information from a network control facility to a process. These commands can be entered by an operator using a network control facility and serve as a tool that operators can use to perform network management functions such as changing certain processing parameters, warmbooting a specific process, monitoring a specific process, and resolving problems encountered by a specific process. For detailed descriptions and the syntax of all text commands, refer to the ***BASE24 Text Command Reference Manual***.

The following network control facility text commands can be used with the MHI module:

DEACTIVATE FLAG — Sets the status of the terminal to activated or deactivated. If the terminal is deactivated, the MHI module declines transactions originated by that terminal. The default (initialized value) is deactivated. If the deactivate flag in the command is 1, the terminal is deactivated unless there are batch, shift, day, or network totals. If any of these totals are greater than or less than zero, the terminal is set to pending deactivation. If the deactivate flag value in the command is 0, the terminal is activated.

LOADFLAGON — Indicates that download data is waiting for the terminal. Upon receipt of the LOADFLAGON command, the MHI module sets the REQUEST-LOAD flag in the device dependent area of the POS Terminal Data File (PTDF) to on.

LOADFLAGOFF — Indicates that download data is not waiting for the terminal. The LOADFLAGOFF command sets the REQUEST-LOAD field in the POS Terminal Data File (PTDF) to off.

PERFORMANCE — Generates a report containing all the request processing activity for a specific DPC or for all DPCs.

READ PTDF — Rereads a particular PTDF record held in memory. This command is only appropriate when the MHI module is operating in host control mode.

RESETSTATUS — Resets the transaction status in the TRAN-STATUS field of the PTDF to NO-TRAN-IN-PROGRESS. This command is intended for use if a transaction abends or some other uncontrollable event leaves the transaction status corrupted in the PTDF.

STATUS — Generates a report containing the status of a DPC or all DPCs associated the specified MHI module. The status command sends miscellaneous information on its current processing parameters to a log.

TRACE ON — Activates the trace function within the MHI module.

TRACE OFF — Deactivates the trace function within the MHI module.

TRACE1 — Activates message trace only.

TRACE2 — Activates transaction information trace only.

TRACE3 — Activates key trace only.

TRACE4 *number* — Activates response time trace only. The *number* value is the number of transactions that are accumulated before the average response time is displayed. This value is entered by the network operator.

9501**** — Directs the MHI module to warmboot.

For a list of the steps the MHI module follows when a 9501**** command is issued, refer to the topic “Warmbooting the MHI Module” earlier in this section.

Log Message Generation

The BASE24-pos Merchant Host Interface (MHI) module and the Merchant Totals process generate log messages to assist network operators in performing their daily tasks. Log messages are unsolicited messages generated by BASE24 processes that provide information regarding the internal condition of the system. They are a network operator's primary link to what is actually occurring in the system. Log messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

ACI has developed a method that allows customers to access log message documentation electronically. Log messages are contained in files on a designated Tandem volume and subvolume. Each file contains log messages for a specific BASE24 process. Files on the designated Tandem volume and subvolume are organized according to product module ID (PMID). One PMID exists for every process in BASE24. PMIDs identify a specific process in the BASE24 system (for example, PSMHI identifies the MHI module). Providing log message documentation in files on the Tandem gives customers greater flexibility by allowing them to print out log message documentation as needed at their location or to electronically manipulate the data to suit their needs.

Documentation is available in file form for the log messages generated by the MHI module and Merchant Totals process. They are located on the Tandem subvolume in files named PSMHI for the MHI module and PSMTTL for the Merchant Totals process. For a complete explanation of log message documentation, refer to the ***BASE24 Log Message Reference Manual***.

Section 4

The BASE24 Standard Message

This section contains the following information concerning the BASE24 Standard Message:

- Standard message header
- Optional data fields
- Request message requirements
- Response message requirements
- Determining transaction codes

Every message sent between the host and the Merchant Host Interface (MHI) module contains a standard message header. The header contains information on the transaction type and the terminal that initiated the transaction.

Although the standard header can stand alone, any number of optional data fields may follow it. The only restrictions on the number of optional data fields that can be included in the message are the protocol limitations, the size of the terminal read buffer, and the size of the MHI read buffer. The MHI read buffer is configurable to a maximum of 4,088 bytes, including all data communication control characters.

The optional data fields to be included in requests from the terminal and responses from the MHI module for each transaction type are specified in the Merchant Terminal Configuration File (MCNF). The MCNF is explained in more detail in sections 5 and 8.

Each optional data field included in the message is identified with a field identifier (FID) preceding the optional data field. In addition, each FID is prefaced with a Field Separator character. The field separator character, prefacing the next FID, indicates the end of the data in the optional field preceding it.

Standard Message Header

The standard header must be in every message sent from the host to the MHI module, and will also be in every response sent from the MHI module to the terminal. It consists of 6 fields totaling 30 bytes and is not configurable. The standard message header identifies information such as the type of message being sent, the ID of the terminal where the transaction was entered, and the transaction code associated with the transaction.

Standard Message Header Structure

The following table describes the structure of the standard message header. This table includes the positions in the standard message header occupied by each field, the length and format of each field, and the name of the field. Each field is described in further detail immediately following the table.

Position	Field Length	Field Name
1–4	4 numeric characters	DPC number
5–20	16 alphanumeric characters	Terminal ID
21	1 alphanumeric character	Message type
22	1 alphanumeric character	Message subtype
23–24	2 numeric characters	Transaction code
25–30	6 numeric characters	Transmission number

Standard Message Header Field Descriptions

Descriptions of the standard message header fields are included on the following pages. Possible values for each of these fields are listed throughout the descriptions. A value of a blank space is indicated with the symbol ∕. Fields not used in requests should be blank filled.

DPC NUMBER — A number used by the MHI module as a key to the MHCF. The value in this field is set by the originating merchant host. Valid values range from 0001 to 9999.

TERMINAL ID — A value that uniquely identifies the terminal. It is used by the MHI module as the key to the appropriate record in the POS Terminal Data File (PTDF) when operating in terminal control mode.

This field is left-justified and blank filled.

MESSAGE TYPE — A code that identifies the type of transaction being processed. This field is used together with the TRANSACTION CODE and MESSAGE SUBTYPE fields. A listing of the various combinations is included later in this section. Valid values are as follows:

A = Administrative transaction
F = Financial transaction
N = Network management transaction
P = Force post transaction
R = Reversal transaction
U = Customer-canceled reversal

MESSAGE SUBTYPE — A code that identifies the message as being either online or store-and-forward. Valid values are as follows:

O = Online
S = Store-and-forward

Note: All valid values are uppercase letters.

The online (O) value indicates the terminal and the merchant host are online to the MHI module.

The store-and-forward (S) value indicates the terminal or the merchant host was offline or otherwise not communicating with the MHI module when the transaction was initiated. The transaction was approved and held in the memory of the terminal or merchant host until communications resumed, at which time the transaction was sent to the MHI module. BASE24-pos must accept and post the transaction.

Type A messages cannot be subtype S. Type A messages must be subtype O.

TRANSACTION CODE — A code that identifies the transaction type associated with the message. Together with the values in the MESSAGE TYPE and MESSAGE SUBTYPE fields, this code identifies the type of transaction sent to the MHI module. A listing of the various combinations is included later in this section. Valid values are as follows:

- 00 = Normal purchase
- 01 = Preauthorization purchase
- 02 = Preauthorization purchase completion
- 03 = Mail or telephone order
- 04 = Merchandise return
- 05 = Cash advance
- 06 = Card verification
- 07 = Balance inquiry
- 08 = Purchase with cash back
- 09 = Check verification
- 10 = Check guarantee
- 11 = Purchase adjustment
- 12 = Merchandise return adjustment
- 13 = Cash advance adjustment
- 14 = Cash back adjustment
- 50 = Logon request
- 51 = Logoff request
- 60 = Close batch request
- 61 = Close shift request
- 62 = Close day request
- 64 = Employee subtotals request
- 65 = Batch subtotals request
- 66 = Shift subtotals request
- 67 = Day subtotals request
- 68 = Retailer/group subtotals request
- 70 = Read mail request
- 71 = Mail delivered request
- 75 = Send mail request
- 90 = Download request
- 95 = Handshake request
- 97 = DPC logon
- 98 = DPC logoff
- 99 = DPC echo test

TRANSMISSION NUMBER — A value used by the MHI module to detect and drop duplicate requests. If this field is equal to the transmission number of the last request, the MHI module is operating in terminal control mode, the transaction is a financial or force-post transaction, and the previous response was an approval, the MHI module drops the request and sends a response indicating that it received a duplicate request and that it was dropped. This field is right-justified and zero filled. Valid values are 000001 through 999999. If this field does not contain a valid value, the transaction is denied.

Standard Message Header Examples

This subsection provides examples of the standard message headers used for normal purchase, merchandise return, and mail or telephone order transactions. These examples are designed to show how the standard message header is formatted in different transaction scenarios. Each example contains three parts. The first part explains the transaction scenario being described. The second part shows a graphic illustrating how the standard message header is formatted in the request from the terminal or merchant host to the MHI module. The third part shows a graphic illustrating how the standard message header is formatted in a response from the MHI module to the terminal or merchant host. Each field and its position in the standard message header are labeled on all graphics.

In these examples, quotation marks (“ ”) are used to delimit the text and are not part of the header.

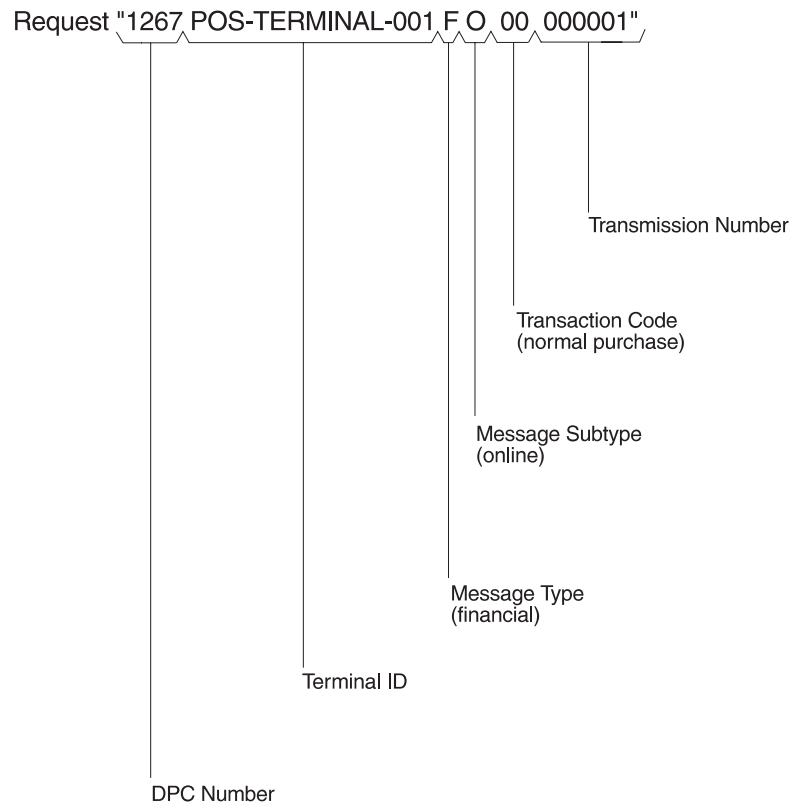
Normal Purchase Transaction

The examples shown below describe the standard message headers used in a typical request and response sequence for a normal purchase transaction. The standard message header field settings are also explained for each example.

Request

The following example is a request for a normal purchase transaction. The graphic illustrates the format of the standard message header when its fields are set as shown in the following table.

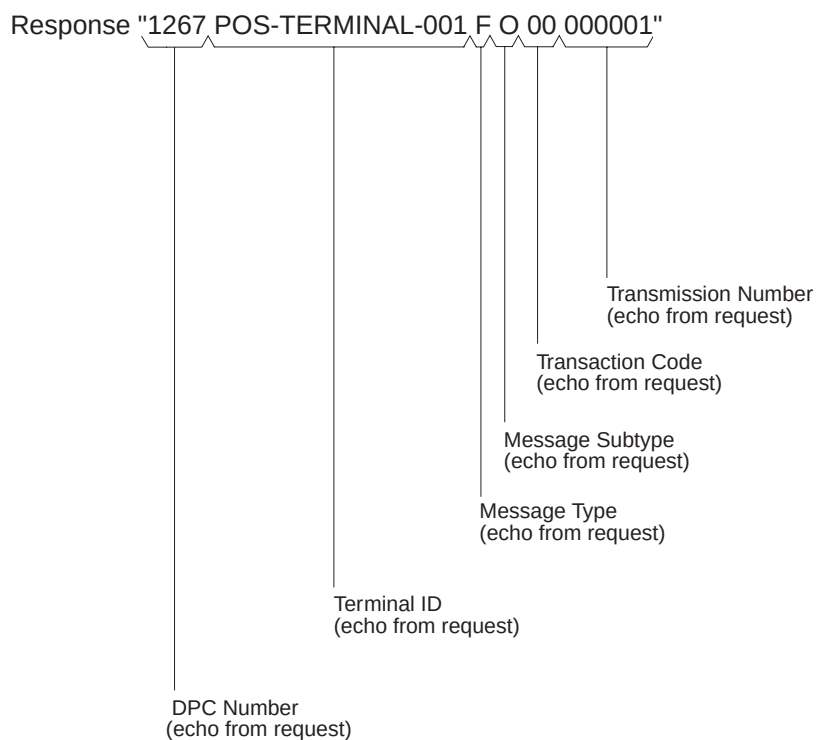
Field	Description
DPC NUMBER	1267
TERMINAL ID	POS-TERMINAL-001
MESSAGE TYPE	F (normal purchases are financial transactions)
MESSAGE SUBTYPE	O (online)
TRANSACTION CODE	00 (normal purchase)
TRANSMISSION NUMBER	000001



Response

The following example is a response to a normal purchase transaction request. The graphic illustrates the format of the standard message header when its fields are set as shown in the following table.

Field	Description
DPC NUMBER	1267 (echoed from request)
TERMINAL ID	POS-TERMINAL-001 (echoed from request)
MESSAGE TYPE	F (echoed from request)
MESSAGE SUBTYPE	O (echoed from request)
TRANSACTION CODE	00 (echoed from request)
TRANSMISSION NUMBER	000001 (echoed from request)



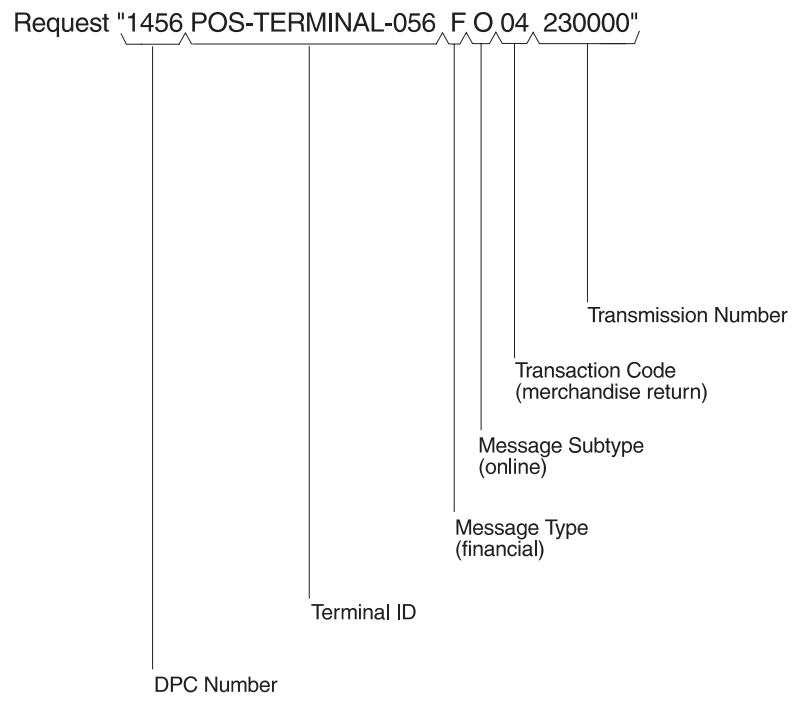
Merchandise Return Transaction

The following examples describe the standard message headers used in a typical request and response sequence for a merchandise return transaction. The standard message header field settings are also explained for each example.

Request

The following example is a request for a merchandise return transaction. The graphic illustrates the format of the standard message header when its fields are set as shown in the following table.

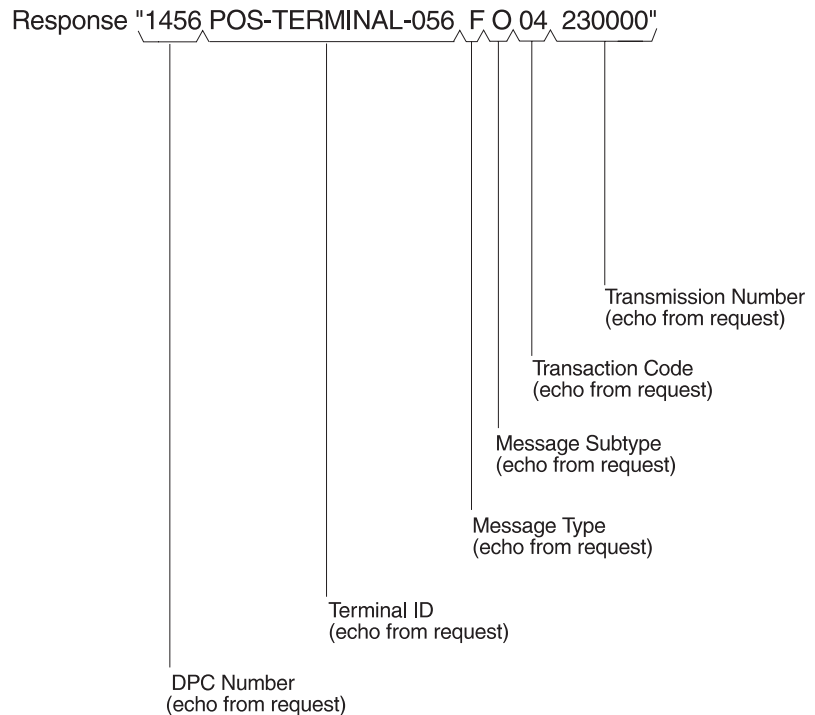
Field	Description
DPC NUMBER	1456
TERMINAL ID	POS-TERMINAL-056
MESSAGE TYPE	F (merchandise returns are financial transactions)
MESSAGE SUBTYPE	O (online)
TRANSACTION CODE	04 (merchandise return)
TRANSMISSION NUMBER	230000



Response

The following example is a response to a merchandise return transaction request. The graphic illustrates the format of the standard message header when its fields are set as shown in the following table.

Field	Description
DPC NUMBER	1456 (echoed from request)
TERMINAL ID	POS-TERMINAL-056 (echoed from request)
MESSAGE TYPE	F (echoed from request)
MESSAGE SUBTYPE	O (echoed from request)
TRANSACTION CODE	04 (echoed from request)
TRANSMISSION NUMBER	230000 (echoed from request)



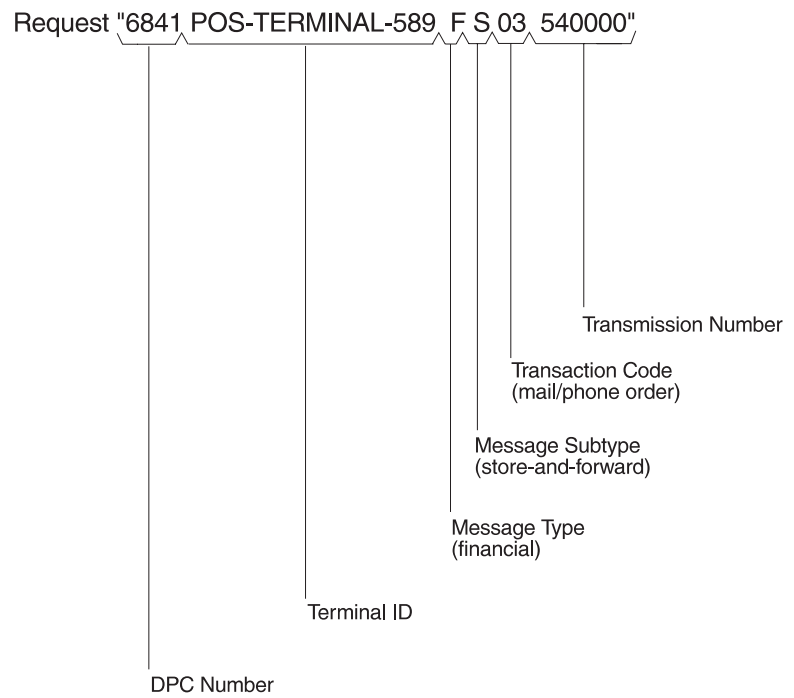
Mail or Telephone Order

The examples shown below describe the standard message headers used in a typical request and response sequence for a mail or telephone order transaction. The standard message header field settings are also explained for each example.

Request

The following example is a request for a mail or telephone order transaction. The graphic illustrates the format of the standard message header when its fields are set as shown in the following table.

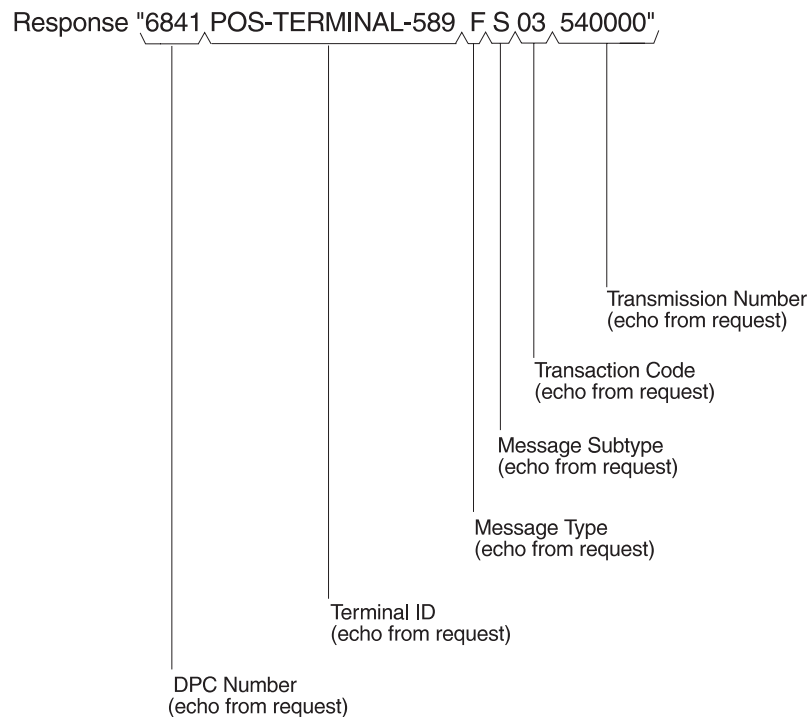
Field	Description
DPC NUMBER	6841
TERMINAL ID	POS-TERMINAL-589
MESSAGE TYPE	F (mail or telephone orders are financial transactions)
MESSAGE SUBTYPE	S (store-and-forward)
TRANSACTION CODE	03 (mail or telephone order)
TRANSMISSION NUMBER	540000



Response

The following example is a response to a mail or telephone order transaction request. The graphic illustrates the format of the standard message header when its fields are set as shown in the following table.

Field	Description
DPC NUMBER	6841 (echoed from request)
TERMINAL ID	POS-TERMINAL-589 (echoed from request)
MESSAGE TYPE	F (echoed from request)
MESSAGE SUBTYPE	S (echoed from request)
TRANSACTION CODE	03 (echoed from request)
TRANSMISSION NUMBER	540000 (echoed from request)



Optional Data Fields

After the standard message header, the remaining portion of the message consists of a series of optional data fields. Optional data fields can be included in requests from the merchant host and in responses from the MHI module for each transaction type. Data fields are identified in the system using field identifiers (FIDs). FIDs are specified by the customer in the Merchant Terminal Configuration File (MCNF).

Optional data fields 6 through 9 contain subfields that supply additional optional information. These subfields are identified by Subfield Identifiers (SFIDs). For more information about these subfields, see the topic “Optional Data Subfields” later in this section.

Optional Data Field Structures

The optional data fields are described below according to their FIDs. The table is organized in alphabetical order with capital letters listed before lowercase letters and numbers listed after letters. The table lists the FID, its associated field name (as specified in the MCNF), and the length of the field in the message. In addition, a Y (Yes, the FID is available), N (No, the FID is not available), or E (Echoed, the FID is taken from the request) in the RQST or RESP fields indicates if the optional data field is available for requests or responses, respectively.

Note: If the MHI module receives a request that contains an invalid FID, the request is denied with a response code of 800 (format error).

FID	Field Name	Picture	Length	RQST	RESP
A	Billing Address	PIC X(20)	1 to 20 bytes	Y	E
B	Amount 1	PIC 9(18)	1 to 18 bytes	Y	E
C	Amount 2	PIC 9(18)	1 to 18 bytes	Y	E
D	Application account type	PIC 9(01)	1 byte	Y	Y
E	Application account number	PIC 9(19)	1 to 19 bytes	N	Y
F	Approval code	PIC X(08)	1 to 8 bytes	Y	E
G	Authentication code	PIC X(08)	8 bytes	Y	Y

FID	Field Name	Picture	Length	RQST	RESP
H	Authentication key	PIC X(16)	16 bytes	N	Y
I	Clerk ID	PIC 9(06)	6 bytes	Y	N
J	Available balance	PIC 9(18)	18 bytes	N	Y
K	Business date	PIC 9(06)	6 bytes	Y	E
L	Check type/Category	PIC X(01)	1 byte	Y	E
M	Communications key	PIC X(16)	16 bytes	N	Y
N	Customer ID	PIC X(40)	1 to 40 bytes	Y	E
O	Customer ID type	PIC 9(02)	2 bytes	Y	E
P	Draft capture flag	PIC 9(01)	1 byte	Y	Y
Q	Echo data	PIC X(16)	1 to 16 bytes	Y	E
R	Card type	PIC X(01)	1 byte	Y	Y
S	Invoice number	PIC X(10)	1 to 10 bytes	Y	E
T	Invoice number/original	PIC X(10)	1 to 10 bytes	Y	E
U	Language code	PIC 9(01)	1 byte	Y	Y
V	Mail/Download key	PIC X(15)	1 to 15 bytes	Y	Y
W	Mail text/Download data	PIC X(449) PIC X(957)	1 to 957 bytes	Y	Y
X	Transaction date/time	PIC 9(12)	12 bytes	Y	Y
Y	Postal (ZIP) code	PIC X(09)	9 bytes	Y	E
Z	Address verification status code	PIC X(01)	1 byte	N	Y
a	Optional data	PIC X(250)	1 to 250 bytes	Y	E
b	PIN/Customer	PIC X(16)	1 to 16 bytes	Y	N
c	PIN/Supervisor	PIC X(16)	1 to 16 bytes	Y	N

FID	Field Name	Picture	Length	RQST	RESP
d	Retailer ID	PIC 9(12)	1 to 12 bytes	Y	E
e	POS condition code	PIC 9(02)	2 bytes	Y	E
f	PIN length—request	PIC 9(2)	1 to 2 bytes	Y	N
f	Receipt data—response (this FID is not supported in standard BASE24-pos product)	PIC X(200)	1 to 200 bytes	N	Y
g	Response display	PIC X(48)	1 to 48 bytes	N	Y
h	Sequence number	PIC X(10)	10 bytes	Y	E
i	Sequence number/original	PIC X(09)	9 bytes	Y	E
j	State code	PIC X(02)	2 bytes	Y	N
k	Drivers license/date of birth (for responses)	PIC 9(6)	6 bytes	Y	N
k	Terminal location (for requests)	PIC X(25)	0 to 25 bytes	N	Y
l	Totals/Batch	PIC X(72)	72 bytes	Y	Y
m	Totals/Day	PIC X(72)	72 bytes	Y	Y
n	Totals/Clerk	PIC X(116) PIC X(98)	116 bytes 98 bytes (if PTDF REL- NUM is not 50)	N	Y
o	Totals/Shift	PIC X(72)	72 bytes	Y	Y
p	Response code	PIC X(03)	3 bytes	N	Y
q	Track 2/Customer	PIC X(40)	1 to 40 bytes	Y	E
r	Track 2/Supervisor	PIC X(40)	1 to 40 bytes	Y	E
s	Transaction description	PIC X(24)	1 to 24 bytes	N	Y
t	Mail pending flag	PIC X(01)	1 byte	N	Y

FID	Field Name	Picture	Length	RQST	RESP
u	Download pending flag	PIC X(01)	1 byte	N	Y
v	Authorization timeout override	PIC 9(06)	6 bytes	Y	N
w	Clerk totals flag	PIC X(01)	1 byte	Y	N
x	Group totals ID	PIC X(10)	10 bytes	Y	N
y	Totals/Retailer or group	PIC X(74)	74 bytes	N	Y
z	Reversal text	PIC X(40)	40 bytes	Y	Y
0	American Express data collection	PIC X(118)	46 to 118 bytes	Y	E
1	PS2000 Data	PIC X(24)	24 bytes	Y	Y
2	Track 1/Customer	PIC X(82)	1 to 82 bytes	Y	Y
3	Track 1/Supervisor	PIC X(82)	1 to 82 bytes	Y	Y
6	Product SubFIDs	PIC X(1043)	1 to 1043 bytes	Y	Y
7	Product SubFIDs				
8	Channel/Distributor SubFIDs				
9	Customer SubFIDs				

Note: The fields required in request and response messages are documented in greater detail later in this section.

Optional Data Field Description Format

Descriptions of FIDs are included on the following pages in the format listed below. Explanations of the headings are included in the format.

FID	FIELD NAME (Associated BASE24-pos field)
-----	--

A brief description of the FID and its use follows the FID and FIELD NAME, when appropriate. The description includes the valid values associated with the FID, if applicable. If the field associated with the FID consists of a group of fields, each field is listed followed by its length.

Request: Indicates whether the field can be included in the request. Optional indicates the customer can configure the field to be included in requests. Not available indicates the field cannot be included in requests. An indication of whether the request field length is fixed or variable is included.

Response: Indicates whether the field can be included in the response. Optional indicates the customer can configure the field to be included in responses. Not available indicates the field cannot be included in responses. Fields that are echoed from the request are also noted. However, fields that are noted as being echoed in responses are not included in responses unless they are configured to be included. An indication of whether the response field length is fixed or variable is included.

The fields associated with the format shown above are described in the following paragraphs.

FID — The field identifier (FID).

Field Name — The name of the field associated with the FID and its data name in the POS Standard Internal Message (PSTM), if applicable.

Optional Data Field Descriptions

The following are the descriptions for the FIDs.

FID A	BILLING ADDRESS (PSTM.ADDR-FLDS.ADDR)
--------------	--

The Billing Address field contains the cardholder's billing address. This address is used for address verification.

Request: Optional. Variable length of 1 to 20 bytes.

Response: Optional. Fixed length of 20 bytes. If this field is included in the request message, it can be echoed in the response.

FID B	AMOUNT 1 (PSTM.TRAN.AMT-1)
--------------	-----------------------------------

The Amount 1 field is the primary amount field. For transactions that involve one amount, this field contains the amount. For transactions that involve more than one amount, this field contains the original or total amount.

Request: Required for the following transactions. Variable length of 1 to 18 bytes.

Normal purchase
Preauthorization purchase
Preauthorization purchase completion
Mail or telephone order
Merchandise return
Cash advance
Purchase with cash back
Check guarantee
Purchase adjustment
Merchandise return adjustment
Cash advance adjustment
Cash back adjustment

Response: Optional. Variable length up to 18 bytes. If this field is included in the request, that value can be echoed in the response for the length of the value.

FID C	AMOUNT 2 (PSTM.TRAN.AMT-2)
--------------	-----------------------------------

The Amount 2 field is the secondary amount field. For transactions with two amounts involved, the Amount 2 field contains the revised amount. For transactions with cash back, the Amount 2 field contains the cash back amount.

Request: Required for the following transactions. Variable length of 1 to 18 bytes.

Purchase with cash back
Purchase adjustment
Merchandise return adjustment
Cash advance adjustment
Cash back adjustment

Response: Optional. Fixed length of 18 bytes. If this field is included in the request, it can be echoed in the response. This field is right-justified and zero filled.

FID D	APPLICATION ACCOUNT TYPE (PSTM.TRAN.TRAN-CDE.AA)
--------------	---

The Application Account Type field contains a code indicating the type of account for the transaction. If this field is not submitted in the request message, BASE24-pos uses the DEFAULT ACCOUNT TYPE field on Card Prefix File (CPF) screen 7. Valid values are as follows:

0 = None (use CPF)
1 = Checking

Request: Optional. Fixed length of 1 byte. Required in reversal messages if the MHI module is running in host control mode.

Response: Optional. Fixed length of 1 byte. If this field is included in the request message, it can be echoed in the response. If this field is not in the request message, it reflects the DEFAULT ACCOUNT TYPE field in the CPF.

FID E	APPLICATION ACCOUNT NUMBER (PSTM.TRAN.ACCT)
--------------	--

The value in the Application Account Number field indicates the account against which the transaction was processed. This field can originate with BASE24-pos or with a host.

Request: Not available.

Response: Optional. Variable length of 1 to 19 bytes. Trailing blanks will be removed.

FID F	APPROVAL CODE (PSTM.TRAN.APPRV-CDE)
--------------	--

The value in the Approval Code field represents a unique number generated by the authorizing agent for the transaction. It can be generated by the terminal, BASE24, an issuer host, or an interchange.

Request: Optional. Fixed length of 8 bytes.

Response: Optional. Fixed length of 8 bytes. If this field is included in the request message, it can be echoed in the response. If this field is not included in the request message, it can be generated by BASE24, a host, or an interchange.

FID G	AUTHENTICATION CODE
--------------	----------------------------

The Authentication Code field contains the 8-byte hexadecimal message authentication code (MAC) used to verify the message when MACs are being used.

Request: Optional. Fixed length of 8 bytes. If this field is included, the MAC is verified.

Response: Optional. If responses are verified using MACs, this field is required. Fixed length of 8 bytes.

FID H	AUTHENTICATION KEY
--------------	---------------------------

The value in the Authentication Key field is generated by the MHI module when the MHI module is operating in terminal control mode and sent to the merchant host. If the MHI module is operating in host control mode, the key from the MHCF is returned in this field. The authentication key is the MAC communications key, encrypted under the MAC Terminal Master communications key.

Request: Not available.

Response: Optional. Fixed length of 16 bytes. This field can be configured in any response and is included if the consecutive MAC error threshold is passed. This field also can be included optionally in downloads to the terminal.

FID I	CLERK ID (PTDF.CLERK-ID)
--------------	---------------------------------

The value in the Clerk ID field is used to identify the clerk operating a particular terminal.

Request: Optional. Fixed length of 6 bytes.

Response: Not available.

FID J	AVAILABLE BALANCE (PSTM.TRAN.AMT-1)
--------------	--

The Available Balance field contains the amount available to the cardholder from the application account against which a debit or credit card transaction was processed. It is available in responses only on balance inquiry transactions. If this field is configured in other responses, it is zero filled.

Request: Not available.

Response: Optional. Fixed length of 18 bytes. The Available Balance can originate at a host or from BASE24. This field is right-justified and zero-filled.

FID K	BUSINESS DATE (PSTM.USER-DATA)
--------------	---------------------------------------

The Business Date field contains the terminal business date and is mapped to the PSTM.USER-DATA field, if it is configured.

Request: Optional. Required for retailer and group totals requests. Fixed length of 6 bytes.

Response: Optional. Fixed length of 6 bytes. If included, the value is echoed from the request.

FID L	CHECK TYPE/CATEGORY (PSTM.TRAN.TRAN-CDE.C)
--------------	---

The value in the Check Type/Category field indicates the type of check involved in a check verification or check guarantee transaction. For other noncheck financial transactions, the Check Type/Category field contains the transaction category. Valid values are as follows:

Check Transactions

- 4 = Personal check for cash
- 5 = Personal check for purchase
- 6 = Personal check for purchase and cash
- 7 = Government
- 8 = Payroll
- 9 = Electronic

Financial Transactions

- 0 = Unspecified check or Normal transaction
- 1 = Sales draft
- 2 = Representation
- 3 = Chargeback

Request: Optional. Required for any transaction in which a category is specified. Fixed length of 1 byte.

Response: Optional. Fixed length of 1 byte. If this field is included, the value is echoed from the request.

FID M	COMMUNICATIONS KEY
--------------	---------------------------

The value in the Communications Key field is the communication key used when working with PINs. The value in the field is generated by the MHI module or the security module if the MHI module is operating in terminal control mode or host control mode.

Request: Not available.

Response: Optional. Fixed length of 16 bytes.

FID N	CUSTOMER ID (Check Guarantee/Verification token)
--------------	---

The value in the Customer ID field identifies the cardholder in a check guarantee or check verification transaction. The value in this field could be a social security number, a drivers license number, or another type of identification.

If the customer ID type specified in FID O is 11 (MICR data), the format of FID N is as follows:

Institution routing number	X(11) right-justified, zero filled to the left
Account number	X(19) left-justified, space filled to the right
Check number	X(8) left-justified

Request: Optional. Variable length of 1 to 40 bytes. Required for a check verification or check guarantee transaction. Trailing blanks will be removed.

Response: Optional. Variable length of 1 to 40 bytes. If this field is included, the value is echoed from the request. Trailing blanks will be removed.

FID O	CUSTOMER ID TYPE (Check Guarantee/Verification token)
-------	---

The Customer ID Type field specifies the type of identification used in the Customer ID field. The values in these fields are logged by BASE24-pos. Valid values are as follows:

00	= None
01	= Credit card
02	= Drivers license
03	= Checking account number
04	= Debit card
05	= Proprietary check cashing card
06	= State ID number
07	= Social security number
08	= Student ID number
09	= Employee ID
10	= Passport number
11	= MICR data (refer to FID N for format)
12–50	= Reserved for national use
51–75	= Reserved for ISO use
76–99	= Reserved for private use

Request: Optional. Fixed length of 2 bytes.

Response: Optional. Fixed length of 2 bytes. If this field is included, the value is echoed from the request.

FID P	DRAFT CAPTURE FLAG (PSTM.TRAN.DFT-CAPTURE-FLG)
--------------	---

The value in the Draft Capture Flag field can be specified by the terminal. However, the transaction profile kept in the POS Terminal Data File (PTDF) overrides the value in this field, unless the PTDF transaction profile is 3 (terminal determines data capture mode for each transaction), in which case the MHI module uses the value in this field. Values for this field would typically originate in data downloaded to the terminal by card prefix ranges. Valid values are as follows:

- 0 = Authorize only
- 1 = Authorize and capture

Request: Optional. Fixed length of 1 byte.

Response: Optional. Fixed length of 1 byte. This field in the response represents the draft capture flag used for transactions by BASE24-pos.

FID Q	ECHO DATA (PSTM.USER-DATA)
--------------	-----------------------------------

The value in the Echo Data field represents data that the terminal requires to be echoed back to it in the response. BASE24-pos does not edit this field. The data is recorded in the transaction log data and returned in the response to the terminal.

Request: Optional. Variable length of 1 to 16 bytes. This field is padded with trailing blanks.

Response: Optional. Variable length of 1 to 16 bytes. If this field is included, the value is echoed from the request. Trailing blanks will be removed.

FID R	CARD TYPE (PSTM.TRAN.TRAN-CDE.T)
--------------	---

The value in the Card Type field allows the terminal to specify the intended usage for a card used as a debit card and a credit card. If this FID is not present, a default value is taken from the DEFAULT COMBO CARD TYPE field on Card Prefix File (CPF) screen 7. Valid values are as follows:

- C = Credit card
- D = Debit card

Request: Optional. Fixed length of 1 byte. If this field is not submitted, BASE24-pos uses a default value from the DEFAULT COMBO CARD TYPE field on CPF screen 7. If this field is submitted for a non-combination type card, BASE24-pos ignores it and uses the card type from the CPF.

Response: Optional. Fixed length of 1 byte. The value in this field is the card type as determined by BASE24-pos.

FID S	INVOICE NUMBER (PSTM.INVOICE-NUM)
--------------	--

The value in the Invoice Number field allows a terminal to submit a unique stamp to further identify a transaction.

Request: Optional. Variable length of 1 to 10 bytes.

Response: Optional. Fixed length of 10 bytes. If this field is included, the value is echoed from the request. This field is left-justified and blank filled.

FID T	INVOICE NUMBER/ORIGINAL (PSTM.ORIG-INVOICE-NUM)
--------------	--

The Invoice Number/Original field allows the terminal to uniquely identify a previous transaction that is now being adjusted.

Request: Optional. Variable length of 1 to 10 bytes. This field may be required by some interfaces and authorizing entities.

Response: Optional. Fixed length of 10 bytes. If this field is included, the value is echoed from the request. This field is left-justified and blank filled.

FID U	LANGUAGE CODE
--------------	----------------------

The value in the Language Code field allows the terminal to override the default language in the POS Terminal Data File (PTDF). It is used in formulating a terminal display response. The terminal can select one of three different language displays. Valid values are as follows:

- 0 = Table 1
- 1 = Table 2
- 2 = Table 3

Request: Optional. Fixed length of 1 byte.

Response: Optional. Fixed length of 1 byte. If this field is included in both the request and response, this field echoes the request. If this field was not included in the request, this field represents the default language code from the POS Terminal Data File (PTDF).

FID V	MAIL/DOWNLOAD KEY
-------	-------------------

The Mail/Download Key field consists of a group of fields that identify mail to be read or marked as delivered, or the type of download to be performed. The structure of FID V is shown below. Included for each field is the position of the field in the element, its field length, and a description of its contents.

Concerning mail, this group identifies a mail category (user-defined), the type of mail access desired (any mail, only undelivered mail, specific mail), a flag used for mail broadcast processing, and a mail ID. The mail ID serves to identify a specific piece of mail or the ID of the last mail read. The format is as follows:

Category Code	X(2)
Access Code	9(1)
Processing Flag	X(2)
Mail Date	9(6)
Mail ID	9(4)

Concerning downloading, this group specifies either a full or partial download. If full, this group identifies whether this is the initial request or a continuation request (when the download spans several request and response pairs). If partial, this group also identifies which download field is requested. The format is as follows:

Category Code	X(2)
Access Code	X
Processing Flag	X(2)
Filler	X(10)

In continuation requests (that is, read next mail or continue download), the terminal should echo the value contained in the previous response.

Request: Optional. Required for a read mail request transaction and a mail delivered transaction. Required for downloads except on initial requests.

Response: Optional. Required for a read mail request transaction and a mail delivered transaction. Required for downloads except on initial requests.

FID W	MAIL/DOWNLOAD TEXT
--------------	---------------------------

The Mail/Download Text field consists of a group of fields representing the text of a send mail request (from the terminal to the MHI module) or the download fields from the Merchant Terminal Configuration File (MCNF).

Concerning mail, this group contains a maximum of 449 bytes and identifies the destination DPC number (when more than one exists) and the text itself. The format is as follows:

Position	Length	Description
01	1	Destination DPC Number The values for the DPC Number Code are 0, 1, and 2. These values map to different four-character DPC codes listed in the Merchant Terminal Configuration File (MCNF).
02–449	448	Mail Text (variable-length, 448 bytes maximum) Text of the mail message.

Concerning downloading, this group contains the fields being downloaded from the MCNF. This FID W contains a maximum of 956 bytes of download data. The format is as follows:

Position	Length	Description
01	1	Destination DPC Number This field is not used for downloads and contains a value of 0.
02–956	955	Download Text (variable-length, 955 bytes maximum) Text of the download text.

For more information about download text, refer to section 5.

Request: Optional. Required for a Send Mail Request transaction. Variable length of 1 to 449 bytes for BASE24-mail transactions.

Response: Optional. Variable length of 1 to 449 bytes for BASE24-mail transactions and 1 to 957 bytes for download text. With responses, only the text of the mail itself is used. If no mail was found, this field is not included in the response, even if the configuration supports read mail responses.

FID X	TRANSACTION DATE/TIME (PSTM.TRAN-DATE, PSTM. TRAN-TIME)
--------------	--

The value in the Transaction Date/Time field represents the terminal local date and time. The terminal local date and time is the BASE24 transaction date and time adjusted with the PTDF offset.

When the MHI module generates a reversal, it should send the transaction date and time it received in the financial transaction response.

Request: Required on host control reversal requests. Optional for all other requests. Fixed length of 12 bytes.

Response: Required in all responses. Fixed length of 12 bytes.

FID Y	POSTAL (ZIP) CODE (PSTM.ZIP-CDE)
--------------	---

The Postal (ZIP) Code field contains the postal code of the cardholder's billing address. The postal code should be either five or nine digits in length. If it is five digits in length, the digits should be left-justified and the remaining positions are blank filled.

Request: Optional. Fixed length of 9 bytes.

Response: Optional. Fixed length of 9 bytes. If included, the value is echoed from the request.

FID Z	ADDRESS VERIFICATION STATUS CODE (PSTM.ADDR-FLDS.ADDR-VRFY-STAT)
--------------	---

The value in the Address Verification Status Code field indicates the results of comparing address verification information received in the transaction and address verification information contained in the processor database. Valid values are as follows:

- A = Address. Addresses matched, but ZIP codes did not match.
- E = The transaction was either not eligible for address verification or an error occurred while attempting to process the message.
- N = Error. Neither the addresses nor the ZIP codes matched.
- R = Retry. The primary and secondary authorizers were either unavailable or they declined the transaction and address verification was not performed by BASE24-pos.
- S = Service not supported. BASE24-pos authorized the transaction, but does not support the BASE24-pos Address Verification product.
- U = Unavailable. Address information was not available to the processor performing address verification.
- W = Whole ZIP. The nine digit ZIP code matched, but the address did not match.
- X = Exact. Both the addresses and the nine digit ZIP codes matched.
- Y = Yes. Both the addresses and the five digit ZIP codes matched.
- Z = ZIP. The five digit ZIP codes matched, but the addresses did not match.
- ␣ = Address verification information was not included in the transaction where ␣ is a blank).
- 0 = Address verification information was included in the transaction, but was not verified. This code is used by transactions to be verified by either a host or an interchange. In addition, transactions to be verified by BASE24-pos that are declined before address verification can be performed also carry this code.

Request: Not available.

Response: Optional. Fixed length of 1 byte.

FID a	OPTIONAL DATA (PSTM.USER-DATA or Optional Data token)
--------------	--

The Optional Data field allows the terminal to forward any type of optional data for BASE24-pos. Typically, this field is used to indicate product codes, quantities, and amounts within a total purchase amount.

When receiving a request from the POS device, the Merchant Host Interface module checks the length of the data in this field. If this field contains 80 bytes or fewer, the data is carried in the USER-DATA field of the PSTM. If the field contains more than 80 bytes, the data is carried in the Optional Data token.

When operating in terminal control mode and formatting a response to return to the device, the Merchant Host Interface module checks the contents of the optional data field in device-dependent data. If the device-dependent data field is not blank filled, the contents of this field are returned in the response. If the device-dependent data is blank filled or not present, the Merchant Host Interface module returns the data in the Optional Data token if present.

When operating in host control mode and formatting a response to return to the device, the Merchant Host Interface module checks the contents of the optional data field in device-dependent data. If the device-dependent data field is not blank filled, the contents of this field are returned in the response. If the device-dependent field is blank filled, the Merchant Host Interface module returns the data in the Optional Data token if present.

Request: Optional. Variable length of 1 to 250 bytes. If the length of this field is 80 bytes or less, this field is padded with up to 80 trailing blanks and is put in the User Data field. If the length of the field is greater than 80 characters, no trailing blanks are added, and it is put in the Optional Data token.

Response: Optional. Variable length of 1 to 250 bytes. If this field is included, the value is echoed from the request. Trailing blanks are removed.

FID b	PIN/CUSTOMER (PSTM.PIN)
--------------	--------------------------------

The PIN/Customer field consists of the cardholder-entered PIN in a clear or encrypted format, depending on whether the terminal supports PIN encryption.

Request: Optional. Variable length of 1 to 16 bytes. This field is in PIN/PAD or PIN/PAN PIN block format. The PIN PAD character is 1 byte.

Response: Optional. Variable length of 1 to 16 bytes.

FID c	PIN/SUPERVISOR (PSTM.PIN)
--------------	----------------------------------

The PIN/Supervisor field consists of the PIN as entered by the supervisor in a clear or encrypted format, depending on whether the terminal supports PIN encryption.

Request: Optional. Variable length of 1 to 16 bytes. Typically, this field is submitted only in certain transaction requests, like returns and adjustments. In these cases, BASE24-pos is also configured to indicate that supervisor security is to be applied to these transactions. This field is in PIN/PAD or PIN/PAN PIN block format.

Response: Not available.

FID d	RETAILER ID (PSTM.USER-DATA)
--------------	-------------------------------------

The Retailer ID field consists of the ID assigned to a merchant group by organizations such as MasterCard, Visa, or American Express.

Request: Optional. Variable length of 1 to 12 bytes. This field is padded with trailing blanks.

Response: Optional. Variable length of 1 to 12 bytes. If this field is included, the value is echoed from the request. Trailing blanks are removed.

FID e	POS CONDITION CODE
--------------	---------------------------

The POS Condition Code field further describes the transaction being submitted. Valid values are as follows:

- 00 = Normal presentment
- 06 = Preauthorization request
- 08 = Mail or telephone order
- 09 = Security alert
- 10 = Customer identity verified
- 11 = Suspected fraud

Request: Optional. Fixed length of 2 bytes.

Response: Optional. Fixed length of 2 bytes. If this field is included, the value is echoed from the request.

FID f	PIN LENGTH or RECEIPT DATA
--------------	-----------------------------------

FID f identifies different data, depending on whether it is associated with a request or a response message. In a request message, FID f identifies the PIN length. In a response message, FID f identifies receipt data.

The PIN Length field contains the actual length of the PIN entered at the terminal.

The Receipt Data field applies only to situations where the institution has commissioned ACI to modify the MHI module to format receipts. In this case, this field contains the receipt data. This FID is not supported for receipt data in the standard BASE24-pos product.

Request: Optional. Variable length of 1 to 2 bytes.

Response: Not available.

FID g	RESPONSE DISPLAY
-------	------------------

The Response Display field represents a 48-character display that explains the response code returned by the MHI module. This field is extracted from one of the three MHI language tables in the Merchant Terminal Response File (MRSP), depending on the default language code stored in the POS Terminal Data File (PTDF) or the language code included in the request.

Request: Not available.

Response: Optional. Variable length 1 to 48 bytes. Trailing blanks are removed.

FID h	SEQUENCE NUMBER (PTDF.SHIFT-NUM, PTDF.BATCH-NUM, PTDF.SEQ-NUM)
-------	---

The Sequence Number field consists of a group of fields used to uniquely identify each transaction. The structure of FID h is shown below. Included for each field is the position of the field in the element, its field length, and a description of its contents.

Position	Length	Description
01-03	3	Shift Number The shift number ranges from 001 to 999. The shift number increases by one digit whenever the shift is closed. It rolls to 001 after the 999th shift is closed.

Position	Length	Description
04–06	3	Batch Number The batch number ranges from 001 to 999. The batch number increases by one digit whenever the batch is closed. It rolls to 001 after the 999th batch is closed or when the shift is closed. Whenever the shift closes, the batch is also expected to close.
07–09	3	Seq # The seq # field ranges from 001 to 999. It is to be a unique number for the current batch. When the batch number changes, this sequence number is expected to return to 001. This portion of FID h is referred to as Seq # so as not to confuse it with the name of FID h which is Sequence Number.
10	1	Reset Flag The Reset Flag indicates whether the terminal or the MHI module is responsible for determining the correct shift number, batch number, and sequence number. Valid values are as follows: 0 = The MHI module determines the shift, batch, and sequence numbers. 1 = The terminal determines the shift, batch, and sequence numbers. When the terminal submits FID h, the MHI module checks the Shift Number field, Batch Number field, and Seq # field against its own expected values. For information on the actions of the MHI module if it receives a shift number, batch number, or sequence number it is not expecting, see section 7.

Request: Required in requests where sequence number checking is required. Fixed length of 10 bytes.

Response: Optional. Fixed length of 10 bytes. When this field is included in the request message and the number is found to be valid, the value is echoed in the response if it is to be included. However, this field is returned in a response, even if it is not configured to be included, if the sequence number is included in the

request and the MHI module finds the number to be invalid. In this case, the MHI module calculates the sequence number that the terminal should have sent and sends it in the response.

FID i	SEQUENCE NUMBER/ORIGINAL (PSTM.USER-DATA)
--------------	--

The Sequence Number/Original field allows the terminal to optionally identify the sequence number of a previous transaction. For more information on this field, see FID h. FID i has the same format as positions 1 through 9 of FID h.

Request: Optional. Fixed length of 9 bytes. Required for preauthorization purchase transactions and preauthorization purchase completion transactions. Without this field, holds are not removed from cardholder accounts efficiently.

Response: Optional. Fixed length of 9 bytes. If this field is included, the value is echoed from the request.

FID j	STATE CODE (PSTM.STATE-CDE (Check Guarantee/ Verification Token)
--------------	---

The State Code field contains the state code associated with the transaction request. The state code is normally taken from the cardholder's drivers license and represents the state where the cardholder resides.

Request: Optional. Fixed length of 2 alphanumeric bytes.

Response: Optional. Fixed length of 2 alphanumeric bytes. If this field is included, the value is echoed from the request.

FID k	BIRTH DATE/DRIVERS LICENSE/ TERMINAL LOCATION (Check Guarantee/Verification token)
--------------	---

In requests, the Birth Date/Drivers License/Terminal Location field contains the birth date (MMDDYY) of the cardholder associated with the transaction. If no day was indicated on the cardholder ID containing the birth date, or if the date is an expiration date, the format is MMY. In responses, this field contains the terminal location as indicated in the POS Terminal Data File (PTDF). This field is used primarily for Regulation E purposes.

Request: Optional. Fixed length of 6 bytes.

Response: Optional. Variable length of 1 to 25 bytes. Trailing blanks will be removed.

FID I	TOTALS/BATCH(PTDF.BATCH,PTDF.SHIFT,PTDF.BATCH-DC, PTDF.SHIFT-DC)
--------------	---

The Totals/Batch field consists of a group of fields representing batch totals as accumulated by the terminal. This field includes a shift and batch ID, along with counts and amounts of debits, credits, and adjustments. BASE24-pos logs these totals and the MHI module-accumulated totals in the POS Terminal Data (PTDF). These totals contain a sign character (+ or -) in the first byte of the amount fields. The format is as follows:

Position	Field	Picture
01-03	Shift number	9(3)
04-06	Batch number	9(3)
07-10	Number of debits in batch	9(4)
11-28	Amount of debits in batch	S9(16)v99
29-32	Number of credits in batch	9(4)
33-50	Amount of credits in batch	S9(16)v99
51-54	Number of adjustments in batch	9(4)
55-72	Amount of adjustments in batch	S9(16)v99

Request: Optional. Fixed length of 72 bytes.

Response: Optional. Fixed length of 72 bytes. Totals can be returned in any response, not just a balancing transaction.

FID m	TOTALS/DAY (PTDF.BATCH, PTDF.BATCH-DC, PTDF.SHIFT, PTDF.SHIFT-DC, PTDF.DAILY, PTDF.DAILY-DC)
--------------	---

The Totals/Day field consists of a group of fields representing terminal day totals as accumulated by the terminal. This field includes a shift and batch count, along with counts and amounts of debits, credits, and adjustments. BASE24-pos logs these totals and the MHI module-accumulated totals in the POS Terminal Data File (PTDF). These totals contain a sign character (+ or -) in the first byte of the amount fields. The format is as follows:

Position	Field	Picture
01-03	Number of shifts in day	9(3)
04-06	Number of batches in day	9(3)
07-10	Number of debits in day	9(4)

Position	Field	Picture
11–28	Amount of debits in day	S9(16)v99
29–32	Number of credits in day	9(4)
33–50	Amount of credits in day	S9(16)v99
51–54	Number of adjustments in day	9(4)
55–72	Amount of adjustments in day	S9(16)v99

Request: Optional. Fixed length of 72 bytes.

Response: Optional. Fixed length of 72 bytes. Totals can be returned in any response, not just a balancing transaction.

FID n	TOTALS/EMPLOYEE (PTDF.BATCH-NUM, PTDF.SHIFT-NUM, PTDF.CLERK-TOTS.DB-CNT, PTDF.CLERK-TOTS.DB-AMT, PTDF.CLERK-TOTS.CR-CNT, PTDF.CLERK-TOTS.CR-AMT, PTDF.CLERK-TOTS.ADJ-CNT, PTDF.CLERK-TOTS.ADJ-AMT, PTDF.CLERK-TOTS.CASH-CNT, PTDF.CLERK-TOTS.CASH-AMT, PTDF.CLERK-TOTS.CHECK-CNT, PTDF.CLERK-TOTS.CHECK-AMT)
--------------	---

The Totals/Employee field consists of a group of fields representing terminal employee totals as accumulated by the terminal. This field includes a shift and batch ID, along with counts and amounts of debits, credits, adjustments, cash outs, and checks. BASE24-pos logs these totals and the MHI module-accumulated totals in the POS Terminal Data File (PTDF). These totals contain a sign character (+ or -) in the first byte of the amount fields. The format of the FID is as follows:

Position	Field	Picture
01–03	Current shift number	9(3)
04–06	Current batch number	9(3)
07–10	Number of debits for employee	9(4)
11–28	Amount of debits for employee	S9(16)v99
29–32	Number of credits for employee	9(4)
33–50	Amount of credits for employee	S9(16)v99
51–54	Number of adjustments for employee	9(4)
55–72	Amount of adjustments for employee	S9(16)v99
73–76	Number of cash outs for employee	9(4)
77–94	Amount of cash outs for employee	S9(16)v99
95–98	Number of checks for employee	9(4)
99–116	Amount of checks for employee	S9(16)v99

Request: Not available.

Response: Optional. Fixed length of 116 bytes. If the REL-NUM field in the PTDF is equal to 50, the length is 98 bytes and the amount of checks for employee is not included in the FID.

FID o	TOTALS/SHIFT (PTDF.BATCH, PTDF.BATCH-DC, PTDF.SHIFT, PTDF.SHIFT-DC)
--------------	--

The Totals/Shift field consists of a group of fields representing terminal shift totals as accumulated by the terminal. This field includes a shift ID and batch count, along with counts and amounts of debits, credits, and adjustments. BASE24-pos logs these totals and the BASE24-accumulated totals in the POS Terminal Data File (PTDF). These totals contain a sign character (+ or -) in the first byte of the amount fields. The format is as follows:

Position	Field	Picture
01-03	Number of shifts in day	9(3)
04-06	Number of batches in shift	9(3)
07-10	Number of debits in shift	9(4)
11-28	Amount of debits in shift	S9(16)v99
29-32	Number of credits in shift	9(4)
33-50	Amount of credits in shift	S9(16)v99
51-54	Number of adjustments in shift	9(4)
55-72	Amount of adjustments in shift	S9(16)v99

Request: Optional. Fixed length of 72 bytes.

Response: Optional. Fixed length of 72 bytes. Totals can be returned in any response, not just a balancing transaction.

FID p	RESPONSE CODE (PSTM.TRAN.RESP-CDE)
--------------	---

Indicates the results of the transaction to the terminal. Valid values are listed on the following pages.

Approved Codes

- 000 = Approved balances available
- 001 = Approved no balances available
- 002 = Approved country club status
- 003 = Approved (maybe more identification is required)
- 004 = Approved pending identification (sign paper draft is required)
- 005 = Approved blind
- 006 = Approved VIP status
- 007 = Approved administrative transaction
- 008 = Approved national NEG hit OK
- 009 = Approved commercial status

Declined Codes

- 050 = Declined
- 051 = Expired card
- 052 = Number of PIN tries exceeded
- 053 = No sharing allowed
- 054 = No security module
- 055 = Invalid transaction
- 056 = Transaction not supported by institution
- 057 = Lost or stolen card
- 058 = Invalid card status
- 059 = Restricted status
- 060 = Account not found on CAF
- 061 = PBF record not found
- 062 = PBF update error
- 063 = Invalid authorization type in IDF
- 064 = Bad track information
- 065 = Adjustment not allowed in IDF
- 066 = Invalid credit card advance increment
- 067 = Invalid transaction date
- 068 = PTLF error
- 069 = Bad message edit
- 070 = No IDF
- 071 = Invalid routing to Authorization
- 072 = Card on National Negative File
- 073 = Invalid routing authorization service
- 074 = Unable to authorize
- 075 = Invalid PAN length
- 076 = Insufficient funds in PBF
- 077 = Preauthorization full
- 078 = Duplicate transaction received

079 = Maximum online refund reached
080 = Maximum offline refund reached
081 = Maximum credit per refund reached
082 = Maximum number of times used
083 = Maximum refund credit reached
084 = Customer selected NEG reason
085 = Inquiry not allowed—no balances
086 = Over floor limit
087 = Maximum number refund credits reached
088 = Place call
089 = CAF status = 0 or 9
090 = Referral file full
091 = NEG file problem
092 = Advance less than minimum
093 = Delinquent
094 = Over limit table*
095 = Amount over maximum
096 = PIN required
097 = Mod 10 check
098 = Force post
099 = Bad PBF

* Response code 094 has two meanings. For parametric authorization, the code indicates the transaction amount exceeded the amount available. For a return or adjustment transaction, the code indicates a PTDF return or adjustment limit (count or amount) was exceeded.

Referral Codes

100 = Unable to process transaction
101 = Unable to authorize—issue call
102 = Call
103 = NEG file problem
104 = CAF problem
105 = Card not supported
106 = Amount over maximum
107 = Over daily limit
108 = CAPF not found
109 = Advance less than minimum
110 = Number times used
111 = Delinquent
112 = Over limit table
113 = Timeout
115 = PTLF full

- 120 = Bad UAF
- 121 = ADMN file problem
- 122 = Unable to validate PIN; security module is down

Service Code

- 150 = Merchant not on file

Transaction Error Codes

- 200 = Invalid account
- 201 = Incorrect PIN
- 202 = Cash advance is less than minimum
- 203 = Administrative card needed
- 204 = Enter lesser amount[†]
- 205 = Invalid advance amount
- 206 = CAF not found
- 207 = Invalid transaction date
- 208 = Invalid expiration date
- 209 = Invalid transaction code
- 251 = Cash back exceeds daily limit
- 800 = Format error
- 801 = Invalid data
- 802 = Invalid employee number
- 809 = Invalid close transaction
- 810 = Transaction timeout
- 811 = System error
- 820 = Invalid terminal identifier
- 821 = Invalid response length
- 830 = Invalid DPC
- 831 = Invalid message type in the request header
- 832 = Invalid message subtype in the request header
- 833 = Invalid transmission number in the request header
- 834 = Invalid amount in the request message
- 835 = Invalid account type in the request message
- 836 = Invalid approval code in the request message
- 837 = Invalid business date in the request message
- 838 = Invalid check type in the request message
- 839 = Invalid customer ID in the request message
- 840 = Invalid customer ID type in the request message
- 841 = Invalid draft capture flag in the request message
- 842 = Invalid echo data in the request message
- 843 = Invalid card type in the request header
- 845 = Invalid language code in the request header

- 851 = Invalid totals format
- 852 = Invalid Authorization timeout override value
- 855 = Invalid PIN length in the request header
- 856 = Invalid reversal context

† Response code 204 has two meanings. This code can also be used when the transaction amount exceeds the POS Retailer Definition File (PRDF) ceiling limits.

Mail And Download Codes

- 870 = Mail delivered
- 871 = Mail stored
- 880 = Mail message has been received in its entirety
- 881 = Mail message received successfully and there is more data for this mail message
- 880 = Download has been received in its entirety
- 881 = Download received successfully and there is more data for this download
- 882 = Download aborted (call for service)

MAC Error Codes

- 898 = Invalid MAC
- 899 = Sequence error—resync

POS Capture Codes

- 900 = Number of PIN tries exceeded
- 901 = Expired card
- 902 = NEG capture code
- 903 = CAF status 3
- 904 = Advance less than minimum
- 905 = Number times used exceeded
- 906 = Delinquent
- 907 = Over limit table
- 908 = Amount over maximum
- 909 = Capture

Decline Administrative Card—Call Required

- 950 = Administrative card not found
- 951 = Administrative card not allowed

Approved Administrative

- 952 = Approved administrative request—in window
- 953 = Approved administrative request—out of window
- 954 = Approved administrative request—anytime

Chargebacks

- 955 = Chargeback—customer file updated
- 956 = Chargeback—customer file updated—acquirer not found
- 957 = Chargeback—incorrect prefix number
- 958 = Chargeback—incorrect response code or CPF configuration
- 960 = Chargeback—approved customer file not updated
- 961 = Chargeback—approved customer file not updated, acquirer not found
- 962 = Chargeback—accepted, incorrect destination

Request: Required for reversal requests from DPCs running in host control mode. Fixed length of 3 bytes.

Response: Required. Fixed length of 3 bytes.

FID q	TRACK2/CUSTOMER (PSTM.TRACK2)
--------------	--------------------------------------

The Track 2/Customer field consists of a group of fields representing the cardholder Track 2 data. This data can be entered manually or obtained from the terminal card swipe reader. The format of Track 2 indicates which of the two methods were used. The format is as follows:

For manually-entered data:	X(29)
Entry ID (M)	X(01)
PAN (variable-length, 19 bytes maximum)	X(19)
Separator character (=)	X(01)
Expiration date (YYMM)	9(04)
Member number	9(03)
End sentinel (?)	X(01)
For swiped data:	X(40)
Start sentinel (;)	X(01)
PAN (19 bytes maximum)	X(19)
Separator character (=)	X(01)
Expiration date (YYMM)	9(04)
Discretionary data (14 bytes maximum)	X(14)
End sentinel (?)	X(01)

Request: Optional. Variable length of 1 to 40 bytes. Either this field or the Track 1/Customer field (FID 2) is required for the following transactions:

Normal purchase
 Preauthorization purchase
 Preauthorization completion
 Mail or telephone order
 Merchandise return
 Cash advance
 Card verification
 Balance inquiry
 Purchase with cash back
 Purchase adjustment
 Merchandise return adjustment
 Cash advance adjustment
 Cash back adjustment

Response: Optional. Variable length of 1 to 40 bytes. If this field is included, the value is echoed from the request.

FID r	TRACK2/SUPERVISOR (PSTM.TRACK2)
-------	---------------------------------

The Track 2/Supervisor field consists of a group of fields representing the supervisor Track 2 data. This data can be entered manually or obtained from the terminal card swipe reader. The format indicates which of the two methods were used. The format is as follows:

For manually-entered data:	X(29)
Entry ID (M)	X(01)
PAN (variable-length, 19 bytes maximum)	X(19)
Separator character (=)	X(01)
Expiration date (YYMM)	9(04)
Member number	9(03)
End sentinel (?)	X(01)
For swiped data:	X(40)
Start sentinel (;)	X(01)
PAN (variable-length, 19 bytes maximum)	X(19)
Separator character (=)	X(01)
Expiration date (YYMM)	9(04)
Discretionary data (14 bytes maximum)	X(14)
End sentinel (?)	X(01)

Request: Optional. Variable length of 1 to 40 bytes. Track 2 is typically submitted only in certain transaction requests such as returns and adjustments. In these cases, BASE24-pos is also configured to indicate that supervisor security is to be applied to these transactions.

Response: Optional. Variable length of 1 to 40 bytes. If this field is included, the value is echoed from the request.

FID s	TRANSACTION DESCRIPTION
--------------	--------------------------------

The Transaction Description field contains a 24-character description of the transaction sent by the MHI module to the terminal for receipt purposes.

Request: Not available.

Response: Optional. Variable length of 1 to 24 bytes.

FID t	MAIL PENDING FLAG
--------------	--------------------------

The value in the Mail Pending Flag field is returned in a response message to indicate whether the terminal has mail to read.

Request: Not available.

Response: Optional. Fixed length of 1 byte.

FID u	DOWNLOAD PENDING FLAG
--------------	------------------------------

The value in the Download Pending Flag field is returned in a response message to indicate that a download has been requested for the terminal.

Request: Not available.

Response: Optional. A fixed length of 1 byte. The Download Pending Flag field is returned in the response if the POS Terminal Data File (PTDF) download pending flag is set to true and the MHI module is running in terminal control mode.

FID v	AUTHORIZATION TIMEOUT OVERRIDE
--------------	---------------------------------------

The value in the Authorization Timeout Override field represents a timeout value that can be used to override the value set in the AUTH TIMEOUT (SECS) field on MCNF screen 2. Valid values are 000001 through 999999.

Request: Optional. Fixed length of 6 bytes.

Response: Not available.

FID w	CLERK TOTALS FLAG
--------------	--------------------------

The value in the Clerk Totals Flag field represents the types of records to be included in the clerk totals on a clerk totals request. Valid values are as follows:

- 0 = All clerk totals for a specific terminal
- 1 = Totals for a clerk at a specific terminal
- 2 = Clerk totals for a specific batch for a terminal
- 3 = Totals for a specific clerk over all terminals

Request: Required when requesting clerk totals. Fixed length of 1 byte.

Response: Not available.

FID x	GROUP TOTALS ID
--------------	------------------------

The value in the Group Totals ID represents an identifier of the totaling group for which group totals are to be calculated. It is used as a key to the Merchant Totals Configuration File (MTCF). If retailer totals are being requested, this field is filled with asterisks (*).

Request: Required when requesting retailer or group totals. Fixed length of 10 bytes.

Response: Not available.

FID y	TOTALS/RETAILER or GROUP
-------	--------------------------

The Totals/Retailer or Group field represents a group of fields that contain the retailer or group totals accumulated by the MHI module and recorded in the POS Transaction Log File (PTLF). The totals calculated by the MHI module are described below. These totals contain a sign character (+ or -) in the first byte of the amount field.

Position	Length	Description
01–04	4	Number of terminals in the terminal group
05–08	4	Number of terminals in the totals
09–12	4	Debit count
13–30	18	Debit amount
31–34	4	Credit count
35–52	18	Credit amount
53–56	4	Adjustment count
57–74	18	Adjustment amount

Request: Not available.

Response: Required on responses for a retailer or group totals request. Fixed length of 74 bytes.

FID z	REVERSAL CONTEXT
-------	------------------

The Reversal Context field represents a group of fields that are required for a reversal request from the merchant host when a host-controlled terminal or merchant host must generate a reversal. The field includes the following:

Position	Length	Description
01–04	4	Card logical network
05–08	4	Card FIID

Position	Length	Description
09–27	19	Account number
28–35	8	Approval code
36	1	Draft capture
37–38	2	Service
39	1	Authorization indicator
40	1	Authorization indicator 2

Request: Required on reversal requests when the MHI module is in host control mode. Fixed length of 40 bytes.

Response: Required on approved financial responses when the MHI module is in host control mode. Fixed length of 40 bytes.

FID 0	AMEX DATA COLLECTION
--------------	-----------------------------

The AMEX Data Collection option allows customers to perform draft capture on transactions originating from American Express cardholders. Generally, American Express has defined categories under which transactions are placed. These categories are as follows:

- Auto rental
- Lodging
- Restaurant
- General retail
- Oil

For each of these categories, American Express requires different data to be sent from the device. Therefore, American Express has set forth standard industry formats for each category to ensure the data they require is captured by the device and sent to BASE24. The MHI module is able to recognize these standard industry formats and, subsequently, capture and process transactions sent using them. In conjunction with this field, the customer must set up Standard Industrial Classification (SIC) codes or Merchant Category codes in the PTDF. SIC and

Merchant Category codes identify the merchant's line of business. For more information on the PTDF, refer to the ***BASE24-pos Files Maintenance Manual***. For the American Express standard industry formats, refer to appendix A.

Request: Optional. Variable length of 46 to 118 bytes, depending on standard industry format being used. The data length for each specific standard industry format is fixed.

Response: Optional. Variable length of 46 to 118 bytes, depending on the standard industry format being used. The data length for each specific standard industry format is fixed. This FID is not returned in a response if the transaction is declined.

FID 1	PS2000 DATA (PS2000 token)
--------------	-----------------------------------

The PS2000 Data field consists of a group of fields used for support of the Visa PS2000 program. This field contains PS2000 data for both request and response messages. The terminal is responsible for correctly filling in any of the fields that are required in a given request, placing default values in the other fields, and parsing any of the fields out of the response message that need to be printed on a receipt or displayed. The format for this field is as follows:

Authorization characteristics indicator	PIC X(1)
Transaction identifier	PIC X(15)
Validation code	PIC X(4)
Market-specific data fields	
Market-specific data identifier	PIC X(1)
Duration	PIC 9(2)
Prestigious property indicator	PIC X(1)

Request: Optional. Fixed length of 24 bytes.

Response: Optional. Fixed length of 24 bytes.

FID 2	TRACK 1/CUSTOMER (PSTM.TRACK2 or Track 1 token)
--------------	--

The Track 1/Customer field consists of a group of fields representing the cardholder Track 1. This data is obtained from the terminal card swipe reader. The format for cardholder Track 1 data, organized in ISO Standard Format, is as follows:

Field	Description
Start sentinel	1 character (%)
Format code	1 character (B for credit cards)

Field	Description
Identification (PAN)	Up to 19 digits (variable length)
Field separator	1 character (^)
Country code	3 digits (only present when the PAN starts with 59)
Name	Up to 26 characters (variable length)
Field separator	1 character (^)
Expiration date	4 digits (YYMM)
Service code	3 digits
Discretionary data	Up to 21 characters (variable length)
End sentinel	1 character (?)
Longitudinal Redundancy	1 character
Check character	

Request: Optional. Variable length of 1 to 82 bytes. Either this field or the Track 2/Customer field (FID q) is required for the following transactions:

Normal purchase
Preauthorization purchase
Preauthorization completion
Mail or telephone order
Merchandise return
Cash advance
Card verification
Balance inquiry
Purchase with cash back
Purchase adjustment
Merchandise return adjustment
Cash advance adjustment
Cash back adjustment

The customer determines when or if Track 1 data is required on Card Prefix File (CPF) screen 1. The MHI module checks that all financial transactions contain either FID q or FID 2. For more information about how the MHI module formats track data, refer to appendix C.

Response: Optional. Variable length of 1 to 82 bytes. If this field is included, the value is echoed from the request.

FID 3	TRACK 1/SUPERVISOR (PSTM.TRACK2 or Track 1 token)
--------------	--

The Track 1/Supervisor field consists of a group of fields representing the supervisor Track 1. This data is obtained from the terminal card swipe reader. The format for supervisor Track 1 data, organized in ISO standard format, is as follows:

Field	Description
Start sentinel	1 character (%)
Format code	1 character (B for credit cards)
Identification (PAN)	Up to 19 digits (variable length)
Field separator	1 character (^)
Country code	3 digits (only present when the PAN starts with 59)
Name	Up to 26 characters (variable length)
Field separator	1 character (^)
Expiration date	4 digits (YYMM)
Service code	3 digits
Discretionary data	Up to 21 characters (variable length)
End sentinel	1 character (?)
Longitudinal Redundancy	1 character
Check character	

Request: Optional. Variable length of 1 to 82 bytes. Supervisor Track 1 is typically submitted only in certain transaction requests such as returns and adjustments. In these cases, BASE24-pos is also configured to indicate that supervisor security is to be applied to these transactions.

Response: Optional. Variable length of 1 to 82 bytes. If included, the value is echoed from the request.

FID 6	PRODUCT SUBFIDS
--------------	------------------------

The Product SubFIDs field consists of the BASE24 original data and manual card verification data for cardholder and administrative transactions. This data is arranged in subordinate fields and identified by Subfield Identifiers (SFIDs). For more information about these subfields, see the topic “Optional Data Subfields” later in this section.

When you use the request and response field maps in the Merchant Terminal Configuration File (MNCF), you are enabling all subfields to be added to the request or response.

Request: Optional. Variable length.

Response: Optional. Variable length.

FID 7	PRODUCT SUBFIDS
--------------	------------------------

This Product SubFIDs field is reserved for future use.

Request: Not applicable.

Response: Not applicable.

FID 8	CHANNEL/DISTRIBUTOR SUBFIDS
--------------	------------------------------------

The Channel/Distributor SubFIDs field is reserved for future use by distribution channels.

Request: Not applicable.

Response: Not applicable.

FID 9	CUSTOMER SUBFIDS
--------------	-------------------------

The Customer SubFIDs field is reserved for future use by customers.

Request: Not applicable.

Response: Not applicable.

Optional Data Subfields

The information in the Product SubFIDs data field (FID 6) is contained in subfields. The subfields are included in the request or response message when FID 6 is specified in the Merchant Terminal Configuration File (MCNF). Specifying FID 6 in the MCNF specifies all subfields for FID 6. The other Product SubFIDs, Channel/Distributor SubFIDs, and Customer SubFIDs (FIDs 7, 8, and 9) fields are reserved for future use.

Product SFIDs Subfield Structures

The subfields for the Product SubFIDs data field (FID 6) are described below according to their subfield identifiers (SFIDs). The table lists the subfield name, its picture clause, the length of the field in the message, and the associated field name. In addition, a check mark (✓) is in the RQST or RESP column if a subfield is available for requests and responses, respectively.

SFID	Picture	Length	Field Name	RQST	RESP
A	PIC X(12)	12 bytes	BASE24 original data	✓	✓
B	PIC X(4)	4 bytes	Manual card verification data (CVD)/customer	✓	✓
C	PIC X(4)	4 bytes	Manual card verification data (CVD)/administrative	✓	✓
D	PIC X(876)	30–876 bytes	Purchasing card/fleet card data	✓	
E	PIC 999	3 bytes	POS entry mode		✓
F	PIC X	1 byte	Electronic commerce data	✓	
G	PIC X	1 byte	Visa commercial card type indicator		✓
H	PIC X(2)	2 bytes	CVD presence indicator and CVD result	✓	✓
I			Currency code indicator (reserved for future use)		

SFID	Picture	Length	Field Name	RQST	RESP
J	PIC X(32)	32 bytes	Cardholder certificate serial number	✓	
K	PIC X(32)	32 bytes	Merchant certificate serial number	✓	
L	PIC X(80)	80 bytes	XID/Trans/Stain	✓	

Subfield Descriptions

The following are the descriptions for the subfields for FID 6. Refer to the topic “Optional Data Field Description Format” previously in this section for an explanation of the documentation format.

SFID A	BASE24 ORIGINAL DATA (PSTM.ORIG-DATA.TRN-TIM and PSTM.ORIG-DATA.TRN-DAT)
---------------	---

The BASE24 Original Data subfield represents the original transaction time (hhmmsshh) and date (MMDD) for information carried in the BASE24-pos Standard Internal Message (PSTM). The format of this data is identical to the subordinate data definitions TRN-TIM and TRN-DAT for the ORIG-DATA structure of the PSTM.

Request: Optional. Fixed length of 12 bytes.

Response: Optional. Fixed length of 12 bytes. If this field is included, the value is echoed from the request.

SFID B	MANUAL CVD—CUSTOMER (BASE24-pos Release 5.1 token)
---------------	---

The Manual CVD—Customer subfield contains the manually entered card verification digits from the cardholder’s card. The format is the same as the MANUAL CVD field of the BASE24-pos Release 5.1 token. Refer to the ***BASE24 Tokens Manual*** for more information about the BASE24-pos Release 5.1 token.

Request: Optional. Fixed length of 4 bytes.

Response: Optional. Fixed length of 4 bytes. If this field is included, the value is echoed from the request.

SFID C	MANUAL CVD—ADMINISTRATIVE (BASE24-pos Release 5.1 token)
---------------	---

The Manual CVD—Administrative subfield contains the manually entered card verification digits from the administrative card. The format is exactly the same as the Manual CVD field of the BASE24-pos Release 5.1 token. Refer to the ***BASE24 Tokens Manual*** for more information about the BASE24-pos Release 5.1 token.

Request: Optional. Fixed length of 4 bytes.

Response: Optional. Fixed length of 4 bytes. If this field is included, the value is echoed from the request.

SFID D	PURCHASING CARD/FLEET CARD DATA (Purchasing Card token)
---------------	--

The Purchasing Card/Fleet Card Data field contains purchasing or fleet card information used to create the Purchasing Card token. The first byte of the field indicates the type of data present. Valid values are as follows:

- C = MasterCard Fleet card
- F = Visa Fleet card
- M = MasterCard Purchasing card level 3
- V = Visa Purchasing card level 3.
- ␣ = Level 2 purchasing card (where ␣ is a blank)

Refer to the ***BASE24 Tokens Manual*** for a description of the format of this field.

Request: Optional. Variable length of 30–876 bytes, depending on the data type.

Response: Not applicable.

SFID E	POS ENTRY MODE (PSTM.PT-SRV-ENTRY-MDE)
---------------	---

A code indicating the manner in which this transaction entered BASE24-pos.
Valid values for the first and second digits are as follows:

- 00 = Unspecified
- 01 = Manual entry
- 02 = Magnetic stripe
- 03 = Bar code
- 04 = OCR
- 05 = Integrated circuit card
- 06–60 = Reserved for ISO use
- 61–80 = Reserved for national use
- 81–99 = Reserved for private use

Valid values for the third digit are as follows:

- 0 = Unspecified
- 1 = PIN entry capability
- 2 = No PIN entry capability
- 3–5 = Reserved for ISO use
- 6–7 = Reserved for national use
- 8–9 = Reserved for private use

Request: Not applicable.

Response: Optional. Fixed length of 3 bytes.

SFID F	ELECTRONIC COMMERCE FLAG (BASE24-pos Release 5.1 token)
---------------	--

The electronic commerce and mail or telephone order flag. Valid values are as follows:

- 0 = Not an electronic commerce or mail or telephone order transaction
- 1 = Single mail or telephone order transaction
- 2 = Recurring mail or telephone order transaction
- 3 = Mail or telephone order installment payment
- 4 = Mail or telephone order unknown classification
- 5 = Secure electronic transaction with cardholder certificate
- 6 = Secure electronic transaction without cardholder certificate
- 7 = Channel-encrypted electronic commerce transaction
- 8 = Nonsecure electronic commerce transaction

Request: Not applicable.

Response: Optional. Fixed length of 1 byte.

SFID G	COMMERCIAL CARD TYPE (BASE24-pos Release 5.1 token)
---------------	--

The commercial card type. Valid values are as follows:

- 0 = Noncommercial card or unknown or unspecified card
- B = Business card
- R = Corporate card
- S = Purchasing card

Request: Not applicable.

Response: Optional. Fixed length of 1 byte.

SFID H	CARD VERIFICATION DIGITS PRESENCE INDICATOR AND RESULT (BASE24-pos Release 5.1 token)
---------------	--

A field indicating whether the card verification digits (CVD) are present, and if so, the result of the CVD check. The first byte is the CVD presence indicator. Valid values for this byte are as follows:

- 0 = Visa Card Verification Value 2 (CVV2) value is deliberately bypassed or is not provided by the merchant.
- 1 = CVV2 value is present
- 2 = CVV2 value is on the card, but is illegible
- 9 = Cardholder states that the card has no CVV2 imprint.

The second byte of SFID H is the CVD result. Valid values for this byte are as follows:

- 0 = Card verification was not performed because the transaction was denied before card verification processing started.
- C = Card verification was performed and the CVD were invalid. The situation was noted, and transaction processing continued. This value is used when the appropriate byte of the CV-BAD-DISP field in the Base segment of the CPF contains the value 0.
- D = Card verification was performed and the CVD was invalid. The transaction was denied and the ERR-FLG field was set to C. This value is used when the appropriate byte of the CV-BAD-DISP field in the Base segment of the CPF contains the value 1, 2, or 3.

- J = CVV checking was not performed. The track length was in error. The BAD TRACK ACTION flag in the CPF indicates that the transaction should be denied.
- K = Card verification was not performed. The track length was in error. The situation was noted and the transaction was referred.
- L = CVV checking was not performed. The track length was in error. The BAD TRACK ACTION flag in the CPF indicates that processing should continue.
- N, 0, or ¨ = Authorizing entity has not attempted card verification or could not verify the CVD due to a security device error. (¨ indicates a blank character.)
- O = CVD checking was not performed because the CVD was not on the card. Not all cards have a CVD value encoded. The card expiration date must be equal to or greater than the date defined in the CPF to ensure that the CVD field has been encoded. If the card expiration date is equal to or greater than the CPF date, the CVD checks are performed.
- P = Card verification was not performed; CVD was not on the card. Not all cards have a CVD value encoded. The card expiration date must be equal to or greater than an expiration date defined on the CPF to ensure that the CVD field has been encoded. If the card expiration date is equal to or greater than the CPF date, the CVD checks are performed.
- R = Card verification was performed and the CVD was invalid. The situation was noted and the transaction should be referred.
- Y = Card verification was performed and the CVD was valid.

Request: Optional. Fixed length of 2 bytes.

Response: Optional. Fixed length of 2 bytes. If this field is included, the first byte is echoed from the request.

SFID I	CURRENCY CODE INDICATOR (Reserved for future use)
---------------	--

This SFID is reserved for future use.

SFID J	CARDHOLDER CERTIFICATE SERIAL NUMBER (Cardholder Serial Number token)
---------------	--

The cardholder certificate serial number for secure electronic commerce.

Request: Optional. Fixed length of 32 bytes.

Response: Not applicable.

SFID K	MERCHANT CERTIFICATE SERIAL NUMBER (Merchant Serial Number token)
---------------	--

The merchant certificate serial number for secure electronic commerce.

Request: Optional. Fixed length of 32 bytes.

Response: Not applicable.

SFID L	XID/TRANS STAIN (Trans Stain XID token)
---------------	--

The transaction identifier and transaction stain. The transaction identifier (XID) is 40 bytes long. The transaction stain is a hash value calculated by applying a secure hash algorithm to the XID and CardSecret (a secret SET-defined value known only to the cardholder and the issuer of the cardholder certificate). It is 40 bytes long.

Request: Optional. Fixed length of 80 bytes.

Response: Not applicable.

Request Message Requirements

The customer has the option of including any optional data fields in requests. When the customer determines that any of these data fields must be included in requests for certain transactions, those fields are configured in the Merchant Terminal Configuration File (MCNF). The MHI module then enforces these fields as being required. However, the terminal cannot use the MCNF to determine the fields to include in its requests. The terminal itself must also be configured to determine the fields it requires for certain transactions.

Optional Data Fields for Requests

The optional data fields available for requests are listed in the “Optional Data Field Structures” topic documented earlier in this section. The fields in request messages can be in any order. If a field other than those listed in the table described in the “Optional Data Field Structures” topic is submitted in a request and the field identifier is an ASCII alphabetic character or a numeric character, the MHI module disregards the field and continues processing. If the unexpected field identifier is anything other than an ASCII alphabetic character or a numeric character, the request is declined with a response code of 800, indicating a format error occurred.

Mandatory Request Fields

Some transaction types require that certain optional fields be used in order for processing to occur. The following list shows the transaction types and the field IDs (FIDs), if any, that are mandatory. This is important information for customers deciding which fields to include in their messages.

The TTC heading in the following list identifies the terminal transaction codes. These codes contain either an F or an A, followed by a two digit transaction code (for example, F00, F04, A50, A65). F indicates the transaction is a financial transaction and A indicates the transaction is an administrative transaction.

Note: Either FID q or FID 2 is required for all financial transactions except mail or telephone order, check verification, and check guarantee transactions. The customer selects the preferred track on Card Prefix File (CPF) screen 1. The MHI module checks each financial transaction for the presence of either FID q or FID 2 and declines those transactions that have neither FID.

TTC	Transaction Description	Mandatory FIDs
F00	Normal purchase	B and q or B and 2
F01	Preauthorization purchase	B and q or B and 2
F02	Preauthorization purchase completion	B and q or B and 2
F03	Mail or telephone order	B and q
F04	Merchandise return	B and q or B and 2
F05	Cash advance	B and q or B and 2
F06	Card verification	q or 2
F07	Balance inquiry	q or 2
F08	Purchase with cash back	B, C, and q or B, C, and 2
F09	Check verification	L
F10	Check guarantee	B and L
F11	Purchase adjustment	B, C, and q or B, C, and 2
F12	Merchandise return adjustment	B, C, and q or B, C, and 2
F13	Cash advance adjustment	B, C, and q or B, C, and 2
F14	Cash back adjustment	B, C, and q or B, C, and 2

TTC	Transaction Description	Mandatory FIDs
A50	Logon request	
A51	Logoff request	
A60	Close batch request	
A61	Close shift request	
A62	Close day request	
A64	Clerk totals request	
A65	Batch totals request	
A66	Shift totals request	
A67	Day totals request	
A68	Retailer/Group totals request	K and x
A70	Read mail request	V
A71	Mail delivered request	V
A75	Send mail request	W
A90	Download request	
A95	Handshake request	

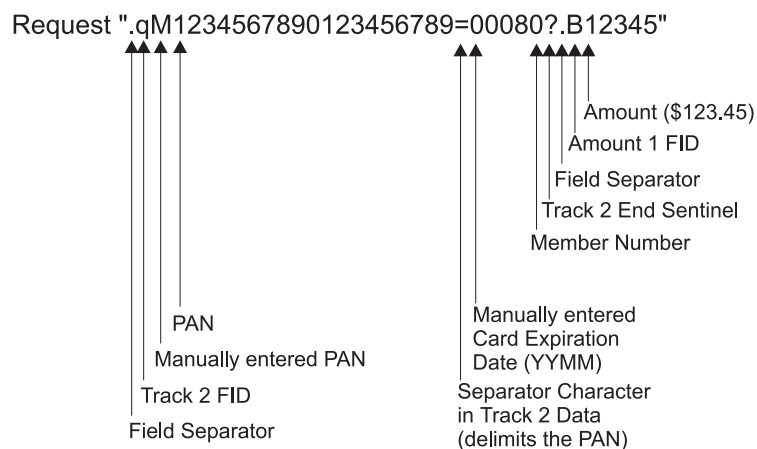
Request Field Examples

The following examples represent various request configurations. In the examples, a period (.) represents a field separator character (hex 1C). The request data shown in the following examples is assumed to follow the standard message header and to be terminated with an ETX in each situation. The quotation marks (“ ”) are used only to delimit the text and are not part of the request. Each example is set up in two parts. The first part identifies and explains the fields included in the request example. The second part provides a graphic to illustrate the request format.

Track 2 with Amount 1

The following is an example where Track 2 data and the transaction amount are included in the request. In the example, Track 2 data was manually entered, so the terminal builds the request as follows:

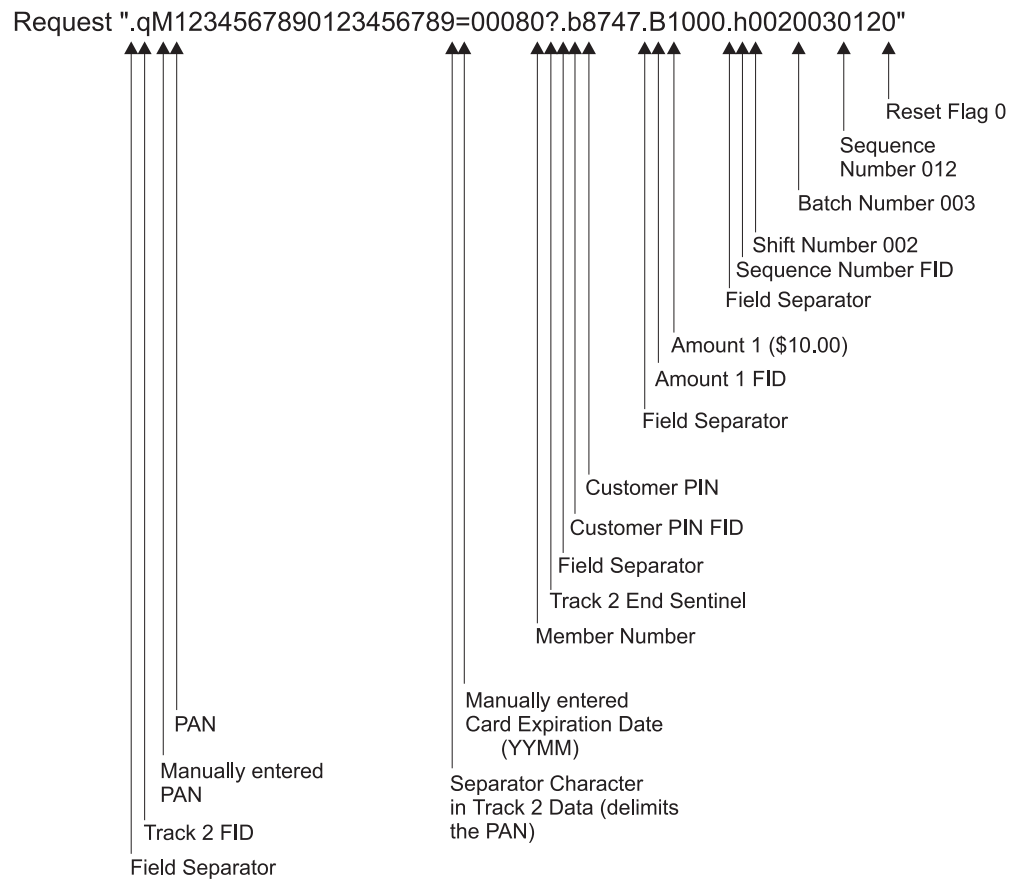
Field	Description
q	FID associated with Track 2 data
M	Entry ID (M = manually entered Track 2 data)
1234567890123456789	Primary account number (PAN)—maximum 19 characters
=	Separator character—delimits the PAN
0008	Card expiration date (YYMM)
0	Member number—optional
?	End sentinel
B	FID associated with the Amount 1 field
12345	The amount of the transaction (\$123.45)



Track 2, Cardholder PIN, Amount 1, and Sequence Number

The following is an example where Track 2 data, the cardholder PIN, the transaction amount, and the transaction sequence number are included in the request. In the example, Track 2 data was manually entered, so the terminal builds the request as follows:

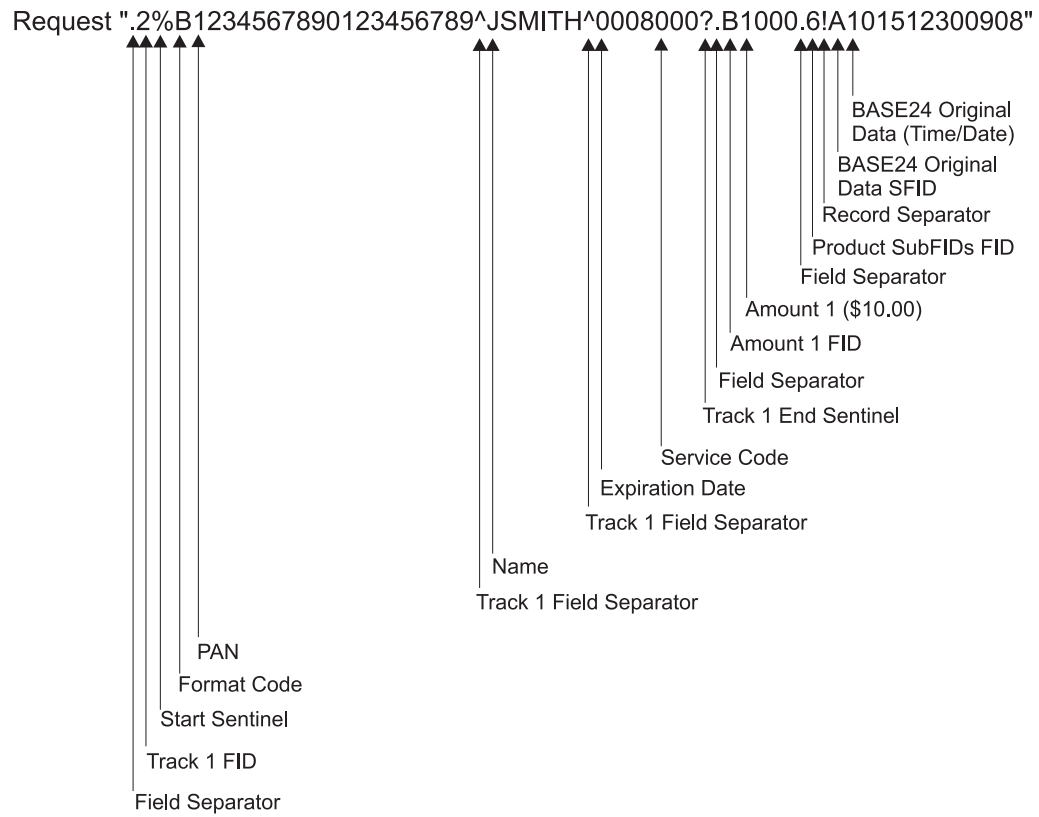
Field	Description
q	FID associated with Track 2 data
M	Entry ID (M = manually entered Track 2 data)
1234567890123456789	Primary account number (PAN)—maximum 19 characters
=	Separator character—delimits the PAN
0008	Card expiration date (YYMM)
0	Member number—optional
?	End sentinel
b	FID associated with the cardholder PIN
8747	Cardholder PIN
B	FID associated with the Amount 1 field
1000	The amount of the transaction (\$10.00)
h	FID associated with the Sequence Number field
0020030120	The sequence number in the following format: 002 = shift number 003 = batch number 012 = sequence number 0 = reset flag



Track 1, Amount 1, and BASE24 Original Data

The following is an example where Track 1 data, the transaction amount, and BASE24 original data are included in the request. In the example, the terminal builds the request as follows:

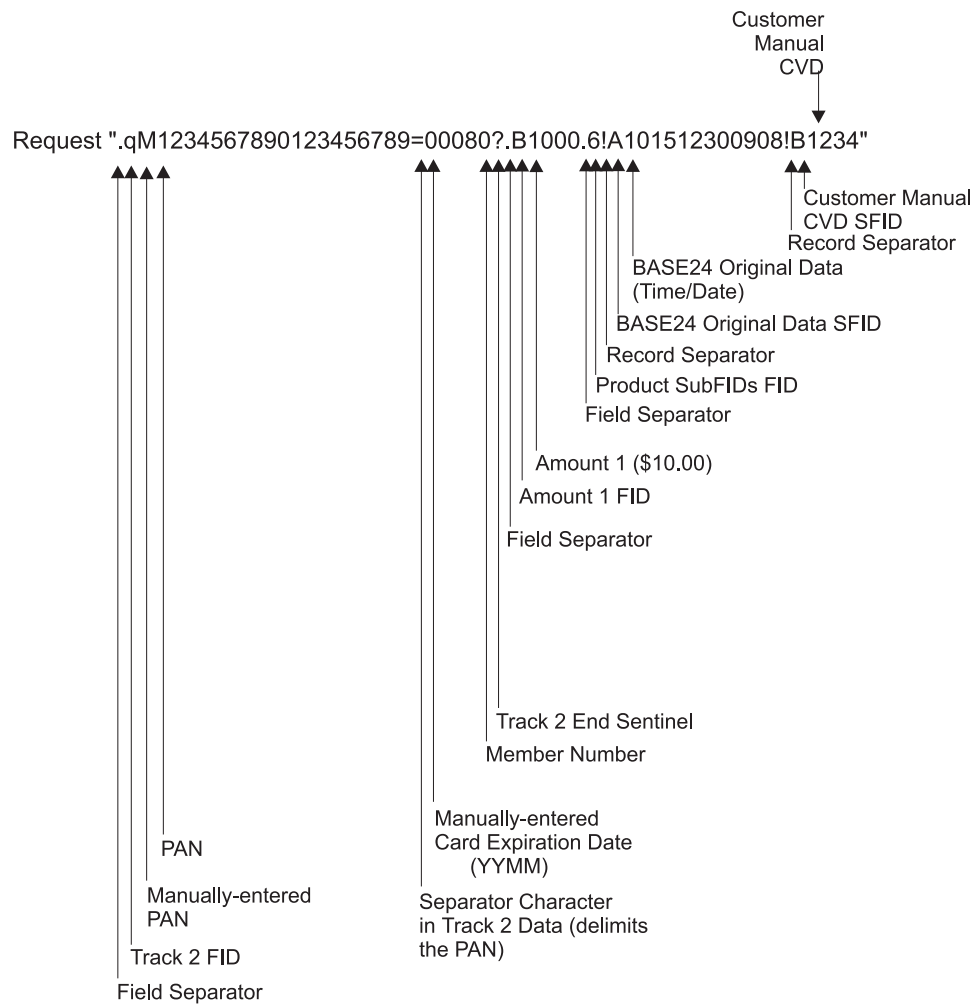
Field	Description
2	FID associated with cardholder Track 1 data
%	Track 1 data start sentinel
B	Format code
1234567890123456789	Primary account number (PAN)
^	Track 1 field separator
JSMITH	Cardholder name
^	Track 1 field separator
0008	Card expiration date (YYMM)
000	Service code
?	Track 1 end sentinel
B	FID associated with the Amount 1 field
1000	The amount of the transaction (\$10.00)
6	FID associated with the product subfields
!	Record separator
A	SFID associated with BASE24 original data
101512300908	BASE24 original data—time (HHMMSShh) and date (MMDD)



Track 2, Amount 1, BASE24 Original Data, and Customer Manual Card Verification Digits (CVD)

The following is an example where Track 2 data, the transaction amount, BASE24 original data, and customer manual card verification digits (CVD) are included in the request. In the example, the terminal builds the request as follows:

Field	Description
q	FID associated with Track 2 data
M	Entry ID (M is for manually entered Track 2 data)
1234567890123456789	Primary account number (PAN)
=	Separator character—delimits the PAN
0008	Card expiration date (YYMM)
0	Member number
?	End sentinel
B	FID associated with the Amount 1 field
1000	The amount of the transaction (\$10.00)
6	FID associated with product subfields
!	Record separator
A	SFID associated with BASE24 original data
101512300908	BASE24 original data—time (HHMMSShh) and date (MMDD)
!	Record separator
B	SFID associated with customer manual CVD
1234	Customer manual CVD



Response Message Requirements

The MHI module formats responses to the merchant host as indicated in the Merchant Terminal Configuration File (MCNF). These fields consist of both mandatory fields (set by BASE24) and those required by the customer. The MHI module formats responses using the fields identified in the MCNF as being required. Any optional data fields to be included in responses are listed in the MCNF.

Optional Data Fields for Responses

The optional data fields available for responses are listed in the “Optional Data Field Structures” topic documented earlier in this section. The order of the fields in the response messages is according to FIDs, with numbered FIDs after alphabetic FIDs, and uppercase FIDs before lowercase letters. Some fields have initialized values. For example, if the card data for the request PAN cannot be located, any related fields such as application account numbers, contain initialized values or zeros.

Some fields may be included in responses regardless of the MCNF data, including the Authentication Key (FID H) and the Mail/Download Key (FID V). However, these are returned only when found to be applicable. The Authentication Key is included when message authentication codes (MACs) are being used and a new MAC communications key is generated due to the configured number of consecutive messages failing to be verified.

If the MHI module is unable to format a response message due to errors in the MCNF, the response message consists of a header and the response code set to 821 in FID p, indicating an invalid response length. A possible error in the MCNF could be that the format specified results in a response longer than the maximum allowable response.

Note: FIDs V and W are the only FIDs allowed in the response message for a download.

Mandatory Response Fields

Some transaction types require certain fields to be present in responses. The following lists show the transaction types and the field IDs (FIDs), if any, that are mandatory when operating in terminal control mode or host control mode. This information is important for customers deciding which fields to include in their messages.

The TTC headings in the following lists identify the terminal transaction codes. These codes contain either an F or an A, followed by a two-digit transaction code (for example, F00, F04, A50, A65). F indicates the transaction is a financial transaction and A indicates the transaction is an administrative transaction.

Terminal Control Mode		
TTC	Description	Mandatory FIDs
F00	Normal purchase	p
F01	Preauthorization purchase	p
F02	Preauthorization purchase completion	p
F03	Mail or telephone order	p
F04	Merchandise return	p
F05	Cash advance	p
F06	Card verification	p
F07	Balance inquiry	p
F08	Purchase with cash back	p
F09	Check verification	p
F10	Check guarantee	p
F11	Purchase adjustment	p
F12	Merchandise return adjustment	p
F13	Cash advance adjustment	p
F14	Cash back adjustment	p

Terminal Control Mode		
TTC	Description	Mandatory FIDs
A50	Logon request	p
A51	Logoff request	p
A60	Close batch request	p
A61	Close shift request	p
A62	Close day request	p
A64	Clerk totals request	n and p
A65	Batch totals request	l and p
A66	Shift totals request	o and p
A67	Day totals request	m and p
A68	Retailer/Group totals request	p and y
A70	Read mail request	p
A71	Mail delivered request	p
A75	Send mail request	p
A90	Download request	p
A95	Handshake request	p

Host Control Mode		
TTC	Description	Mandatory FIDs
F00	Normal purchase	D,R, X, p, and z
F01	Preauthorization purchase	D, R, X, p, and z
F02	Preauthorization purchase completion	D, R, X, p, and z
F03	Mail or telephone order	D, R, X, p, and z
F04	Merchandise return	D, R, X, p, and z

Host Control Mode		
TTC	Description	Mandatory FIDs
F05	Cash advance	D, R, X, p, and z
F06	Card verification	p
F07	Balance inquiry	p
F08	Purchase with cash back	D, R, X, p, and z
F09	Check verification	D, R, X, p, and z
F10	Check guarantee	D, R, X, p, and z
F11	Purchase adjustment	D, R, X, p, and z
F12	Merchandise return adjustment	D, R, X, p, and z
F13	Cash advance adjustment	D, R, X, p, and z
F14	Cash back adjustment	D, R, X, p, and z
A50	Logon request	p
A51	Logoff request	p
A60	Close batch request	p
A61	Close shift request	p
A62	Close day request	p
A64	Clerk totals request	n and p
A65	Batch totals request	l and p
A66	Shift totals request	o and p
A67	Day totals request	m and p
A68	Retailer/Group totals request	p and y
A70	Read mail request	p
A71	Mail delivered request	p

Host Control Mode		
TTC	Description	Mandatory FIDs
A75	Send mail request	p
A90	Download request	p
A95	Handshake request	p

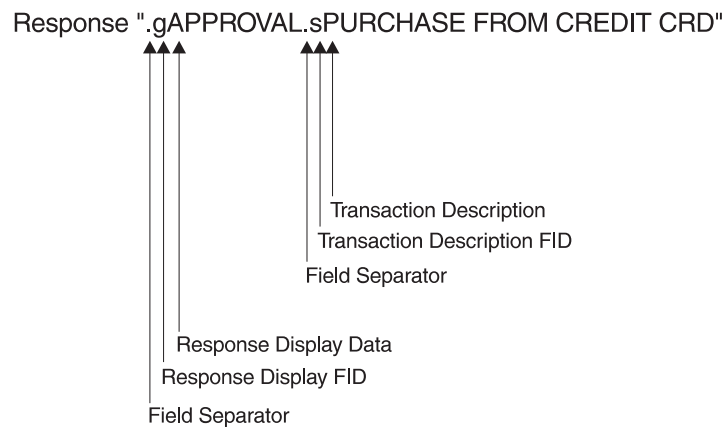
Response Field Examples

The following examples represent various response configurations. In the examples, a period (.) represents a field separator character (hex 1C). The response data shown below is assumed to follow the standard message header and to be terminated with an ETX in each situation. The quotation marks (“ ”) are used only to delimit the text and are not part of the response. Each example is set up in two parts. The first part identifies and explains the fields included in the response example. The second part provides a graphic to illustrate the response format.

Response Display and Transaction Description

The following is an example where response display data and a description of the transaction are included in the response. In the example, the response is formatted as follows:

Field	Description
g	FID associated with the Response Display field.
APPROVAL	The text associated with the response display set up in the Merchant Terminal Response File (MRSP)—maximum 48 characters.
s	FID associated with the Transaction Description field.
PURCHASE FROM CREDIT CRD	A description of the transaction developed for the MHI module for receipt purposes—maximum 24 characters.

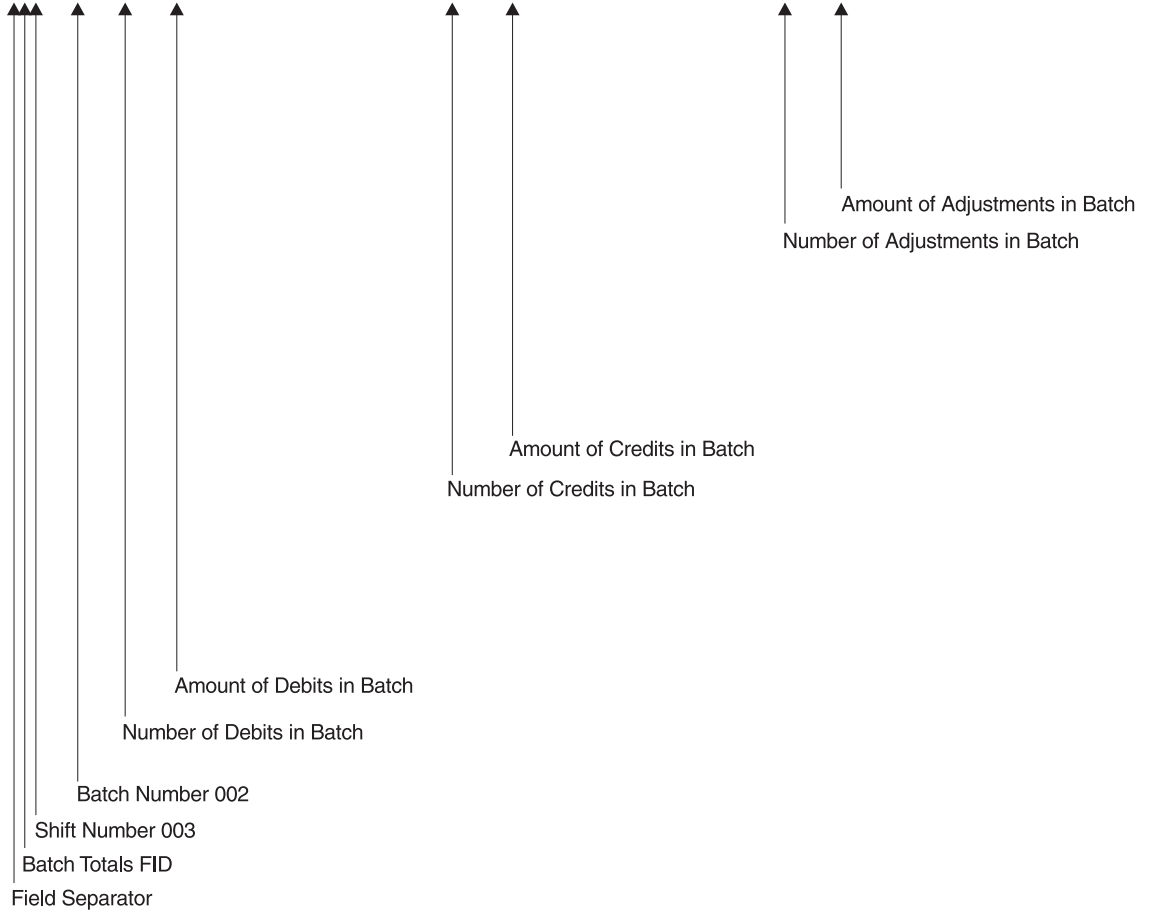


Batch Totals

The following is an example where batch totals are included in the response. In the example, the response is formatted as follows:

Field	Description
1	FID associated with the Totals/Batch field (PTDF.SHIFT, PTDF.BATCH, PTDF.SHIFT-DC, PTDF.BATCH-DC).
003	Shift number portion of the batch totals.
002	Batch number portion of the batch totals.
0023	Number of debits portion of the batch totals.
+123456789012345678	Amount of debits portion of the batch totals. All amount fields are preceded by a sign (+ or -).
0003	Number of credits portion of the batch totals.
+123456789012345678	Amount of credits portion of the batch totals. All amount fields are preceded by a sign (+ or -).
0001	Number of adjustments portion of the batch totals.
+123456789012345678	Amount of debits portion of the batch totals. All amount fields are preceded by a sign (+ or -).

Response ".I0030020023+1234567890123456780003+1234567890123456780001+123456789012345678"



Determining Transaction Codes

The MHI module determines the type of BASE24-pos transaction associated with a message using two fields in the standard header. These two fields are the MESSAGE TYPE and the TRANSACTION CODE (described previously in this section in the topic “Standard Message Header”). The MESSAGE TYPE field specifies the transaction as being either a financial transaction (F) or an administrative transaction (A). The TRANSACTION CODE field identifies the type of transaction, as determined by the terminal.

The BASE24-pos transaction code, in addition to indicating the type of transaction, identifies whether a debit or a credit card initiated the transaction, and the type of application account used when the transaction was initiated by a debit card. In the case of debit cards, the type of application account is required in the transaction code because debit cards can be associated with checking, savings, or credit card accounts.

The list below identifies the message types and transaction codes sent in the standard message header and the BASE24 transaction codes subsequently computed by the MHI module. The BASE24 transaction codes listed below include the letter *t* to indicate the position of the code identifying the type of card that generated the transaction. Also, the BASE24 transaction codes include the letters *aa* to indicate the position of the code that identifies the type of application account used in transactions initiated by debit cards. Refer to the ***BASE24-pos Transaction Processing Manual*** for a list of valid card type and account type codes.

Message Type	Terminal Transaction Code	Description	BASE24 Transaction Code
F	00	Normal purchase	10taa
F	01	Preauthorization purchase	11taa
F	02	Preauthorization purchase completion	12taa
F	03	Mail or telephone order	13taa
F	04	Merchandise return	14taa
F	05	Cash advance	15taa

Message Type	Terminal Transaction Code	Description	BASE24 Transaction Code
F	06	Card verification	16taa
F	07	Balance inquiry	17taa
F	08	Purchase with cash back	18taa
F	09	Check verification	19taa
F	10	Check guarantee	20taa
F	11	Purchase adjustment	21taa
F	12	Merchandise return adjustment	22taa
F	13	Cash advance adjustment	23taa
F	14	Cash back adjustment	24taa
A	50	Logon request	Not applicable
A	51	Logoff request	Not applicable
A	60	Close batch request	50taa
A	61	Close shift request	51taa
A	62	Close day request	52taa
A	64	Employee subtotals request	Not applicable
A	65	Batch subtotals request	Not applicable
A	66	Shift subtotals request	Not applicable
A	67	Day subtotals request	Not applicable
A	68	Retailer/Group totals request	Not applicable
A	70	Read mail request	56taa
A	71	Mail delivered request	Not applicable
A	75	Send mail request	58taa

Message Type	Terminal Transaction Code	Description	BASE24 Transaction Code
A	90	Download request	Not applicable
A	95	Handshake request	Not applicable

Section 5

Download Data

The Merchant Host Interface (MHI) module supports full and partial downloads to terminals. Downloads provide a way to transport and load the data necessary to configure a terminal. Configuration data downloaded to terminals is set up in the Merchant Terminal Configuration File (MCNF), the POS Terminal Data File (PTDF), and the POS Retailer Definition File (PRDF). This section provides the following information on MHI module download support:

- Downloading records
- Data elements and download field IDs (DIDs)
- Data element records
- Full download requests and responses
- Partial download requests and responses

Downloading Data to Terminals

Terminals must request download information from the Merchant Host Interface (MHI) module before they can receive any of the configuration data required for processing. Upon receiving a request for download information, the MHI module accesses the Merchant Terminal Configuration File (MCNF). This file contains up to 12 records for each terminal group. Eleven of these records contain configuration data that can be downloaded to the terminal. One of these records contains POS Terminal Data File (PTDF) fields and one field from the POS Retailer Definition File (PRDF) that can be included in the download.

Configuration data downloaded to the terminal from the MCNF records depends on the request sent from the terminal. Terminals can request a full download, which results in all of the configuration data contained in the MCNF records being sent to the terminal. Terminals can also request a partial download, asking for a single download field.

The data elements to be included within the MCNF records are determined by the customer. Information that can be included within these records includes telephone numbers, floor limits, and receipt data.

The information in these records is applicable for a group of terminals. There is one MCNF record for each terminal group. Terminal groups are identified in the TERM GROUP field on PTDF screen 1. When a download request is received from a terminal, the MHI module reads the PTDF to determine the group to which the terminal belongs. The MHI module then uses this value to access the appropriate records in the MCNF from which configuration data is to be taken.

Download Records

Each terminal group has up to 12 download records in the MCNF. They are designated with record identifiers of 00, 01, 02, 03, 04, 05, 06, 07, 08, 09, 10, and 11. Records 00, 01, and 02 contain 26 40-byte fields. Records 03, 04, 05, 07, 08, 09, 10, and 11 contain processing parameters for 30 card prefix ranges. Record 06 contains 26 flags, indicating what fields from the PTDF and PRDF are downloaded to the terminal. The download fields are set up in the MCNF using files maintenance functions. Refer to section 8 for more information on the download fields in the MCNF.

Downloading MCNF Records

The information in the MCNF can be downloaded to the terminal. If the RELEASE # field on screen 1 of the POS Terminal Data File (PTDF) is set to 50 (BASE24-pos release 5.0 or 5.1), the MHI module attempts to retrieve MCNF records in the following order: 00, 01, 02, 03, 04, 05, 07, 08, 09, 10, 11, and 06. Data is not required in each record in order for the MHI module to perform downloads. If the RELEASE # field is set to 34, the MHI module attempts to retrieve MCNF records 00, 01, 02, 03, 04, 05, and 06.

The type of information that an MCNF record contains determines which records the MHI module attempts to retrieve. For example, the MHI module always attempts to retrieve records 00, 01, and 02. If record 00 is found, the data from this record is downloaded and the MHI module attempts to retrieve record 01. If record 00 is not found, the MHI module does not download any records for record 00 and attempts to retrieve record 01. The MHI module also attempts to retrieve record 02 regardless of whether it finds record 00 or record 01.

After downloading records 00, 01, and 02, the MHI module downloads card prefix range data (record IDs 03, 04, 05, 07, 08, 09, 10, and 11). The MHI module reads these records until it finds the first blank card prefix range in the MCNF. The MHI module then stops searching for card prefix range data and moves to record 06.

For example, the MHI module locates record ID 03 and all four card prefixes are defined in the record. The MHI module formats this data in the download data message and attempts to locate record ID 04. If record ID 04 is not found, the MHI module attempts to locate record ID 06 (that is, the MHI module does not attempt to locate record IDs 05 and 07 through 11).

As another example, assume the MHI module locates record ID 03 and the third card prefix range in the record contains blanks. In this case, the MHI module downloads only the first two card prefix ranges, then stops attempting to locate card prefix ranges and moves to record ID 06.

Download Record Format

When a terminal requests a download, the response from the MHI module must include the Download Key field (FID V) and the Download Data field (FID W). The download key and the download data provide the processing mechanisms required in order for the MHI module to perform downloads. The functions of these fields include initiating the download, notifying the terminal that download data exists, and transporting the download text to the terminal. The following paragraphs describe the download key and the download text.

Note: The Merchant Host Interface module can place only the Download Key field (FID V) and the Download Data field (FID W) in the response message. The Merchant Host Interface module does not send the Authentication Code field (FID G) in the response, so you cannot use message authentication with a download response.

Download Key

The download key (FID V) is the first block of information formatted in the response to a download request. The download key is used to determine whether a full or partial download was requested.

If a full download is requested, the download key specifies whether this is an initial request or a continuation request. Continuation requests occur when the download is too large to fit into a single message and requires several request and response message exchanges in order to complete. If a partial download is requested, the download key identifies the download field requested.

Note: The fields associated with the download key are further explained later in this section in the topical discussions of “Full Downloads” and “Partial Downloads.”

The Download Key field precedes the Download Data field (FID W) in download responses from the MHI module to the terminal. A field separator (FS) of a period (.) is used to identify where the Download Key field begins and ends. The download key is formatted in the following manner:

F S	FID V	Category Code	Access Code	Processing Flag	Filler	F S
----------------	--------------	--------------------------	------------------------	----------------------------	---------------	----------------

The download key consists of the following fields:

Category Code — Indicates an initial download or whether the terminal requests additional download data.

The Category Code consists of two bytes. The first byte is used in responses and the second byte is used in requests.

In responses, the Category Code indicates whether all of a particular download field has been loaded or if more download data exists. If the value in the first byte of the Category Code is 1, this indicates to the terminal that more data exists for a particular download field. If the first byte of the Category Code is 0, this indicates to the terminal that no more data exists for the particular download field. In this case, the next field will be downloaded.

In requests, the Category Code indicates whether the terminal requests the remaining download data (as indicated in the response) or to begin receiving information from the next field to be downloaded. If the second byte of the Category Code is set to a value of 1, this indicates that the terminal is requesting the remaining data be sent for the download field which is identified as incomplete in the response (indicated by a value of 1 in the first byte of the Category Code). The terminal can continue to receive the balance of the data for the incomplete download field until no more data exists as long as the second byte of the Category Code is 1. When no more data exists, the next field will be downloaded. If the second byte of the Category Code is 0, this indicates that the terminal is requesting information from the next field to be downloaded.

Refer to appendix B for examples of how the category code works.

Access Code — Indicates initial or continuation download request. Valid values for downloads are as follows:

- 1 = Initial download request
- 2 = Download continuation request
- 5 = Read specific download field

Processing Flag — The download field ID (DID) of the current or next download field included in the response. For full downloads, the DID is echoed. For partial downloads, the DID is requested.

Filler — Ten bytes filled with zeroes. The value in this field is not used for downloading.

Download Text

The Download Data field (FID W) contains the download field ID (DID) and the download text associated with the DID. This is the configuration data sent to the terminal. The download text follows the download key in download responses from the MHI module to the terminal. Configuration data is taken from the appropriate MCNF records and formatted as shown below. A total of 82 data elements can be included within the format, although only 68 are currently supported (14 DIDs are reserved for future use). A field separator (FS) of a period (.) is used to identify where download text begins and ends. Within the Download Data field, every data element included is prefixed with a group separator (GS) character of a comma (,) and a download field ID (DID) character. All these data elements are included within the Download Data field.

F S	FID W	Destination DPC Number	Download Text	F S
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The download text consists of the following fields:

Destination DPC Number — This field is not used in downloads, but contains a value of 0.

Download Data Text — Contains the data elements from the MCNF being downloaded to the terminal. This includes the DID and the associated text.

Defining Data Elements

Data elements are defined and set up by customers in the MCNF. Each data element that can be downloaded to the terminal is identified with a Download Field ID (DID). The range of identified DIDs is A through Z, 0 through 29, and a through z.

Instructions for entering information in these download fields using files maintenance are included in section 8. The table shown below identifies the DIDs, specifies their location within the MCNF records, and provides a brief description of the type of information that can be entered for each DID.

DIDs	Record	Description
A, B, C, D, E, F, G, H, I, J	00	The DIDs (A through J) associated with record 00 can consist of any information the terminal owner and operator want to include or that the terminal vendor requires. This record is optional. Overall, DIDs A through Z can contain any type of data that the terminal requires.
K, L, M, N, O, P, Q, R, S, T	01	The DIDs (K through T) associated with record 01 can consist of any information the terminal owner and operator want to include or that the terminal vendor requires. This record is optional. Overall, DIDs A through Z can contain any type of data that the terminal requires.
U, V, W, X, Y, Z	02	The DIDs (U through Z) associated with record 02 can consist of any information the terminal owner and operator want to include or that the terminal vendor requires. This record is optional. Overall, DIDs A through Z can contain any type of data that the terminal requires.
0, 1, 2, 3	03	Record 03 contains card prefix ranges and associated card prefix range information. Up to 30 card prefix ranges can be defined in records 03, 04, 05, 07, 08, 09, 10, and 11. Record 03 contains data for up to four card prefixes.

DIDs	Record	Description
4, 5, 6, 7	04	Record 04 contains card prefix ranges and associated card prefix range information. Up to 30 card prefix ranges can be defined in records 03, 04, 05, 07, 08, 09, 10, and 11. Record 04 contains data for up to four card prefixes.
8, 9	05	Record 05 contains card prefix ranges and associated card prefix range information. Up to 30 card prefix ranges can be defined in records 03, 04, 05, 07, 08, 09, 10, and 11. Record 05 contains data for two card prefixes.
a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z	06	The DIDs (lower case a through z) associated with record 06 contain flags that indicate whether specific corresponding fields in the POS Terminal Data File (PTDF) and POS Retailer Definition File (PRDF) should be downloaded to the terminal. Presently, there are 26 flags, although 14 of these are reserved for future use.
10, 11, 12, 13	07	Record 07 contains card prefix ranges and associated card prefix range information. Up to 30 card prefix ranges can be defined in records 03, 04, 05, 07, 08, 09, 10, and 11. Record 07 contains data for up to four card prefixes.
14, 15, 16, 17	08	Record 08 contains card prefix ranges and associated card prefix range information. Up to 30 card prefix ranges can be defined in records 03, 04, 05, 07, 08, 09, 10, and 11. Record 08 contains data for up to four card prefixes.
18, 19, 20, 21	09	Record 09 contains card prefix ranges and associated card prefix range information. Up to 30 card prefix ranges can be defined in records 03, 04, 05, 07, 08, 09, 10, and 11. Record 09 contains data for up to four card prefixes.

DIDs	Record	Description
22, 23, 24, 25	10	Record 10 contains card prefix ranges and associated card prefix range information. Up to 30 card prefix ranges can be defined in records 03, 04, 05, 07, 08, 09, 10, and 11. Record 10 contains data for up to four card prefixes.
26, 27, 28, 29	11	Record 11 contains card prefix ranges and associated card prefix range information. Up to 30 card prefix ranges can be defined in records 03, 04, 05, 07, 08, 09, 10, and 11. Record 11 contains data for up to four card prefixes.

Records 00, 01, and 02

The data elements contained in records 00, 01, and 02 can consist of any information the terminal owner and operator want to include or that the terminal vendor requires. All of these records are optional.

The data elements are identified with the DIDs of A through Z. DIDs A through J are stored in record 00. DIDs K through T are stored in record 01. DIDs U through Z are stored in record 02.

Each of the DIDs can be associated with up to 40 alphanumeric characters entered in the Merchant Terminal Configuration File (MCNF) through BASE24 files maintenance. The data elements contained in records 00, 01, and 02 are only sent if they are requested by the terminal and if they contain data. If the terminal requests these data elements, but the data elements contain no data, they are not sent to the terminal. Refer to section 8 for more information about setting up download records 00, 01, and 02.

Records 03, 04, 05, 07, 08, 09, 10, and 11

The data elements contained in records 03, 04, 05, 07, 08, 09, 10, and 11 consist of card prefix range information defined by the terminal owner and operator. Up to 30 card prefix ranges can be defined in these records. Each card prefix range is identified with a DID of 0 through 29. The DIDs are stored in the following manner:

- DIDs 0 through 3 are stored in record 03
- DIDs 4 through 7 are stored in record 04
- DIDs 8 through 9 are stored in record 05
- DIDs 10 through 13 are stored in record 07
- DIDs 14 through 17 are stored in record 08
- DIDs 18 through 21 are stored in record 09
- DIDs 22 through 25 are stored in record 10
- DIDs 26 through 29 are stored in record 11

Data concerning the card prefix ranges is entered in the Merchant Terminal Configuration File (MCNF) through BASE24 files maintenance. The data elements in records 03, 04, 05, 07, 08, 09, 10, and 11 are only sent to the terminal if they are requested by the terminal and if they contain data. Refer to section 8 for more information about setting up records 03, 04, 05, 07, 08, 09, 10, and 11.

Note: For BASE24-pos release 5.0 and later, the MHI module can support up to 30 card prefix ranges. In prior releases, only 10 card prefix ranges were supported. In order for the MHI module to support up to 30 card prefixes, the RELEASE # field on PTDF screen 1 must be set to 50 (release 5.0). The RELEASE # field on PTDF screen 1 identifies the BASE24-pos message format used by the terminal. This field is used by the MHI module to determine the BASE24-pos release number of the message format the terminal understands. If the RELEASE # field on PTDF screen 1 is set to 50, the MHI module can download up to 30 card prefix ranges. If the RELEASE # field on PTDF screen 1 is not set to 50, only 10 card prefix ranges can be downloaded. In this situation, if an operator attempts to download more than 10 card prefix ranges (records 07 through 11), only the first 10 will be successful. In addition, a message will be logged indicating that 30 card prefix ranges cannot be downloaded for the release number specified in the RELEASE # field on PTDF screen 1. An 880 response code is also sent indicating no more data exists.

Card Prefix Data Element Structures

The following table describes the data element structure of the card prefix information included in DIDs 0 through 29. The table includes the positions in the DID occupied by each card prefix data element, a picture clause containing the field length and format of the card prefix data element, and the name of the card prefix data element. The card prefix information associated with records 03, 04, 05, 07, 08, 09, 10, and 11 consists of 108 alphanumeric characters per prefix, organized in the following format.

CARD PREFIX DATA ELEMENT STRUCTURES FOR RECORDS 03, 04, 05, 07, 08, 09, 10, AND 11		
Position	Picture	Data Element Name
1–11	PIC X(11)	Low Prefix
12–22	PIC X(11)	High Prefix
23–42	PIC X(20)	Main Network Telephone Number
43–62	PIC X(20)	Backup Network Telephone Number
63–82	PIC X(20)	Referrals Telephone Number
83–94	PIC X(12)	Retailer ID
95	PIC X(01)	Draft Capture
96	PIC X(01)	Totals
97	PIC X(01)	PIN
98	PIC X(01)	Receipt
99	PIC X(01)	Mod-Check
100	PIC X(01)	PAN Fraud Check
101–108	PIC X(08)	User-Defined Area

The card prefix data elements listed above are taken from records 03, 04, 05, 07, 08, 09, 10, and 11 in the MCNF. For descriptions of the card prefix data elements, refer to the “Screens for Download Records 03, 04, 05, 07, 08, 09, 10, and 11” topic documented in section 8.

Record 06

The data elements in record 06 consist of flags corresponding to fields in the POS Terminal Data File (PTDF) and the POS Retailer Definition File (PRDF). These flags are used to determine the PTDF and PRDF fields to be downloaded. Record 06 can store 26 download flags. DIDs a through z identify the 26 data elements that can be included in record 06, although only 12 of the data elements are currently supported. The other 14 DIDs are reserved for future use.

Flags concerning the data elements are set in the Merchant Terminal Configuration File (MCNF) through BASE24 files maintenance. The data elements in record 06 are sent to the terminal only if they are requested by the terminal and if they contain data. Refer to section 8 for more information about setting up download record 06.

Data Element Structures

The following table describes the structure of the data elements associated with DIDs a through z. The table includes the DID associated with each data element, a picture clause containing the field length and format of the data element, and the name of the data element. The data elements contained in record 06 are organized in the following format.

DID	Picture	Data Element Name
a	PIC X(25)	TERM-NAM-LOC
b	PIC X(16)	TERM-CITY-ST
c	PIC X(22)	TERM-OWNER-NAM
d	Not applicable	RESERVED
e	Not applicable	RESERVED
f	PIC X(88)	SRV-REP
g	PIC X(16)	P-KEY
h	PIC X(16)	A-KEY
i	PIC X(01)	PIN-PAD-CHAR
j	Not applicable	RESERVED

DID	Picture	Data Element Name
k	PIC X(1170)	SRV (OCCURS 30 TIMES)
l	PIC X(44)	FLOOR-LIMITS
m	Not applicable	RESERVED
n	Not applicable	RESERVED
o	Not applicable	RESERVED
p	PIC X(30)	ALLOWED-TRANS
q	PIC X(19)	RETAILER-ID
r	PIC X(40)	MERCHANT-NAME
s	PIC X(20)	REFERRAL PHONE #
t	Not applicable	RESERVED
u	Not applicable	RESERVED
v	Not applicable	RESERVED
w	Not applicable	RESERVED
x	Not applicable	RESERVED
y	Not applicable	RESERVED
z	Not applicable	RESERVED

Data Element Description Format

Descriptions of DIDs are included on the following pages in the format listed below. Explanations of the headings are included in the format.

DID	DATA ELEMENT NAME
------------	--------------------------

A brief description of the DID and its use follows the DID and DATA ELEMENT NAME, when appropriate. If the data element associated with the DID consists of a group of fields, each field is listed followed by its length.

The fields associated with the format shown above are described in the following paragraphs.

DID — The data element identifier (FID).

DATA ELEMENT NAME — The name of the data element associated with the DID.

Data Element Descriptions

The following pages provide descriptions of the data elements that can be taken from record 06 in the MCNF. The descriptions include any values associated with the data element. For elements that consist of multiple fields, the structure of the elements are shown below. Included for each field is the position of the field in the element, the field length, and a description of its contents if necessary.

a	TERM-NAM-LOC
----------	---------------------

The terminal name/description for printing on receipts and to comply with Regulation E. DID a is taken from the PTDF.

b	TERM-CITY-ST
----------	---------------------

The city and state in which this terminal is located for printing receipts and to comply with Regulation E. These elements are taken from the PTDF. The following fields are associated with DID b.

Position	Length	Description
01–13	13	CITY The city in which the terminal is located.
14–16	3	ST The state in which the terminal is located.

c	TERM-OWNER-NAM
----------	-----------------------

The name of the retailer who owns the terminal. DID c is taken from the PTDF.

d	RESERVED
----------	-----------------

DID d is reserved for future use.

e	RESERVED
----------	-----------------

DID e is reserved for future use.

f	SRV-REP
----------	----------------

The name, address, and telephone number of the service representative to contact when problems occur with this terminal. DID f is taken from the PTDF. The following fields are associated with this data element:

Position	Length	Description
01–25	25	NAM The name of the service representative.
26–50	25	ADDRESS The address of the service representative.
51–66	16	CITY-ST The city and state in which the service representative is located.
67–68	2	CNTRY The country in which the service representative is located.
69–88	20	PHONE The telephone number of the service representative.

g	P-KEY
----------	--------------

The PIN encryption key used by this terminal. DID g is taken from the PTDF if the MHI module is operating in terminal control mode. If the MHI module is operating in host control mode, the key is taken from the MHCF.

h	A-KEY
----------	--------------

The message authentication code (MAC) generation key used by this terminal. DID h is taken from the PTDF if the MHI module is operating in terminal control mode. If the MHI module is operating in host control mode, the key is taken from the MHCF.

i	PIN-PAD-CHAR
----------	---------------------

The character used to pad the PIN block. DID i is taken from the PTDF if the MHI module is operating in terminal control mode. If the MHI module is operating in host control mode, the key is taken from the MHCF.

j	RESERVED
----------	-----------------

DID j is reserved for future use.

k	SRV (OCCURS 30 TIMES)
----------	------------------------------

The card processing parameters for this terminal. These parameters are taken from the PTDF.

Note: Release 3.4 and release 5.0 and later releases of the MHI module support functionality that allows downloading of fields whose lengths typically exceed the maximum response length of the MHI module. The only field to which this option currently applies is DID k. The following paragraph describes the way in which the MHI module processes fields whose lengths exceed the maximum response length allowed by the MHI module.

When DID k is configured to be included in a full download to a terminal using release 3.4, release 5.0, or greater, the MHI module adds as many occurrences as will fit to the current response. The MHI module then sends the response to the terminal with the first byte of the Download Key field (FID V) set as an ASCII 1 to indicate to the terminal that more data remains in this field. Next, the terminal can

request the remaining portion of this field by changing the second byte of the Download Key field to an ASCII 1 and using the resulting download key in its next request, or the terminal can proceed to the next DID by echoing the download key unaltered. The following structure represents one occurrence of this data element. This data element can occur up to 30 times.

Position	Length	Description
01-02	2	TYP Indicates the type of card accepted at this terminal. Codes used in this field are either reserved by BASE24 or user-defined. Refer to the <i>BASE24-pos Files Maintenance Manual</i> for definitions of reserved card types.
03-11	9	NP-FLR-LMT The floor limit as used by BASE24, in whole currency units, for normal purchase transactions performed at this terminal for this card type.
12-20	9	CA-FLR-LMT The floor limit as used by BASE24, in whole currency units, for cash advance transactions performed at this terminal for this card type.
21-29	9	MO-FLR-LMT The floor limit as used by BASE24, in whole currency units, for mail or telephone order transactions performed at this terminal for this card type.
30-38	9	TRAN-LMT The transaction amount limit for this card type as used by BASE24, in whole currency units, for this terminal. Transactions for amounts exceeding this limit are denied. This limit does not apply to cards with VIP status.

Position	Length	Description
39	1	TRAN-PROFILE A value indicating the transaction profile. Valid values are as follows: 0 = Authorize only. 1 = Authorize and capture. 2 = Authorize and expect electronic follow up. 3 = Terminal determines data capture mode for each transaction.
1	FLOOR LIMITS	

The limit parameters for the terminal. DID 1 is taken from the PTDF. The following fields are associated with this data element:

Position	Length	Description
01-04	4	ADJ-LMT-CNT The maximum number of adjustment transactions allowed to be performed on this terminal for each terminal cutover-to-cutover period. When this limit is exceeded, a log message is generated.
05-22	18	ADJ-LMT-AMT The maximum amount, in whole and fractional currency units, that can be accepted at this terminal using adjustment transactions. This limit is invoked for each terminal cutover-to-cutover period. When this limit is exceeded, a log message is generated.
23-26	4	RETURN-LMT-CNT The maximum number of merchandise return transactions allowed to be performed on this terminal for each terminal cutover-to-cutover period. When this limit is exceeded, a log message is generated.

Position	Length	Description
27-44	18	RETURN-LMT-AMT The maximum amount, in whole and fractional currency units, that can be accepted at this terminal using merchandise return transactions. This limit is invoked for each terminal cutover-to-cutover period. When this limit is exceeded, a log message is generated.

m	RESERVED
----------	-----------------

DID m is reserved for future use.

n	RESERVED
----------	-----------------

DID n is reserved for future use.

o	RESERVED
----------	-----------------

DID o is reserved for future use.

p	ALLOWED-TRANS
----------	----------------------

The following fields are used to determine if a transaction is allowed at this terminal. Whenever a transaction is allowed, a 1 is placed in the appropriate field shown below. A zero indicates the transaction is not allowed. These fields are taken from the PTDF. The following list contains the name of the field controlling the transaction. The length of each field shown below is one byte.

Position	Transaction Description
1	NORMAL PURCHASE
2	PREAUTHORIZATION PURCHASE
3	PREAUTHORIZATION PURCHASE COMPLETION
4	MAIL/PHONE ORDER
5	MERCHANDISE RETURN

Position	Transaction Description
6	CASH ADVANCE
7	CARD VERIFICATION
8	BALANCE INQUIRY
9	PURCHASE WITH CASH BACK
10	CHECK VERIFICATION
11	CHECK GUARANTEE
12	PURCHASE ADJUSTMENT
13	MERCHANDISE RETURN ADJUSTMENT
14	CASH ADVANCE ADJUSTMENT
15	CLOSE BATCH
16	CLOSE SHIFT
17	CLOSE DAY
18	RESERVED
19	REQUEST RETAILER MAIL
20	SEND RETAILER MAIL (PASSTHRU)
21	SEND RETAILER MAIL (STORED)
22	SALES DRAFTS
23	CLERK TOTALS
24	RETAILER/GROUP TOTALS
25	CASH BACK ADJUSTMENT
26	ADJUSTMENTS WHEN AMT2>AMT1
27	RESERVED
28	RESERVED

Position	Transaction Description
29	RESERVED
30	RESERVED

q	RETAILER-ID
----------	--------------------

The retailer identifier of the retailer who owns this terminal, as defined in the RETAILER ID field on PTDF screen 1.

r	MERCHANT-NAME
----------	----------------------

The merchant name, as defined in the RETAILER NAME field on POS Retailer Definition File (PRDF) screen 1.

S	REFERRAL PHONE #
----------	-------------------------

The retailer referral telephone number, as defined in the REFERRAL PHONE field on PRDF screen 1.

t	RESERVED
----------	-----------------

DID t is reserved for future use.

u	RESERVED
----------	-----------------

DID u is reserved for future use.

v	RESERVED
----------	-----------------

DID v is reserved for future use.

w	RESERVED
----------	-----------------

DID w is reserved for future use.

x	RESERVED
----------	-----------------

DID x is reserved for future use.

y	RESERVED
----------	-----------------

DID y is reserved for future use.

z	RESERVED
----------	-----------------

DID z is reserved for future use.

Full Downloads

A full download allows a terminal to receive all of the configuration data it requires at once. The terminal can request a full download in order to receive all the configuration data specified in the 12 MCNF records. Responses to a full download request have as much of the 12 records included as will fit in a response. When a response to a download does not contain all the configuration data possible, a response code is included in the response message, indicating more download data exists. When the terminal receives a response with this type of response code, the terminal must send a continuation download request to the MHI module in order to receive more configuration data.

A continuation download request uses the same transaction code as the initial download request, but some of the fields associated with the Download Key field (FID V) are modified. Download key data consists of the following fields:

Category Code — Indicates an initial download or whether the terminal requests additional download data. For additional information and examples of how the Category Code field works, refer to appendix B.

Access Code — Indicates initial or continuation download request. Note that initial download can also be indicated by the absence of FID V. A DID is defined as a download field. Valid values for downloads are as follows:

- 1 = Initial download request
- 2 = Download continuation request
- 5 = Read specific download field

Processing Flag — The DID of the last download field included in the previous response.

Filler — Ten bytes filled with zeroes. The value in this field is not used for downloading.

Terminal Requests a Full Download

When the terminal requests a full download, The Download Key field (FID V) should contain the information shown in the table below. The download key consists of four fields, totaling 15 bytes. A value of 1 in the third byte of the Download Key field, the Access Code field, indicates that the terminal is requesting a full download for the first time. Fields A through Z are downloaded first, followed by the card prefix ranges and fields a through z.

FID V—DOWNLOAD KEY			
Field	Format	Value	Meaning/Origin
Category Code	PIC X(02)	Not applicable	Not applicable
Access Code	PIC X(01)	1	Read first download record
Processing Flag	PIC X(02)	Not applicable	Not applicable
Filler	PIC X(10)	0000000000	Not applicable

The example below illustrates the release 5.0 message format for a full download request from the terminal.

```
.V00100000000000000
```

The key describing the information contained in the example is as follows:

```
.      = Field separator
V      = FID V
00     = Category code
1      = Access code (1 indicates an initial download request)
00     = Processing flag
0000000000 = Filler
```

MHI Module Response to a Full Download Request

When the terminal sends a download request to the MHI module, the MHI module responds by sending a specific set of information in the download key back to the terminal. The Download key field (FID V) consists of four fields, totaling 15 bytes.

If the last field in the message response cannot fit into a single message based on the constraints imposed by the maximum response length configured in the MCNF, the MHI module changes the first byte of the Download Key field (FID V), the Category Code field, to a value of 1. This indicates to the terminal that the field must be sent in multiple messages. If all of the information in the last field of the message response has been downloaded to the terminal, the MHI module sets the Category Code field to 0. The second byte of the Category Code field is used in requests.

A value of 2 in the third byte of the Download Key field (FID V), the Access Code field, indicates that this is a response to a download request.

The fourth and fifth bytes, the Processing Flag field, contain the last DID included in the response. Following the download key, the response contains the download text, with up to 956 characters of download text from the MCNF.

The following tables show the information contained in the Download Key field (FID V) and the Download Data field (FID W) for a response from the MHI module to the terminal.

FID V—DOWNLOAD KEY			
Field	Format	Value	Meaning/Origin
Category Code	X(02)	00, 10	00 = Download for a DID completed. 10 = More information exists.
Access Code	X(01)	2	Continue full download (not initial request)
Processing Flag	X(02)	<i>DID</i>	Last DID in response; DIDs must be left-justified and blank filled (for example, “4 ”)

FID V—DOWNLOAD KEY			
Filler	X(10)	0000000000	Not applicable

FID W—DOWNLOAD TEXT			
Field	Format	Value	Meaning/Origin
DPC #	N/A	N/A	Not used
Download text	X(956)	N/A	Maximum = 956 characters

The example below illustrates the release 5.0 message format for a full download response from the MHI module to the terminal. Card prefixes are downloaded after DID Z.

.V0024 0000000000.W0,4b prefix 4b data

The key describing the information contained in the example is as follows:

.	=	Field separator
V	=	FID V
00	=	Category code (download for a DID completed)
2	=	Access code (2 indicates a download continuation request)
4	=	Processing flag
0000000000	=	Filler
,	=	Field separator
W	=	FID W
0	=	Fill byte (always ASCII 0)
,	=	Group separator
4b	=	DID 4, where b is a blank
<i>prefix 4b data</i>	=	Download data for card prefix range 4

Continuation of a Full Download Request

Continuation requests from the terminal are sent when the response received from the MHI module indicates that more download information needs to be retrieved. Continuations mean that the last field in the message is incomplete, and that the field cannot fit into a single message based on the constraints imposed by the maximum response length configured in the MCNF. Therefore, the field or remaining fields must be sent in multiple messages.

Upon receipt of a response with a response code indicating that more download information needs to be retrieved, the terminal sends a specific set of information in the download key.

If the last field in the message response cannot fit into a single message based on the constraints imposed by the maximum response length configured in the MCNF, the MHI module changes the Category Code field in the Download Key field (FID V.) The first byte of the Category Code is used in responses from the MHI module to the terminal. The second byte of the Category Code field is used in requests from the terminal to the MHI module. The following processing transpires during a continuation request from the terminal:

1. The MHI module receives a request from the terminal to send download data.
2. The MHI module changes the first byte of The Download Key field (FID V), the Category Code field, to a value of 1 in the response to the terminal. This indicates to the terminal that the field must be sent in multiple messages.
3. If the terminal wants the remaining information to be sent, it changes the second byte of the Category Code field to 1. If the terminal does not require the remaining information for the last field in the response, the second byte of the Category Code field is 0. It formats the request and sends it to the MHI module.
4. The MHI module receives the request. If the terminal wants the remaining download information, the MHI module checks to determine if the remaining information can fit into the response. If it can, the MHI module changes the value of the Category Code to 0, indicating no more download data exists. If it cannot, the MHI module leaves the Category Code value at 1, indicating to the terminal that more download data exists.

This request/response sequence continues until all of the download information requested by the terminal has been sent.

Once all of the information has been downloaded for the DID, the MHI module changes the Category Code to 00, indicating the download for the DID is complete. In cases where the message response can fit into a single message, the download key is echoed in the request from the terminal.

A value of 2 in the third byte of the Download Key field, the Access Code field, indicates that this is not an initial download request, but rather a continuation request. The last DID included in the response is returned to the MHI module. This information is echoed from the response from the MHI module.

For more information on the Category Code and an example of how it works during a continuation request, refer to appendix B.

The following table shows the information contained in the Download Key field for a continuation request from the terminal.

FID V—DOWNLOAD KEY			
Field	Format	Value	Meaning/Origin
Category Code	PIC X(02)	00, 10, 11	00 = Download for a DID completed. 10 = No more data exists, or more information exists but the terminal does not request it. 11 = The terminal requests the remaining information.
Access Code	PIC X(01)	2	Read next
Processing Flag	PIC X(02)	<i>DID</i>	Last DID downloaded; DID must be left-justified and blank filled (for example “Z ”)
Filler	PIC X(10)	0000000000	Not applicable

The following example illustrates the release 5.0 message format for a full download continuation request from the terminal. The last field downloaded to the terminal in its entirety, as indicated by the Category Code, was DID Z.

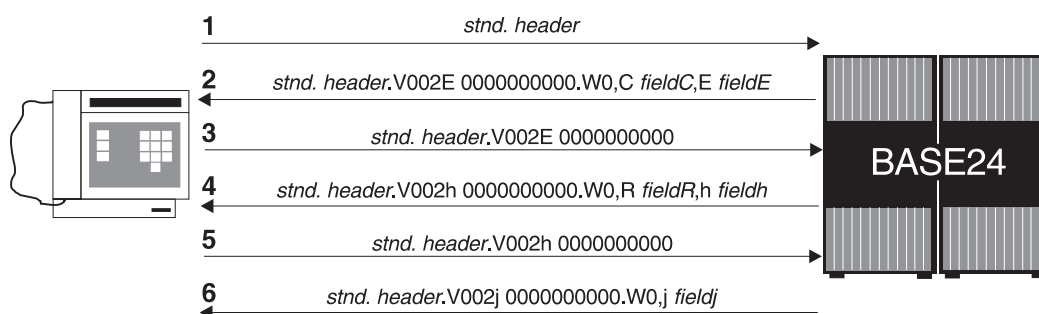
```
.V002Z 0000000000
```

The key describing the information contained in the example is as follows:

.	= Field separator
V	= FID V
00	= Category code (download for a DID completed)
2	= Access code (2 indicates a download continuation request)
Z ␣	= Processing flag, where ␣ is a blank
0000000000	= Filler

Full Download Example

The example shown below is of a full download. The data shown is assumed to contain a standard header and be terminated with an ETX. The period (.) represents a field separator character that is a hexadecimal representation of 1C. The comma (,) represents a group separator character that is a hexadecimal representation of 1D. Explanations of the messages are included below.



1. The terminal makes a download request with no optional fields.
2. The MHI module, finding no download key (that is, FID V), assumes this to be a full download request. The MHI module formats (into the download text field) as much download data as fits in one response message. In the example, not all data fits in one response. The MHI module sets the Response Code field (FID p) in the standard message header to 881, indicating more data exists.

3. The terminal processes the response. The response code of 881 in the standard message header indicates to the terminal that more download data exists. Therefore, the terminal makes another download request, this time including the download key as received in the prior response.
4. The MHI module uses the DID in the Processing Code to reinstate itself into the download process at the point it left off, and formats another response with download text and key.
5. The terminal processes the response. The response code of 881 in the standard message header indicates to the terminal that more download data exists. Therefore, the terminal makes another download request, this time including the download key as received in the prior response.
6. The above cycle continues until the remaining download data fits within the response message to the terminal. When the last data has been included, the MHI module sets the response code to 880, indicating no more data exists. Only those download fields containing values are downloaded. Download fields that do not contain information are not sent to the terminal. In this example, DIDs C, E, R, h, and j were downloaded to the terminal. The fields downloaded depend on the configuration in the Merchant Terminal Configuration File (MCNF). Download fields are set up using record 06 in the MCNF and only fields that contain data are downloaded to the terminal.

Partial Downloads

The terminal can also make a request to the MHI module for a partial download. A partial download consists of sending one item to the terminal. This capability is intended for use whenever a single field, such as a telephone number, has changed. In this case, operators of the affected terminals can be notified by mail that they should make a partial download request for the modified field.

The difference between a full and a partial download request is that the partial download request must contain the Download Key field (FID V) to indicate the following:

Category Code — A partial load may be continued in the same manner as a full download if the field exceeds the maximum response length. For more information on how the Category Code is used with partial downloads and an example, refer to appendix B.

Access Code — A value of 5 is placed in the Access Code to request to read a specific download field. This indicates a partial download.

Processing Flag — The DID of the download field requested. The DID must be left-justified and blank filled.

Filler — Ten bytes filled with zeroes. The value in this field is not used for downloading.

The response to a valid partial download request looks just like the response to a full download request with a response code of 880, indicating that no more data exists. In addition, partial download requests can require multiple response messages to return all of the requested data.

Partial Download Request

The terminal formats a download request that includes the Download Key field (FID V) indicating which download field is desired. The following table shows the information contained in the download key for a partial download request from the terminal.

FID V—DOWNLOAD KEY			
Field	Format	Value	Meaning/Origin
Category Code	PIC X(02)	00 or 11	00 = Download for DID completed. 11 = Terminal requests remaining information for a DID.
Access Code	PIC X(01)	5	Read specific FID
Processing Flag	PIC X(02)	<i>DID</i>	DID to be downloaded; DID must be left-justified and blank filled (for example, “r ”)
Filler	PIC X(10)	0000000000	Not applicable

The example below illustrates the release 5.0 message format for a partial download request from the terminal.

```
.V005r 0000000000
```

The key describing the information contained in the example is as follows:

```
.      = Field separator
V      = FID V
00     = Category code (download for a DID completed)
5      = Access code (5 indicates to read a specific download field)
r/     = Processing flag (DID r), where / is a blank
0000000000 = Filler
```

MHI Module Response to Partial Download Request

The MHI module uses the Download Key (FID V) to determine that the request is for a partial download. The MHI module accesses the requested field and responds to the terminal with that field and a response code of 880, indicating that no more data exists. The following tables show the information contained in the download Key field and the Download Data field (FID W) for a response from the MHI module to the terminal.

FID V—DOWNLOAD KEY (RESPONSE KEY)			
Field	Format	Value	Meaning/Origin
Category Code	PIC X(02)	00, 10, 11	00 = Download for a DID completed. 10 = No more data exists, or more information exists but the terminal does not request it. 11 = The terminal requests the remaining information for a DID.
Access Code	PIC X(01)	5	Read next (not initial request)
Processing Flag	PIC X(02)	<i>DID</i>	Last DID in response; DID must be left-justified and blank filled (for example “r ”)
Filler	PIC X(10)	0000000000	Not applicable

FID W—DOWNLOAD TEXT			
Field	Format	Value	Meaning/Origin
DPC #	0000000000	N/A	Not used
Text	PIC X(956)	N/A	Maximum = 956 Characters

The example below illustrates the release 5.0 message format for a partial download response from the MHI module to the terminal.

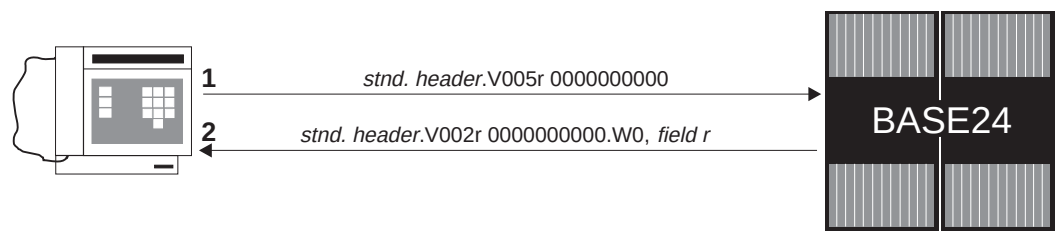
```
.V005r 0000000000.W0,r field r
```

The key describing the information contained in the example is as follows:

.	= Field separator
V	= FID V
00	= Category code (download for a DID completed)
5	= Access code (5 indicates to read a specific download field)
r 	= Processing flag (DID r), where is a blank
0000000000	= Filler
.	= Field separator
W	= FID W
0	= Fill byte (always ASCII 0)
,	= Group separator
r 	= DID r (requested DID), where is a blank
field r	= Download data for DID r

Partial Download Example

The example shown below is of a partial download. The data shown is assumed to contain a standard header and be terminated with an ETX. The period (.) represents a field separator character, with a comma (,) representing a group separator character. Explanations of the messages are included below.



1. The terminal makes a download request that includes the download key field, indicating which download field is desired. In this example, DID r is requested.
2. The MHI module uses the download key to indicate that this is a partial download. The MHI module accesses the requested field and responds to the terminal with just that field and a response code of 880 indicating that no more data is to be downloaded. The response code is carried in the message in the Response Code field (FID p).

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Section 6

BASE24-pos Merchant Totals Processing

The Merchant Host Interface (MHI) module supports the accumulation of retailer and group totals for terminals connected to the merchant host. Totals accumulation is performed by the Merchant Totals process. The Merchant Totals process is separate from the MHI module and its exclusive purpose is to provide retailer and group totals to the merchant host.

When the MHI module receives a valid request for retailer or group totals, it accesses the appropriate Merchant Host Configuration File (MHCF) record to retrieve the name of the Merchant Totals process. The MHI module then routes the request directly to the specified Merchant Totals process. When the Merchant Totals process receives the request, it calculates the totals and includes them in the response the Merchant Totals process sends, by way of the MHI module, to the merchant host.

The remainder of this section discusses the following topics concerning the Merchant Totals process:

- File usage
- Initialization and warmboot procedures
- Totals processing

File Usage

The Merchant Totals process uses the following files during processing. Each file listed is followed by a brief description explaining how it is used by the Merchant Totals process.

Log Message Configuration File — The Log Message Configuration File (LMCF) contains log message information such as text, routing code, and severity that have been modified by the institution. At initialization, the Merchant Totals process builds a table in memory to store the modified LMCF records. When a log message is generated, the Merchant Totals process accesses the table to verify whether the generated log message has been modified. If so, the Merchant Totals process logs the modified log message. Refer to the ***XPNET BASE24 Configuration Guidelines*** for information on modifying the LMCF.

Merchant Totals Configuration File — The Merchant Totals Configuration File (MTCF) contains records defining totaling groups for the purpose of obtaining group totals on a retailer or group totals request transaction. One record exists for each terminal within a totaling group. This allows an unlimited number of terminals to be included in one totaling group. Refer to section 8 for information about modifying the MTCF.

POS Retailer Definition File — The Merchant Totals process accesses the POS Retailer Definition File (PRDF) when processing transactions to retrieve the posting date. The posting date is compared with the business date in the request message to determine whether the totals should be extracted from the POS Transaction Log File (PTLF) or the POS Terminal Data File (PTDF). Refer to the ***BASE24-pos Files Maintenance Manual*** for information about modifying the PRDF.

POS Terminal Data File — At initialization, the Merchant Totals process builds a table of all the merchant host terminals assigned to each retailer. During transaction processing, the Merchant Totals process may accumulate the PTDF totals if the business date in the request matches the current posting date in the PRDF. Refer to the ***BASE24-pos Files Maintenance Manual*** for information about modifying the PTDF.

POS Transaction Log File — The POS Transaction Log File (PTLF) contains the totals records for each terminal if that terminal has been cut over. These totals are used by the Merchant Totals process to accumulate the totals for a retailer and group subtotals request. Refer to the ***BASE24-pos Files Maintenance Manual*** for information about modifying the PTLF.

Merchant Totals process Initialization and Warmbooting

The following procedure describes the steps taken by the Merchant Totals process to initialize:

1. Opens the Logical Network Configuration File (LCONF) and retrieves the following assigns:
LMCF
POS-PTLF
POS-PTDF
POS-PRDF
POS-MHI-MTCF
2. Initializes the logging process with the name of the specific Merchant Totals process being initialized and reads the Log Message Configuration File (LMCF).
3. Opens the POS Retailer Definition File (PRDF) and sets the file mode so any process attempting to lock or read a PRDF record while the Merchant Totals process has the record locked will receive an error. If the Merchant Totals process is unable to open the PRDF, the Merchant Totals process abends.
4. Opens the POS Terminal Data File (PTDF) and sets the file mode so any process attempting to lock or read a PTDF record while the Merchant Totals process has the record locked will receive an error. If the Merchant Totals process is unable to open the PTDF, the Merchant Totals process abends.
5. Opens the Merchant Totals Configuration File (MTCF) and sets the file mode so any process attempting to lock or read a MTCF record while the Merchant Totals process has the record locked will receive an error. If the Merchant Totals process is unable to open the MTCF, the Merchant Totals process abends.
6. Allocates extended memory space for the number of PTDF and MTCF records.
7. Allocates extended memory for the retailer and totaling group tables.
8. Sets up extended memory pointers used to retrieve totals from the PTDF and MTCF.
9. Builds a table of all the merchant host terminals for each retailer by sequentially reading the PTDF.

The retailer table is built by reading the PTDF. Terminal IDs that belong to the same retailer are grouped together with their associated retailer as PTDF records are read.

10. Builds a totaling group table for subsets of terminals for each retailer by sequentially reading the MTCF. The totaling group table links terminals with the Retailer ID and TOTALLING GROUP ID defined in the MTCF.

Warmbooting the Merchant Totals process

When the Merchant Totals process receives a WARMBOOT command, it performs the initialization steps described earlier with two exceptions.

- Step 2 in the initialization procedures indicates the Merchant Totals process initializes the logging process with the name of the specific Merchant Totals process being initialized. This step is not performed by the Merchant Totals process when it receives a WARMBOOT command.
- When the Merchant Totals process receives a WARMBOOT command, the Merchant Totals process deallocates the extended memory that was previously allocated for the PRDF, PTDF, and MTCF. The Merchant Totals process then closes these files. The MHI module then performs steps 3 through 10 described in the initialization procedures.

Totals Processing

Totals processing is performed in two stages: request validation and totals accumulation. The following topics describe how the Merchant Totals process handles retailer and group subtotals requests, accumulates the appropriate totals, and responds to the retailer and group subtotals request.

Request Processing

The Merchant Host Interface (MHI) module supports retailer/group subtotals request transactions. This transaction is sent from the merchant host to the MHI module and then routed to the Merchant Totals process. The request is received by the Merchant Totals process in the following format:

Position	Field Name	Description
1–4	MSG-TYP	Identifies the message type. Valid values are as follows: 0600 = Request message 0610 = Response message
5–23	RETAILER-ID	Identifies the retailer ID. The retailer ID is required in request messages and included in response messages.
24–39	TERMINAL-ID	Identifies the terminal ID. The terminal ID is required in request messages and included in response messages.
40–49	TTL-GRP-ID	Identifies the totaling group. The totaling group is required in request messages and included in response messages.
50–55	BUS-DATE	Specifies the business date. The business date is required in request messages and included in response messages.

Position	Field Name	Description
56–58	RESP-CDE	Identifies the response code present in the response message. Valid values are as follows: 001 = Request processed okay 150 = Invalid retailer ID 837 = Invalid business date 860 = Invalid totaling group 861 = Invalid terminal ID 862 = Partial totals received 863 = Totals overflowed 864 = Invalid PTLF name 865 = Open error on PTLF
59–62	NUM-TERMS-IN-GRP	Indicates the number of terminals in the totaling group. The NUM-TERMS-IN-GRP field is set in response messages.
63–66	NUM-TERMS-IN-TTL	Indicates the number of terminals that were processed for this request. If the value does not match the NUM-TERMS-IN-GRP value, this is a partial totals response (that is, not all of the PTDF records could be accessed to calculate the correct value). The NUM-TERMS-IN-TTL field is set in response messages.
67–70	DB-CNT	Indicates the number of debit transactions performed for this totaling group. The DB-CNT field is set in response messages.
71–88	DB-AMT	Indicates the amount of debit transactions performed for this totaling group. The DB-AMT field is set in response messages.
89–92	CR-CNT	Indicates the number of credit transactions performed for this totaling group. The CR-CNT field is set in response messages.

Position	Field Name	Description
93–110	CR-AMT	Indicates the amount of credit transactions performed for this totaling group. The CR-AMT field is set in response messages.
111–114	ADJ-CNT	Indicates the number of adjustment transactions performed for this totaling group. The ADJ-CNT field is set in response messages.
115–132	ADJ-AMT	Indicates the amount of adjustment transactions performed for this totaling group. The ADJ-AMT field is set in response messages.

Only the message type, terminal ID, associated retailer, totaling group, and business date are set in request messages, but are included in both requests and responses. All of the other fields are set in response messages.

Request Validation

In order for a request to be processed, the totaling group must exist in the Merchant Totals Configuration File (MTCF) and it must be associated with the retailer specified in the request. If it is not, response code 860 (Invalid totaling group for the retailer) is returned. If the totaling group is set to ***** in the request, the request is for all the terminals associated with the retailer.

In addition, the terminal requesting the subtotals request must belong to the totaling group. If it is not, response code 861 (Invalid totaling group for the terminal) is returned.

Totals Accumulation

The totals accumulated by the Merchant Totals process represent the end-of-day draft capture totals for the specified totaling group and business date. The method used to accumulate the totals depends on whether the business date in the request message matches the current posting date in the POS Retailer Definition File (PRDF). The methods for accumulating totals are described in the following paragraphs.

Method 1: Requested Business Date Does Not Equal Current PRDF Posting Date

Under this method, the business date in the request does not equal the posting date in the PRDF. In this case, the Merchant Totals process is able to process all terminal totals (that is, since the dates do not match, the Merchant Totals process assumes cutover has taken place for all terminals) and the Merchant Totals process accumulates totals in the following manner:

1. Initializes the response totals to zero.
2. Locates the totaling group record in the extended memory tables.
3. Locates the PRDF record for the requested retailer and retrieves the current posting date.
4. Opens the POS Transaction Log File (PTLF) for the requested business date.
5. Reads the network record for the first terminal ID in the totaling group record in extended memory. The key used to read the record in the PTLF is as follows:

Terminal ID = The terminal ID taken from the terminal ID table in extended memory
Record type = 3 (network totals)
Retailer ID = Retailer ID from the request
6. If the network totals record is found, the Merchant Totals process adds the draft capture totals to the response totals and updates the number of terminals in the totals field.
7. Repeats steps 5 and 6 for all terminal IDs included in the totaling group specified.

Method 2: Requested Business Date Equals Current PRDF Posting Date

Under this method, the business date in the request equals the posting date in the PRDF. In this case, the Merchant Totals process may be unable to process all terminal totals (that is, since the dates match, the Merchant Totals process assumes cutover has not taken place for all terminals and totals may be pending) and the Merchant Totals process accumulates totals in the following manner:

1. Initializes the response totals to zero.
2. Locates the totaling group record in the extended memory tables.

3. Locates the PRDF record for the requested retailer. Retrieves the current posting date.
4. Opens the PTLF for the requested business date.
5. Reads the network record for the first terminal ID in the totaling group record in extended memory. The key used to read the record in the PTLF is as follows:

Terminal ID = The terminal ID taken from the terminal ID table in extended memory
Record type = 3 (network totals)
Retailer ID = Retailer ID from the request
6. If the network totals record is found, the Merchant Totals process adds the draft capture totals to the response totals and updates the number of terminals in the totals field.
7. If the record for the network totals is not found in the PTLF, the Merchant Totals process assumes the MHI module is still collecting the network totals in the PTDF (that is, the terminal has not been cut over). The following steps are then taken:
 - a. Reads the PTDF record for the specified terminal ID. If the record is successfully read, the Merchant Totals process adds the PTDF current network draft capture totals to the response totals and updates the number of terminals in the totals field.
 - b. Adds the totals if the record is read successfully. However, if the record is locked, the Merchant Totals process waits two seconds and attempts to read the record again. If the record is still locked, the Merchant Totals process waits two more seconds and attempts to read the record again. If the record still cannot be read, neither the totals nor the number of terminals in the totals field are updated.

If not all the terminals were processed, the response message indicates the totals returned are partial totals for the specified group.
8. Repeats steps 5, 6, and 7 for all terminal IDs included in the specified totaling group.

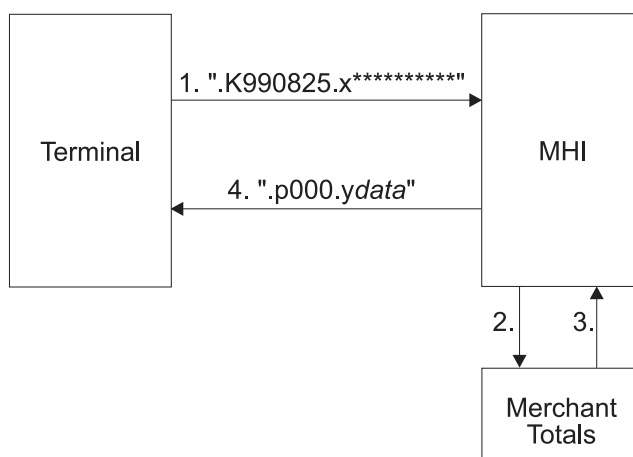
Response Processing

After all of the totals for the specified totaling group have been accumulated, the Merchant Totals process sends a response with a 0610 message type to the MHI module. The response contains the retailer ID, totaling group ID, and terminal ID along with the amounts and counts calculated by the Merchant Totals process.

In addition, the response contains a response code indicating the outcome of the processing. Response codes identify conditions such as validation errors, PTDF record read errors, or successful accumulation of all totals for the specified totaling group.

Example

The following example is a retailer subtotals request for all terminals associated with the specified retailer. The Merchant Totals process is responsible for accumulating the totals.



1. The terminal makes a request for the retailer subtotals for the business date (in the example, 990825 is August 25, 1999). The business date is used in accumulating retailer or group totals. In this example, retailer totals are accumulated for all terminals. This is specified by ***** in FID x.
2. The MHI module reads the POS Terminal Data File (PTDF) to retrieve retailer information, and formats a request to the Merchant Totals process with a message type of 0600. The MHI module sends the request to the Merchant Totals process specified in the Merchant Host Configuration File (MHCF) record for the DPC. The PTDF record should not be locked.

3. The Merchant Totals process uses the retailer ID in the request to read the POS Retailer Definition File (PRDF) for retrieval of the appropriate network records to be totaled. The Merchant Totals process then responds to the MHI module with a message type of 0610.
4. The MHI module formats the totals returned by the Merchant Totals process and returns the totals to the terminal in optional FID y. FID y contains the following data:

Number of terminals in the group
Number of terminals in the totals
Debit count
Debit amount
Credit count
Credit amount
Adjustment count
Adjustment amount

Section 7

MHI Module Configuration Considerations

Besides setting up files, customers are responsible for making decisions concerning several additional configuration issues. These issues determine the way the MHI module processes transactions. This section discusses the following configuration decisions the customer must make when using the MHI module:

- Configurable receipts
- Draft capture
- Message sequencing
- Transaction accumulation totals
- PIN encryption
- Message authentication codes (MACs)
- Handshaking
- BASE24-mail support

Configurable Receipts

The MHI module does not format receipts. Instead, it is the customer's responsibility to configure each response in the terminal configuration data to include sufficient data for the terminal to format and print a receipt.

However, the MHI module has the capability of returning a response determined by the customer in up to three different languages as well as returning a free format 48-character response. This is set up in the Merchant Terminal Response File (MRSP).

This subsection provides information on the following topics concerning receipts:

- Receipt information
- Response considerations
- Optional responses

Receipt Information

Although it is the customer's responsibility to determine the information to be included on receipts, several suggestions follow.

Standard Message Header Fields

The standard header fields typically printed on a receipt include the following:

- Terminal ID
- Employee ID
- Date
- Time
- Transaction code
- Response code

Optional Data Fields

The optional data fields typically printed on a receipt include the following:

Field Identifier (FID)	Field Name
B	Amount 1
C	Amount 2
E	Application account number
F	Approval code
J	Available balance (on balance inquiry transactions)
K	Business date (Terminal)
N	Customer ID
Q	Echo data
S	Invoice number
T	Original invoice number
d	Retailer ID
h	Sequence number
k	Terminal location
s	Transaction description

With the exception of FID s (Transaction Description), any of the FIDs described above can be configured in the Merchant Terminal Response File (MRSP) to be returned to the terminal in FID g (Response Display). FID g is an optional, free-format field of up to 48 characters. It can be used to return either text or other FID information. For example, it could be used to return the terminal business date and terminal location. Since the contents of FID g can be configured differently for each response code, it can serve as a tool to easily provide the necessary receipt information.

For more information on optional data fields, refer to section 4.

Response Considerations

Although the customer and the terminal vendor are responsible for determining what the receipt looks like, several aspects of the receipts must be set up through BASE24 software. These include setting parameters so the MHI module knows which response to return to the terminal and developing the exact text of the responses to be sent.

Language Code

The MHI module uses the request first to determine the language in which to return responses. Specifically, the MHI module checks for the presence of the language code field identified with FID U. If present, the MHI module checks the value in this field. If this field is not included in the request, the MHI module uses the default language code set in the POS Terminal Data File (PTDF).

Therefore, the customer is required to set a default language code in the PTDF. If a specific language code is required for certain transactions or for any other reason, the customer needs to include FID U in the request message. For consistency, the Merchant Terminal Configuration File (MCNF) should also be set appropriately, indicating when FID U is going to be included in requests.

Language Index and Responses

The MHI module uses the language code, along with the BASE24-pos response code returned from the Authorization module, to determine the response to send to the terminal.

When the MHI module receives the response code from the Authorization module, the MHI module accesses the MRSP using the terminal group identified in the PTDF or the default terminal group (****) and record number 00. An index in record 00 of the MRSP links each BASE24-pos response code to a number. The MHI module then uses this number as an offset into the appropriate language table to determine the correct response. The language tables are located in records 01, 02, and 03 of the MRSP.

Customers must enter the appropriate responses in the language tables and the correct numbers in the index in the MRSP for the response to be accurate. Instructions on entering this information are included in section 8.

Optional Responses

If the MCNF is set to include FID g in responses to the terminal, the MHI module returns a customer-configured response to the terminal. Although these responses are typically used to inform the terminal operator of processing results, the response text is optional and up to the discretion of the customer. The only restriction on these responses is that they not exceed 48 characters. If the response is fewer than 48 characters, trailing spaces are removed by the MHI module.

These customer-configured responses are based on the response code. For example, different responses can be made for each transaction type. Information on entering these responses is included in section 8. BASE24 also provides for the insertion of variable data fields into the response display using the MRSP. Up to the maximum field length, one or more of the following optional or download data fields can be included in any display. Refer to section 4 for more information about the length of each field.

FID	FID Name
B	Amount 1
C	Amount 2
E	Application account number
F	Approval code
J	Available balance (on balance inquiry transactions)
K	Business date (Terminal)
N	Customer ID
Q	Echo data
S	Invoice number
T	Original invoice number
d	Retailer ID
h	Sequence number
k	Terminal location

Draft Capture

BASE24-pos allows for processing retailer transactions as draft capture or non-draft capture transactions. Non-draft capture transactions are transactions for which the paper draft represents the actual settlement instrument. Paper drafts must be physically presented for deposit in order for the transaction to be settled. While the transaction may have been authorized by BASE24-pos, the BASE24-pos record cannot be used to settle the item.

Draft capture transactions are electronically captured at the time of the transaction and are sufficient in themselves to enact a funds transfer for the transaction amount. BASE24-pos identifies draft capture transactions in its files and includes draft capture totals on its retailer reports so that the retailer sponsor can settle with the retailer.

Draft capture is a settlement tool only. It has no impact on the authorization of transactions. Cardholders are still checked against the same authorization criteria without regard for whether a transaction is draft capture or non-draft capture. Thus, the retailer is not incurring additional risk of cardholder fraud by using or not using draft capture.

Draft Capture Options

Customers using the Merchant Host Interface (MHI) module have three draft capture options available to them. These are as follows:

- Authorization only with paper draft follow-up
- Authorize and draft capture
- Terminal-defined draft capture

Different draft capture options can be selected for each terminal and card type. Customers set the TP (transaction profile) field on PTFD screens 8, 9, and 10 to indicate the draft capture option they have selected for each terminal and card type. Valid values are as follows:

- 0 = Authorize only with paper draft follow-up.
- 1 = Authorize and draft capture.
- 3 = Terminal determines draft capture profile.

Although the value of 2 is generally acceptable in this field, the MHI module does not support the value. The value in the PTDF is used by the MHI module, unless the value in the PTDF is a 3, indicating the terminal is to determine the draft capture value.

If the terminal is configured to determine the draft capture option, the MHI module uses the value sent in FID P (Draft Capture Flag) of the request. Valid values are as follows:

- 0 = Authorize only with paper follow-up
- 1 = Authorize and draft capture

FID P is the TP (transaction profile) field taken from the PTDF. If the value in FID P is set to a 1, then one-step transactions with financial impact are processed in an authorization and draft capture mode. Check guarantee, check verification, card verification and balance inquiry transactions have a draft capture profile of 0, if FID P is set to a value of 1.

The draft capture profile code used by the MHI module is the one returned to the terminal in the response.

Authorization Only With Paper Draft Follow-up

The authorization only with paper draft follow-up method of draft capture obtains online authorization for the amount of the customer purchase or other type of data, depending on the transaction. In an authorization only environment, the card issuer waits for paper follow-up before posting the transaction and settling with the retailer. If a card type is set up to use this option, none of the transactions for the card type from the specified terminal are draft capture.

With this method, BASE24 updates the cardholder usage and activity totals, logs transactions to the PTLF, and updates the service totals and authorization totals in the PTDF.

The host waits for paper to post and settle with retailer and on-us cardholder accounts.

The value in the PTDF must be set to 0 or 3 with the terminal sending 0, indicating to authorize with paper follow-up.

Authorization and Draft Capture

The authorization and draft capture option is a one-step process that allows settlement to occur without paper follow-up. This option performs an authorization on the transaction and at the same time automatically captures the draft electronically. Using electronic draft capture, no paper follow-up is required by the card issuer to post the transaction and settle with the retailer. The procedure is completed in one transaction.

With this method, BASE24-pos updates the cardholder usage and activity totals, logs the transaction to the PTLF, and updates service totals, authorization totals, and draft capture totals on the PTDF.

The host posts and settles with the retailer and on-us cardholder accounts without paper follow-up.

The value in the PTDF must be set to 1 or 3 with the terminal sending 1, indicating to use the authorization and draft capture method.

Terminal-Defined Draft Capture

The terminal-defined draft capture option allows the terminal to determine the draft capture mode for each transaction based on card type. If a card type is set up with this option, the terminal selects one of the draft capture options described previously (that is, authorization only with paper follow-up or authorization and draft capture) for each transaction involving the card type. The MHI module uses the value sent in FID P of the request as described previously.

Setting the Transaction Profile

The transaction profile indicates the draft capture option under which a particular card type is processed at a particular terminal. The transaction profile is specified for the MHI module in the TP field on PTDF screens 8, 9, and 10. Each type of terminal has specific requirements for setting the TP field. The recommended values for the TP field for the MHI module are 0 (authorize only), 1 (authorize and draft capture), or 3 (terminal-defined draft capture).

For request or force-post transactions (PSTM message types 0200 and 0220, respectively) from a terminal/merchant host, the transaction profile from the terminal can be 0 or 1. For a reversal transaction (PSTM message type 0420) from a terminal/merchant host, the transaction profile must be the same as that of the original request.

Setting the Draft Capture Flag in the PSTM

The draft capture flag indicates whether BASE24-pos balances should immediately reflect transaction activity. The MHI module sets this draft capture flag using the TP field on PTDF screens or the transaction profile value set by the terminal (using FID P). The Router/Authorization module then sets the DFT-CAPTURE-FLG field in the PSTM. The valid values for the DFT-CAPTURE-FLG field in the PSTM for the MHI module are as follows:

- 0 = Authorize only with paper follow-up
- 1 = Authorize and draft capture

The MHI module uses the TP field on PTDF screens or the transaction profile sent by the terminal to set the SRV.TRAN-PROFILE field in the PSTM. The Router module then moves the value in the SRV.TRAN-PROFILE field of the PSTM to the DFT-CAPTURE-FLG field in the PSTM. The transaction profile values and the corresponding draft capture flag settings for the MHI module are shown in the following table.

TRAN-PROFILE	DFT-CAPTURE-FLG	Transaction Type
0	0	All transactions.
1	0	Balance inquiry, check guarantee, check verification, card verification, and preauthorization purchase transactions.
	1	Adjustment, cash advance, mail/phone order, normal purchase, purchase with cash back, preauthorization purchase completion, and refund transactions.

TRAN- PROFILE	DFT-CAPTURE- FLG	Transaction Type
2	1	A value of 2 is an invalid TRAN- PROFILE value for the MHI module. The DFT-CAPTURE-FLG defaults to a value of 1.
3	Value sent from the terminal	All transactions.

Message Sequencing

A terminal that conforms to the MHI module can request the MHI module to validate transmission numbers and sequence numbers by including these in the requests. Transmission number checking is intended only to avoid duplicates. Sequence number checking is intended to ensure that BASE24-pos has received and processed each transaction only once. Any combination of these checking techniques can be implemented.

Transmission Number Checking

Transmission numbers occur in the Transmission Number (HDR.XMIT-NUM) field in the standard message header. The MHI module expects a unique transmission number to be sent with every transaction originating at a terminal connected to the merchant host. The value received in the request is compared to the last transmission number recorded in the PTDF for terminal control mode. If the numbers match, the transmission number check fails and the transaction is declined. A response code of 078 (duplicate transaction received and dropped) is returned to the terminal. If the current transmission number is different than the previous transmission number, the request is accepted and the MHI module continues processing the transaction. ACI requires transmission numbers to be used in the manner specified below.

- The terminal must assign a new transmission number to each new transaction. The MHI module expects the terminal or merchant host to increment the number from 1 to 999999 and then roll back to 1 again. If the MHI module is operating in terminal control mode, the terminal is responsible for setting the transmission number. If the MHI module is operating in host control mode, the merchant host is responsible for setting the transmission number, although the MHI module does not perform duplicate checks in host control mode.
- When the MHI module receives a message having the same transmission number as the last message received by the MHI module, and the last message received by the MHI module was approved, the message is declined with a response code of 078, indicating the transaction is a duplicate and is dropped. When the MHI module receives a message having the same transmission number as the last message received by the MHI module, and the last message received by the MHI module was declined, the message is processed. Duplicate checks are only performed if the MHI module is operating in terminal control mode.
- When a new PTDF record is initialized, the transmission number is set to 000000, so the terminal can be configured to send 000001 as an initial value.

Store-and-Forward Considerations

For store-and-forward transactions (message subtype, HDR.REQ-SUBTYPE field, set to S) no message resynchronization processing is done. The transaction is processed without checking its sequence number. Sequence numbers are resynchronized if the HDR.REQ-SUBTYPE field is set to O, the SEQ NUM CHECKING field on MHCF screen 1 is set to 1, and the MHI module is operating in terminal control mode.

With store-and-forward transactions, the possibility exists for the terminal or the merchant host to transmit the transaction, but not receive a response from the MHI module. The terminal or merchant host has no way of determining whether the MHI module actually received the transaction and has no alternative but to retransmit until a valid response is received.

Because the MHI module does not support sequence number resynchronization for store-and-forward transactions, it processes any retransmission as a new transaction unless there is some way to determine that the transaction has already been processed. If the transaction is a request or force-post, the transmission number cannot be the same as the previous transaction.

Force-post Considerations

The MHI module offers an alternate solution to the duplicate detection problem. Transactions that are offline-authorized by the terminal or merchant host can be sent as force post transactions (message subtype, HDR.MESSAGE-SUBTYPE field, set to O). In this case, sequence number checking logic is applied, exactly as with online transactions, before force posting the transaction.

This alternate solution is less efficient than the transmission number solution because it compromises the distinction that the MHI module draws between force post and store-and-forward transactions.

A force post transaction is technically a transaction for which prior authorization was received from the host (for example, a preauthorization purchase completion transaction that received prior authorization with a preauthorization purchase transaction). Another example of a force post transaction is a transaction that received authorization from a CRT authorization center.

A store-and-forward transaction is one for which the terminal is standing-in without communicating with an outside authorizer. In some cases, if offline-authorized transactions are sent to the host as force post transactions, BASE24-pos reporting is not able to identify them as authorized by a stand-in POS terminal.

Message Sequence Number Checking

Message sequence number checking is intended to ensure that BASE24-pos has received and processed each transaction only once. When the MHI module encounters FID h in a request, the MHI module verifies that the sequence number in this field is the one the MHI module is expecting. The MHI module checks the sequence number sent in the request with the sequence number stored in the PTDF.

Message sequence number checking is only performed if the MHI module is operating in terminal control mode. If the MHI module is operating in host control mode, the current sequence number is not updated in the PTDF.

Message sequence number checking is done by the MHI module only when FID h is included in the request from the terminal. This field consists of 10 numeric characters. The first three characters indicate the shift number and range from 001 to 999, rolling back to 001. The next three characters identify the current batch and range from 001 to 999, rolling back to 001. The next three characters consist of a unique sequence number (seq #) within the shift and batch, which ranges from 001 to 999, rolling back to 001. This sequence number is always referred to as *seq #* so it is not confused with the name of the field name of FID h. The last character is a reset flag, indicating whether the terminal or the MHI module is responsible for determining the correct sequence number (a value of 0 indicates the MHI module determines the correct sequence number; a value of 1 indicates the terminal determines the correct sequence number).

The MHI module is initialized to expect sequence number 001001001, with the shift number set to 001, the batch number set to 001, and the seq # set to 001. From this point on, the MHI module expects the seq # to increment by one. When 999 is reached, the MHI module expects 001 as the next seq #.

Unexpected Seq #, Batch Number, or Shift Number

When the MHI module receives an unexpected seq #, batch number, or shift number and the reset flag in the Sequence Number field (FID h) is set to 0, indicating the MHI module is to determine the seq #, the MHI module responds to the terminal with a denial response code of 899 and FID h. Response code 899 indicates a sequence error has occurred and resynchronization is to be performed. FID h contains the expected sequence number.

The terminal has two options when it receives a response code of 899 with FID h. It can adjust its sequence number, batch number, or shift number to correspond with the seq #, batch number, or shift number received from the MHI module. This implies the terminal is able to adjust its sequence numbers, batch numbers,

and shift numbers backward or forward in its internal queue and send subsequent transactions that correspond to the adjusted sequence number, batch number, or shift number. The terminal can also direct the MHI module to adjust its seq #, batch number, or shift number to match the sequence number, batch number, or shift number maintained by the terminal. To do this, the terminal needs to put the seq #, batch number, and shift number that the terminal determines to be correct in FID h, set the reset flag to 1, and send the transaction that originally received a response code of 899 back to the MHI module. Once the transaction is resubmitted to the MHI module, the MHI module accepts the transaction.

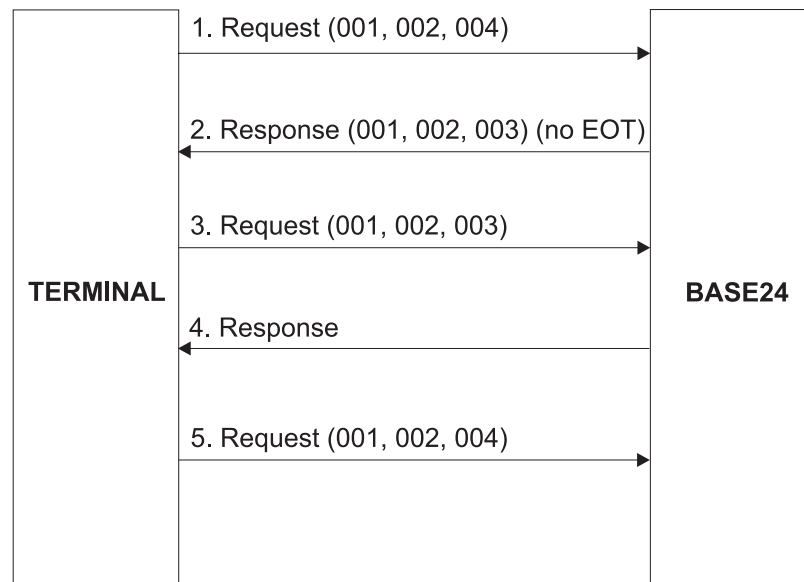
In addition, if the batch or shift number has been changed from the batch or shift number in the original transaction, the MHI module determines that the transaction being resubmitted contains a new batch or shift number and the MHI module performs an implied batch close or an implied shift close. The MHI module then processes the transaction and responds to the terminal.

Sequence Number Checking Examples

The following examples show the processing that occurs between the terminal and the MHI module during various sequence number checking scenarios. Each scenario includes a graphic illustrating the processing flows between the terminal and the MHI module, and each graphic is followed by descriptions of each step exchanged in the flow.

Terminal Backward-Adjusted Sequence Number

The following example describes a situation in which the MHI module is expecting a sequence number other than the one requested by the terminal. This example is applicable only in situations where the terminal is able to search backwards in its transaction queue to locate a sequence number.



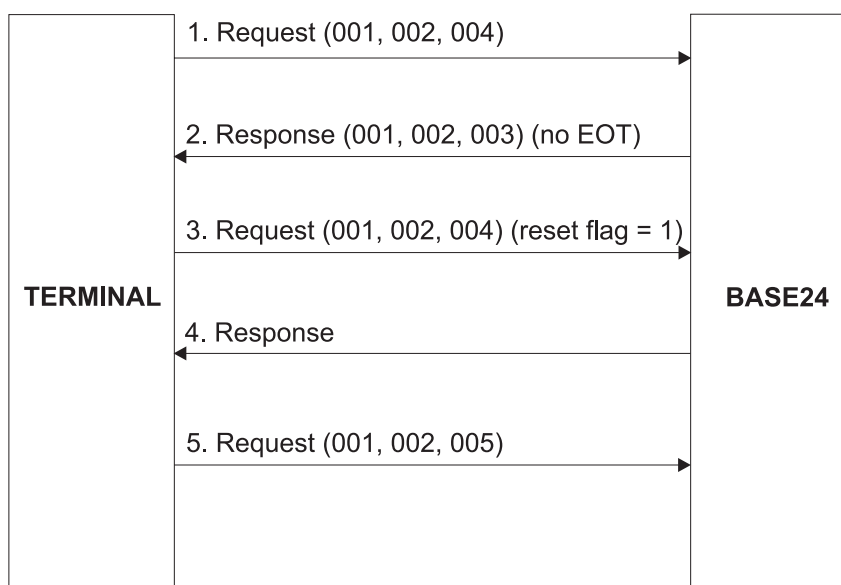
1. The terminal sends a transaction request to the MHI module, and includes the Sequence Number field (FID h) with a value of 001,002, 004 (shift 001, batch 002, sequence number 004 within current batch).
2. For example purposes, it is assumed that the MHI module is expecting 001, 002, 003. The MHI module responds to the terminal request with response code 899 (Sequence error—resync), and includes FID h to indicate a sequence number of 001, 002, 003.

Note: The MHI module does not issue a disconnect command with response code 899.

3. The terminal receives the response from the MHI module and searches its internal transaction queue until it locates sequence number 001, 002, 003. The terminal then sends the transaction with sequence number 001, 002, 003 to the MHI module.
4. The MHI module processes the transaction with sequence number 001, 002, 003 and responds to the terminal.
5. The terminal then sends the next transaction with sequence number 001, 002, 004, which the MHI module is now expecting.

Terminal Forward-Adjusted Sequence Number

The following example describes a situation in which the MHI module is expecting a sequence number other than the one requested by the terminal. This example is applicable only in situations where the terminal is able to search forward in its transaction queue to locate a sequence number.



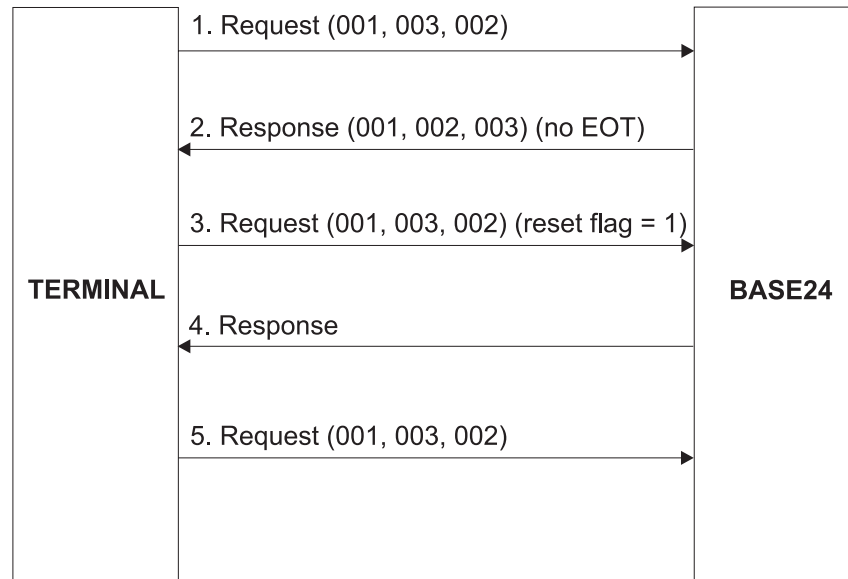
1. The terminal sends a transaction request to the MHI module, and includes the Sequence Number field (FID h) with a value of 001,00 2, 004 (shift 001, batch 002, sequence number 004 within current batch).
2. For example purposes, it is assumed that the MHI module is expecting 001, 002, 003. The MHI module responds to the terminal request with response code 899 (Sequence error—resync), and includes FID h to indicate a sequence number of 001,00 2, 003.

Note: The MHI module does not issue a disconnect command with response code 899.

3. The terminal is unable to search backwards in its internal transaction queue to locate sequence number 001, 002, 003. Thus, the terminal resubmits the transaction to the MHI module with sequence number 001, 002,00 4 and the Reset Flag set to 1.
4. The MHI module accepts and processes the transaction with sequence number 001, 002, 004 and responds to the terminal.
5. The terminal then sends the next transaction with sequence number 001, 002, 005, which the MHI module is now expecting.

Terminal Forward-Adjusted Sequence Number (Implicit Batch Close)

The following example describes a situation in which the MHI module is expecting a sequence number other than the one requested by the terminal. This example is applicable only in situations where the terminal is able to search forward in its transaction queue to locate a sequence number.



1. The terminal makes a transaction request to the MHI module, and includes the Sequence Number field (FID h) with a value of 001, 003, 002 (shift 001, batch 003, sequence number 002 within current batch).
2. For example purposes, it is assumed that the MHI module is expecting 001, 002, 003. The MHI module responds to the terminal request with response code 899 (Sequence error—resync), and includes FID h to indicate a sequence number of 001, 002, 003.

Note: The MHI module does not issue a disconnect command with response code 899.

3. The terminal is unable to search backwards in its internal transaction queue to locate sequence number 001, 002, 003. For example purposes, it is assumed that the terminal is unable to search backwards any further than 001, 003, 002. The terminal thus resubmits the transaction with a sequence number of 001, 003, 002 with the Reset Flag set to 1.

4. The MHI module accepts the transaction with sequence number 001, 003, 002. At this point, the MHI module determines that this is a new batch and performs an implied batch close. The MHI module then processes the transaction and sends a response to the terminal.
5. The terminal then sends the next transaction with sequence number 001, 003, 003, which the MHI module is now expecting.

Additional Store-and-Forward Considerations

When considering how to implement message sequencing, the processing of store-and-forward transactions must be considered. The interspersing of online transactions with store-and-forward transactions subverts duplicate message detection logic in the MHI module. ACI recommends that manufacturers design their terminals to process store-and-forward transactions in a first-in, first-out (FIFO) sequence. As new transactions occur at the terminal, they should be added to the end of the stored transactions so that in all cases the oldest transactions are sent to the MHI module first.

In addition, ACI recommends assigning the sequence number as the transaction is initialized and the transmission number as the transaction is sent to the MHI module.

Transaction Accumulation Totals

The customer can choose to have the terminal accumulate any set of batch, shift, and day totals and forward them to the MHI module with the corresponding close request. If the totals differ from MHI module totals, they are recorded in the POS Transaction Log File (PTLF). No attempt is made to indicate or reconcile any differences.

Transaction accumulation totals are only supported when the MHI module is operating in terminal control mode because MHI module totals are stored in the PTDF and terminal control mode maintains a PTDF record for each terminal. Host control mode does not require a PTDF record for each terminal.

When operating in terminal control mode, the MHI module always accumulates batch, shift, and day totals. When an explicit or implicit close for the corresponding period is received, the MHI module records these totals in the PTLF and resets the totals as appropriate. The MHI module also supports reconciliation with a series of subtotals request transactions prior to an explicit or implicit close.

The BALANCE SUPPORT field in the PTDF indicates to the MHI module which totals are supported by the terminal by means of close transactions. Totals specified in the PTDF are allowed to accumulate until reset by means of a close transaction.

The MHI module records clerk totals as required by the terminal. It logs these totals to the PTLF. When a clerk totals request is received from the terminal, the MHI module accumulates the clerk totals by employee ID or terminal ID and returns them to the terminal.

PIN Encryption

Customers are not required to support PIN encryption, but if they do decide to support it, the MHI module offers several methods of PIN encryption depending on whether the MHI module is operating in host control or terminal control mode. It is up to customers to determine which method best suits their particular requirements. The UPDATE PTDF field on Merchant Host Configuration File (MHCF) screen 1 determines whether the MHI module is operating in terminal control mode (set to 1) or host control mode (set to 0).

If the MHI module is operating in terminal control mode and depending on how the terminal is configured in the POS Terminal Data File (PTDF), FID b and FID c can be encrypted using DES software or DES hardware methods. If PIN encryption is selected, the terminal must support the manual entry or injection of a master key. For more information on the PIN fields contained for terminals in the PTDF, refer to the ***BASE24-pos Files Maintenance Manual***.

If the MHI module is operating in host control mode, PIN information is set up in the MHCF. The MHCF contains fields equivalent to those set up in the PTDF. In addition, the customer must manually update the PIN MASTER KEY field on MHCF screen 2 to contain the new communications key. Then all affected MHI module processes must be sent a warmboot command from a network control facility to read the updated MHCF record into extended memory. When the communications key is requested in a download or as part of the response, the MHI module obtains the value stored in the device dependent data for the communications key. For information on the PIN fields in the MHCF, refer to section 8.

The MHI module also uses data in the Merchant Terminal Configuration File (MCNF) to determine when to provide the terminal with a new communications key. The customer can configure the MHI module to return a new communications key in its response to a full or partial download request when the MHI module is operating in terminal or host control mode. The customer can also configure the MHI module to provide a new communications key in all responses to requests with a specific transaction code. For example, the MHI module can be directed to provide a new communications key in any response to a batch, shift, or day close transaction whether the transaction was denied or not. If the MHI module is required to perform PIN decryption and the communications key is all spaces, the MHI module assumes the terminal has never received a communications key from the MHI module and rejects the PIN as invalid.

For more information on PIN encryption, refer to the ***BASE24 Transaction Security Manual***.

Message Authentication Codes (MACs)

The MHI module support of message authentication codes (MACs) uses the standards documented in the ANSI X9.19-1986 standard, ***Financial Institution Retail Message Authentication***. The ACI MAC, along with the ANSI standard, is designed to protect financial transaction messages against accidental or deliberate alteration. In addition, it protects against the fraudulent insertion of messages.

The remainder of this subsection discusses how MACs are used specifically with the MHI module. For more information on using software and hardware MACs, refer to the ***BASE24 Transaction Security Manual***.

The institution determines whether or not to use MACs and, if so, how they are implemented. The options for using MACs with the MHI module consist of software or hardware MAC generation and verification.

If MACs are supported and the MHI module is operating in terminal control mode, the MAC TYPE field on PTDF screen 7 indicates the form of MACs to be used. A value of 1 in the MAC TYPE field indicates software MACs are supported, and a value of 3 in the MAC TYPE field indicates hardware MACs are supported.

If MACs are supported and the MHI module is operating in host control mode, the MAC ENCRYPTION TYPE field on MHCF screen 2 indicates the form of MACs to be used. A value of 1 in the MAC TYPE field indicates software MACs are supported, and a value of 3 in the MAC TYPE field indicates hardware MACs are supported. If the MHI module is operating in host control mode, MAC information is set up in the MHCF. The MHCF contains fields equivalent to those set up in the PTDF. For information on the MAC fields in the MHCF, refer to section 8.

If MACs are used in any form, the following four fields in the standard message header are included in the MAC text in every request:

- DPC Number
- Transmission Number
- Terminal ID
- Transaction Code

Setting Up MACs

Any number of additional optional data fields can be included in the MAC text, but BASE24-pos does not support authenticating the entire message. While the data within the fields are used in the MAC text, the FIDs identifying the optional data fields are not included in the MAC. The number of bytes being verified depends on whether authentication is performed in hardware or software. The MHI module can authenticate up to 1000 bytes in a message. Refer to the ***BASE24 Transaction Security Manual*** for message limits when authentication is performed in hardware.

The fields to be included in the MAC text are set in the MCNF for requests and responses for every transaction type. However, the customer and the vendor must agree on the fields using MACs in order for the messages to be verified.

ACI recommends that several optional data fields be included when MACs are used. These fields, including their field identifiers (FIDs), follow.

FID	Field Name
B	Amount 1
C	Amount 2
F	Approval Code
b	PIN/Customer
q	Track 2/Customer
2	Track 1/Customer

Whenever MACs are used at a terminal that conforms to the MHI module, FID G (Authentication Code) must be included in both the request and the response. In addition, the MAC terminal master key must be manually entered or injected into the terminal. If the MHI module is operating in host control mode, this key is stored in the MHCF. If the MHI module is operating in terminal control mode, this key is stored in the PTDF. The same key also must be entered in the MAC TERMINAL MASTER KEY field in the MHCF or PTDF. The other key used with MACs, the MAC communications key, is randomly generated by the MHI module if the MHI module is operating in terminal control mode. If the MHI module is operating in host control mode, the key is taken from the MHCF. Although this key is written to the MHCF or PTDF record by the MHI module, the value is not available through BASE24 files maintenance. The MAC

communications key, generated by the MHI module, is encrypted under the MAC terminal master key and downloaded to the terminal. In addition, the MAC communications key, encrypted under the MAC terminal master key, is also generated and included in all responses that include FID H (Authentication Key).

Generating a New MAC Communications Key

The MHI module generates a new MAC communications key under the following three conditions:

- When a response message is configured to contain FID H (Authentication Key). This is set in the Field Map Data record in the MCNF.
- After the MHI module is unable to verify using MACs in a configured number of consecutive requests from a single terminal. The MHI module maintains a counter in the device dependent data in the PTDF for this purpose. Whenever a message is successfully verified using MACs, this counter is reset to zero. The number of consecutive MAC errors that can occur before a new MAC communications key is generated is configured using the POS-DH-CONS-MAC-ERR-LMT param in the Logical Network Configuration File (LCONF). A value of 0 in this param indicates that a new MAC communications key is **not** generated, regardless of the number of consecutive MAC errors. If no value or an invalid value is placed in this param, a default value of 3 is used.

The new MAC communications key is sent in the next response to the terminal, regardless of whether the response is configured to contain FID H (Authentication Key).

- During a download if download field ID (DID) h, Authentication Key (A-KEY), is to be sent. In order to include the authentication key in a download, record 06 in the MCNF must have the DID h (A-KEY) flag set.

Whenever a new MAC communications key is sent to the terminal in a response, it is encrypted under the MAC terminal master key. Once a new key is generated, the MHI module updates the PTDF.

Failed MAC Procedure

When the MHI module receives a request that cannot be verified using MACs, it formats a response message with a response code of 898, indicating an invalid MAC request. The MHI module also increments the invalid MAC counter in the PTDF.

When a terminal is unable to verify a MAC received from the MHI module in a transaction response with monetary impact, the terminal must generate a reversal. The reversal is identical to the response message with two exceptions. The response code is set to 898, indicating an invalid MAC response. Also the MESSAGE TYPE field in the standard message header contains an R, indicating a reversal. The reversal is not verified using a MAC.

When the MHI module receives a MAC reversal, the MHI module sends a warning message to EMS. The MHI module also sends a 0420 reversal message to the authorizer of the original request.

Handshaking

The customer must decide whether to support text-level handshake requests. The purpose of this request is to allow the terminal or merchant host to verify the status of the communications link and, optionally, validate the communications key and authentication key for the terminal or merchant host.

When the MHI module receives a handshake request, it formats a response using the Merchant Terminal Configuration File (MCNF) field map. The response is then sent to the merchant host.

If the handshake request includes FID b (PIN/Customer field), the MHI module decrypts the PIN and compares the decrypted PIN to 16 zeros. If they are not equal, the response returned to the merchant host includes response code 201, indicating an invalid PIN.

If the handshake request includes FID G (Authentication Code), the MHI module validates the MAC, using the specified MAC FIDs. If the MAC is invalid, the MHI module sends a response message with a response code value of 898, indicating an invalid MAC, to the merchant host.

BASE24-mail Support

The MHI module is fully compatible with release 5.3 of the BASE24-mail product. BASE24-mail can be purchased separately from BASE24-pos to provide customers with the added advantage of a fully functional electronic mail system. It is the customer's decision whether to support BASE24-mail, and to determine the degree to which the product is supported. For example, the customer can choose to allow the terminal only to receive mail, or the terminal can be allowed only to send mail. As another option, the customer can choose to support mail message transmission in both directions. The terminal also can be configured to accept unsolicited mail, or it can be configured to not allow mail to be sent to the terminal without the terminal first initiating a Read Mail Request transaction. In short, customers can choose from several options of sending and receiving mail electronically using BASE24-mail.

MHI Module Integration With BASE24-mail

BASE24-mail allows host processors to send informational messages to POS and CRT terminals throughout a network. Mail messages can also be transmitted from these terminals to the host processor.

The host processor has the ability to attach an expiration date and time to every mail message, as well as specify that it is to be notified when a particular message is delivered. The message can be displayed on the terminal immediately, or stored by BASE24-mail on the Tandem system until the terminal requests receipt of any outstanding mail.

Broadcast lists can be created that allow a series of terminals to receive the same message at the same time.

An extracted tape image of all mail messages that were stored and have expired is available for the host. The image shows the message that was transmitted, and indicates if the message was successfully delivered to the terminal. All mail messages that were unable to be transmitted to the host are also included on an extract tape.

A mail message received by a user terminal can be read again by the user until the message expires. The user can specify which message is to be received, or can ask for the next message within the system. However, messages destined for a host are transmitted successfully to the host once and are then purged.

Role of the MHI Module During BASE24-mail Processing

With the MHI module in place along with BASE24-mail, the terminal owner and operator can choose to support mail in any form offered by the product. This includes allowing terminals to send messages or only receive messages, depending on the customer. In addition, customers must determine whether they want to support unsolicited mail messages. Unsolicited mail messages are messages sent to the terminal without a request to read mail messages being sent to the MHI module.

Unsolicited Mail

When the MHI module receives an unsolicited mail message from the BASE24-mail process intended for a terminal, the MHI module checks the PTDF to determine if the terminal can accept unsolicited mail. The MHI module also checks if the terminal is a dial-up terminal.

The MHI module sends unsolicited mail messages to the terminal if the terminal is not a dial-up terminal and the PTDF indicates the terminal can accept unsolicited mail. If both of these conditions are met, the MHI module sends unsolicited mail messages to the terminal even if the terminal has not sent a Read Mail request to the MHI module. The only time the MHI module does not send unsolicited mail messages to the terminal when these two conditions are met is if the terminal has outstanding requests being processed. Unsolicited mail is sent to the terminal containing a standard message header that appears as though the MHI module actually received a Read Mail request. FID V is also included in the message and is configured so the terminal can return it in a subsequent Read Mail request to retrieve the next piece of undelivered mail. If mail exists for a terminal and the terminal is unable to accept the mail (that is, the terminal is a dial-up terminal) or is not configured to support unsolicited mail, the MHI module sets the mail pending flag in the device dependent data in the PTDF and includes FID t (Mail Pending Flag) in each successive response to the terminal. This indicates to the terminal that mail is waiting. The MHI module continues to notify the terminal that mail exists in this manner until the terminal makes any Read Mail request (typically, a Read First Undelivered Mail request).

In situations where the terminal is configured to support unsolicited mail, only one unsolicited delivery attempt per piece of mail is made by the MHI module. When the terminal receives a message with the PTDF mail pending flag set, the terminal should send a Read Mail Request to retrieve the mail messages that are waiting.

Send Mail Request

The terminal can send mail to a data processing center (DPC) using the Send Mail Request. In order to support Send Mail Request transactions, the terminal must be able to compose a piece of mail.

The only optional request field required by the MHI module for this transaction is FID W. FID W is the Mail Text field, which is composed of a destination DPC and a maximum of 449 characters of mail text.

Read Mail Request

The terminal can request any outstanding mail messages using the Read Mail request. Typically, the FID V (Mail Key) is required in Read Mail requests other than initial requests. If the terminal does not include FID V in an initial Read Mail request, the MHI module assumes the terminal is required to read the first of any mail of any category. Thus, the initial Read Mail request that the terminal makes for any mail can be made without FID V. However, all subsequent Read Mail requests should contain FID V. FID V contains the following four types of information:

Category Code — The category code is included whenever the terminal is making a generalized read request. The category code is a 2-position alphanumeric field. Category 99 is reserved for download alerts. The category code is a user-defined value.

Access Code — The type of access desired. The access code is a 1-position numeric field. The valid values for this field are as follows:

- 1 = Read first message. This value directs the MHI module to read the first piece of any mail existing for the terminal.
- 2 = Read any next message. This value directs the MHI module to retrieve the next piece of any mail existing for the terminal.
- 3 = Read first undelivered message. This value directs the MHI module to retrieve the first piece of mail existing for the terminal that has not already been delivered.
- 4 = Read next undelivered message. This value directs the MHI module to retrieve the next piece of mail existing for the terminal that has not already been delivered.
- 5 = Read specific message. This value directs the MHI module to retrieve a specific piece of mail existing for the terminal. A valid Mail ID field must be present to use this value.

Processing Flag — Reserved for future use, 2-position field.

Mail ID — The date and identification for the message. The mail ID is a 10-position alphanumeric field. The first six bytes contain the date (YYMMDD) of the message. The last four bytes contain the mail message number. This field specifies the starting point for a Read Next Request or the specific piece for a Read Specific Request.

If the MHI module receives a request from the terminal to read mail, it formats the request and sends it to the BASE24-mail process. The BASE24-mail process then responds and sends the appropriate message back to the MHI module, which sends it back to the terminal. When the MHI module receives an acknowledgment from the terminal, the MHI module sends a completion message to the BASE24-mail process.

Read Mail Response

The MHI module responds to a Read Mail request after requesting the appropriate mail message from the BASE24-mail process. The mail message received from BASE24-mail is placed in FID W, if it is equal to or less than 440 bytes and fits in the field. If the mail fits in FID W, the response code is set to 880, indicating that no more data exists, and the response is returned to the merchant host.

If the mail message response does not fit, the response code is set to 881, indicating that more data exists. The MHI module saves the message context for the anticipated Read Next Request. The terminal echoes FID V from the previous response in Read Next Requests.

When returning a Read Mail Response, the MHI module also translates the value in the Access Code field to the value it anticipates will follow.

An 880 response, indicating that no more data exists for a particular piece of mail, does not mean that no more mail exists for the terminal. In order to determine if no more mail exist for the terminal, the terminal must make another Read Next request. No more mail is implied when the response code is 880 and no mail text is included in the response. In this case, FID W is not included in the response.

Mail Delivered Request

Terminals are allowed to mark mail as delivered so that subsequent Read Undelivered Mail Requests ignore the mail and move on to the next undelivered piece. In order to perform this function, the terminal must submit a Mail Delivered Request containing FID V (Mail Key) of the Read Mail Request (the previous response). In this case, the Mail ID field in FID V identifies the mail to be marked as delivered and the Processing Flag field in FID V is used to mark the mail as delivered.

Section 8

Configuration Files Maintenance

The BASE24-pos Merchant Host Interface (MHI) module is the exclusive reader of the following five files:

- The Merchant Terminal Configuration File (MCNF) containing configuration data for the MHI module and download data to be sent to the terminal.
- The Merchant Terminal Response File (MRSP) containing a table of terminal response codes in up to three different languages. The MHI module uses the MRSP to develop terminal response displays that describe the results of transaction processing to the terminal.
- The Merchant Host Configuration File (MHCF) containing information about each data processing center (DPC) that communicates with the MHI module. This information includes processing flags and station tables.
- The Merchant Totals Configuration File (MTCF) containing records defining totaling groups for the purpose of obtaining group totals for retailer and group subtotals transactions.
- The Merchant Regulation E File (MREG) containing Regulation E data for a specific terminal when the MHI module is running in host control mode. The data in the MREG overrides any Regulation E data contained in the POS Transaction Data File (PTDF).

These files are accessed through BASE24-pos files maintenance. BASE24-pos files maintenance manipulates files through the use of formatted screens. This section describes the formatted screens used to access these files for maintenance and configuration.

The MHI module configuration files are set up in the same manner as most other files in the BASE24-pos system. Additional information on BASE24-pos files maintenance can be found in the **BASE24 CRT Access Manual**, the **BASE24 Base Files Maintenance Manual**, and the **BASE24-pos Files Maintenance Manual**. The information provided in this section, however, should be sufficient to set up the necessary records in the MHI files for persons familiar with BASE24 files maintenance.

Prerequisites for Configuration

The MHI module requires a Device Handler-specific module and files, as well as modifications to standard BASE24-pos and Base files. The files required to use the MHI module are outlined in section 2. Before a customer site can be configured to use the MHI module, the MHI module must be properly installed, the module needs to be added to the system, and the BASE24-pos network must be enabled and available for files maintenance.

Merchant Terminal Configuration File

The Merchant Terminal Configuration File (MCNF) contains configuration data used for processing by the MHI module and downloading to the terminal.

The data in this file is divided into three categories defined as processing data, format data, and download data. The processing data contains timeout values and other processing parameters. The format data consists of up to 30 card prefix ranges used to define processing parameters by card prefix range. The download data contains all the information sent to the terminal in a download.

MCNF records are added and maintained through BASE24-pos files maintenance. To access the MCNF, select POS from the BASE24 Virtual Menu and MCNF from the POS Product Menu or enter MCNF in the FILE DESTINATION field at the bottom of any screen. There are four types, or sets, of screens associated with the MCNF. These screens are used to access the seven records that can be stored in the MCNF. These screen types are as follows:

- Selection
- Processing Record
- Field Map
- Download Record

Selecting MCNF Records

The first screen associated with the MCNF allows you to choose the record type. This screen is your main menu to MCNF records. The selection screen is used to determine whether MHI module processing parameters, formatting parameters, or download parameters are being configured. The terminal group is used to identify the terminals to which the configuration information applies once it is set up. The terminal group corresponds to all terminals that have the same value in the TERM-GRP field in the POS Terminal Data File (PTDF).

Selection Screen

Merchant Terminal Configuration File (MCNF) screen 1 is shown below, followed by its field descriptions and a procedure for using the screen.

```

BASE24-POS      MHI MCNF MAINT      LLLL      YY/MM/DD  HH:MM  01 OF 01
      TERMINAL GROUP: 0000      RECORD TYPE:      RECORD ID:

      RECORD      RECORD      RECORD      RECORD      DOWNLOAD
      TYPE      ID      DESCRIPTION      TYPE      ID      PACKETS
-----
Processing P    00      Requests      Download D    00      A - J
                                         01      K - T
Field map  F    00      Requests      02      U - Z      plus 2
                                         03      0 - 3
                                         04      4 - 7
                                         05      8 - 9      plus 2
                                         06      a - z
                                         07      10 - 13
PLEASE FILL IN KEYS AND PRESS KEY DESIRED      08      14 - 17
                                         09      18 - 21
                                         10      22 - 25
                                         11      26 - 29

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP

```

TERMINAL GROUP — The terminal or group of terminals to which the configuration information set up in the MCNF applies. The configuration information is used by all terminals that have a value corresponding to this field in the TERM GROUP field in the POS Terminal Data File (PTDF). The TERMINAL GROUP combined with the RECORD TYPE and RECORD ID is the primary key into the MCNF.

Field Length:	4 alphanumeric characters
Required Field:	Yes
Default Value:	0000
Data Name:	MCNF.KEY-S.GRP

RECORD TYPE — A value identifying whether the record is a processing record, a formatting record, or a download record. This field is used in conjunction with the RECORD ID field. Valid values are as follows:

D = Download record
F = Field map record
P = Processing record

Field Length: 1 alphanumeric character
Required Field: Yes
Default Value: No default value
Data Name: MCNF.KEY-S.REC-TYP

RECORD ID — A value identifying the type of record being accessed. Only certain values are allowed for the different record types. For example, 00 must be entered in this field when P is entered in the RECORD TYPE field. The valid combinations are listed below.

RECORD TYPE Field	RECORD ID Field
D	00, 01, 02, 03, 04, 05, 06, 07, 08, 09, 10, 11
F	00 or 01
P	00

The combinations along with the associated download packets are listed on the screen.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: No default value
Data Name: MCNF.KEY-S.REC-ID

Selecting Records

The MCNF Selection screen is the customer's primary link to accessing MCNF records. To access the MCNF Selection screen, perform the following steps. For more information on screen access, refer to the ***BASE24 CRT Access Manual***.

1. Select POS from the BASE24 Virtual Menu and MCNF from the POS Product Menu or enter MCNF in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key. MCNF screen 1 is displayed.
2. Enter the appropriate values in the following fields on MCNF screen 1, based on the parameters you want to configure (that is, processing, formatting, or download), and the terminal group to which you want the parameters to apply:

TERMINAL GROUP
RECORD TYPE
RECORD ID

The screen you access is determined by the values placed in the RECORD TYPE and RECORD ID fields.

Field descriptions for these fields are provided in the following topic, together with a graphic illustration of MCNF screen 1.

3. Move to the appropriate MCNF screen by pressing the **F9** key.

Processing Records

The Processing Record screen allows customers to set up general processing parameters used by the MHI module when handling messages. From this screen, you can configure several timeout options, including the amount of time the MHI module waits for responses from the Authorization and Mail processes. You can also determine the maximum length of responses from the MHI module to the terminal and whether the MHI module is to format user data (PSTM.USER-DATA) from the POS Standard Internal Message (PSTM) with every financial transaction. In addition, you can determine if totals returned to the terminal in responses are draft capture totals only or if the totals include all transactions.

Configuring Processing Records

The MCNF Processing Record screen is accessed through BASE24 files maintenance much in the same manner as the MCNF Selection screen. Once accessed, you can set up the processing parameters to meet your specific requirements. The following steps describe the procedures to set up the Processing Record screen. These procedures assume you have already accessed the MCNF Selection screen. To set up the MCNF Processing Record screen, perform the following steps:

1. Enter the appropriate terminal group in the **TERMINAL GROUP** field on screen 1 of the MCNF (Selection screen).
2. Enter **P** in the **RECORD TYPE** field on screen 1 of the MCNF. The **P** in this field identifies the record type as processing records.
3. Enter **00** in the **RECORD ID** field on screen 1 of the MCNF. The **00** in this field identifies the record as information used in requests from the merchant host. A value of **00** is the only value allowed in the **RECORD ID** field when the **RECORD TYPE** is **P**.
4. Press the **F9** key from the MCNF Selection screen to display the MCNF Processing Record screen.
5. Enter the appropriate values in the following fields on the MCNF Processing Record screen, based on the processing parameters you want to configure:

TOTALS TYPE
IMPLICIT CLOSES
EXPIRED MSGS
FORMAT PSTM
MAX RESP LEN (BYTES)
AUTH TIMEOUT (SECS)
LOAD TIMEOUT (SECS)
MAIL TIMEOUT (SECS)
OLD MSG TIMEOUT (SECS)
GRP TTL TIMEOUT (SECS)
DPC #0
DPC #1
DPC #2

Field descriptions for these fields are provided in the following topic, together with a graphic illustration of MCNF screen 2.

6. Press the **F3** key to add the record.

Processing Record Screen

Merchant Terminal Configuration File (MCNF) screen 2 is shown below, followed by its field descriptions.

```

BASE24-POS  MHI MCNF MAINT      LLLL      YY/MM/DD HH:MM 02 OF 02
      TERMINAL GROUP: 0000      RECORD TYPE: P  RECORD ID: 00
      PROCESSING RECORD SCREEN

      ( N = NO, Y = YES )
TOTALS TYPE:      N  IMPLICIT CLOSES:      N  EXPIRED MSGS:      N
FORMAT PSTM:      N  RESERVED:      N  RESERVED:      N
RESERVED:      N  RESERVED:      N  RESERVED:      N
RESERVED:      N  RESERVED:      N  RESERVED:      N
RESERVED:      N  RESERVED:      N  RESERVED:      N
RESERVED:      N  RESERVED:      N  RESERVED:      N
RESERVED:      N  RESERVED:      N  RESERVED:      N

MAX RESP LEN( BYTES ):      0  AUTH TIMEOUT( SECS ):      0
LOAD TIMEOUT( SECS ):      0  MAIL TIMEOUT( SECS ):      0
OLD MSG TIMEOUT( SECS ):      0  GRP TTL TIMEOUT ( SECS ):      0

DPC #0:      0      DPC #1:      0      DPC #2:      0

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
      F12-HELP

```

TOTALS TYPE — Indicates if totals returned to the terminal in responses are draft capture totals or if the totals include all transactions. Valid values are as follows:

Y = All totals

N = Draft capture totals

Field Length: 1 alphabetic character

Required Field: Yes

Default Value: N

Data Name: MCNFG.PRO-DATA.FLAGS

IMPLICIT CLOSES — Indicates whether the terminal is allowed to perform implicit closes. Valid values are as follows:

Y = Implicit closes allowed

N = No implicit closes allowed

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: N
Data Name: MCNFG.PRO-DATA.FLAGS

EXPIRED MSGS — Indicates the procedure to be taken with messages that have expired. Valid values are as follows:

Y = Log an error message and drop the expired message
N = Drop the expired message

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: N
Data Name: MCNFG.PRO-DATA.FLAGS

FORMAT PSTM — Indicates whether the MHI module is to format user data in the POS Standard Internal Message (PSTM) for financial transactions. Valid values are as follows:

Y = Yes
N = No

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: N
Data Name: MCNFG.PRO-DATA.FLAGS

RESERVED — The fields marked RESERVED are included to allow for future expansion. All of these fields default to N.

MAX RESP LEN (BYTES) — Defines the maximum length, in bytes, for a transaction response message that can be sent to a merchant host. This value varies depending on the protocol being used. The length set in this field must be 3 to 5 bytes less than the value specified in the XPNET XMITLEN attribute. Valid values are 26 to 1024 bytes.

Field Length: 1–4 numeric characters
Required Field: Yes
Default Value: 0
Data Name: MCNFG.PRO-DATA.MAX-RESP-LEN

AUTH TIMEOUT (SECS) — Defines the length of time, in seconds, the MHI module waits for a response from the Authorization process before responding to the terminal with a timeout response code. This value should be less than the value specified in the XPNET TIMEOUT attribute. The MHI module checks all timeout fields for a valid value of greater than 5 seconds.

Field Length: 1–4 numeric characters
Required Field: Yes
Default Value: 0
Data Name: MCNFG.PRO-DATA.AUTH-TIMEOUT

LOAD TIMEOUT (SECS) — Defines the length of time, in seconds, the MHI module waits for a download response from the terminal before aborting the download. The MHI module checks all timeout fields for a valid value of greater than 5 seconds.

Field Length: 1–4 numeric characters
Required Field: Yes
Default Value: 0
Data Name: MCNFG.PRO-DATA.LOAD-TIMEOUT

MAIL TIMEOUT (SECS) — Defines the length of time, in seconds, the MHI module waits for a response from the Mail process before responding to the terminal with a timeout response code. The MHI module checks all timeout fields for a valid value of greater than 5 seconds.

Field Length: 1–4 numeric characters
Required Field: Yes
Default Value: 0
Data Name: MCNFG.PRO-DATA.MAIL-TIMEOUT

OLD MSG TIMEOUT (SECS) — Defines how old, in seconds, a message can be before it is considered to be an expired message. If this field is set to 0, no expired message processing is performed. Instead, expired messages go through the system like any other message. If this field is set greater than 0, expired message processing is performed based on the value in the EXPIRED MSGS field on this screen.

Expired messages are messages received by the MHI module after the terminal has already timed out. This occurs when an MHI module is stopped and the merchant host is still sending transaction messages. These messages are placed in the queue and are processed when the MHI module is restarted.

Field Length: 1–4 numeric characters
Required Field: Yes
Default Value: 0
Data Name: MCNFG.PRO-DATA.OLDMSG-TIMEOUT

GRP TTL TIMEOUT (SECS) — Defines the length of time, in seconds, the MHI module waits for a response from the Merchant Totals process before declining a group totals transaction request. The MHI module checks this timeout field for a valid value of greater than 1 second.

Field Length: 1–4 numeric characters
Required Field: Yes
Default Value: 0
Data Name: MCNFG.PRO-DATA.GRP-TTL-TIMEOUT

DPC #0 — Defines the DPC to which mail sent to DPC #0 is delivered. This field is used by Mail Text optional data field (field identifier (FID) W) when sending BASE24-mail messages.

Field Length: 1–4 numeric characters
Required Field: Yes
Default Value: 0
Data Name: MCNFG.PRO-DATA.DPC-0

DPC #1 — Defines the DPC to which mail sent to DPC #1 is delivered. This field is used by the Mail Text optional data field (FID W) when sending BASE24-mail messages.

Field Length: 1–4 numeric characters
Required Field: Yes
Default Value: 0
Data Name: MCNFG.PRO-DATA.DPC-1

DPC #2 — Defines the DPC to which mail sent to DPC #2 is delivered. This field is used by the Mail Text optional data field (FID W) when sending BASE24-mail messages.

Field Length: 1–4 numeric characters
Required Field: Yes
Default Value: 0
Data Name: MCNFG.PRO-DATA.DPC-2

Field Map Records

Field map records allow customers to determine the optional data fields sent in requests and responses. Separate maps are used for request and response messages. Besides allowing different fields in requests and responses, the field maps allow customers to include different fields based on transaction type. For example, different fields can be included in a normal transaction request message than in a preauthorization request message. See section 4 for more information about optional data fields and associated field identifiers (FIDs).

In addition, the field maps allow you to determine which fields in any request or response message should be included in message authentication codes (MACs). The MHI module allows individual message fields to be included in MACs. Customers are responsible for identifying these fields on this screen.

Although changes to this data can be made online through BASE24 files maintenance, the customer and merchant host must coordinate this data and not rely on the file to determine the fields to be included in messages or the fields to be included in MACs. The field maps do not determine the data sent in requests. Terminals do not receive the field map data. It is the responsibility of the terminal software to determine the data in requests. The purpose of the field map data in the MCNF is to allow the MHI module to know what to expect from the terminal and how to format the responses to the terminal.

Field Map Screens

The Field Map screens contain information for each optional data field. You must determine whether or not each optional data field is required for a transaction message and a message authentication code (MAC) for each transaction type. The optional data fields are described in section 4.

You can access the Field Map screens by performing the following steps:

1. Enter the appropriate terminal group in the TERMINAL GROUP field on screen 1 of the MCNF (Selection screen).
2. Enter F in the RECORD TYPE field on screen 1 of the MCNF. The F in this field identifies the record type as Field Map.
3. Enter 00 or 01 in the RECORD ID field on screen 1 of the MCNF. A value of 00 in this field identifies the record as the field map used to determine the fields to include in request messages. A value of 01 in this field allows you to configure the fields to include in response messages.
4. Press the **F9** key from the MCNF Selection screen to display the first MCNF Field Map screen for requests or responses, and press the **F2** key to read the record. The field maps for requests and responses each consist of six screens. Each screen contains five transaction types. Pressing the **F9** key again allows you to page through all of the field map screens associated with this TERMINAL GROUP, RECORD ID, and RECORD TYPE.

Field Map Screen for Requests

The first Field Map screen for requests (00 in the RECORD ID field) is shown below, followed by its field descriptions.

```

BASE24-POS   MHI MCNF MAINT           LLLL           YY/MM/DD HH:MM 02 OF 07
              TERMINAL GROUP: 0000     RECORD TYPE: F   RECORD ID: 00

      TRANSACTION                      FIELD MAP ( REQUEST )

                                ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789
NORM PURCH MSG NYNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
MAC NNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
PRE AUTH  MSG NYNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
MAC NNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
PRE AUTH C MSG NYNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
MAC NNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
PH/MAIL   MSG NYNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
MAC NNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
MERCH RET MSG NYNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
MAC NNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                      F12-HELP

```

TRANSACTION

The following fields specify the required FIDs in requests. You can specify different FIDs for transaction messages and MACs for each type of transaction supported by the MHI module. Refer to section 1 for a description of MHI transaction support. The following table lists the supported transactions by screen number. Each screen displays five transaction types that must be configured separately. Press the **F9** key to access subsequent Field Map screens. Generally, screens 02 through 04 are used to configure financial transactions and screens 05 through 07 are used to configure administrative transactions.

Field Map Screen Number	Transaction Type
02 of 07	Normal purchase Preauthorization Preauthorization completion Mail/phone order Merchandise return

Field Map Screen Number	Transaction Type
03 of 07	Cash advance Card verification Balance inquiry Purchase with cash back Check verification
04 of 07	Check guarantee Purchase adjustment Merchandise return adjustment Cash advance adjustment Cash back adjustments
05 of 07	Logon Logoff Close batch Close shift Close day
06 of 07	Clerk totals Batch totals Shift totals Day totals Group totals
07 of 07	Read mail Mail delivery Send mail request Download Handshake

MSG — Indicates the FIDs that are required in transaction message requests for the corresponding transaction type. Valid values are as follows:

Y = Yes, the FID is required for a transaction message.

N = No, the FID is not required for a transaction message.

Field Length: 1 alphabetic character

Occurs: 62 times

Required Field: Yes

Default Value: N for all FIDs except B and q; Y for FIDs B and q

Data Name: MCNFG.FLD-DATA.REQ-FLDS

Y = Yes, the FID is required for a transaction MAC.
N = No, the FID is not required for a transaction MAC.

The remaining five Field Map request screens are formatted in the same manner as screen 2.

The first Field Map screen for responses (01 in the RECORD ID field) is shown below, followed by its field descriptions.

11/1998 PS-DH020-01
Applied Communications, Inc.

TRANSACTION

The following fields specify the required FIDs in responses. You can specify different FIDs for transaction messages and MACs for each type of transaction supported by the MHI module. Refer to section 1 for a description of MHI module transaction support. Refer to the description of the Field Map screen for requests earlier in this section for a list of supported transactions and screen numbers.

MSG — Indicates the FIDs that are required in transaction message responses for the corresponding transaction type. Valid values are as follows:

Y = Yes, the FID is required for a transaction message.

N = No, the FID is not required for a transaction message.

Field Length: 1 alphabetic character
Occurs: 62 times
Required Field: Yes
Default Value: N for all FIDs except X and p; Y for FIDs X and p
Data Name: MCNFG.FLD-DATA.REQ-FLDS

MAC — Indicates the FIDs that are required in MAC responses for the corresponding transaction type. Valid values are as follows:

Y = Yes, the FID is required for a transaction MAC.

N = No, the FID is not required for a transaction MAC.

Field Length: 1 alphabetic character
Occurs: 62 times
Required Field: Yes
Default Value: N
Data Name: MCNFG.FLD-DATA.MAC-FLDS

The remaining five Field Map response screens are formatted in the same manner as screen 2.

Field Map Help Screen

The Field Map Help screen displays the optional data fields and associated FIDs allowed for requests or responses. You can access the Field Map Help screen by pressing the **F7** key from the Field Map screen for requests or the Field Map screen for responses. The Field Map Help screen for requests is shown below, followed by a procedure for printing the screen.

```

BASE24-POS   MHI MCNF MAINT           LLLL           YY/MM/DD HH:MM 02 OF 07
      TERMINAL GROUP: 0000      RECORD TYPE: F  RECORD ID: 00
      FIELD MAP INFORMATION FOR SCREENS 2 THRU 7 OF FIELD TYPE F
A- Address          Q- Echo Data          g- Response Dsply   w- Clerk Flg
B- Amount 1         R- Card Type           h- Sequence Num     x- Group ID
C- Amount 2         S- Invoice Num          i- Orig. Seq. Num    y- Ttls/Grp
D- Appl Acct Type   T- Orig Invoice Num     j- Drvrs Lic ST      z- Reserved
E- Appl Acct Num    U- Language Code        k- Term Loc/DL DOB   0- Amex Data
F- Approval Code    V- Mail/DLL Key          l- Totals/Batch      1- PS2000 Data
G- Authen. Code     W- Mail/DLL Text          m- Totals/Day        2- Trk1/Customer
H- Authen. Key      X- Tran Date & Time        n- Totals/Clerk      3- Trk1/Super
I- Clerk ID         Y- Zip Code           o- Totals/Shift      4- Reserved
J- Avail Bal        Z- Addr Vrfy Stat        p- Response Code     5- Reserved
K- Business Date    a- Optional Data          q- Trk2/Customer     6- Product SFIDs
L- Check Typ/Ctgry b- PIN/ Customer          r- Trk2/Supervisor   7- Product SFIDs
M- Comm key         c- PIN/ Supervisor        s- Tran Dscrption    8- Chnl/Dst SFIDs
N- Customer ID      d- Retailer ID           t- Mail Pend Flg     9- Customer SFIDs
O- Customer ID Typ  e- POS Condition Cde       u- Download Pend
P- Draft Captr Flg  f- Receipt Data           v- Auth Timeout
***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
ANY FUNCTION KEY EXCEPT SF9-SF16, F10, OR F16 RETURNS. F10 PRINTS AND RETURNS

```

It may be helpful to print this screen for use as a reference tool when setting up the Field Map screens. To print this screen, perform the following steps:

1. Specify a printer location in the SCREEN PRINTER field located on the BASE24 Logon Menu.
2. Access the MCNF screens.
3. Press the **F7** key to display the Field Map Help screen.
4. Press the **F10** key to print the screen. For more information on printing screens, refer to the **BASE24 CRT Access Manual**.

Configuring Field Map Records

There are six Field Map screens, listing all the possible transactions supported by the MHI module. Each screen has two pages, one for requests and one for responses.

For each transaction type, the FIDs associated with the optional data fields are listed across the top of the screen. The fields listed on the Field Map screens allow optional data fields to be included in requests and responses for each type of transaction. A detailed explanation of each FID is included in section 4.

The transaction types are listed down the left side of the screen, along with two rows of codes for each transaction type. The first row indicates which optional data fields you want to include in the message and the second row indicates which optional data fields in the message you want to be verified using MACs.

If an optional data field is to be included in a message, a Y is placed in the one-byte MSG field corresponding to the appropriate FID for the specific transaction type. If a field is to be included in a MAC, a Y is placed in the one-byte MAC field corresponding to the appropriate FID for the specific transaction type. Refer to the topic “Optional Data Field Requirements by Transaction Type” for information about the optional data fields required for each type of transaction.

The MCNF Field Map screens are accessed through BASE24 files maintenance much in the same manner as the MCNF Processing Record screen. Once accessed, you can set up the field maps to meet your specific requirements. The following steps describe the procedures to set up the Field Map screens. These procedures assume you have already accessed the Field Map screen for requests. To set up the MCNF Field Map screen for requests, perform the following steps:

1. Determine which optional data fields to include for each transaction type. For a list of FIDs and their corresponding optional data field names, press the **F7** key to display the Field Map Help screen. The Field Map Help screen displays the optional data fields allowed for requests. Press any function key other than **Shift-F9** through **Shift-F16**, **F10**, or **F16** to return to the Field Map screen for requests.
2. Enter a Y or N in the MSG row after each transaction type for each optional data field listed across the top of the screen. Optional data fields are identified across the top of the screen by their corresponding FIDs. A value of Y indicates the FID is included in requests and a value of N indicates the FID is not included in requests.

3. Determine which optional data fields should be included in MACs and enter a Y or N in the MAC row after each transaction type for each optional data field listed across the top of the screen. A value of Y indicates the optional data field should be included in MACs, and a value of N indicates the optional data field should not be included in MACs.
4. Verify that the optional data fields set up for each transaction type are correct, and that the appropriate optional data fields are being included in the MAC text.
5. Repeat steps 2 through 4 for each of the remaining request screens.
6. Press the **F3** key to add the record.
7. Change the RECORD ID to 01 and repeat steps 2 through 6 for each of the response screens.

Optional Data Field Requirements by Transaction Type

Although much of the information contained in message requests and responses is configurable using the MCNF Field Map screens, some of the optional data fields are required for certain transaction types. These optional data fields must be used (i.e., a setting of Y on the Field Map screens) in order for processing to occur for a specific transaction type. The following tables outline the minimum optional data fields required in requests and responses for each transaction type. Other optional data fields can also be included, but the fields shown in the tables must be present for the specified transaction type. Where applicable, requirements for terminal control mode and host control mode are given.

Purchases, Preauthorization Purchases, Preauthorization Completions, Merchandise Returns, and Cash Advances

The following tables show the MCNF settings for the MHI module when it is operating in terminal mode and in host mode.

Terminal Control Mode Requirements. The following table contains the settings for optional data fields and MAC values on the Field Map screens for purchases, preauthorization purchases, completions, merchandise returns, and cash advances when the MHI module is operating in terminal control mode. The MAC values have a default of N, but this value is not required and can be changed if the customer wants the field to be included in the MAC calculation.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
B	AMOUNT 1 (PSTM.TRAN.AMT-1)	Y	N	N	N
p	RESPONSE CODE (PSTM.TRAN.RES-CDE)	N	N	Y	N
q	TRACK2/CUSTOMER (PSTM.TRACK2)	N*	N	N	N
2	TRACK1/CUSTOMER (Track 1 token)	N*	N	N	N

* Either FID q or 2 is required for request messages.

Host Control Mode Requirements. The following table contains the settings for optional data fields and MAC values on the Field Map screens for purchases, preauthorization purchases, completions, merchandise returns, and cash advances when the MHI module is operating in host control mode. The MAC values have a default of N, but this value is not required and can be changed if the customer wants the field to be included in the MAC calculation.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
B	AMOUNT 1 (PSTM.TRAN.AMT-1)	Y	N	N	N
D	APPLICATION ACCOUNT TYPE (PSTM.TRAN.TRAN-CDE.AA)	N	N	Y	N
R	CARD TYPE (PSTM.TRAN.TRAN-CDE.T)	N	N	Y	N
X	TRANSACTION DATE/TIME	N	N	Y	N
p	RESPONSE CODE	N	N	Y	N
q	TRACK2/CUSTOMER (PSTM.TRACK2)	N*	N	N	N

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
z	REVERSAL CONTEXT	N	N	Y	N
2	TRACK1/CUSTOMER (Track 1 token)	N*	N	N	N

* Either FID q or 2 is required for request messages.

Mail or Telephone Orders

The following tables show the MCNF settings for mail or telephone order transactions for the MHI module when it is operating in terminal control mode and in host control mode.

Terminal Control Mode. The following table contains the settings for optional data fields and MAC values for mail or telephone order transactions when the MHI module is operating in terminal control mode.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
B	AMOUNT 1 (PSTM.TRAN.AMT-1)	Y	N	N	N
p	RESPONSE CODE (PSTM.TRAN.RES-CDE)	N	N	Y	N
q	TRACK2/CUSTOMER (PSTM.TRACK2)	Y	N	N	N

Host Control Mode. The following table contains the settings for optional data fields and MAC values for mail or telephone order transactions when the MHI module is operating in host control mode.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
B	AMOUNT 1 (PSTM.TRAN.AMT-1)	Y	N	N	N
D	APPLICATION ACCOUNT TYPE (PSTM.TRAN.TRAN-CDE.AA)	N	N	Y	N
R	CARD TYPE (PSTM.TRAN.TRAN-CDE.T)	N	N	Y	N
X	TRANSACTION DATE/TIME	N	N	Y	N
p	RESPONSE CODE	N	N	Y	N
q	TRACK2/CUSTOMER (PSTM.TRACK2)	Y	N	N	N
z	REVERSAL CONTEXT	N	N	Y	N

Card Verification and Balance Inquiries

The following tables show the MCNF settings for card verification and balance inquiry transactions for the MHI module when it is operating in terminal control mode and in host control mode.

Terminal Control Mode. The following table contains the settings for optional data fields and MAC values for card verification and balance inquiry transactions when the MHI module is operating in terminal control mode.

The MAC values have a default of N, but this value is not required and can be changed if the customer wants the field to be included in the MAC calculation.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
p	RESPONSE CODE (PSTM.TRAN.RES-CDE)	N	N	Y	N
q	TRACK2/CUSTOMER (PSTM.TRACK2)	N*	N	N	N
2	TRACK1/CUSTOMER (Track 1 token)	N*	N	N	N

* Either FID q or 2 is required for request messages.

Host Control Mode. The following table contains the settings for optional data fields and MAC values on the Field Map screens for card verification and balance inquiries when the MHI module is operating in host control mode. The MAC values have a default of N, but this value is not required and can be changed if the customer wants the field to be included in the MAC calculation.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
p	RESPONSE CODE	N	N	Y	N
q	TRACK2/CUSTOMER (PSTM.TRACK2)	N*	N	N	N
2	TRACK1/CUSTOMER (Track 1 token)	N*	N	N	N

* Either FID q or 2 is required for request messages.

Purchases-with-Cash-Back and Adjustments

The following tables show the MCNF settings for purchase-with-cash-back and adjustment transactions for the MHI module when it is operating in terminal control mode and in host control mode.

Terminal Control Mode. The following table contains the settings for optional data fields and MAC values for purchase-with-cash-back, purchase adjustment, merchandise return adjustment, cash advance adjustment, and cash-back adjustment transactions when the MHI module is operating in terminal control mode.

The MAC values have a default of N, but this value is not required and can be changed if the customer wants the field to be included in the MAC calculation.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
B	AMOUNT 1 (PSTM.TRAN.AMT-1)	Y	N	N	N
C	AMOUNT 2 (PSTM.TRAN.AMT-2)	Y	N	N	N
p	RESPONSE CODE (PSTM.TRAN.RES-CDE)	N	N	Y	N
q	TRACK2/CUSTOMER (PSTM.TRACK2)	N*	N	N	N
2	TRACK1/CUSTOMER (Track 1 token)	N*	N	N	N

* Either FID q or 2 is required for request messages.

Host Control Mode. The following table contains the settings for optional data fields and MAC values on the Field Map screens for purchase-with-cash-back, purchase adjustment, merchandise return adjustment, cash advance adjustment, and cash-back adjustment transactions when the MHI module is operating in host control mode. The MAC values have a default of N, but this value is not required and can be changed if the customer wants the field to be included in the MAC calculation.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
B	AMOUNT 1 (PSTM.TRAN.AMT-1)	Y	N	N	N
C	AMOUNT 2 (PSTM.TRAN.AMT-2)	Y	N	N	N
D	APPLICATION ACCOUNT TYPE (PSTM.TRAN.TRAN-CDE.AA)	N	N	Y	Y
R	CARD TYPE (PSTM.TRAN.TRAN-CDE.T)	N	N	Y	Y
X	TRANSACTION DATE/TIME	N	N	Y	N
p	RESPONSE CODE	N	N	Y	N
q	TRACK2/CUSTOMER (PSTM.TRACK2)	N*	N	N	N
z	REVERSAL CONTEXT	N	N	Y	Y
2	TRACK1/CUSTOMER (Track 1 token)	N*	N	N	N

* Either FID q or 2 is required for request messages.

Check Verification

The following tables show the MCNF settings for check verification transactions for the MHI module when it is operating in terminal control mode and in host control mode.

Terminal Control Mode. The following table contains the settings for optional data fields and MAC values for check verification transactions when the MHI module is operating in terminal control mode. The MAC values have a default of N, but this value is not required and can be changed if the customer wants the field to be included in the MAC calculation.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
L	CHECK TYPE	Y	N	N	N
N	CUSTOMER ID	N*	N	N	N
O	CUSTOMER ID TYPE	N	N	N	N
j	STATE CODE	N	N	N	N
k	BIRTH DATE or TERMINAL LOCATION	N	N	N	N
p	RESPONSE CODE (PSTM.TRAN.RES-CDE)	N	N	Y	N
q	TRACK2/CUSTOMER (PSTM.TRACK2)	N*	N	N	N
2	TRACK1/CUSTOMER (PSTM.TRACK2)	N*	N	N	N

* Either FID q or 2 is required if the customer ID type (FID O) is defined as a debit or credit card. Otherwise, FID N contains the cardholder ID.

Host Control Mode. The following table contains the settings for optional data fields and MAC values on the Field Map screens when the MHI module is operating in host control mode. The MAC values have a default of N, but this value is not required and can be changed if the customer wants the field to be included in the MAC calculation.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
D	APPLICATION ACCOUNT TYPE (PSTM.TRAN.TRAN-CDE.AA)	N	N	Y	Y
L	CHECK TYPE	Y	N	N	N
N	CUSTOMER ID	N*	N	N	N
O	CUSTOMER ID TYPE	N	N	N	N

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
R	CARD TYPE (PSTM.TRAN.TRAN-CDE.T)	N	N	Y	Y
X	TRANSACTION DATE/TIME	N	N	Y	N
j	STATE CODE	N	N	N	N
k	BIRTH DATE or TERMINAL LOCATION	N	N	N	N
p	RESPONSE CODE	N	N	Y	N
q	TRACK2/CUSTOMER (PSTM.TRACK2)	N*	N	N	N
z	REVERSAL CONTEXT	N	N	Y	Y
2	TRACK1/CUSTOMER (Track 1 token)	N*	N	N	N

* Either FID q or 2 is required if the customer ID type (FID O) is defined as a debit or credit card. Otherwise, FID N contains the cardholder ID.

Check Guarantee

The following tables show the MCNF settings for check guarantee transactions for the MHI module when it is operating in terminal control mode and in host control mode.

Terminal Control Mode. The following table contains the settings for optional data fields and MAC values for check guarantee transactions when the MHI module is operating in terminal control mode. The MAC values have a default of N, but this value is not required and can be changed if the customer wants the field to be included in the MAC calculation.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
B	AMOUNT 1 (PSTM.TRAN.AMT-1)	Y	N	N	N
L	CHECK TYPE	Y	N	N	N
N	CUSTOMER ID	N*	N	N	N
O	CUSTOMER ID TYPE	N	N	N	N
j	STATE CODE	N	N	N	N
k	BIRTH DATE or TERMINAL LOCATION	N	N	N	N
p	RESPONSE CODE (PSTM.TRAN.RES-CDE)	N	N	Y	N
q	TRACK2/CUSTOMER (PSTM.TRACK2)	N*	N	N	N
2	TRACK1/CUSTOMER (Track 1 token)	N*	N	N	N

* FID q or 2 is required if the customer ID type (FID O) is defined as a debit or credit card. Otherwise, FID N contains the cardholder ID.

Host Control Mode. The following table contains the settings for optional data fields and MAC values on the Field Map screens for check guarantee transactions when the MHI module is operating in host control mode. The MAC values have a default of N, but this value is not required and can be changed if the customer wants the field to be included in the MAC calculation.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
B	AMOUNT 1 (PSTM.TRAN.AMT-1)	Y	N	N	N
D	APPLICATION ACCOUNT TYPE (PSTM.TRAN.TRAN-CDE.AA)	N	N	Y	Y
L	CHECK TYPE	Y	N	N	N
N	CUSTOMER ID	N*	N	N	N
O	CUSTOMER ID TYPE	N	N	N	N
R	CARD TYPE (PSTM.TRAN.TRAN-CDE.T)	N	N	Y	Y
X	TRANSACTION DATE/TIME	N	N	Y	N
j	STATE CODE	N	N	N	N
k	BIRTH DATE or TERMINAL LOCATION	N	N	N	N
p	RESPONSE CODE	N	N	Y	N
q	TRACK2/CUSTOMER (PSTM.TRACK2)	N*	N	N	N
z	REVERSAL CONTEXT	N	N	Y	Y
2	TRACK1/CUSTOMER (Track 1 token)	N*	N	N	N

* Either FID q or 2 is required if the customer ID type (FID O) is defined as a debit or credit card. Otherwise, FID N contains the cardholder ID.

Read-Mail and Mail-Delivered Requests

The following table contains the settings for optional data fields and MAC values on the Field Map screens for read-mail and mail-delivered requests. The MAC values have a default of N, but this value is not required and can be changed if the customer wants the field to be included in the MAC calculation.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
V	MAIL/DOWNLOAD KEY	Y	N	N	N
p	RESPONSE CODE	N	N	Y	N

Send-Mail Requests

The following table contains the settings for optional data fields and MAC values on the Field Map screens for send-mail requests. The MAC values have a default of N, but this value is not required and can be changed if the customer wants the field to be included in the MAC calculation.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
W	MAIL/DOWNLOAD TEXT	Y	N	N	N
p	RESPONSE CODE	N	N	Y	N

Clerk Totals

The following table contains the settings for optional data fields and MAC values on the Field Map screens for clerk-totals transactions. The MAC values have a default of N, but this value is not required and can be changed if the customer wants the field to be included in the MAC calculation.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
n	TOTALS/CLERK	N	N	Y	N
p	RESPONSE CODE	N	N	Y	N
w	CLERK TOTALS FLAG	N	N	Y	N

Batch Totals

The following table contains the settings for optional data fields and MAC values on the Field Map screens for batch-totals transactions. The MAC values have a default of N, but this value is not required and can be changed if the customer wants the field to be included in the MAC calculation.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
l	TOTALS/BATCH	N	N	Y	N
p	RESPONSE CODE	N	N	Y	N

Shift Totals

The following table contains the settings for optional data fields and MAC values on the Field Map screens for shift-totals transactions. The MAC values have a default of N, but this value is not required and can be changed if the customer wants the field to be included in the MAC calculation.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
o	TOTALS/SHIFT	N	N	Y	N
p	RESPONSE CODE	N	N	Y	N

Day Totals

The following table contains the settings for optional data fields and MAC values on the Field Map screens for day-totals transactions. The MAC values have a default of N, but this value is not required and can be changed if the customer wants the field to be included in the MAC calculation.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
m	TOTALS/DAY	N	N	Y	N
p	RESPONSE CODE	N	N	Y	N

Group Totals

The following table contains the settings for optional data fields and MAC values on the Field Map screens for group-totals transactions. The MAC values have a default of N, but this value is not required and can be changed if the customer wants the field to be included in the MAC calculation.

FID	Field Name	Request		Response	
		MSG	MAC	MSG	MAC
K	BUSINESS DATE	Y	N	N	N
p	RESPONSE CODE	N	N	Y	N
y	TOTALS/RETAILER OR GROUP	N	N	Y	N
x	GROUP TOTALS ID	Y	N	N	N

Download Records

The MHI module supports full and partial downloads to terminals. The information to be sent to terminals during a download is configured using the MCNF. Download records 00, 01, and 02 allow customers to download up to 26 fields of 40 characters each. The data to be included is left to the discretion of the customer. Download records 03, 04, 05, 07, 08, 09, 10, and 11 contain card prefix range processing parameters for up to 30 card prefix ranges. Download record 06 is used to identify which fields from the POS Terminal Data File (PTDF) and POS Retailer Definition File (PRDF) are downloaded to the terminal. Additional information on download records is included in section 5.

Configuring Download Records 00, 01, and 02

Download records 00, 01, and 02 allow customers to configure free format download data to be sent to the terminal. This information can consist of any information the terminal owner and operator find useful in performing day-to-day functions or that the terminal vendor requires. All of these records are optional.

The data elements are identified with download field IDs (DIDs) A through Z. DIDs A through J are stored in record 00. DIDs K through T are stored in record 01. DIDs U through Z are stored in record 02.

The data elements in records 00, 01, and 02 are sent to the terminal only if they are requested by the terminal and if they contain data.

To access MCNF Download records 00, 01, and 02, perform the following steps:

1. Enter the appropriate terminal group in the **TERMINAL GROUP** field on screen 1 of the MCNF (Selection screen).
2. Enter **D** in the **RECORD TYPE** field on screen 1 of the MCNF. The **D** in this field identifies the record type as a download.
3. Enter 00, 01, or 02 in the **RECORD ID** field on screen 1 of the MCNF, depending on which screen you want to access.

The MHI module reads all three download records, so record 00 does not have to be full before the record 01 is used. The same is true for record 01. Record 01 does not have to be full before record 02 is used.

4. Press the **F9** key from the MCNF Selection screen to display the appropriate MCNF Download Data screen. The three possible screens that can be displayed are shown on the following pages. The DIDs are listed in a column on the left side of the screens, followed by fields of 40 characters each. Free-format data can be entered in any of the 40-character fields.
5. Determine the download data to be included in DIDs A through Z.
6. Enter the download data in the appropriate DIDs in fields A through J on the first download data record screen.
7. Verify the data you entered is correct.
8. Press the **F9** key to move to the next download record screen, which contains download data for DIDs K through T.
9. Repeat steps 6 and 7 to add download data for DIDs K through T.
10. Press the **F9** key to move to the next download record screen, which contains download data for DIDs U through Z.
11. Repeat steps 6 and 7 to add download data for DIDs U through Z.
12. Press the **F3** key to add the records.

Screens for Download Records 00, 01, and 02

You can access the Download Data screens by entering a terminal group name in the **TERMINAL GROUP** field; a **D** in the **RECORD TYPE** field; and 00, 01, or 02 in the **RECORD ID** field on MCNF screen 1, and by pressing the **F9** key. A Data Download screen is shown below, followed by a description of its fields.

```

BASE24-POS   MHI MCNF MAINT           LLLL           YY/MM/DD  HH:MM  02 OF 02
              TERMINAL GROUP: 0000      RECORD TYPE: D  RECORD ID: 00

              DOWNLOAD DATA RECORD

              A
              B
              C
              D
              E
              F
              G
              H
              I
              J

*****
NEW PAGE:           FILE DESTINATION:           NEW LOGICAL NETWORK ID:
                      F12-HELP
  
```

DOWNLOAD DATA RECORD

The **DOWNLOAD DATA RECORD** fields correspond to **DIDs** A through Z. Each occurrence of the field can contain information the terminal owner and operator find useful in performing day-to-day tasks. You can enter up to 40 characters of free format text. The value in the **RECORD ID** field determines the **DIDs** that appear on the screen. The following table shows the **DIDs** by record ID:

RECORD ID Field	DIDs
00	A through J
01	K through T
02	U through Z

Field Length: 1 to 40 alphanumeric characters
Occurs: 6 to 10 times per screen
Required Field: No
Default Value: No default value
Data Name: MCNFG.FORMAT1.FLD.INFO

Configuring Download Records 03, 04, 05, 07, 08, 09, 10, and 11

Download records 03, 04, 05, 07, 08, 09, 10, and 11 contain card prefix range processing parameters. Up to 30 ranges can be defined within this set of records. Card prefix information is defined by the terminal owner and operator.

Note: For customers who have not upgraded their firmware to support 30 card prefixes (release 5.0 and later releases), only records 03, 04, and 05 are used. These records support up to 10 card prefixes.

Records 03, 04, 07, 08, 09, 10, and 11 can contain parameters for four separate card prefix ranges. Record 05 can contain parameters for two separate card prefix ranges.

To access MCNF Download Records 03, 04, 05, 07, 08, 09, 10, and 11, perform the following steps:

1. Enter the appropriate terminal group in the TERMINAL GROUP field on screen 1 of the MCNF (Selection screen).
2. Enter D in the RECORD TYPE field on screen 1 of the MCNF. The D in this field identifies the record type as a download.
3. Enter one of the following in the RECORD ID field on screen 1 of the MCNF. Note that the MHI module starts with the first card prefix on record 03 and stops with the first unused card prefix, so blank, invalid, or unused card prefixes are not allowed.

Record 03 (DIDs 0 through 3) is associated with two screens that are accessed by placing 03 in the RECORD ID field on the MCNF Selection screen.

Record 04 (DIDs 4 through 7) is associated with two screens that are accessed by placing 04 in the RECORD ID field on the MCNF Selection screen.

Record 05 (DIDs 8 through 9) is associated with one screen that is accessed by placing 05 in the RECORD ID field on the MCNF Selection screen.

Record 07 (DIDs 10 through 13) is associated with two screens that are accessed by placing 07 in the RECORD ID field on the MCNF Selection screen.

Record 08 (DIDs 14 through 17) is associated with two screens that are accessed by placing 08 in the RECORD ID field on the MCNF Selection screen.

Record 09 (DIDs 18 through 21) is associated with two screens accessed by placing 09 in the RECORD ID field on the MCNF Selection screen.

Record 10 (DIDs 22 through 25) is associated with two screens that are accessed by placing 10 in the RECORD ID field on the MCNF Selection screen.

Record 11 (DIDs 26 through 29) is associated with two screens that are accessed by placing 11 in the RECORD ID field on the MCNF Selection screen.

4. Enter information in the following fields for each card prefix range to identify the parameters:

LOW PREFIX
HIGH PREFIX
NETWORK PH
NETWORK BKUP PH
REFERRAL PH
RETAILER ID
DRAFT CAP FLG
TOTAL FLG
PIN VALIDATION FLAG
RECEIPT FLG
MOD-CK-FLG
PAN FRAUD CK FLG
USER-DEF

For a description of the MCNF fields, refer to the following topic “Screens for Download Records 03, 04, 05, 07, 08, 09, 10, and 11” where the fields displayed on the screens are described.

5. Press the **F3** key to add the record.

Screens for Download Records 03, 04, 05, 07, 08, 09, 10, and 11

Merchant Terminal Configuration File (MCNF) screens for download record 03 are shown below, followed by their field descriptions. The screens for records 04, 05, 07, 08, 09, 10, and 11 are formatted in the exact same manner as record 03. Download record 03 has two screens and defines information for up to four card prefix ranges. Download records 04, 05, 07, 08, 09, 10, and 11 also have two screens and define information for up to four card prefix ranges. Download record 05 does not have the second screen because it only defines information for two card prefix ranges. The first screen associated with download record 03 is formatted as follows:

```

BASE24-POS  MHI MCNF MAINT          LLLL          YY/MM/DD  HH:MM  02 OF 03
              TERMINAL GROUP: 0000      RECORD TYPE: D  RECORD ID: 03
              DOWNLOAD DATA RECORD

LOW PREFIX(1)                      HIGH PREFIX(1)
NETWORK PH(1)                      NETWORK BKUP PH(1)
REFERRAL PH(1)                     RETAILER ID(1)
DRAFT CAP FLG(1)                   TOTAL FLG(1)      PIN VALIDATION FLAG(1)
RECEIPT FLG(1)                     MOD-CK FLG(1)   PAN FRAUD CK FLG(1)
USER DEF(1)

LOW PREFIX(2)                      HIGH PREFIX(2)
NETWORK PH(2)                      NETWORK BKUP PH(2)
REFERRAL PH(2)                     RETAILER ID(2)
DRAFT CAP FLG(2)                   TOTAL FLG(2)      PIN VALIDATION FLAG(2)
RECEIPT FLG(2)                     MOD-CK FLG(2)   PAN FRAUD CK FLG(2)
USER DEF(2)

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:  NEW LOGICAL NETWORK ID:
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```

The second screen associated with download record 03 is formatted as follows:

```

BASE24-POS   MHI MCNF MAINT           LLLL           YY/MM/DD  HH:MM  03 OF 03
              TERMINAL GROUP:  0000           RECORD TYPE:  D   RECORD ID:  03
              DOWNLOAD DATA RECORD

LOW PREFIX(3)                HIGH PREFIX(3)
NETWORK PH(3)                NETWORK BKUP PH(3)
REFERRAL PH(3)              RETAILER ID(3)
DRAFT CAP FLG(3)            TOTAL FLG(3)        PIN VALIDATION FLG(3)
RECEIPT FLG(3)              MOD-CK FLG(3)        PAN FRAUD CK FLG(3)
USER DEF(3)

LOW PREFIX(4)                HIGH PREFIX(4)
NETWORK PH(4)                NETWORK BKUP PH(4)
REFERRAL PH(4)              RETAILER ID(4)
DRAFT CAP FLG(4)            TOTAL FLG(4)        PIN VALIDATION FLG(4)
RECEIPT FLG(4)              MOD-CK FLG(4)        PAN FRAUD CK FLG(4)
USER DEF(4)

***** BASE24 *****
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```

LOW PREFIX — The lowest card prefix value defined by this range. Any card prefix values greater than or equal to the value specified in this field, and less than or equal to the value specified in the HIGH PREFIX field are included within the card prefix range defined.

Field Length: 1–11 alphanumeric characters
 Required Field: Yes
 Default Value: No default value
 Data Name: MCNFG.FORMAT2.FLD.INFO

HIGH PREFIX — The highest card prefix range defined by this range. Any card prefix range values equal to or less than the value specified in this field, and greater than or equal to the value specified in the LOW PREFIX field are included within the card prefix range defined.

Field Length: 1–11 alphanumeric characters
 Required Field: Yes
 Default Value: No default value
 Data Name: MCNFG.FORMAT2.FLD.INFO

NETWORK PH — The telephone number of the primary network associated with transactions initiated with cards in the defined card prefix range.

Field Length: 1–20 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: MCNFG.FORMAT2.FLD.INFO

NETWORK BKUP PH — The telephone number of the backup network associated with transactions initiated with cards in the defined card prefix range.

Field Length: 1–20 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: MCNFG.FORMAT2.FLD.INFO

REFERRAL PH — The telephone number of the network designated to receive referrals concerning transactions initiated with cards in the defined card prefix range.

Field Length: 1–20 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: MCNFG.FORMAT2.FLD.INFO

RETAILER ID — An identifier for the retailer associated with this card prefix range. The retailer ID is assigned by MasterCard, Visa, American Express, or another card issuer.

Field Length: 1–12 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: MCNFG.FORMAT2.FLD.INFO

DRAFT CAP FLG — A flag indicating whether cards within the defined range use draft capture processing. This field is taken from the MCNF and overrides the TP (transaction profile) field in the POS Terminal Data File (PTDF), if the PTDF field is set so the terminal determines the draft capture mode for each transaction. Valid values are as follows:

- 0 = Authorize only
- 1 = Authorize and capture

Field Length: 1 alphanumeric character
Required Field: Yes
Default Value: No default value
Data Name: MCNFG.FORMAT2.FLD.INFO

TOTAL FLG — A flag indicating whether the terminal maintains totals for the card prefix range. The terminal must have the capability of maintaining totals if the value entered in this field is 1. Valid values are as follows:

- 0 = Do not maintain totals for the card prefix range.
- 1 = Maintain totals for the card prefix range

Field Length: 1 alphanumeric character
Required Field: Yes
Default Value: No default value
Data Name: MCNFG.FORMAT2.FLD.INFO

PIN VALIDATION FLAG — A flag indicating whether cards within the card prefix range require PINs. Valid values are as follows:

- 0 = PINs are not required for the card prefix range.
- 1 = PINs are required for the card prefix range.

Field Length: 1 alphanumeric character
Required Field: Yes
Default Value: No default value
Data Name: MCNFG.FORMAT2.FLD.INFO

RECEIPT FLG — A flag indicating whether transactions initiated with cards within the card prefix range require a receipt. Valid values are as follows:

- 0 = Receipts are not required for the card prefix range.
- 1 = Receipts are required for the card prefix range.

Field Length: 1 alphanumeric character
Required Field: Yes
Default Value: No default value
Data Name: MCNFG.FORMAT2.FLD.INFO

MOD-CK FLG — A flag indicating whether MOD-10 checks are used. Valid values are as follows:

- 0 = No terminal MOD-10 checks are used for the card prefix range.
- 1 = Terminal MOD-10 checks are used for the card prefix range.

Field Length: 1 alphanumeric character
Required Field: Yes
Default Value: No default value
Data Name: MCNFG.FORMAT2.FLD.INFO

PAN FRAUD CK FLG — A flag indicating whether the terminal performs PAN fraud checks on cards within the card prefix range. If PAN fraud checks are performed, this field also indicates the type. Valid values are as follows:

- 0 = No fraud checks are used for the card prefix range.
- 1 = Visually check the last four digits of the PAN.
- 2 = Visually check Track 2.

Field Length: 1 alphanumeric character
Required Field: Yes
Default Value: No default value
Data Name: MCNFG.FORMAT2.FLD.INFO

USER DEF — Up to eight characters of user-defined data can be entered in this field. This data is used by the terminal for cards within the card prefix range.

Field Length: 8 alphanumeric characters
Required Field: No
Default Value: No default value
Data Name: MCNFG.FORMAT2.FILLER

Configuring Download Record 06

The data elements in download record 06 consist of flags corresponding to fields in the POS Terminal Data File (PTDF) and POS Retailer Definition File (PRDF). These flags are used to determine the fields to be downloaded to the terminal. Record 06 can store up to 26 download flags. DIDs a through z identify the 26 data elements that can be included in record 06, although only 12 of the data elements are currently supported. The data elements in download record 06 are sent to the terminal only if they are configured to be sent and are requested by the terminal.

To access MCNF download record 06, perform the following steps:

1. Enter the appropriate terminal group in the **TERMINAL GROUP** field on screen 1 of the MCNF (Selection screen).
2. Enter **D** in the **RECORD TYPE** field on screen 1 of the MCNF. The **D** in this field identifies the record type as a download.
3. Enter **06** in the **RECORD ID** field on screen 1 of the MCNF.
4. Press the **F9** key from the MCNF Selection screen to display the MCNF Download Record screen.
5. Enter **Y** for each field to be included in download record 06. Enter **N** for each field to be excluded from download record 06.
6. Press the **F3** key to add the record.

Download Record 06 Screen

Merchant Terminal Configuration File (MCNF) screens for download record 06 is shown below. Refer to section 5 for descriptions of the data elements shown on the screen.

```

BASE24-POS    MHI MCNF MAINT          LLLL          YY/MM/DD  HH:MM  02 OF 02
              TERMINAL GROUP: 0000    RECORD TYPE:  D   RECORD ID: 06

```

DOWNLOAD DATA RECORD

a	TERM-NAM-LOC	N	b	TERM-CITY-ST	N	c	TERM-OWNER-NAM	N
d	RESERVED	N	e	RESERVED	N	f	SRV-REP	N
g	P-KEY	N	h	A-KEY(DEV-DEP-DATA)	N	i	PIN-PAD-CHAR	N
j	RESERVED	N	k	SRV	N	l	FLOOR LIMITS	N
m	RESERVED	N	n	RESERVED	N	o	RESERVED	N
p	ALLOWED-TRANS	N	q	RETAILER-ID	N	r	MERCHANT NAME	N
s	REFERRAL PHONE #	N	t	RESERVED	N	u	RESERVED	N
v	RESERVED	N	w	RESERVED	N	x	RESERVED	N
y	RESERVED	N	z	RESERVED	N			

(VALID ENTRIES ARE Y OR N)

```

***** BASE24 *****
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```

DOWNLOAD DATA RECORD

The DOWNLOAD DATA RECORD fields correspond to DIDs a through z. Each occurrence of the field indicates whether the corresponding DID is to be included in record 06.

Field Length:	1 alphabetic character
Occurs:	26 times
Required Field:	No
Default Value:	N
Data Name:	MCNFG.FORMAT3.FLD.PRESENT

Merchant Terminal Response File

The Merchant Terminal Response File (MRSP) allows customers to alter the responses sent to the terminal, depending on the transaction being sent and the language being used.

The MRSP contains a map that links BASE24 response codes to response code descriptions. This map is one record within the MRSP. Up to three more records can be added to the file. These additional records can contain the wording to send a response explanation in different languages. The wording can be sent to the terminal in the response and used for printing receipts and displaying messages to the terminal.

Selecting MRSP Records

The first screen associated with the MRSP allows you to choose the record type. This screen is your main menu for selecting the appropriate MRSP record to configure. The MRSP contains various types of information required to configure responses sent from the MHI module to the terminal.

There are four types, or sets, of screens associated with the MRSP. These are used to access five records that can be stored in the MRSP. These screen types are the following:

- Selection
- Response Display Map
- Language Display
- Transaction Description

Each of the above screen groups is described on the following pages.

Accessing the Selection Screen

The MRSP Selection screen is your primary link to accessing MRSP records. To access the MRSP Selection screen, perform the following steps:

1. Enter MRSP in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key. MRSP screen 1 is displayed.

2. Enter the appropriate values in the **TERMINAL GROUP** and **RECORD NUMBER** fields on MRSP screen 1, based on the parameters you want to configure (that is, responses, language display, or transaction descriptions) and the terminal group to which you want the parameters to apply.

The Response Display Map record, accessed by placing 00 in this field, links each response code to a response display.

The three Language Display records, accessed by placing 01, 02, or 03 in this field, can contain responses in different languages.

The Transaction Description record, accessed by placing 04 in this field, allows customers to determine the name of transactions.

3. Once values are entered in these fields, press the **F9** key to access the record.
4. Configure the selected record and press the **F3** key to add the record.

Selection Screen

Merchant Terminal Response File (MRSP) screen 1 is shown below, followed by descriptions of its fields.

```

BASE24-POS   MHI MRESP MAINT           LLLL       YY/MM/DD  HH:MM  01 OF 07
      TERMINAL GROUP:                RECORD NUM:  00

                                DIRECTIONS:

                                ENTER THE DESIRED TERMINAL GROUP AND RECORD NUM.

                                PLEASE PRESS DESIRED FUNCTION KEY TO CONTINUE

                                RECORD NUM 00 = RESPONSES
                                RECORD NUM 01 = DISPLAY LANGUAGE #1
                                RECORD NUM 02 = DISPLAY LANGUAGE #2
                                RECORD NUM 03 = DISPLAY LANGUAGE #3
                                RECORD NUM 04 = DESCRIPTION OF TRANSACTIONS

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
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```

TERMINAL GROUP — The terminal group identifies the terminal or group of terminals to which the configuration information in the MRSP applies. The TERMINAL GROUP combined with the RECORD NUMBER is the primary key into the MRSP. The configuration information is used by all terminals that have a value corresponding to this field in the TERM GROUP field in the POS Terminal Data File (PTDF).

Field Length: 4 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: MRESP.PRIKEY.TERM-GRP

RECORD NUMBER — The number associated with the record to be accessed. The value entered in this field determines the type of screens displayed. Valid values are as follows:

00 = Response display map
01 = Display language table 1
02 = Display language table 2
03 = Display language table 3
04 = Transaction description

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: MRESP.PRIKEY.REC-NUM

Response Display Map Record

The MRSP Response Display Map record allows you to provide a text description for the response code contained in the standard message header. The Response Display Map screens are used in conjunction with the Language Display records (described later in this section) to provide customized response code descriptions in up to three languages. The Response Display Map screens link response codes to response code descriptions contained in the Language Display records. For example, you may prefer certain wording for a response code or require response code descriptions in different languages. Records can be entered for all three languages.

The main purpose of the Response Display Map screens is to provide an index into the Language Display screens. The Response Display Map record consists of six screens that list the BASE24-pos response codes and their corresponding

descriptions as defined in the POS Standard Internal Message (PSTM). From these screens, you can enter an index number for each response code. The index number corresponds to text contained in the Language Display records.

MHI Module Response Code Processing

Two optional data fields control the MRSP Response Display Map record: Response Display (FID g) and Language Code (FID U). FID g must be configured in the Merchant Terminal Configuration File (MCNF) for inclusion in messages if the MRSP Response Display Map record is used. FID g contains up to 48 characters of alphanumeric text that you define in this MRSP record to describe the response code contained in the standard message header. If FID g is configured in the MCNF for a particular transaction type, a description corresponding to a specific response code contained in the standard message header is included in the message.

If FID g is not configured in the MCNF for a particular transaction type, no response code description is included in the message. You can determine whether to include FID g in messages for each transaction type when configuring the Field Map screens in the MCNF. For more information on setting up the MCNF Field Map screens, refer to the MCNF documentation earlier in this section.

FID U is used to determine which of the three language displays to use for responses. If FID U is included in the request, this language code overrides the default language code defined in the POS Terminal Data File (PTDF) and is echoed in responses. If FID U is not included in the request, the default language code from the PTDF identifies the language display used for the response.

Configuring Response Display Map Record 00

Response Display Map record 00 consists of six screens that list all of the BASE24-pos response codes and their corresponding descriptions as defined in the PSTM. From these screens, you can enter a number for each response code. This number corresponds to an index entry in the Language Display record. Each index entry in the Language Display record contains text describing the response code. The MHI module determines which Language Display record to use based on the language code it receives from the terminal in the request. If no language code is

contained in the request, the MHI module uses the default language code contained in the PTDF. To access and set up the MRSP Response Display Map record 00, perform the following steps:

1. Check the MCNF Field Map screens to ensure that FID g (Response Display) has been configured to be included in messages for the transaction types for which you want to have customized response messages. If FID g is not configured for the transaction types you want, add it to the configuration using the steps described in the “Configuring Field Map Records” topic documented earlier in this section.
2. Enter MRSP in the FILE DESTINATION field at the bottom of the BASE24 screen. If the current screen is a menu, press the **F1** key. Otherwise, press the **F16** key to access the MRSP Selection screen.
3. Enter the appropriate value in the TERMINAL GROUP field, based on the terminal group to which you want the MRSP parameters to apply.
4. Enter 00 in the RECORD NUMBER field on the MRSP Selection screen and press the **F9** key to access the first screen associated with the MRSP Response Display Map record.

Record 00 consists of six screens of response codes. Continue to press the **F9** key to access subsequent screens.

Each screen associated with record 00 is divided into two columns, each containing three fields. The first field in each column contains the BASE24-pos standard response code number. Response codes are listed on the screens in numeric order from left to right across the screen.

The second field in each column contains the standard BASE24-pos response code description that corresponds to the response code number identified in the first field. The response code number and description are taken from the COBNAMES table.

The third field in each column is the only field on the screens that you can alter. This field contains the number used by the MHI module to index the response code description entered in the Language Display record and placed in FID g of the response.

5. Select the response code you want to index by pressing the **Tab** key until the cursor is positioned directly to the right of the response code on the screen.

The **Tab** key automatically positions the cursor in the second column on the screens where you can enter your modified response code descriptions.

6. Enter a two-digit number to index the Language Display record.
7. Repeat the previous two steps until you have completed indexing the response codes on one screen.

8. Verify the response codes are indexed as you want them on the screen, make any corrections, and display the next screen by pressing the **F9** key. The **F9** key is used to access subsequent pages associated with the record.
9. Repeat the procedures described above on each MRSP Response Display screen for each response code you want to index.

Note: The MRSP Response Display record consists of six screens, each containing response codes. You must enter information on each screen for each response code you want to index.
10. When you have completed indexing the response codes on all of the MRSP Response Display screens, press the **F3** key to add the record.
11. Press the **F2** key to read the record to verify it has been added correctly.

Response Display Map Screens

The MRSP Response Display Map screens are shown below and on the following pages. Since the fields on the screens associated with the Response Display Map record do not have field names, the data on the screens is described according to the column in which it is located. The screens consist of two sets of three columns each. The first column in each set contains a three-digit number. Column descriptions are provided following all the sample screens.

```

BASE24-POS  MHI MRESP MAINT          LLLL      YY/MM/DD  HH:MM  02 OF 07
      TERMINAL GROUP:                RECORD NUM: 00
<rc><description>                   <ld>  <rc><description>          <ld>
000 APPROVED(BALANCES)              00   001 APPROVED(NO BALANCES)      00
002 APPROVED(COUNTRY CLUB)          00   003 APPROVED(MAYBE MORE ID)    00
004 APPROVED(PEND. ID. SIGN)        00   005 APPROVED(BLIND)             00
006 APPROVED(VIP)                   00   007 APPROVED(ADMIN TRAN)       00
008 APPROVED(NATL NEG HIT OK)       00   009 APPROVED(COMMERCIAL CRD)    00
050 DECLINED                        00   051 EXPIRED CRD                 00
052 PIN TRIES EXCEEDED               00   053 NO SHARING W/USER          00
054 NO ATALLA BOX                   00   055 INVALID TRAN              00
056 TRAN NOT SUPPORTED BY FI        00   057 LOST OR STOLEN CRD         00
058 INVALID CRD STAT                00   059 INQUIRY ONLY              00
060 ACCT NOT ON CAF                 00   061 PBF REC NOT FOUND          00
062 PBF UPDATE ERROR                00   063 INVALID AUTH TYPE IN IDF    00
064 BAD TRACK INFO                  00   065 ADJ NOT ALLOWED IN IDF     00
066 INVALID CCD ADV INCREMENT       00   067 INVALID TRAN DATE         00
068 TLF ERROR                       00   069 BAD MSG IN PSTM           00

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
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```

```
BASE24-POS  MHI MRESP MAINT      LLLL      YY/MM/DD  HH:MM  03 OF 07
      TERMINAL GROUP:      RECORD NUM:  00
<rc><description>      <ld>  <rc><description>      <ld>
070 NO IDF      00      071 INVALID ROUT TO AUTH      00
072 CRD ON NATL NEG      00      073 INVALID ROUT TO AUTH-SER FLD 00
074 UNABLE TO AUTHORIZE      00      075 INVALID PAN LENGTH      00
076 INSUF FNDS IN PBF      00      077 PRE-AUTH FULL      00
078 DUP TRAN RECVD AND DROPPED      00      079 MAX ONLIN REFND REACHED      00
080 MAX OFFLIN REFND REACHED      00      081 MAX CRED % REFND REACHED      00
082 MAX NUM TIMES USED      00      083 MAX REFND CRED REACHED      00
084 CUST SELECT NEG REASN      00      085 INQ NOT ALLOWED-NO BAL      00
086 OVER FLOOR LIMIT      00      087 MAX NUM REFND REACHED      00
088 PLACE CALL      00      089 CAF STAT 0-9      00
090 REFERRAL FILE FULL      00      091 NEG FILE PROB      00
092 ADV LESS THAN MINIMUM      00      093 DELINQUENT      00
094 OVER TBL LIMIT      00      095 AMT OVER MAX      00
096 PIN REQUIRED      00      097 MOD 10 CHECK      00
098 DECLINE FORCE POST      00      099 BAD PBF      00
```

```
***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP
```

```
BASE24-POS  MHI MRESP MAINT      LLLL      YY/MM/DD  HH:MM  04 OF 07
      TERMINAL GROUP:      RECORD NUM:  00
<rc><description>      <ld>  <rc><description>      <ld>
100 UNABLE TO PROCESS TRAN      00      101 UNABLE TO AUTHORIZE-CALL      00
102 CALL      00      103 NEG FILE ERROR      00
104 CAF FILE PROBLEM      00      105 CARD NOT SUPPORTED      00
106 AMT OVER MAX      00      107 OVER DAILY LIMIT      00
108 CAPF NOT FOUND      00      109 ADV LESS THAN MIN      00
110 NUM TIMES USED      00      111 DELINQUENT      00
112 OVER LIMIT TBL      00      113 TIMEOUT      00
115 PTLF/PRF FILE PROBLEM      00      120 RESERVED FOR FUTURE USE      00
150 MERCHANT NOT ON FILE      00      121 RESERVED FOR FUTURE USE      00
200 INVALID ACCT      00      201 INCORRECT PIN      00
202 CASH ADV LESS THAN MIN      00      203 ADMIN CRD NEEDED      00
204 ENTER LESSER AMT      00      205 INVALID CCD ADV AMT      00
206 CAF NOT FOUND      00      207 INVALID TRAN DATE      00
208 INVALID EXPIRATION DATE      00      209 INVALID TRAN CODE      00
      RESERVED FOR FUTURE USE      00      251 ENTER LESSER CASH BACK AMT 00
```

```
***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP
```

```
BASE24-POS  MHI MRESP MAINT      LLLL      YY/MM/DD  HH:MM  05 OF 07
      TERMINAL GROUP:      RECORD NUM: 00
<rc><description>      <ld>  <rc><description>      <ld>
800 FORMAT ERROR      00      801 INVALID DATE      00
802 INVALID CLERK NUM      00      809 INVALID CLOSE TRANSACTION      00
810 TIMEOUT      00      811 SYSTEM ERROR      00
820 INVALID TERM ID      00      821 INVALID RESP LENGTH      00
830 INVALID DPC #      00      831 INVALID MSG TYPE      00
832 INVALID MSG SUBTYPE      00      833 INVALID TRANSMISSION #      00
834 INVALID AMT FORMAT      00      835 INVALID ACCT TYPE      00
836 INVALID APPROVAL CDE      00      837 INVALID BUSINESS DATE      00
838 INVALID TRAN CATEGORY      00      839 INVALID CUSTOMER ID      00
840 INVALID CUSTOMER ID TYPE      00      841 INVALID DRAFT CAPTURE FLG      00
842 INVALID ECHO DATA      00      843 INVALID CARD TYPE      00
844 INVALID INVOICE NUM      00      845 INVALID LANGUAGE CODE      00
846 INVALID OPTIONAL DATA      00      847 INVALID RETAILER ID      00
848 INVALID CONDITION CODE      00      849 INVALID DRIVERS LIC. STATE      00
850 INVALID DRIVERS LIC. D.O.B.      00      851 INVALID TOTALS FORMAT      00
```

```
***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP
```

```
BASE24-POS  MHI MRESP MAINT      LLLL      YY/MM/DD  HH:MM  06 OF 07
      TERMINAL GROUP:      RECORD NUM: 00
<rc><description>      <ld>  <rc><description>      <ld>
852 INVALID AUTH TIMEOUT OVERRIDE      00      853 INVALID CLERK TOTALS FLG      00
854 INVALID GRP TOTALS ID      00      855 INVALID PIN LENGTH      00
856 INVALID REVERSAL CONTEXT      00      RESERVED FOR FUTURE USE      00
860 INVALID TOTAL GRP FOR RETAIL      00      861 INVALID TTL GRP FOR TERM      00
862 PARTIAL TTLS RET FOR TTL GRP      00      863 INVALID TTLS/FLD OVERFLOW      00
864 INVALID PTLF FILENAME      00      865 PTLF OPEN ERROR      00
870 OK - DELIVERED      00      871 OK - STORED      00
880 OK - NO MORE DATA      00      881 OK - MORE DATA EXISTS      00
882 LOAD ABORTED      00      883 INVALID MAIL/DOWNLOAD KEY      00
884 INVALID MAIL/DOWNLOAD TEXT      00      RESERVED FOR FUTURE USE      00
898 AUTHENTICATE ERR- INVALID MAC      00      899 SEQ ERR / BEGIN RESYNC      00
900 PIN TRIES EXCEEDED      00      901 EXPIRED CRD      00
902 NEG CAPTURE CODE      00      903 CAF STATUS 3      00
904 ADV LESS THAN MIN      00      905 NUM TIMES USED      00
906 DELINQUENT      00      907 OVER LIMIT TBL      00
```

```
***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP
```

```

BASE24-POS  MHI MRESP MAINT      LLLL      YY/MM/DD  HH:MM  07 OF 07
      TERMINAL GROUP:      RECORD NUM:  00
<rc><description>      <ld>  <rc><description>      <ld>
908 AMT OVER MAX      00    909 CAPTURE      00
950 ADMIN CRD NOT ON FILE      00    951 ADMIN CRD NOT ALLOWED      00
952 ADMIN REQ / IN WINDOW      00    953 ADMIN REQ / OUT OF WINDOW      00
954 ADMIN REQ / ANYTIME      00      RESERVED FOR FUTURE USE      00

```

```

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
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```

<rc> — The values in this column are BASE24-pos response codes and cannot be altered by the user. All BASE24-pos response codes are listed in the record. The screens on which particular response codes are listed are noted below. The screen numbers are in the upper-right corner of the screen.

Several of the spaces in the response code column are blank, indicating that numbers have been reserved for future use.

The BASE24-pos response codes listed here correspond to the RESP-CDE field in the POS Standard Internal Message (PSTM). Additional information about the response codes can be found in the ***BASE24-pos Transaction Processing Manual***.

Response Codes	Screen Number
000 through 069	02
070 through 099	03
100 through 251	04
800 through 851	05

Response Codes	Screen Number
852 through 907	06
908 through 954	07

Field Length: System-protected
Occurs: Up to 30 times per screen

<description> — The values in this column are descriptions of the BASE24-pos response codes listed in the first column and cannot be modified.

Field Length: System-protected
Occurs: Up to 30 times per screen

<ld> — The values in this column relate to the various language displays. The numbers in this column are used as an index into the Language Display records. The range of valid values is 00 through 39.

Field Length: 2 numeric characters
Occurs: Up to 30 times per screen
Required Field: Yes
Default Value: 00
Data Name: MRESP.RESP.TBL

Language Display Records

The MRSP Language Display records contain response code descriptions that correspond to index entries contained in the Response Display Map record. There are three separate Language Display records, each consisting of four screens. Each Language Display record is organized by index entry numbers of 0 through 39. You can enter response code descriptions following each one of these index entry numbers, and you can enter response code descriptions in up to three different languages in these records. If you enter a response code description following one of the index entry numbers, you must link the description to the corresponding response code in the Response Display Map record.

MHI Module Language Code Processing

FID U controls the language code used to access the appropriate Language Display record. FID U must be configured in the MCNF if it is to be included in messages. FID U is used to determine which of the three language displays to use for responses. If FID U is included in the request from the terminal, its value overrides the default language code defined in the POS Terminal Data File (PTDF) and is echoed in responses. If FID U is not included in the request, the MHI module uses the default language code set in the PTDF to determine which record to access.

The MHI module reads the language code it receives in the request from the terminal. The MHI module uses the language code (that is, FID U) in the request to determine which Language Display record to access. If no language code is included in the request (that is, FID U is absent), the MHI module uses the default language code set in the PTDF to determine which record to access. The language codes that can be included in the request from the terminal are as follows:

- 0 = Language Display Record 1
- 1 = Language Display Record 2
- 2 = Language Display Record 3

The MHI module reads the Response Display Map record and determines which response codes have corresponding response code descriptions indexed in the Language Display records. The MHI module accesses the appropriate Language Display record, matches the correct response code description to the response code number it is indexed to in the Response Display record, and sends the information to the terminal in the response.

Configuring Language Display Records

The MRSP Language Display records consist of three separate records, each containing four screens. Record 01 retrieves the record for Language Display Table 1 responses; record 02 retrieves the record for Language Display Table 2 responses; and record 03 retrieves the record for Language Display Table 3 responses.

To access the MRSP Language Display Records screen, perform the following steps:

1. Check the MCNF Field Map screens to ensure that FID U (Language Code) has been configured to be included in messages for the transaction types you want. If FID U is not configured for the transaction types you want, add it to

the configuration using the steps described in the “Configuring Field Map Records” topic documented earlier in this section. An alternative to this step is to use the default language code set up in the PTDF. If the default language code is being used, skip this step and move to step 2.

2. Enter MRSP in the FILE DESTINATION field at the bottom of the MCNF screen. If the current screen is a menu, press the **F1** key. Otherwise, press the **F16** key to access the MRSP Selection screen.
3. Enter the appropriate value in the TERMINAL GROUP field, based on the terminal group to which you want the MRSP parameters to apply.
4. Enter 01, 02, or 03 in the RECORD NUMBER field on the MRSP Selection screen and press the **F9** key to access the first screen associated with the MRSP Language Display record.
5. Select the index number to which you want to add a response code by pressing the **Tab** key until the cursor is positioned directly to the right of the index number on the screen.

The **Tab** key automatically positions the cursor in the second column on the screens where you can enter your modified response code descriptions.

6. Enter a response code description of up to 48 characters in length.
7. Repeat the previous two steps until you have completed indexing the response code descriptions on one screen.
8. Verify the response code descriptions are entered as you want them on the screen, make any corrections, and move to the next screen by pressing the **F9** key. The **F9** key is used to access subsequent pages associated with the record.
9. Repeat the procedures described above on each MRSP Language Display screen for each response code description required.

Note: The response code descriptions indexed in the MRSP Language Display record correspond to response code numbers in the MRSP Response Display records. You should verify that response code descriptions match their corresponding response codes. In addition, you must enter response code descriptions on each screen for each response code number indexed in the MRSP Response Display records.

10. When you have completed indexing the response code descriptions on all of the MRSP screens, press the **F3** key to add the record.
11. Press the **F2** key to read the record to verify it has been added correctly.

Language Display Screens

Since all of the Language Display records are formatted in exactly the same manner, only the screens for record 01 are included in this section. The screens for record 01 are shown on the following pages, followed by descriptions of their fields.

```

BASE24-POS   MHI MRESP MAINT      LLLL      YY/MM/DD  HH:MM  02 OF 05
      TERMINAL GROUP:                RECORD NUM:  01

      INDEX      <response code description>
      0
      1
      2
      3
      4
      5
      6
      7
      8
      9

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                   F12-HELP

```

INDEX — The number corresponding to the Response Display Map record. When a number is listed in the <Id> column of the Response Display Map screens, this column is searched for the same number. The text on the line containing the matching number (the corresponding <response code description> field on the Response Display Map screens) is then sent to the terminal in the response.

Field Length: System-protected
Occurs: 10 times per screen

<response code description> — The description to be sent to the terminal when the number in the INDEX column is referenced on the Response Display Map record.

BASE24-POS MHI MRESP MAINT LLLL YY/MM/DD HH:MM 03 OF 05
 TERMINAL GROUP: RECORD NUM: 01

INDEX	<response code description>
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	

***** BASE24 *****
 NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
 F12-HELP

BASE24-POS MHI MRESP MAINT LLLL YY/MM/DD HH:MM 04 OF 05
 TERMINAL GROUP: RECORD NUM: 01

INDEX	<response code description>
20	
21	
22	
23	
24	
25	
26	
27	
28	
29	

***** BASE24 *****
 NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
 F12-HELP

```
BASE24-POS  MHI MRESP MAINT      LLLL      YY/MM/DD  HH:MM  05 OF 05
      TERMINAL GROUP:                RECORD NUM:  01
```

```
      INDEX      <response code description>
      30
      31
      32
      33
      34
      35
      36
      37
      38
      39
```

```
***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP
```

Optional data may be included in the responses listed on this screen, if certain codes are included in the response. In addition to a text description, you can use the following codes to allow variable data to be included in the response. The codes must be uppercase or lowercase as shown.

\B = Amount 1
\C = Amount 2
\E = Application account
\F = Approval code
\J = Available balance
\K = Business date
\N = Customer ID
\Q = Echo data
\S = Invoice number
\T = Original invoice number
\Z = Address verification status code
\d = Retailer ID
\h = Sequence number
\k = Terminal location

Example:	\K\d
Field Length:	1–48 alphanumeric characters
Occurs:	10 times per screen
Required Field:	Yes
Default Value:	No default value
Data Name:	MRESP.DISPLAYS.TEXTS

Transaction Description Records

The MRSP Transaction Description Record screens allow you to provide transaction descriptions for each transaction type performed by the MHI module. You can either use the descriptions provided in this record or enter text for each transaction type and create your own customized transaction descriptions.

The MRSP Transaction Description record consists of four screens, each containing two columns. The first column provides descriptions of all transaction types known to the MHI module. In the second column, you can enter up to 24 characters of text to create your own transaction description for a particular transaction type. Transaction descriptions you enter in the second column correspond to transaction descriptions displayed in the first column.

MHI Module Transaction Description Processing

Transaction descriptions are controlled by FID s (Transaction Description). FID s must be configured in the Merchant Terminal Configuration File (MCNF) for each transaction type that should have a transaction description included in its messages. If FID s is configured in the MCNF for a particular transaction type, a transaction description corresponding to the transaction code contained in the standard message header is included in the message. For example, if the transaction code in the standard message header is 00 (normal purchase), the corresponding transaction description might be NORM PURCH in the MRSP record.

If FID s is not configured in the MCNF for a particular transaction type, no transaction description is included in the message. You can determine whether to include FID s in messages for each transaction type when configuring the Field Map screens in the MCNF. For more information on setting up the MCNF Field Map screens, refer to the MCNF documentation earlier in this section.

When the MHI module is configured to include FID s in its response, the MHI module uses the transaction descriptions contained in this record when formatting its response. The MHI module checks the transaction code contained in the standard message header of the request. It then uses the transaction code to retrieve the correct transaction description from the MRSP record.

Two conditions can exist when the MHI module is retrieving the transaction description from the MRSP. The first condition occurs if FID s is configured to be included in responses and you have entered your own transaction description in the MRSP record for a particular transaction type. In this case, the MHI module checks the second column of the MRSP record, retrieves the transaction description you have entered, and places it in FID s in the response.

The second condition exists if FID s is configured to be included in responses and you have not entered a transaction description in the second column on the MRSP Transaction Description screens. In this case, the MHI module checks the second column of the record and finds no transaction description. Since no transaction description has been entered in the second column, the MHI module uses the transaction description displayed in the first column of the screen and places it in FID s in the response.

In both cases, the transaction code contained in the standard message header of the request is the index to determine the transaction description the MHI module uses when formatting the response. If you have entered your own transaction description in the second column, the MHI module uses it. If you have not, the MHI module uses the transaction description in the first column.

Configuring Transaction Description Records

The following instructions explain how you can access and set up the MRSP Transaction Description Record screens. Setting up this record may require you to use information contained in the Merchant Terminal Configuration File (MCNF), which is explained earlier in this section. To access and set up the MRSP Transaction Description records, perform the following steps:

1. Check the MCNF Field Map screens to ensure that FID s (Transaction Description) has been configured to be included in messages for the transaction types you want. If FID s is not configured for the transaction types you want, add it to the configuration using the steps described in the “Configuring Field Map Records” topic documented earlier in this section.
2. Enter MRSP in the FILE DESTINATION field at the bottom of the MCNF screen. Press the **F16** key to access the MRSP Selection screen.

3. Enter the appropriate value in the TERMINAL GROUP field, based on the terminal group to which you want the MRSP parameters to apply.
4. Enter 04 in the RECORD NUMBER field on the MRSP Selection screen and press the **F9** key to access the first screen associated with the MRSP Transaction Description record.
5. Select the transaction description you want to alter by pressing the **Tab** key until the cursor is positioned to the right of the transaction description you want to alter.

You cannot alter the transaction descriptions displayed in the first column on the screens. The **Tab** key automatically positions the cursor in the second column on the screens where you can enter your modified transaction descriptions.

6. Enter up to 24 characters of text to create your own transaction description.
7. Repeat the previous two steps until you have completed altering transaction descriptions on one screen.
8. Verify the transaction descriptions are as you want them on the screen, make any corrections, and move to the next screen by pressing the **F9** key. The **F9** key is used to access subsequent pages associated with the record.
9. Repeat steps 5 through 8 described above on each MRSP Transaction Description screen for each transaction description you want to alter.

Note: The MRSP Transaction Description record consists of four screens, each containing transaction descriptions. You must enter information on each screen for each transaction description you want to alter.

10. When you have completed altering the transaction descriptions on all of the MRSP Transaction Description screens, press the **F3** key to add the record.
11. Press the **F2** key to read the record to verify it has been added correctly.

Transaction Description Record Screens

The Merchant Terminal Response File (MRSP) Transaction Description Record screens are shown below, followed by descriptions of their fields.

```
BASE24-POS  MHI MRESP MAINT      LLLL      YY/MM/DD  HH:MM  02 OF 05
          TERMINAL GROUP:          RECORD NUM:  04
```

DESCRIPTION RECORD

<original description>	<customer description>
NORMAL PURCH	
PRE-AUTH PURCH	
PRE-AUTH PURCH/COMP	
PHONE/MAIL ORDER	
MERCH RETURN	
CASH ADVANCE	
CARD VERIFICATION	
BALANCE INQ	

```
***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
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```

```
BASE24-POS  MHI MRESP MAINT      LLLL      YY/MM/DD  HH:MM  03 OF 05
          TERMINAL GROUP:          RECORD NUM:  04
```

DESCRIPTION RECORD

<original description>	<customer description>
PURCH/CASH BACK	
CHK VERIFICATION	
CHK GUARANTEE	
PURCH ADJUSTMENT	
MERCH RETURN ADJUSTMENT	
CASH ADVANCE ADJUSTMENT	
RESERVED	

```
***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                   F12-HELP
```

BASE24-POS MHI MRESP MAINT LLLL YY/MM/DD HH:MM 04 OF 05
TERMINAL GROUP: RECORD NUM: 04

DESCRIPTION RECORD

<original description>	<customer description>
LOGON REQUEST	
LOGOFF REQUEST	
CLOSE BATCH REQUEST	
CLOSE SHIFT REQUEST	
CLOSE DAY REQUEST	
EMPLOYEE SUBTOT REQUEST	
BATCH SUBTOT REQUEST	

***** BASE24 *****
NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

BASE24-POS MHI MRESP MAINT LLLL YY/MM/DD HH:MM 05 OF 05
TERMINAL GROUP: RECORD NUM: 04

DESCRIPTION RECORD

<original description>	<customer description>
SHIFT SUBTOT REQUEST	
DAY SUBTOT REQUEST	
RETAIL GRP SUBTOT REQUEST	
READ MAIL REQUEST	
MAIL DELIVERED REQUEST	
SEND MAIL REQUEST	
DOWNLOAD REQUEST	
HANDSHAKE REQUEST	

***** BASE24 *****
NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

DESCRIPTION RECORD

The following fields describe supported transactions with original descriptions and customer descriptions.

<original description> — The transaction description as it is known by the MHI module.

Field Length: System-protected
Occurs: Up to 8 times per screen

<customer description> — The transaction description as defined by the customer for the purpose of responses.

Field Length: 1–24 alphanumeric characters
Occurs: Up to 8 times per screen
Required Field: No
Default Value: No default value
Data Name: MRESP.DESCR.TBL

Merchant Host Configuration File

The Merchant Host Configuration File (MHCF) contains one record for each data processing center (DPC) that communicates with the MHI module. The record contains processing information specific to that DPC, including processing flags and station tables. The MHI module uses the information contained in the MHCF for processing requests from the merchant host. The processing flags configured in the MHCF are used to determine the processing parameters supported by the MHI module such as message authentication codes (MACs), message synchronization, and whether the MHI module is running in terminal or host control mode.

Accessing MHCF Screens

The MHCF contains only one record type. Therefore, the MHCF does not require a selection screen. To access the MHCF screens, perform the following steps:

1. Enter MHCF in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key. MHCF screen 1 is displayed.
2. Enter the appropriate values in the DPC NUMBER and HISF NAME fields to retrieve an existing record. These two fields are the primary key to the MHCF records. The HISF NAME field is the symbolic name of the MHI module and the DPC NUMBER is the 4-digit DPC number.
3. Press the **F2** key to retrieve the record.

Merchant Host Configuration File Screen 1

Merchant Host Configuration File (MHCF) screen 1 is shown below, followed by descriptions of its fields.

```

BASE24-POS      MHI HOST CONFIG      LLLL      YY/MM/DD  HH:MM  01 OF 03
DPC NUMBER:      1                  HISF NAME: P1A^MHI

      GROUP TOTALS PROCESS NAME: P1A^MHITTL

DPC PERFORMANCE MONITORING PERIOD:  20  (MINUTES)

      ***  PROCESSING FLAGS  ***

      STORE AND FORWARD:  0
      DPC TYPE:           0
      MAC TYPE:           0
      SEQ NUM CHECKING:   0
      LOGON/LOGOFF MSG:   0
      UPDATE PTFD:        1

      PTFD TERMINAL ID:
      USE MESSAGE TERM ID: N  (Y/N)
      RETURN TOKEN DATA: N  (Y/N)

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP

```

DPC NUMBER — The number of the institution data processing center (DPC). This is the host computer that processes messages received from BASE24. The values in the DPC NUMBER and HISF NAME fields combine to make the primary key to the MHCF. Valid values are 0001 to 9999.

Field Length: 4 numeric characters
 Required Field: Yes
 Default Value: 0001
 Data Name: MHCF.PRIKEY.DPC-NUM

HISF NAME — The symbolic name of the BASE24 Merchant Host Interface process that communicates with the specified DPC.

Field Length: 1–16 alphanumeric characters
 Required Field: Yes
 Default Value: P1A^MHI
 Data Name: MHCF.PRIKEY.HISF-PRO

GROUP TOTALS PROCESS NAME — The symbolic name of the Merchant Totals process responsible for processing the retailer or group totals requests from this DPC.

Field Length: 16 alphanumeric characters
Required Field: No
Default Value: P1A^MHITTL
Data Name: MHCF.GRP-TTL-PRO

DPC PERFORMANCE MONITORING PERIOD — The number of minutes the MHI module uses as an interval for tracking performance of the DPC.

BASE24 enables network operators to request performance statistics using a network control facility text command. The value in this field specifies the number of minutes the MHI module uses as an interval for tracking DPC performance.

Field Length: 1–4 numeric characters
Required Field: Yes
Default Value: 20 minutes
Data Name: MHCF.PERFORMANCE

PROCESSING FLAGS

The following six fields contain flags used in processing.

STORE AND FORWARD — Specifies the method the MHI module uses for sending store-and-forward messages to the DPC. Store-and-forward processing is invoked whenever a DPC is available and there are store-and-forward messages to be sent. Valid values are as follows:

- 0 = Intersperse store-and-forward messages with real-time messages going to a merchant host for approval. Under this method, real-time transactions are processed normally, with the exception of reversals, which are placed in the Store-and-Forward File (SAF) in case the message being reversed has not yet been processed.
- 1 = Transmit all store-and-forward messages prior to any real-time transactions. Under this method, if there are any messages for a DPC in the SAF, these messages are sent prior to any real-time transactions.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: MHCF.PROCESSING-FLG.SAF-METHOD

DPC TYPE — Specifies whether the DPC stations for a merchant host are single-threaded or multithreaded. Valid values are as follows:

- 0 = Multithreaded. Multiple messages can be outstanding to a single station at one time. A response is not required to each message before another message is sent to the station.
- 1 = Single-threaded. Only one message can be outstanding to a station at one time. A response is required to each message before another message can be sent to the station.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: MHCF.PROCESSING-FLG.DPC-TYP

MAC TYPE — Indicates the type of message authentication codes (MACs) used by the DPC. Valid values are as follows:

- 0 = No message authentication performed.
- 1 = BASE24-to-host MACs supported. The MACs are used when the MHI module is operating in host or terminal control mode.
- 2 = BASE24-to-terminal MACs supported. The MACs are used when the MHI module is operating in terminal control mode.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: MHCF.PROCESSING-FLG.MAC-FLG

SEQ NUM CHECKING — Indicates the type of sequence number checking performed by the MHI module. If the UPDATE PTDF field on this screen is set to 0, this field must also be set to 0. Valid values are as follows:

- 0 = No batch sequence number checking
- 1 = Batch sequence number checking

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: MHCF.PROCESSING-FLG,SEQ-NUM-FLG

LOGON/LOGOFF MSG — Indicates whether the MHI module should expect the specified DPC to send network management messages to log the DPC on and off. Valid values are as follows:

- 0 = No network management messages are sent.
- 1 = Network management messages are required.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: MHCF.PROCESSING-FLG.LOGON-FLG

UPDATE PTDF — Indicates whether the DPC is running in terminal or host control mode. If the DPC is running in host control mode, the PTDF on disk is not impacted on every transaction (i.e., only the version stored in memory is updated when the transaction is processed). If the DPC is running in terminal control mode, the PTDF on disk is impacted on every transaction. The PTDF is updated with transaction-specific data such as terminal totals and sequence numbers. Valid values are as follows:

- 0 = Host control mode (do not update PTDF on disk)
- 1 = Terminal control mode (update PTDF on disk)

Field Length: 1 numeric character
Required Field: Yes
Default Value: 1
Data Name: MHCF.PROCESSING-FLG.PTDF-UPDT-FLG

PTDF TERMINAL ID — Identifies the terminal ID of the PTDF record read for transactions received from a DPC running in host control mode.

Field Length: 1–16 alphanumeric characters
Required Field: Yes, if the UPDATE PTDF field is set to 0.
Default Value: No default value
Data Name: MHCF.PROCESSING-FLG.DFLT-TERM-ID

USE MESSAGE TERM ID — Indicates if the MHI module should map the terminal ID contained in the message header to the TERM-ID field in the POS Standard Internal Message (PSTM). This field is used only when the MHI module is running in host control mode.

Valid values for this field are as follows:

Y = Yes, map the terminal ID field in the standard message header to the PSTM.
N = No, do not map the terminal ID in the standard message header to the PSTM.

If this field is set to Y and the message header contains a terminal ID that is not blanks or zeros, the message header terminal ID is mapped to the TERM-ID field in the PSTM. If the MHI Regulation E File (MREG) exists, the MHI module locates the MREG record using the terminal ID from the message header. The terminal ID in the message header is used to read the MREG to obtain certain Regulation E data for the specified terminal. The data in the MREG (for example, terminal location, terminal city, etc.) overrides the corresponding Regulation E data contained in the PTDF. The information in the MREG is only mapped to the PSTM and is not included in the response from the MHI module to the host or terminal.

If an MREG record is not found, the message header terminal ID is blanks or zeros, or this field is set to N, the default terminal ID from the PTDF TERMINAL ID field in the MHCF is used to process the transaction (that is, Regulation E data is taken from the PTDF for the terminal).

Note: This field does not affect how the PTDF is read. The PTDF is still read using the terminal ID in the PTDF TERMINAL ID field of the MHCF, regardless of the setting in this field.

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: N
Data Name: MHCF.USE-MSG-TERM-ID

RETURN TOKEN DATA — Indicates whether token data should be returned in FID z in the response message when the merchant host generates a reversal transaction. This field is only used if the MHI module is running in host control mode. Valid values are as follows:

Y = Yes, return token data.
N = No, do not return token data.

Field Length: 1 alphabetic character
 Required Field: Yes
 Default Value: N
 Data Name: MHCF.RETURN-TKN-DATA

Merchant Host Configuration File Screen 2

Merchant Host Configuration File (MHCF) screen 2 is shown below, followed by descriptions of its fields.

```

BASE24-POS      MHI HOST CONFIG      LLLL      YY/MM/DD  HH:MM  02 OF 03
DPC NUMBER:     1                      HISF NAME: P1A^MHI

      ****  MAC ENCRYPTION CONFIGURATION  ****

MAC ENCRYPTION TYPE: 00
      MASTER KEY:
MAC KEY VARIANT: 0

      ****  PIN ENCRYPTION CONFIGURATION  ****

PIN ENCRYPTION TYPE: 00
      MASTER KEY:
PIN PAD CHARACTER:
PIN KEY VARIANT: 0

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP
  
```

MAC ENCRYPTION CONFIGURATION

The following fields contain message authentication code (MAC) configuration information.

MAC ENCRYPTION TYPE — Indicates the MAC generation method used between the terminal and BASE24. Valid values are as follows:

- 00 = No message authentication
- 01 = Software message authentication
- 02 = Proprietary software message authentication
- 03 = Security module message authentication

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: MHCF.MAC-DATA.MAC-ENCRYPT-TYP

MASTER KEY — The MAC master key for the specified DPC.

Field Length: 16 hexadecimal characters
Required Field: No
Default Value: No default value
Data Name: MHCF.MAC-DATA.ENCR-KEYS.MAS-KEY

MAC KEY VARIANT — Specifies the key variant to be applied to the MAC key exchange key (KEK) when translating MAC keys. Valid values are as follows:

0 = Key exchange key variant
3 = MAC encryption key variant

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: MHCF.MAC-KEY-VARIANT

PIN ENCRYPTION CONFIGURATION

The following fields contain PIN encryption configuration information.

PIN ENCRYPTION TYPE — Indicates the PIN generation method used between the terminal and BASE24. This field is required if the MHI module is running in host control mode. Valid values are as follows:

00 = No encryption (default)
01 = Software (PIN/PAD block)
02 = Software (PIN/PAN block)
03 = Hardware (PIN/PAD block)
04 = Hardware (PIN/PAN block)

Field Length: 2 numeric characters
Required Field: Yes, if the MHI module is running in host control mode.
Default Value: 00
Data Name: MHCF.PIN-ENCRYPT-TYP

MASTER KEY — The PIN master key for the specified DPC. This field is required when the MHI module is operating in host control mode.

Field Length: 16 hexadecimal characters
Required Field: Yes, if the MHI module is running in host control mode.
Default Value: No default value
Data Name: MHCF.PIN-DATA.ENCN-KEYS.M-KEY

PIN PAD CHARACTER — Specifies the PAD character used in the formation of the external message PIN block if PIN/PAD PIN block encryption is used. Valid values are A through F and 0 through 9.

Field Length: 1 alphabetic character
Required Field: No
Default Value: No default value
Data Name: MHCF.PIN-PAD-CHAR

PIN KEY VARIANT — Specifies the key variant to be applied to the PIN key exchange key (KEK) when translating the PIN keys. Valid values are as follows:

0 = Key exchange key variant
1 = PIN encryption key variant

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: MHCF.PIN-KEY-VARIANT

Merchant Host Configuration File Screen 3

Merchant Host Configuration File (MHCF) screen 3 is shown below, followed by descriptions of its fields.

```
BASE24-POS      MHI HOST CONFIG      LLLL      YY/MM/DD  HH:MM  03 OF 03
DPC NUMBER:      1                  HISF NAME: P1A^MHI
STATION          TYPE  DESCRIPTION    STATION          TYPE  DESCRIPTION
```

```
***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                   F12-HELP
```

STATION — A symbolic name identifying the station associated with this DPC. The entry in the STATION field must match a symbolic station name in the BASE24-pos environment and be unique within the logical network.

Example: S1AHOST
Field Length: 1–16 alphanumeric characters
Occurs: 32 times
Required Field: No
Data Name: MHCF.STA.STA-SYM-NAME

TYPE — A code identifies the type of station named in the STATION field on the same line of the table. The type determines whether the MHI module can send all messages to this station or only receive messages from this station. Receive-only station types are not allowed for single-threaded DPCs. Valid values are as follows:

- 0 = Unrestricted send and receive station
- 1 = Receive-only station, no output

Field Length: 1 numeric character
Occurs: 32 times
Required Field: Yes, if an entry has been made in the STATION field on the same line of the table.
Default Value: No default value
Data Name: MHCF.STA.STA-TYP

DESCRIPTION — The description of the type of station identified in the TYPE field. If 0 was entered in the TYPE field, the DESCRIPTION field contains SEND/RCV. If 1 was entered in the TYPE field, the DESCRIPTION field contains RCV/NOOUTPUT.

Field Length: System-protected
Occurs: 32 times
Data Name: Not applicable

Merchant Totals Configuration File

The Merchant Totals Configuration File (MTCF) contains records defining which terminals are included in a totaling group for retailer or group subtotals request transactions. One record exists for each terminal within a totaling group. This allows for an unlimited number of terminals to be in a single totaling group.

The Merchant Totals process uses the data in the MTCF to determine which terminal ID records are included in the group totals in response to a retailer or group subtotals request transaction. The Merchant Totals process reads this data into extended memory at initialization and upon receipt of a WARMBOOT command from a network control facility.

Accessing an MTCF Record

The MTCF contains only one record type. Therefore, it does not have a selection screen. To access the MTCF, perform the following steps:

1. Enter MTCF in the FILE DESTINATION field at the bottom of the BASE24 files maintenance screen. From a menu, press the **F1** key and from a file screen, press the **F16** key to display MTCF screen 1.
2. Enter the appropriate values in the RETAILER ID and TOTALLING GROUP fields to retrieve an existing record.

Merchant Totals Configuration File Screen 1

The MTCF screen is shown below, followed by descriptions of its fields.

```

BASE24-POS  MERCHANT TOTALS CNFG  LLLL  YY/MM/DD  HH:MM  01 OF 01
RETAILER ID:  TOTALLING GROUP:

```

```

TERMINAL IDS:

```

```

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:  NEW LOGICAL NETWORK ID:
F2-READ-FIRST-PAGE  F4-DELETE-GROUP  F5-UPDT-CURR-PAGE  F12-HELP

```

RETAILER ID — Identifies the retailer ID to which the terminals belong. This field together with the TOTALLING GROUP field make up the primary key to the MTCF. This field must match the RETAILER NAME field in the POS Retailer Definition File (PRDF).

Field Length: 1–19 alphanumeric characters
 Required Field: Yes
 Default Value: No default value
 Data Name: MTCF.PRI-KEY.PART-PRI-KEY.RETAILER-ID

TOTALLING GROUP — Identifies the totaling group to which the terminal belongs. This field together with the RETAILER ID field make up the primary key to the MTCF.

Field Length: 1–10 alphanumeric characters
 Required Field: Yes
 Default Value: No default value
 Data Name: MTCF.PRI-KEY.PART-PRI-KEY.TTL-GRP-ID

TERMINAL IDS — Identifies the terminal IDs belonging to the totaling group and retailer ID defined in the primary key fields. Up to 48 terminal IDs can be entered on this screen. This field links the terminal IDs to a totaling group and retailer ID for the purpose of maintaining group and retailer totals.

Field Length: 1–16 alphanumeric characters
Occurs: 48 times
Required Field: Yes
Default Value: No default value
Data Name: MTCF.PRI-KEY.TERM-ID

Merchant Regulation E File

The Merchant Regulation E File (MREG) contains Regulation E data for the specified terminal. Data in the MREG must be in compliance with Regulation E requirements. The Regulation E data in the MREG overrides the corresponding data in the POS Terminal Data File (PTDF) when **all** of the following conditions are met:

- The MHI module is running in host control mode. The MREG is only used when the MHI module is running in host control mode. The MHI module cannot use the MREG when it is running in terminal control mode.
- The standard message header in the transaction request contains a terminal ID. The terminal ID in the standard message header cannot be blanks or zeros.
- The USE MESSAGE TERM ID field in the Merchant Host Configuration File (MHCF) is set to Y. This field is used to determine whether the MHI module maps the terminal ID from the message header to the PSTM TERM-ID field to access data from the MREG. If the USE MESSAGE TERM ID field in the MHCF is set to N, the MREG is not used and the value in the PTDF TERMINAL ID field in the MHCF is used to access Regulation E data from the PTDF.
- The MREG contains a record for the specified terminal.

Accessing an MREG Record

The MREG contains only one record type. Therefore, it does not have a selection screen. To access the MREG, perform the following steps:

1. Enter MREG in the FILE DESTINATION field at the bottom of the BASE24 files maintenance screen. From a menu, press the **F1** key, and from a file screen, press the **F16** key to display MREG screen 1.
2. Enter the appropriate values in the TERM ID field to retrieve an existing record. This field makes up the primary key to the MREG.

Merchant Regulation E File Screen 1

The MREG screen is shown below, followed by descriptions of its fields.

```
BASE24-POS  MHI REG-E          LLLL      YY/MM/DD  HH:MM  01 OF 01
TERM ID:                FIID:
```

CITY:

STATE:

LOCATION:

OWNER NAME:

```
***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                   F12-HELP
```

TERM ID — Identifies the terminal ID taken from the standard message header in the transaction request. The MHI module maps the terminal ID to the TERM-ID field in the PSTM to access the MREG. If the terminal ID is not included in the standard message header, or if it contains blanks or zeros, the Regulation E data from the PTDF is used. The terminal ID from the standard message header makes up the primary key to the MREG.

Field Length: 1–16 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: MREG.TERM-ID

FIID — Identifies the financial institution to which the terminal specified in the TERM ID field belongs.

Field Length: 1–4 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: MREG.FIID

CITY — Identifies the city where the specified terminal is located.

Field Length: 1–18 alphanumeric characters
Required Field: No
Default Value: No default value
Data Name: MREG.TERM-CITY

STATE — Identifies an abbreviation of the state or province where the specified terminal is located.

Field Length: 1–3 alphanumeric characters
Required Field: No
Default Value: No default value
Data Name: MREG.TERM-ST

LOCATION — Identifies the name of the terminal and a description of where it is located.

Field Length: 1–25 alphanumeric characters
Required Field: No
Default Value: No default value
Data Name: MREG.TERM-NAME-LOC

OWNER NAME — Identifies the name of the retailer who owns the terminal.

Field Length: 1–22 alphanumeric characters
Required Field: No
Default Value: No default value
Data Name: MREG.TERM-OWNER-NAME

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Section 9

Transaction Message Flows

This section contains examples of the messages flows specific to the MHI module. These examples are not designed to provide detailed descriptions of MHI module processing. They are designed, however, to allow you to follow the logical flow of messages that occurs during transaction processing. The examples, with the exception of the MAC reversal scenario, assume that all message validation is successful. Each example includes a diagram illustrating the message flow, followed by a description of each step.

The transaction message flows listed below are documented in this section. For more information on general BASE24-pos transaction message flows, refer to the ***BASE24-pos Transaction Processing Manual***.

- Approved normal purchase received by the device
- Approved normal purchase; response not returned to the merchant host
- Approved normal purchase; response does not reach the terminal
- Declined normal purchase received by the device
- Declined normal purchase; response does not reach the terminal
- Declined normal purchase; response not returned to the merchant host
- Normal purchase approved offline
- Terminal logon by a clerk
- Terminal logoff
- Network management message logon request
- Network management message echo test request
- Network management message logoff request
- MAC reversal
- Merchant host reversal
- Customer-cancellation reversal

- Unsolicited mail
- Terminal send-mail request
- Mail pick-up request—single response
- Mail pick-up request—multiple responses
- Mail pick-up request—no mail stored

BASE24-pos Transaction Flows

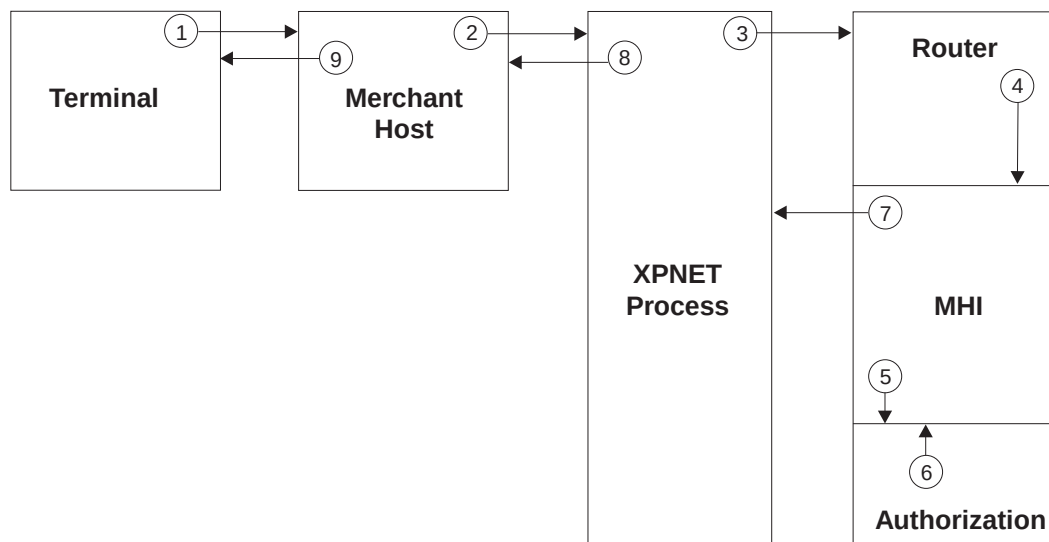
When the merchant host sends a message to BASE24, it is received by the XPNET process, which forwards the message to the Merchant Host Interface (MHI) module.

The MHI module is bound into a BASE24-pos Router/Authorization process. Any messages coming into the bound process from the merchant host are first received by the Router module which, then routes the message to the MHI module.

The MHI module is fully compatible with release 5.3 of BASE24-pos. The following diagrams illustrate the exchange of messages between the MHI module and the terminal or merchant host during transaction processing using BASE24-pos.

Approved Normal Purchase Received by Device

The diagram below illustrates the transaction message flow in a scenario containing a normal purchase transaction with a response. In this example the transaction was approved.



Steps

Processing

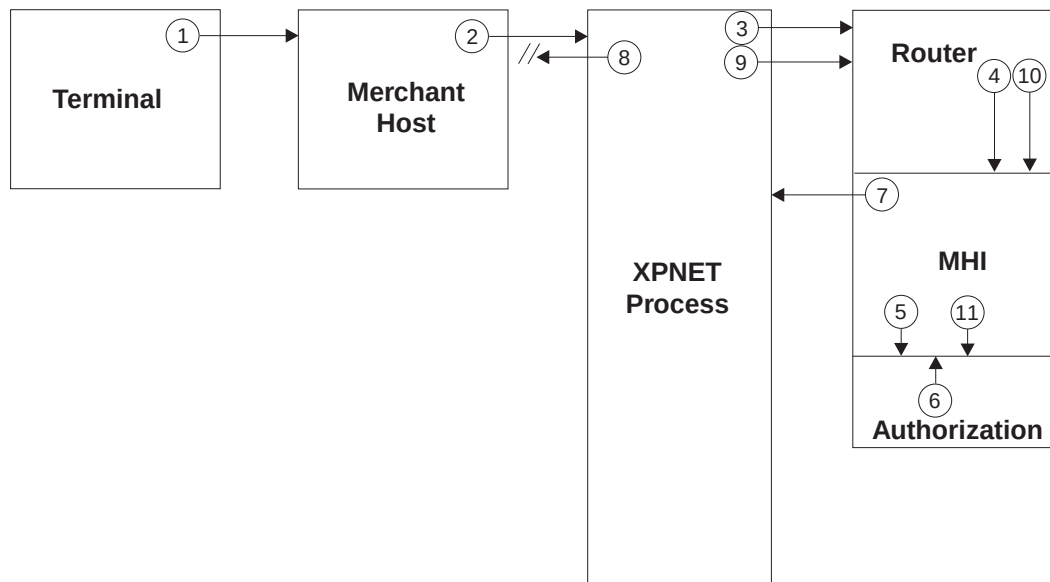
- 1 The terminal sends its native message for a normal purchase to the merchant host.
- 2, 3, 4 The merchant host formats an MHI external message and sends the message to the MHI module. This message travels through the XPNET process and the Router module.
- 5 The MHI module validates the request and formats a POS Standard Internal Message (PSTM) for a normal purchase. The MHI module then sends the message to the Authorization module.
- 6 The Authorization module approves the transaction and returns a response to the MHI module indicating the transaction was approved.
- 7, 8 The MHI module updates the PTDF totals, translates the response from PSTM format into an MHI external message, and sends the response to the merchant host. This message travels through the XPNET process.

Steps**Processing**

- 9 The merchant host formats the response into the native terminal format and sends the response to the terminal.

Approved Normal Purchase; Response Not Returned to the Merchant Host

The diagram below illustrates the transaction message flow in a scenario containing an approved transaction that was not returned to the merchant host because communications between BASE24 and the merchant host were down in the response.

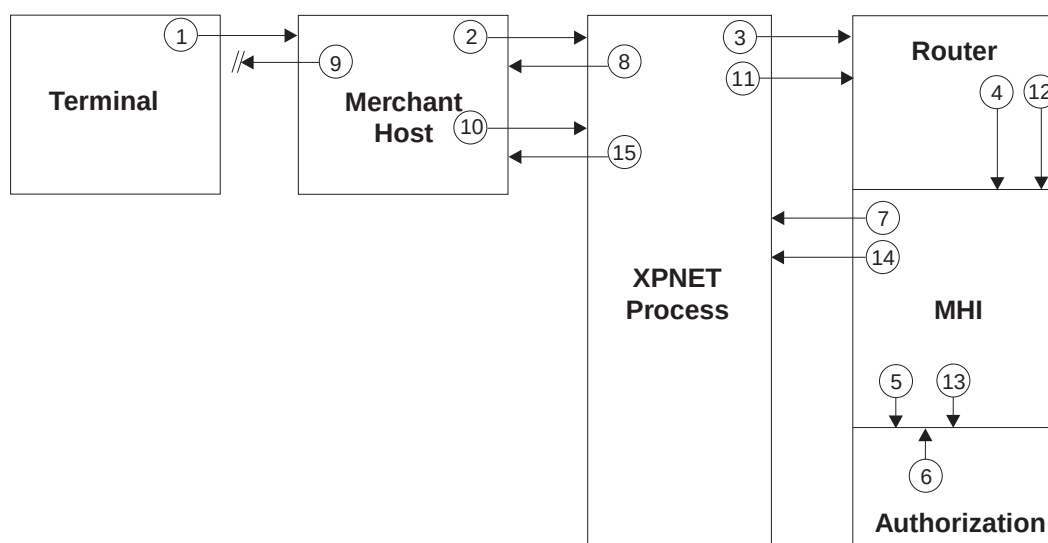


Steps	Processing
1	The terminal sends its native message for a normal purchase to the merchant host.
2, 3, 4	The merchant host formats an MHI external message and sends the message to the MHI module. This message travels through the XPNET process and the Router module.
5	The MHI module validates the request and formats a PSTM for a normal purchase. The MHI module then sends the message to the Authorization module.
6	The Authorization module approves the transaction and returns a response to the MHI module indicating the transaction was approved.

Steps	Processing
7, 8	The MHI module updates the PTDF totals, translates the response from the PSTM format into an MHI external message and sends the response to the merchant host. This message travels through the XPNET process, but it does not reach its destination.
9, 10	The XPNET process returns a failed message to the MHI module. This message travels through the Router module.
11	The MHI module formats a reversal message using the context saved for a reversal. If this is a terminal control DPC, the reversal context is saved in the device dependent data in the PTDF. If this is a host control DPC, the reversal context should be contained in the response message sent in step 7. The MHI module sends the reversal to the Authorization module.

Approved Normal Purchase; Response Does Not Reach the Terminal

The diagram below illustrates the transaction message flow in a scenario containing a normal purchase transaction that was approved. The response, however, was not returned to the terminal since the link between the terminal and the merchant host was down when the response was processed. In this example, the transaction was reversed.



Steps

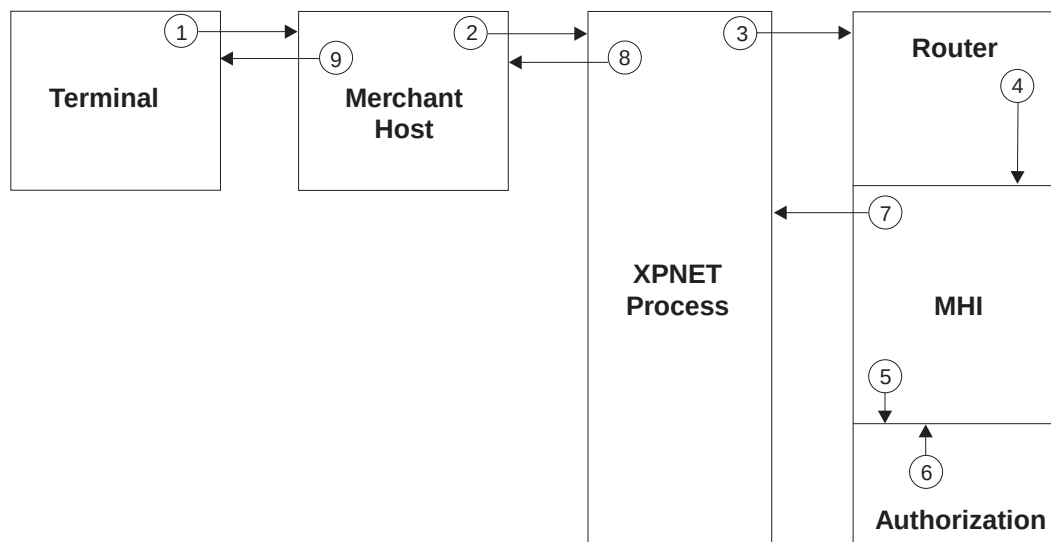
Processing

- 1 The terminal sends its native message for a normal purchase to the merchant host.
- 2, 3, 4 The merchant host formats an MHI external message and sends the message to the MHI module. This message travels through the XPNET process and the Router module.
- 5 The MHI module validates the request and formats a PSTM for a normal purchase. The MHI module then sends the message to the Authorization module.
- 6 The Authorization module approves the transaction and returns a response to the MHI module indicating the transaction was approved.

Steps	Processing
7, 8	The MHI module updates the PTDF totals, translates the response from PSTM format into an MHI external message, and sends the response to the merchant host. This message travels through the XPNET process.
9	The merchant host tries to send the response to the terminal, but the communications link is down.
10, 11, 12	Because the link between the merchant host and the terminal is down, the merchant host formats a reversal (message type R) and sends it to the MHI module. This message travels through the XPNET process and the Router module.
13	The MHI module validates the reversal and formats a reversal message using the context saved for a reversal. If this is a terminal control DPC, the reversal context information is saved in the device dependent data in the POS Terminal Data File (PTDF). If this is a host control DPC, the reversal context should be contained in the response message sent in step 7. The reversal is sent to the Authorization module.
14, 15	The MHI module echoes the reversal response back to the merchant host. This message travels through the XPNET process.

Declined Normal Purchase Received by Device

The diagram below illustrates the transaction message flow in a scenario containing a normal purchase transaction with a response. In this example, the transaction was declined.



Steps

Processing

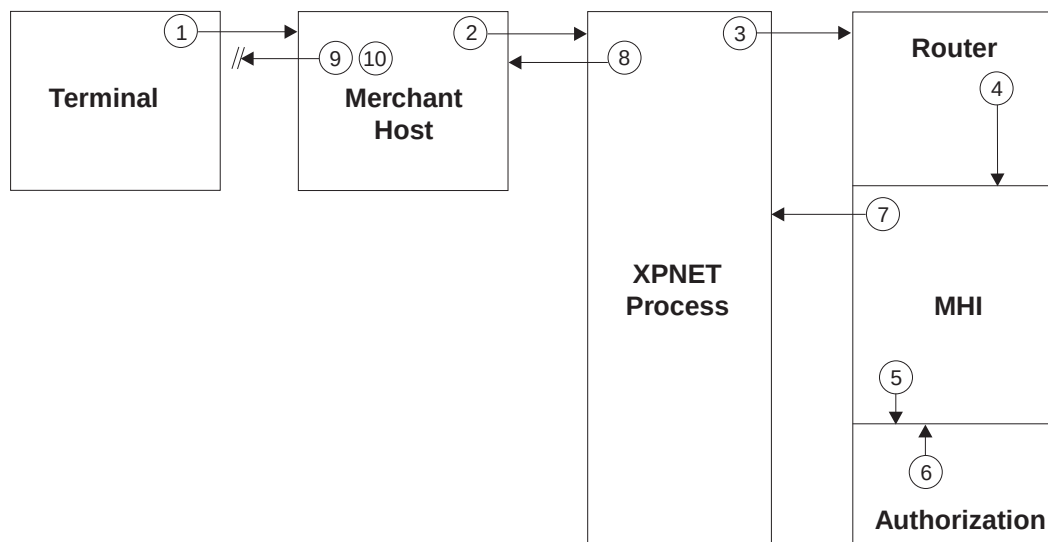
- 1 The terminal sends its native message for a normal purchase to the merchant host.
- 2, 3, 4 The merchant host formats an MHI external message and sends the message to the MHI module. This message travels through the XPNET process and the Router module.
- 5 The MHI module validates the request and formats a POS Standard Internal Message (PSTM) for a normal purchase. The MHI module then sends the message to the Authorization module.
- 6 The Authorization module declines the transaction and returns a response to the MHI module indicating the transaction has been declined.
- 7, 8 The MHI module translates the response from PSTM format into an MHI external message. The MHI module then sends the response to the merchant host. This message travels through the XPNET process.

Steps**Processing**

- 9 The merchant host formats the response into the native terminal format and sends the response to the terminal.

Declined Normal Purchase; Response Does Not Reach the Terminal

The diagram below illustrates the transaction message flow in a scenario containing a normal purchase transaction that was not received by the terminal because communications between the terminal and the merchant host were down. In this example, the transaction was declined.

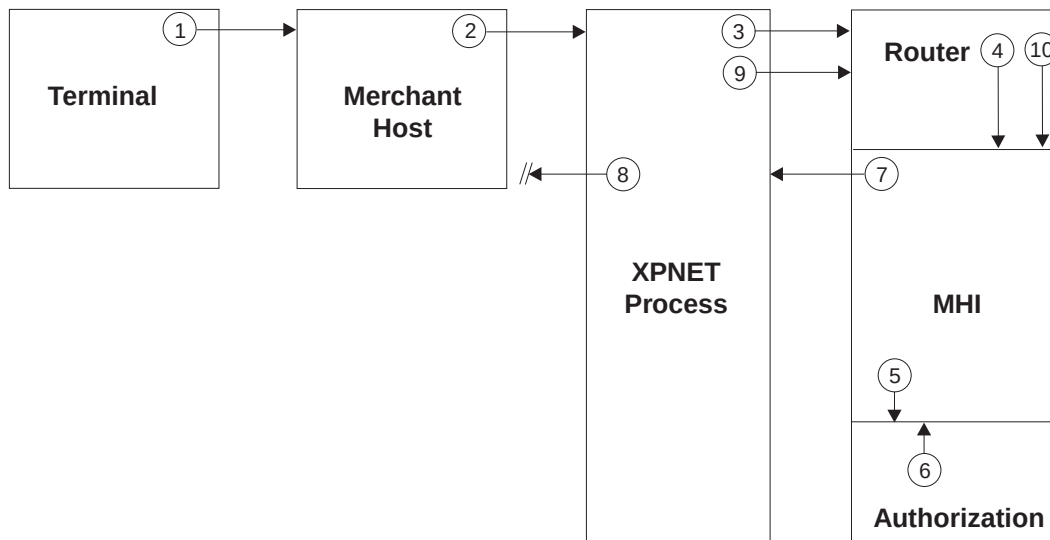


Steps	Processing
1	The terminal sends its native message for a normal purchase to the merchant host.
2, 3, 4	The merchant host formats an MHI external message and sends the message to the MHI module. This message travels through the XPNET process and the Router module.
5	The MHI module validates the request and formats a PSTM for a normal purchase. The MHI module then sends the message to the Authorization module.
6	The Authorization module declines the transaction and returns a response to the MHI module indicating the transaction was declined.
7, 8	The MHI module translates the response from PSTM format into an MHI external message and sends the response to the merchant host. This message travels through the XPNET process.

Steps	Processing
9	The merchant host tries to send the response to the terminal, but the communications link is down.
10	Because the communications link between the merchant host and the terminal is down, the merchant host is unable to send a response to the terminal. Since the transaction was declined, no further processing is necessary. The terminal will eventually time out and take the appropriate actions.

Declined Normal Purchase; Response Not Returned to the Merchant Host

The diagram below illustrates the transaction message flow in a scenario containing a normal purchase transaction that was not returned to the merchant host because communications between BASE24 and the merchant host were down in the response. In this example, the transaction was declined.

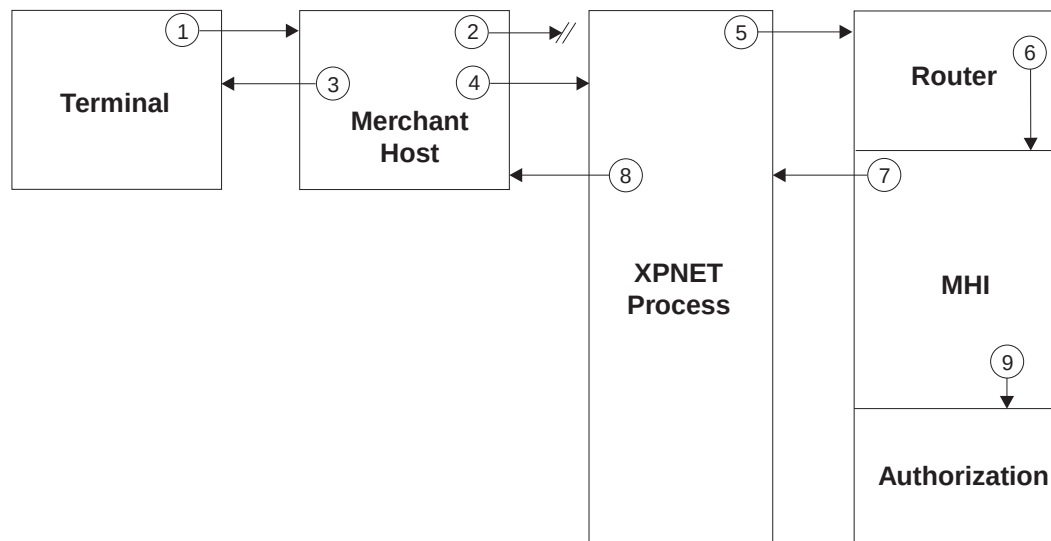


Steps	Processing
1	The terminal sends its native message for a normal purchase to the merchant host.
2, 3, 4	The merchant host formats an MHI external message and sends the message to the MHI module. This message travels through the XPNET process and the Router module.
5	The MHI module validates the request and formats a PSTM for a normal purchase. The MHI module then sends the message to the Authorization module.
6	The Authorization module declines the transaction and returns a response to the MHI module indicating the transaction was declined.

Steps	Processing
7, 8	The MHI module translates the response from the PSTM format into an MHI external message and sends the response to the merchant host. This message travels through the XPNET process, but it does not reach its destination.
9, 10	The XPNET process returns a failed message to the MHI module. This message travels through the Router module. Since the transaction was declined, no further processing is required. The terminal or merchant host will time out and take the appropriate steps.

Normal Purchase Approved Offline

The diagram below illustrates the transaction message flow in a scenario containing a normal purchase transaction that is authorized offline because the communications link between the merchant host and BASE24 is down. In this example, the transaction is approved.



Steps

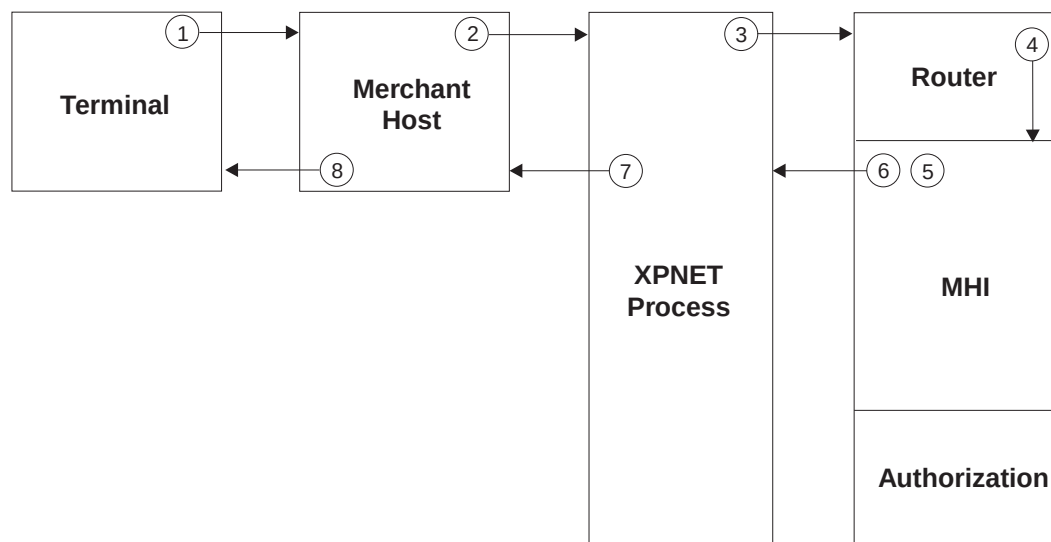
Processing

- 1 The terminal sends its native message for a normal purchase to the merchant host.
- 2 The merchant host formats an MHI external message and sends the message to the XPNET process. However, because the communications link between the merchant host and BASE24 is down, the merchant host must stand in and authorize the transaction. The merchant host formats an advice that is sent to BASE24 to notify the issuer that the transaction has been authorized. This advice is placed in the Store-and-Forward File (SAF).
- 3 The merchant host sends a response to the terminal indicating the transaction was approved.
- 4, 5, 6 When the communications link between BASE24 and the merchant host is restored, the advice stored in the SAF is sent to the MHI module. This message travels through the XPNET process and the Router module.

Steps	Processing
7, 8	The MHI module validates the request, determines that the request is an advice, and formats an acknowledgment. Since the transaction has already been approved there is no need to wait for the Authorization module to approve the transaction. The MHI module sends the acknowledgment to the merchant host. This message travels through the XPNET process.
9	The MHI module formats a PSTM for an advice and sends the message to the Authorization module. The Authorization module logs the advice to the POS Transaction Log File (PTLF).

Terminal Logon by a Clerk

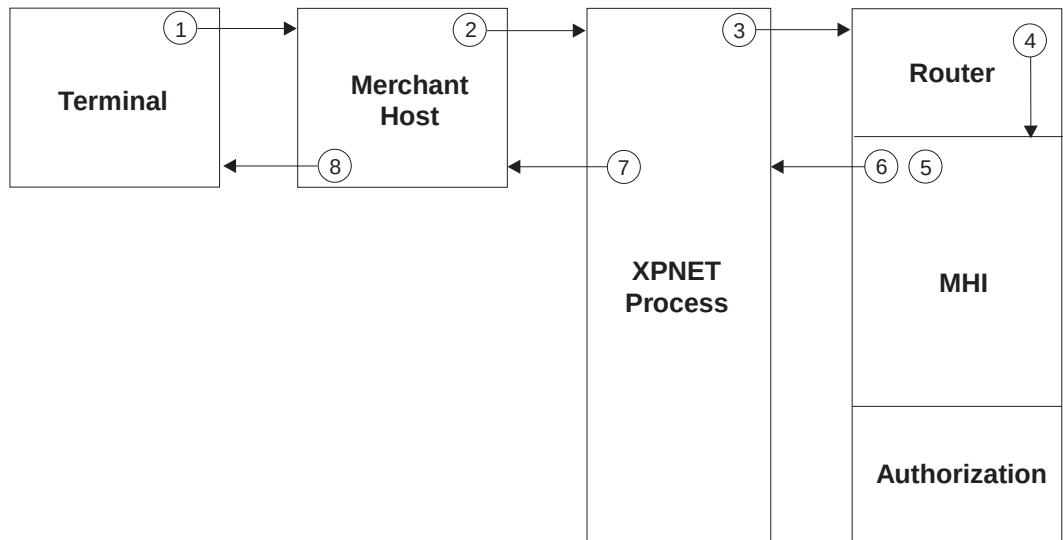
The diagram below illustrates the transaction message flow in a scenario where the terminal sends a request to log on with a different clerk ID.



Steps	Processing
1	The terminal sends a request to log on using a new clerk ID.
2, 3, 4	The merchant host formats an MHI external message and sends the message to the MHI module. This message travels through the XPNET process and the Router module.
5	The MHI module validates the new clerk ID and then examines the CLERK ID field in the POS Terminal Data File (PTDF). If this field is filled, the MHI module logs off the old clerk and logs the old clerk totals. Old clerk totals are zeroed out. The new clerk ID replaces the old clerk ID in the CLERK ID field in the PTDF.
6, 7	The MHI module sends a response to the merchant host. This message travels through the XPNET process.
8	The merchant host formats the message into a terminal response message and sends it to the terminal.

Terminal Logoff

The diagram below illustrates the transaction message flow in a scenario where the terminal sends a request to log off with the current clerk ID.



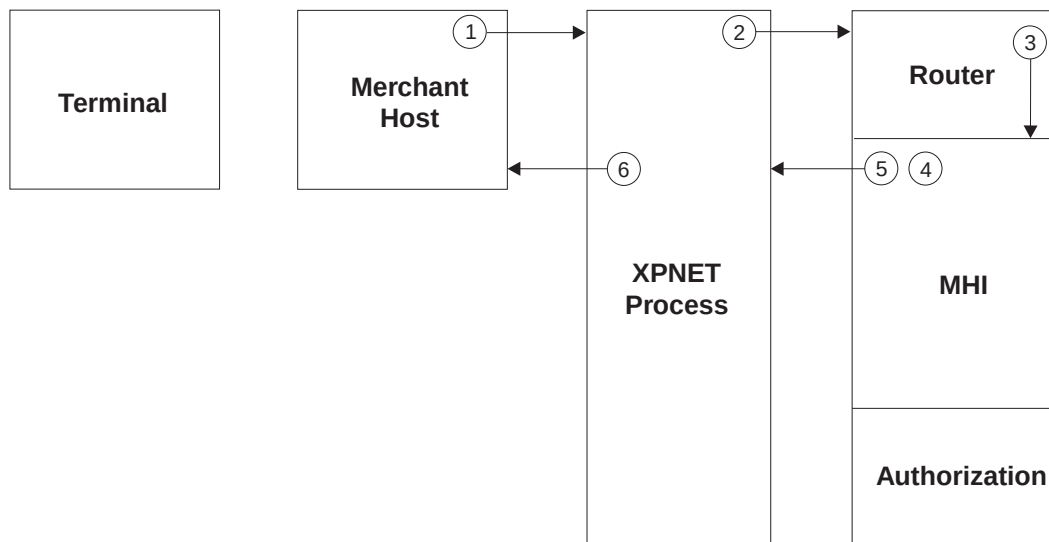
Steps

Processing

- | | |
|---------|--|
| 1 | The terminal sends a request to log off the current clerk. |
| 2, 3, 4 | The merchant host formats an MHI external message and sends the message to the MHI module. This message travels through the XPNET process and the Router module. |
| 5 | The MHI module logs off the clerk and logs the clerk totals to the POS Transaction Log File (PTLF). The totals for the clerk are zeroed out. |
| 6, 7 | The MHI module sends a response to the merchant host. This message travels through the XPNET process. |
| 8 | The merchant host formats the message into a terminal response message and sends it to the terminal. |

Network Management Message Logon Request

The diagram below illustrates the transaction message flow in a scenario where the merchant host is requesting to log on using a network management message.



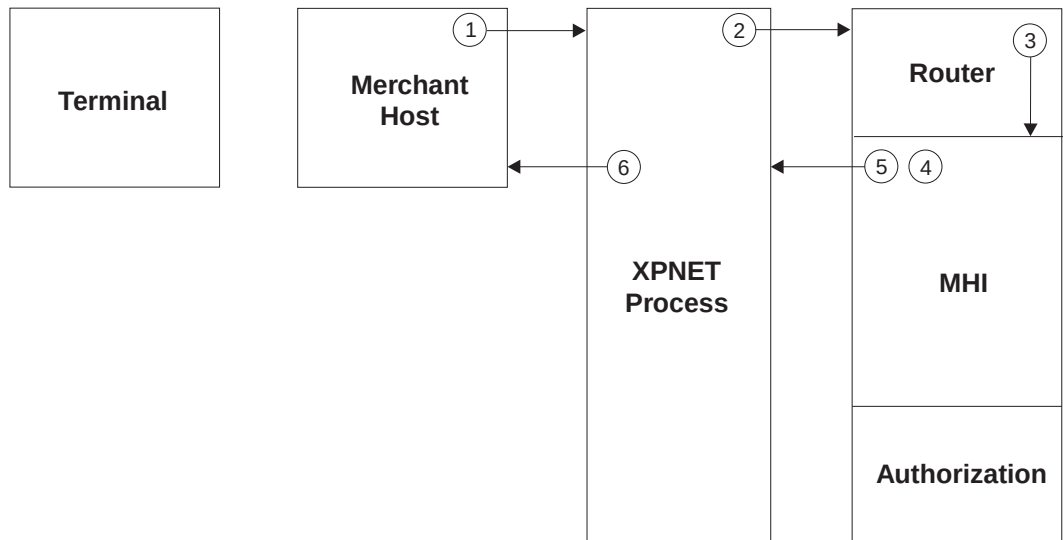
Steps

Processing

- | | |
|---------|---|
| 1, 2, 3 | The merchant host sends a network management logon request to the MHI module. This message travels through the XPNET process and the Router module. |
| 4 | The MHI module marks the DPC status as up. |
| 5, 6 | The MHI module responds to the merchant host with the message header only. This message travels through the XPNET process. |

Network Management Message Echo Test Request

The diagram below illustrates the transaction message flow in a scenario where the merchant host is requesting an echo test using a network management message.



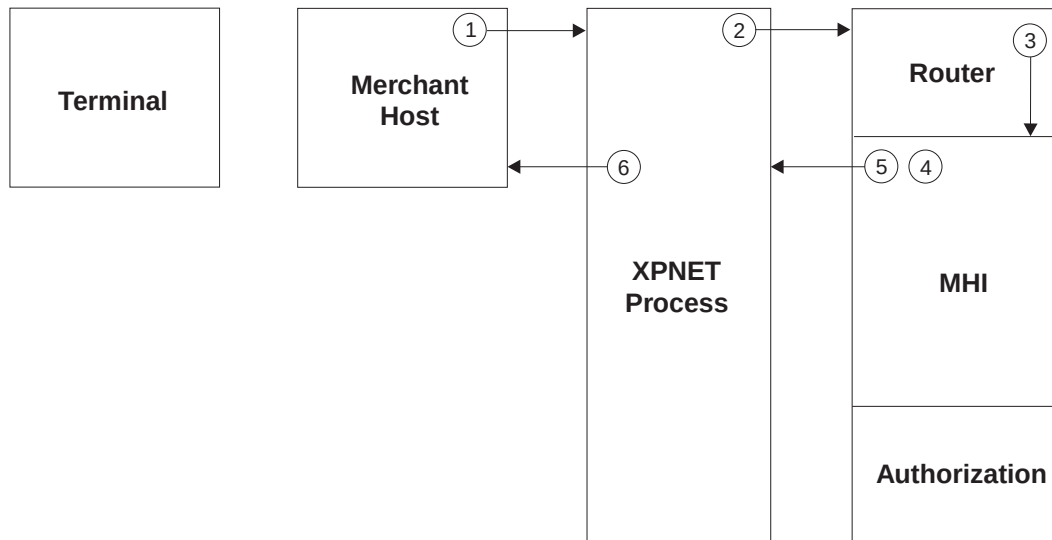
Steps

Processing

- | | |
|---------|---|
| 1, 2, 3 | The merchant host sends a network management echo test request to the MHI module. This message travels through the XPNET process and the Router module. |
| 4 | The MHI module marks the DPC status as up. |
| 5, 6 | The MHI module responds to the merchant host with the message header only. This message travels through the XPNET process. |

Network Management Message Logoff Request

The diagram below illustrates the transaction message flow in a scenario where the merchant host is requesting to log off using a network management message.



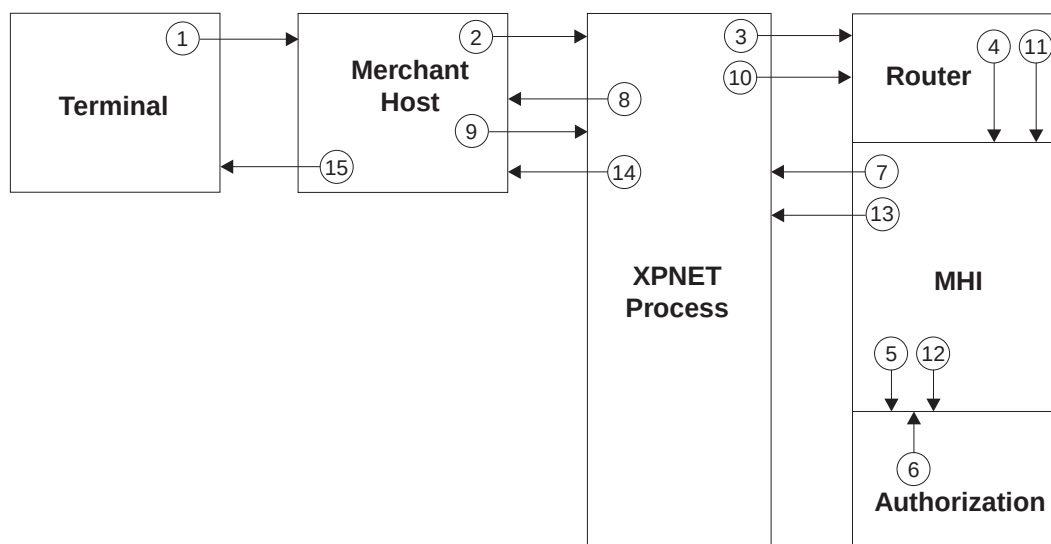
Steps

Processing

- | | |
|---------|--|
| 1, 2, 3 | The merchant host sends a network management logoff request to the MHI module. This message travels through the XPNET process and the Router module. |
| 4 | The MHI module marks the DPC status as down. |
| 5, 6 | The MHI module responds to the merchant host with the message header only. This message travels through the XPNET process. |

MAC Reversal

The diagram below illustrates the transaction message flow in a scenario containing a reversal of a message with an invalid message authentication code (MAC).



Steps

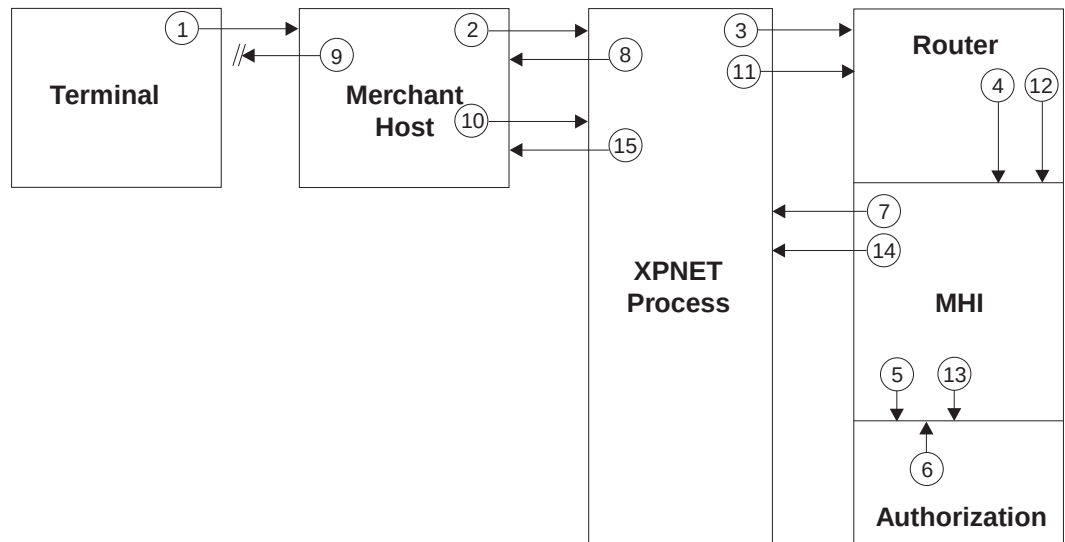
Processing

- 1 The terminal sends its native message for a normal purchase to the merchant host.
- 2, 3, 4 The merchant host formats an MHI external message and sends the message to the MHI module. This message travels through the XPNET process and the Router module.
- 5 The MHI module validates the request and formats a POS Standard Internal Message (PSTM) for a normal purchase. The MHI module sends the message to the Authorization module.
- 6 The Authorization module returns a response to the MHI module indicating the transaction was approved.
- 7, 8 The MHI module updates the PTDF totals, translates the response from PSTM format into an MHI external message, and sends the response to the merchant host. This message travels through the XPNET process.

Steps	Processing
9, 10, 11	The merchant host suspects a corrupted message, generates a reversal with a message type of R and sends it to the MHI module. This message travels through the XPNET process and the Router module.
12	The MHI module updates totals in the PTDF, sets the reversal code to 21 (MAC failure) and generates a 0420 reversal message and sends it to the Authorization module. The MHI module checks the POS-DH-CONS-MAC-ERR-LMT parameter in the LCONF. If the limit has been reached, the MHI module sets a flag that indicates the MHI module needs to send a new MAC key on its next response to the terminal. If the limit has not been reached, the MHI module increases the consecutive MAC error counter by one.
13, 14	The MHI module echoes the request back to the merchant host. This message travels through the XPNET process.
15	The merchant host formats the message into the native terminal format and sends it to the terminal.

Merchant Host Reversal

The diagram below illustrates the transaction message flow in a scenario containing a transaction reversed from a merchant host. In this transaction, the lines of communication between the terminal and the merchant host go down, and the merchant host detects this after it receives a response from the MHI module.



Steps

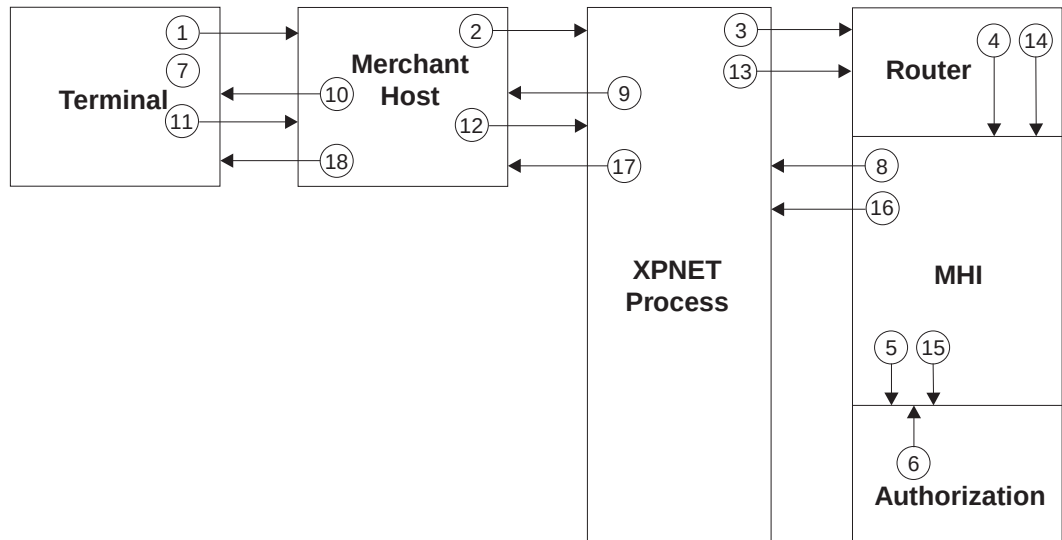
Processing

- 1 The terminal sends its native message for a normal purchase to the merchant host.
- 2, 3, 4 The merchant host formats an MHI external message and sends the message to the MHI module. This message travels through the XPNET process and the Router module.
- 5 The MHI module validates the request and formats the message in POS Standard Internal Message (PSTM) for a normal purchase. The MHI module then sends the message to the Authorization module.
- 6 The Authorization module returns a response to the MHI module indicating the transaction was approved.
- 7, 8 The MHI module updates the PTFDF totals, translates the response from the PSTM format into the MHI external message, and sends it to the merchant host. This message passes through the XPNET process.

Steps	Processing
9	The merchant host tries to send the response to the terminal, but the line is down.
10, 11, 12	The merchant host generates an MHI external message reversal with a message type of R and sends it to the MHI module. This message travels through the XPNET process and the Router module.
13	The MHI module updates totals in the PTDF, sets the reversal code to 03 (destination not available) and generates a 0420 reversal message and sends it to the Authorization module.
14, 15	The MHI module echoes the reversal request back to the merchant host. This message passes through the XPNET process.

Customer-Cancellation Reversal

The diagram below illustrates the transaction message flow in a scenario containing a normal purchase transaction that is canceled by the clerk at the terminal before the MHI module sends a response.



Steps

Processing

- 1 The terminal sends its native message for a normal purchase to the merchant host.
- 2, 3, 4 The merchant host formats an MHI external message and sends the message to the MHI module. This message travels through the XPNET process and the Router module.
- 5 The MHI module validates the request and formats the message in POS Standard Internal Message (PSTM) for a normal purchase. The MHI module then sends the message to the Authorization module.
- 6 The Authorization module returns a response to the MHI module indicating the transaction was approved.
- 7 The clerk cancels the transaction at the terminal. The terminal waits to send the reversal to the MHI module until after it receives the response to the original transaction.

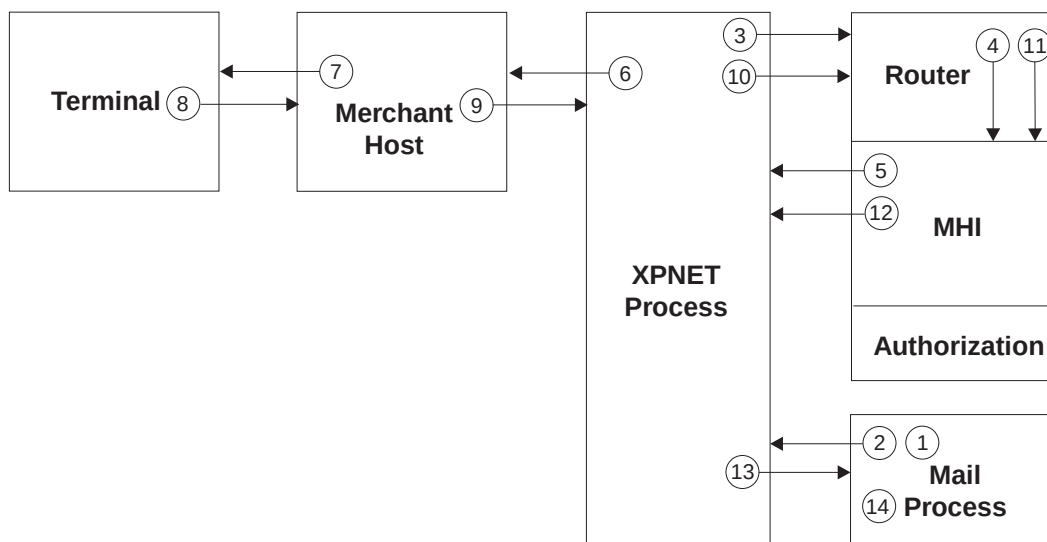
Steps	Processing
8, 9	The MHI module updates the PTDF totals, translates the response from the PSTM format into the MHI external message, and sends it to the merchant host. This message passes through the XPNET process.
10	The merchant host formats the response into the native terminal format and sends the response to the terminal.
11	The terminal generates a reversal with a message type of U (customer cancellation) and sends it to the merchant host.
12, 13, 14	The merchant host formats an MHI external message and sends it to the MHI module. This message travels through the XPNET process and the Router module.
15	The MHI module updates totals in the PTDF, sets the PTLF reversal code to 08 (customer canceled), generates a 0420 reversal message, and sends it to the Authorization module.
16, 17	The MHI module echoes the reversal back to the merchant host. This message travels through the XPNET process.
18	The merchant host formats the message into the native terminal format and sends it to the terminal.

BASE24-mail Transaction Flows

The MHI module is fully compatible with BASE24-mail. The following diagrams illustrate the exchange of messages between the MHI module and the terminal or merchant host during transaction processing using BASE24-mail.

Unsolicited Mail

The diagram below illustrates the transaction message flow in a scenario where the MHI module receives unsolicited mail for a terminal.



Note: Dial-up terminals cannot receive unsolicited mail.

Steps

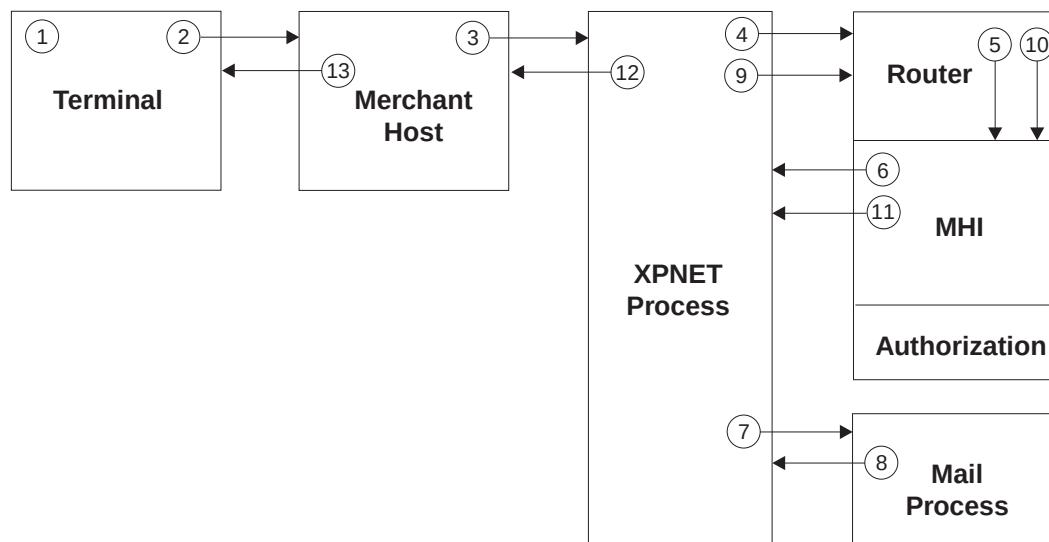
Processing

- 1 The Mail process receives a message indicating that it must send a mail message to the terminal for immediate delivery. This message can come from the host, a Pathway screen, or another terminal.
- 2, 3, 4 The Mail process sends an MB641 message to the MHI module. This message travels through the XPNET process and Router module.

Steps	Processing
5, 6	If the terminal can receive unsolicited mail and there is no transaction in progress, the MHI module formats a message containing the mail message and sends it to the merchant host. This message travels through the XPNET process. If the terminal cannot receive unsolicited mail or there is a transaction in progress, the MHI module sets a flag indicating that there is mail waiting. The next time the terminal receives a response, the response code will indicate that there is mail waiting. The terminal can then request the mail. Refer to “Mail Pick-up Request—Single Response” later in this section for more information about how a terminal can receive this mail.
7	The merchant host translates the message into the terminal message format and sends it to the terminal.
8	The terminal responds with a “mark mail as delivered” message to the merchant host.
9, 10, 11	The merchant host translates the message into the MHI external message and sends it to the MHI module. This message travels through the XPNET process and the Router module.
12, 13	The MHI module translates the message into an MB643 message and sends it to the Mail process. This message travels through the XPNET process.
14	The Mail process receives the MB643 message and marks the mail message as delivered.

Terminal Send Mail Request

The diagram below illustrates the transaction message flow in a scenario where the terminal allows the operator to compose a mail message and route it to one of the three DPCs set up in the Merchant Terminal Configuration File (MCNF).



Steps

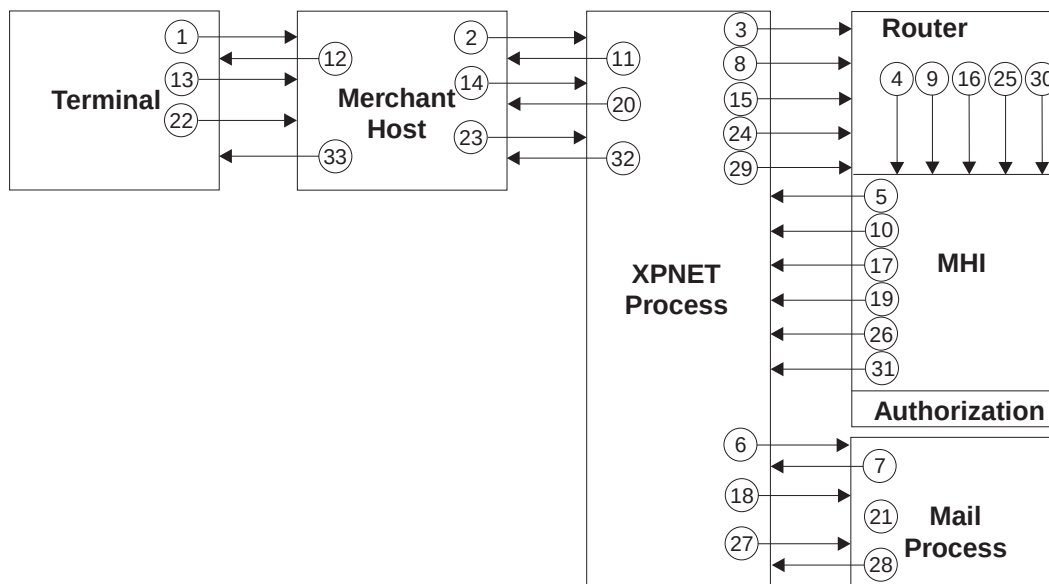
Processing

- 1 The terminal allows the operator to compose a mail message and then formats a Send Mail Request containing the mail text. This is included in the Mail/Download Text field (FID W) along with the number of the DPC to which the mail is to be delivered. A terminal can route mail to one of three possible DPCs. Valid DPC values are 1, 2, and 3. These values are mapped to corresponding DPCs set up in the Merchant Terminal Configuration File (MCNF). The request is formatted as ".W1text".
- 2 The terminal sends the Send Mail Request to the merchant host.
- 3, 4, 5 The merchant host formats an MHI external message and sends it to the MHI module. The message travels through the XPNET process and the Router module.
- 6, 7 The MHI module translates the Send Mail Request to an MB640 message and sends the message to the Mail process. This message travels through the XPNET process.

Steps	Processing
8, 9, 10	The Mail process attempts to send the mail to the indicated DPC. If it cannot, it stores the mail in the Mail Box File (MBF) for later delivery. It then responds to the MHI module indicating whether the mail was delivered or stored. This message travels through the XPNET process and Router module.
11, 12	If the attempt is successful, the MHI module responds to the merchant host with a response code of 870 (OK, delivered). If the attempt is unsuccessful, the MHI module responds with a response code of 871 (OK, mail stored). The response is in the format of an MHI external message. This message travels through the XPNET process.
13	The merchant host translates the response into terminal message format and sends it to the terminal.

Mail Pick-up Request—Single Response

The diagram below illustrates the transaction message flow in a scenario where the terminal is requesting to read mail. In this example, available mail is contained in a single response from the MHI module.



Steps

Processing

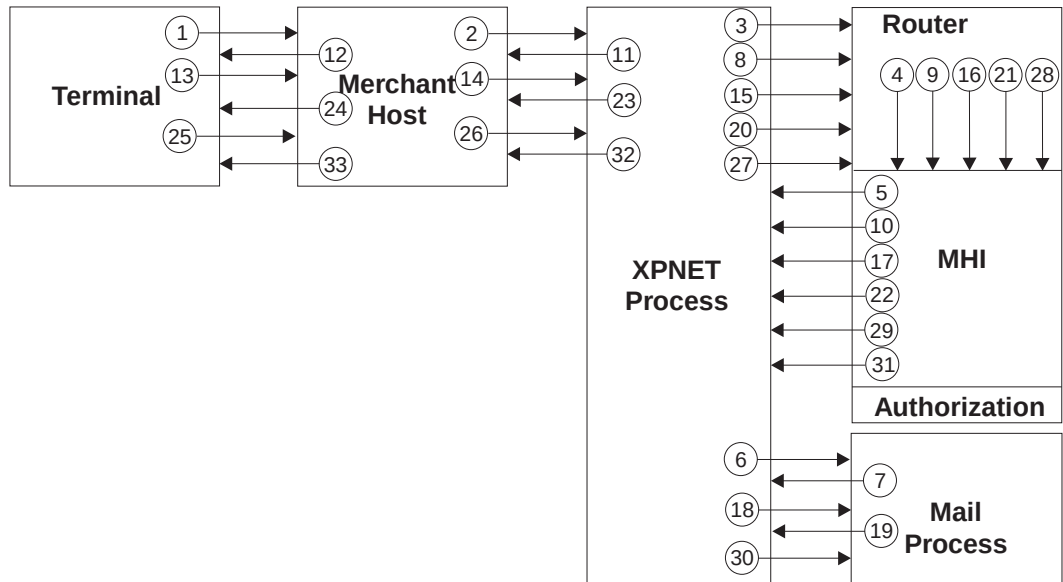
- 1 The terminal formats a Read Mail Request and sends it to the merchant host.
The terminal is asking for the first of any undelivered mail using the Mail/Download Key field (FID V). The message is formatted as "V00300000000000". Instead, the terminal could have requested the first of any mail available. In the latter case, the presence of FID V is not required.
- 2, 3, 4 The merchant host formats an MHI external message and sends the message to the MHI module. This message travels through the XPNET process and the Router module.
- 5, 6 The MHI module sends an MB640 request to the Mail process. This message travels through the XPNET process.

Steps	Processing
7, 8, 9	The Mail process determines from the MB640 request that the terminal is requesting mail. The Mail process reads the Mail Box File (MBF), formats an MB641 mail response message, and sends the message to the MHI module. This message travels through the XPNET process and the Router module.
10, 11	The MHI module formats the response into an MHI external message and sends it to the merchant host. This message travels through the XPNET process. The MHI module responds with the first piece of undelivered mail. The mail includes the mail key and mail text. If the mail fits into a single response, as it does in this example, the response code is 880 (OK, no more data). The response is formatted as “.V00419202170068”. Note: In this example, the Processing Flag in the mail key is set to 1. The terminal must ensure this value is echoed as received.
12	The merchant host translates the response into the terminal message format and sends the response to the terminal.
13	The terminal sends a message to the merchant host to mark the mail as delivered.
14, 15, 16	The merchant host formats an MHI external message and sends it to the MHI module. This message travels through the XPNET process and the Router module.
17, 18	The MHI module formats an MB643 completion and sends it to the Mail process. This message travels through the XPNET process.
19, 20	The MHI module then formats and sends an MHI external message to the merchant host that includes response code 880 (OK, no more data) and the mail key for the delivered message. The request is formatted as “.V00419202170068”. This message travels through the XPNET process.
21	The Mail process updates the mail record in the MBF and marks the mail as read. Information in the MBF concerning the terminal is also updated at this time by the Mail process.

Steps	Processing
22	Since the Mail Text Field (FID W) was included in the previous mail response, the terminal operator sends a Read Next Request to the merchant host to read the next mail message. The request is formatted as “.V00419202170068”.
23, 24, 25	The merchant host formats an MHI external message and sends it to the MHI module. This message travels through the XPNET process and the Router module.
26, 27	The MHI module formats and sends an MB640 request to the Mail process. This message travels through the XPNET process. The MHI module always sends the mail key from the previous mail message to the Mail process when it requests the next mail message. When the last packet of a mail message is delivered, the Mail process uses the mail key to update the MBF. Any subsequent Read Next Requests would then cause the Mail process to retrieve the next mail message from the MBF starting after the key of the last mail message read.
28, 29, 30	The Mail process reads the MBF, finds no mail, and responds with MB641 (no mail). This message travels through the XPNET process and the Router module.
31, 32	The MHI module formats an MHI external message and responds to the merchant host with a response code of 880 (OK, no more data) and the absence of the Mail Text field. This indicates that all mail has been read. The response is formatted as “.V00419202170068”. This message travels through the XPNET process. Note: The terminal operator can continue to send Read Next Requests to the merchant host as long as the MHI module responses contain a response code of 880 and information in the Mail Text field (FID W). The absence of FID W indicates that all mail has been read.
33	The merchant host formats the response in the terminal message format sends the response to the terminal.

Mail Pick-up Request—Multiple Responses

The diagram below illustrates the transaction message flow in a scenario where the terminal is requesting to read mail. In this example, available mail is contained in multiple responses from the MHI module.



Steps

Processing

- 1 The terminal formats a Read Mail Request and sends it to the merchant host.

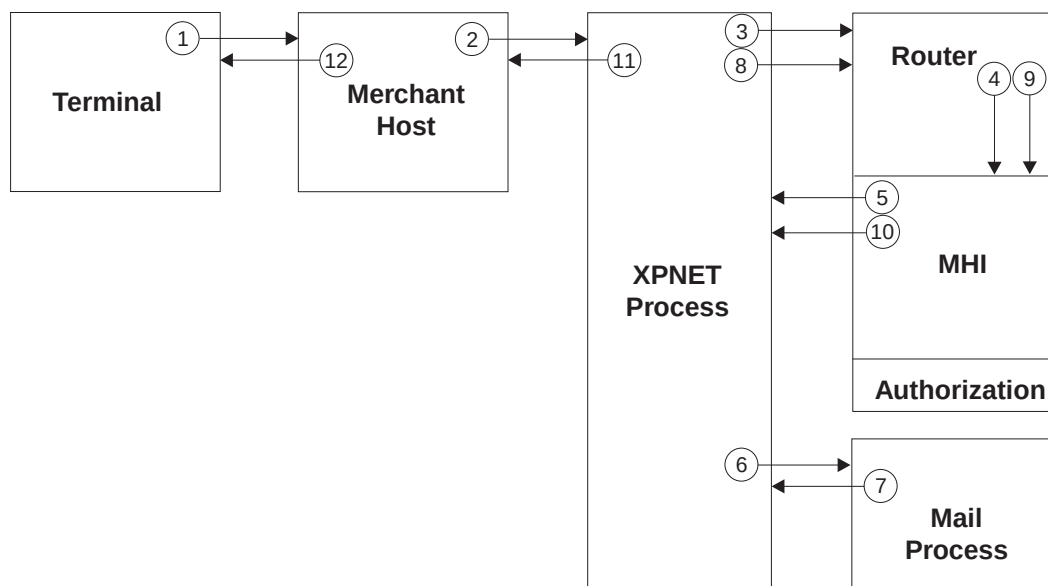
The terminal is asking for the first of any undelivered mail using the Mail/Download Key field (FID V). The message is formatted as "V00300000000000". Instead, the terminal could have requested the first of any mail available. In the latter case, the presence of FID V is not required.
- 2, 3, 4 The merchant host formats an MHI external message and sends it to the MHI module. This message travels through the XPNET process and the Router module.
- 5, 6 The MHI module formats and sends an MB640 request to the Mail process. This message travels through the XPNET process.

Steps	Processing
7, 8, 9	The Mail process determines from the MB640 request that the terminal is requesting mail. The Mail process reads the Mail Box File (MBF), formats an MB641 mail response message, and sends the message to the MHI module. This message travels through the XPNET process and Router module.
10, 11	The MHI module translates the response into MHI external message format and sends it to the merchant host. The response code is set to 881 (OK, more data exists). This message travels through the XPNET process.
12	The merchant host translates the response into the terminal message format and sends it to the terminal.
13	The terminal sends a request to the merchant host for the next packet of information.
14, 15, 16	The merchant host formats an MHI external message and sends it to the MHI module. This message travels through the XPNET process and Router module.
17, 18	The MHI module sends an MB641 request to the Mail process for more data. This message travels through the XPNET process.
19, 20, 21	The Mail process responds to the MHI module with the mail text. This message travels through the XPNET process and the Router module.
22, 23	The MHI module translates the response into an MHI external message and sends it to the merchant host. The response contains the next packet of information. This message travels through the XPNET process.
24	The merchant host translates the response into the terminal message format and sends it to the terminal. Steps 13–24 repeat until no more packets are needed. Along with the last packet the MHI module responds with a response code of 880 (OK, no more data) to the terminal.
25	The terminal sends a message to the merchant host to mark the mail as delivered.

Steps	Processing
26, 27, 28	The merchant host formats an MHI external message and sends it to the MHI module. This message travels through the XPNET process and Router module.
29, 30	The MHI module formats an MB643 completion and sends it to the Mail process. This message travels through the XPNET process.
31, 32	The MHI module then formats and sends an MHI external message to the merchant host that includes response code 880 (OK, no more data) and the mail key for the delivered message. The request is formatted as “V00419202170068”. This message travels through the XPNET process.
33	The merchant host translates the response into the terminal message format and sends it to the terminal. Steps 1–33 repeat until, in step 4, the Mail process reads the MBF, finds no mail, and responds with an MB641 (no mail) response message.

Mail Pick-Up Request—No Mail Stored

The following diagram illustrates a scenario where the terminal initiates a request to receive mail. In this example, no mail is stored for the specified terminal.



Steps	Processing
1	The terminal sends a mail pick-up request to the merchant host.
2, 3, 4	The merchant host translates the request into an MHI external message and sends it to the MHI module. This message travels through the XPNET process and Router module.
5, 6	The MHI module formats and sends an MB640 request to the Mail process. This message travels through the XPNET process.
7, 8, 9	The Mail process determines from the MB640 request that the terminal is requesting mail. The Mail process reads the Mail Box File (MBF), determines there is no mail for the specified terminal, formats an MB641 (no mail) response message, and sends the message to the MHI module. This message travels through the XPNET process and Router module.
10, 11	The MHI module translates the response into an MHI external message and sends it to the merchant host. The response code 880 (OK, no more data) is included. This message travels through the XPNET process.

Steps**Processing**

- | | |
|----|--|
| 12 | The merchant host translates the response into the terminal message format and sends it to the terminal. |
|----|--|

Applied Communications, Inc.

Appendix A

American Express Standard Industry Formats

The AMEX Data Collection option allows customers to perform draft capture on transactions originating from American Express cardholders. Generally, American Express has defined categories under which transactions are placed. These categories are as follows:

- Auto rental
- Lodging
- Restaurant
- General retail
- Oil

For each of these categories, American Express requires different data to be sent from the device. Therefore, American Express has set forth standard industry formats for each category to ensure the data they require is captured by the device and sent to BASE24.

American Express also supports a Descriptive Billing File (DBF) that is used to submit charges to American Express without submitting paper. The format of the DBF conforms to American Express standards. The MHI module is able to recognize these standard industry formats and, subsequently, capture and process transactions sent using them. In conjunction with this file, the customer must set up Standard Industrial Classification (SIC) codes or Merchant Category codes in the PTDF. SIC and Merchant Category codes identify the merchant's line of business. For more information, refer to the ***BASE24-pos Files Maintenance Manual***.

These formats are for FID 0. The data elements for the industry formats described in this appendix must be included in FID 0 in the order shown. Refer to section 4 for more information about FID 0.

Auto Rental

The following format is used for the automobile rental industry.

Field	Length	Description
frmt-cde	9(2)	The American Express Industry Format Code identifies which standard industry format should be used for a particular transaction. This field contains a value of 05 for auto rental transactions.
audit-adj-amt	9(18)	The amount of charges added to or subtracted from the contract after the vehicle was returned (that is, mileage, damages, discounts, etc.). This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0.
agreement-number	X(9)	The invoice number of the rental agreement as issued by the rental agency and signed by the cardholder.
reference-number	9(8)	A reference number used by American Express to identify the transaction and obtain supporting data from the service establishment for a charge. The range of valid values is 000000000–999999999.
rental-city	X(18)	The city where the rental originated. This field is user-defined.
rental-st	X(2)	The valid two-character alphabetic postal code for the state or province where the rental originated.
rental-dat	9(6)	The date (YYMMDD) that the rental occurred.
rental-tim	9(4)	The time, in 24-hour time (hhmm), that the rental occurred.
return-city	X(18)	The city where the auto was returned. This value is user-defined.

Field	Length	Description
return-st	X(2)	The valid two-character alphabetic postal code for the state or province where the rental was returned.
return-dat	9(6)	The date (YYMMDD) on which the auto was returned.
return-tim	9(4)	The time, in 24-hour time (hhmm), at which the auto was returned.
renter-name	X(20)	The name of the person renting the auto. This value is user-defined.

Lodging

The following format is used for the lodging industry.

Field	Length	Description
frmt-cde	9(2)	The American Express Industry Format Code identifies which standard industry format should be used for a particular transaction. This field contains a value of 11 for lodging transactions.
reference-cde	9(9)	A reference number used by American Express to identify the transaction and obtain supporting data from the service establishment for a charge. The range of valid values is 000000000–999999999.
charge-type	9(1)	A code that indicates the type of charge made. Valid values are as follows: 1 = Lodging 2 = Restaurant 3 = Gift shop 0, 4–9 = Reserved
tab-roc-id	X(10)	The original Record of Charge (ROC), invoice, or another number used to identify the original transaction. This is an internal American Express code.
tax-amt	9(18)	The amount of sales tax charged for the transaction. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0.
arrival-dat	9(6)	The scheduled arrival date (MMDDYY).
depart-dat	9(6)	The scheduled departure date (MMDDYY).
room-rate	9(18)	The per diem rate charged for the cardholder's stay at the establishment. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0.

Field	Length	Description
program-cde	9(1)	A code indicating the reason for the charge. If there are no special circumstances that require a code, a value of 1 is placed in this field. Valid values are as follows: 1 = Used if no other code pertains 2 = Assured reservation—no show 3 = CARDeposit 4 = Delayed charge 5 = Express service 6 = Assured reservation 0, 7–9 = Reserved

Restaurant

The following format is used by the restaurant industry.

Field	Length	Description
frmt-cde	9(2)	The American Express Industry Format Code identifies which standard industry format should be used for a particular transaction. This field contains a value of 12 for restaurant transactions.
reference-cde	9(9)	A reference number used by American Express to identify the transaction and obtain supporting data from the service establishment for a charge. The range of valid values is 000000000–999999999.
charge-cde1	X(4)	A code mapped by the host to a description that identifies the transaction associated with the charge. This value is defined by American Express.
tab-roc-id	X(10)	The original Record of Charge (ROC), invoice, or another number used to identify the original transaction. This is an internal American Express code.
description-cde	X(1)	A code that describes the nature of the transaction. Valid values are as follows: 0 = Food 1 = Food and beverage 2 = Gift certificate 3–9 = User-defined A–Z = User-defined
tax-amt	9(18)	The amount of sales tax charged for the transaction. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0.

Field	Length	Description
food-amt	9(18)	The total cost of food, or food and beverage if these charges are combined on the original record of charge (ROC). This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0.
beverage-amt	9(18)	The total cost of beverages, if itemized in the DESCRIPTION-CDE field of the American Express token. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0. Refer to the BASE24 Tokens Manual for additional information about this token.
tip1-cde	X(1)	A code mapped by the host to a description that designates an employee receiving a tip.
tip2-cde	X(1)	A code mapped by the host to a description that designates a second tip itemized on the original record of charge (ROC).
tip1-amt	9(18)	The total amount of the tip described in the TIP1-CDE field of the American Express token. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0. Refer to the BASE24 Tokens Manual for additional information about this token.
tip2-amt	9(18)	The total amount of the second tip described in the TIP2-CDE field of the American Express token. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0. Refer to the BASE24 Tokens Manual for additional information about this token.

General Retail

The following format is used by the general retail industry.

Field	Length	Description
frmt-cde	9(2)	The American Express Industry Format Code identifies which standard industry format should be used for a particular transaction. This field contains a value of 20 for general retail transactions.
tax-amt	9(18)	The amount of sales tax charged for the transaction. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0.
reference-cde	9(9)	A reference number used by American Express to identify the transaction and obtain supporting data from the service establishment for a charge. The range of valid values is 000000000–999999999.
charge-cde1	X(4)	A code mapped by the host to a description that identifies the transaction associated with the charge. This value is defined by American Express.
tab-roc-id	X(10)	The original record of charge (ROC), invoice, or another number used to identify the original transaction. This is an internal American Express code.
charge-cde2	X(4)	A code mapped by the host to a description that identifies the transaction associated with the charge. This value is defined by American Express.
charge-cde3	X(4)	A code mapped by the host to a description that identifies the transaction associated with the charge. This value is defined by American Express.

Field	Length	Description
charge-cde4	X(4)	A code mapped by the host to a description that identifies the transaction associated with the charge. This value is defined by American Express.
charge-cde5	X(4)	A code mapped by the host to a description that identifies the transaction associated with the charge. This value is defined by American Express.

Oil

The following format is used by the oil industry.

Field	Length	Value
frmt-cde	9(2)	The American Express Industry Format Code identifies which standard industry format should be used for a particular transaction. This field contains a value of 21 for oil company transactions.
reference-cde	9(12)	A reference number used by American Express to identify the transaction and obtain supporting data from the service establishment for a charge. The range of valid values is 000000000000–999999999999.
charge-cde1	X(4)	A code mapped by the host to a description that identifies the transaction associated with the charge. This value is defined by American Express.
tab-roc-id	X(10)	The original record of charge (ROC), invoice, or another number used to identify the original transaction. This is an internal American Express code.
tax-amt	9(18)	The amount of sales tax charged for the transaction. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0.

Appendix B

Category Code Examples

This appendix describes how the Category Code works. The Category Code field is contained in the first two bytes of the Mail/Download Key field (FID V). The Category Code serves the following purposes:

- Notifies the MHI module of initial download requests.
- Indicates to the terminal if a download field (DID) being sent from the MHI module to the terminal does not fit in a single message.
- Indicates to the MHI module whether the terminal requests the additional information if a download field does not fit into a single response.
- Indicates to the terminal that the download is complete.

The Category Code consists of two bytes. The first byte is used in responses and the second byte is used in requests.

For more information about the Mail/Download Key field, refer to section 4. For more information about download requests, refer to section 5.

Category Code Overview

The category code serves different purposes in responses and requests. It is reset when the download is complete.

Responses

In responses, the Category Code indicates whether the last field in the Download/Mail Text field (FID W) is complete.

A value of 1 in the first byte of the Category Code indicates the last field downloaded in the Download/Mail Text field is incomplete. This means the download field does not fit in a single message given the constraints imposed by the maximum response length configured in the Merchant Terminal Configuration File (MCNF). In this case, the download field must be sent in multiple messages. A value of 1 in the first byte of the Category Code indicates to the terminal that more data exists for a particular download field.

A value of 2 in the first byte of the Category Code indicates to the terminal that no more data exists for the particular download field. In this case, the next field will be downloaded.

Requests

In requests, the Category Code indicates whether the terminal is required to receive the remaining information from the last field downloaded in the Download/Mail Text field (if applicable) or begin receiving information from the next field to be downloaded.

A value of 1 in the second byte of the Category Code indicates that the terminal is requesting to receive the remaining data for the download field identified as incomplete in the response (indicated by 1 in the first byte of the Category Code). This is known as a continuation request. The terminal can continue to receive the balance of the data for the incomplete download field until no more data exists as long as the second byte of the Category Code is 1. When no more data exists, the next field will be downloaded.

A value of 0 in the second byte of the Category Code indicates that the terminal is requesting to begin receiving information from the next field to be downloaded. This can occur in the following situations:

- If the terminal is not requesting a continuation of a current download field identified in the response as being incomplete
- If an incomplete download field is now identified in the response as complete
- If there was no incomplete download field identified in the response (that is, the download field fit into a single message)

Downloading Complete

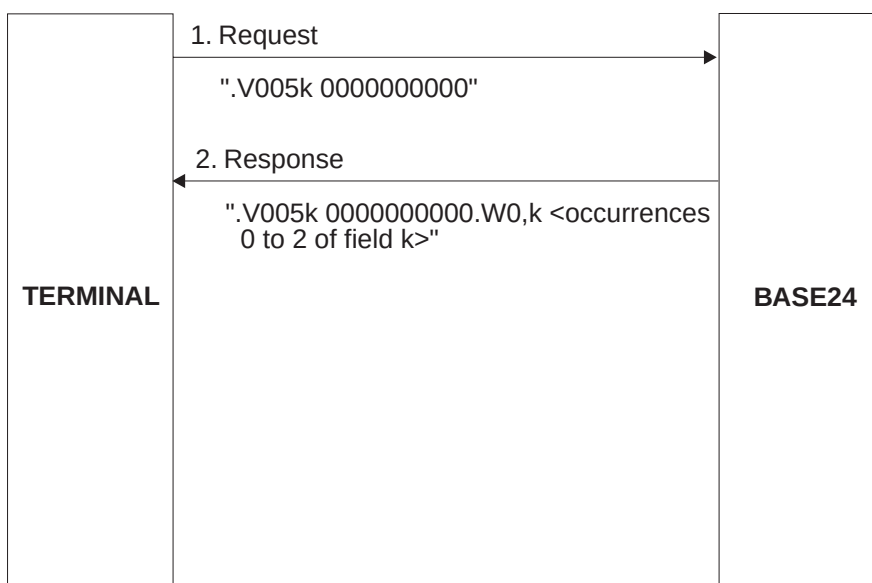
When the download is complete, the Category Code is reset to 00 by the MHI module.

Category Code Examples

The remainder of this appendix provides examples explaining how the Category Code is used in various scenarios. Each example includes a brief summary of the scenario, a graphic illustrating the message exchanges between the MHI module and the terminal during the scenario, and a step-by-step description of each message exchange shown on the graphic.

Entire DID Fits into Response

The following example shows a terminal request where the entire DID fits into a single response sent to the terminal from the MHI module. In this scenario, the terminal requests a partial download of DID k, the MHI module responds with the entire contents of DID k. In this example, there are three services defined in the POS Terminal Data File (PTDF).

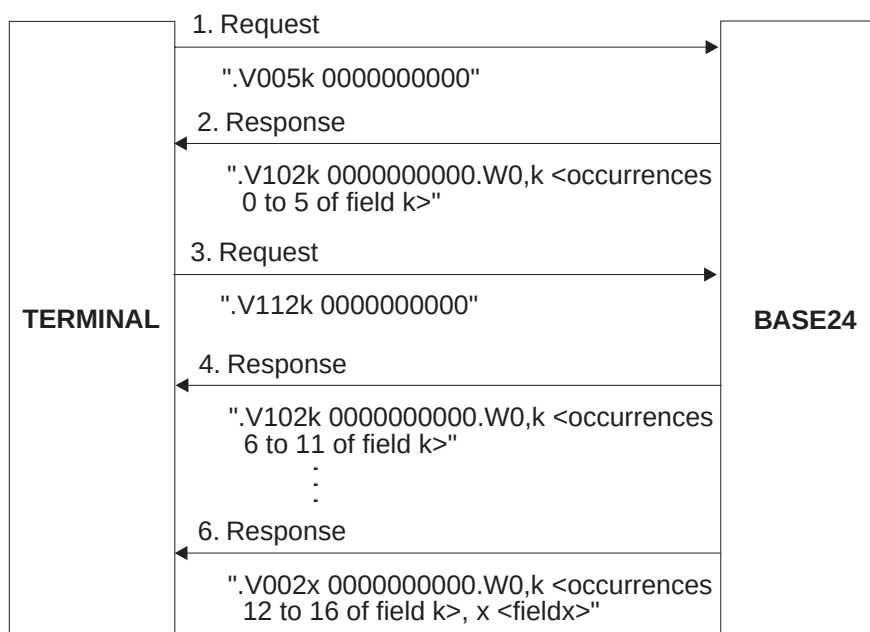


1. The terminal requests a partial download of DID k (PTDF Services). The Category Code is 00 since this is an initial request. For example purposes, it is assumed that the MHI module has already notified the terminal that a download is in order by sending the terminal a 99 in the Category Code field.

2. The MHI module responds to the request by sending as many occurrences of DID k as will fit in a single response message. In this example, the Category Code remains as 00 in the response because the entire contents of DID k is able to be sent in a single response message. The response code in the standard message header is 880 (OK, no more data).

Entire DID Does Not Fit into Response (Full Download)

The following example shows a terminal request where the entire DID does not fit into a single response during a full download. In this case, the DID is sent across multiple message responses. In this example, there are 17 services defined in the PTDF.

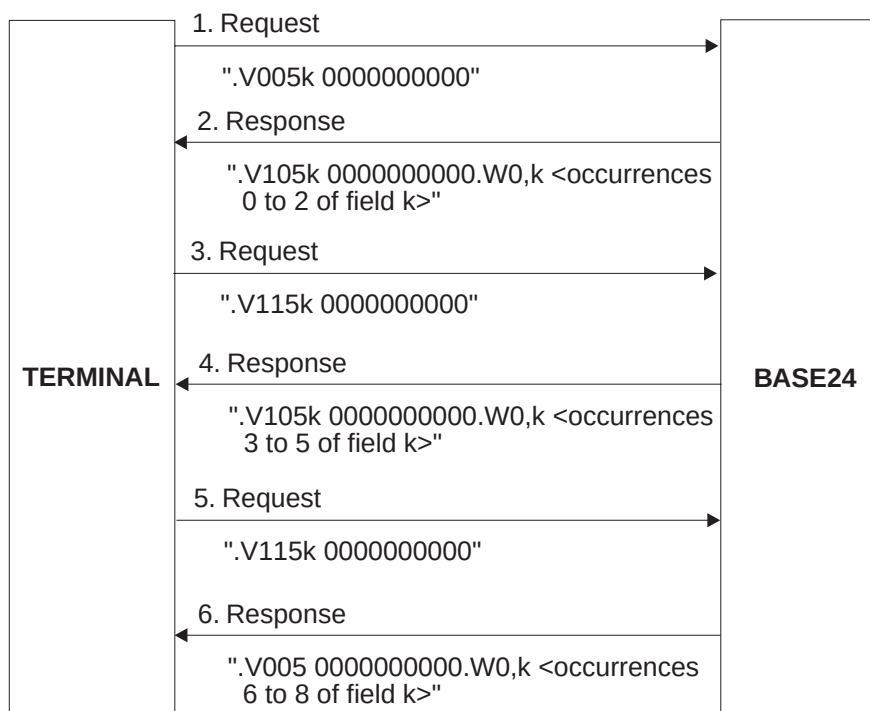


1. The terminal requests a download with no optional data fields. At this point, the Category Code is 00.
2. The MHI module formats a message with six occurrences of DID k, given the message response length constraints configured in the MCNF. In the example, DID k does not completely fit into the response, so the MHI module changes the first byte of the Category Code to 1. This indicates to the terminal that more information for DID k exists. At this point, the Category Code is 10.

3. The terminal requests more occurrences of DID k by echoing the Download/Mail Key field (FID V), changing the second byte of the Category Code to 1, and sending the resulting request to the MHI module. This notifies the MHI module to send more occurrences of DID k to the terminal. At this point, the Category Code is 11. The terminal could also have ignored the balance of DID k and moved to the next DID (DID x in the example) by echoing the Download/Mail Key field unaltered.
4. The MHI module sends the next six occurrences of DID k in the response. DID k still does not fit into a single response, so the first byte of the Category Code remains as 1. This indicates to the terminal that more information for DID k exists. The second byte of the Category Code is changed back to 0 by the MHI module. The terminal can then change the second byte to 1 if the terminal is requesting more DID k information. However, at this point, the Category Code is 10.
5. The terminal requests more occurrences of DID k by formatting a third new request, changing the second byte of the Category Code to 1, and sending the resulting request to the MHI module.
6. The MHI module sends the last five occurrences of DID k, plus any additional DIDs that will fit into the response. At this point, the MHI module changes the Category Code to 00. The response code in the standard message header is 880 (OK, no more data).

Entire DID Does Not Fit into Response (Partial Download)

During partial downloads, the terminal requests a single DID. The following example shows a terminal request where the entire DID does not fit into a single response during a partial download. In this example, the DID requested exceeds the maximum response message length configured in the MCNF and, therefore, is sent in multiple message responses. In this example, there are nine services defined in the PTDF.



1. The terminal requests a download for DID k. At this point, the Category Code is 00.
2. The MHI module formats a message with three occurrences of DID k, given the message response length constraints configured in the MCNF. In the example, DID k does not completely fit into the response, so the MHI module changes the first byte of the Category Code to 1. This indicates to the terminal that more information for DID k exists. The response code in the standard message header is 881 (OK, more data). At this point, the Category Code is 10.

3. The terminal requests more occurrences of DID k by formatting a new request, changing the second byte of the Category Code to 1, and sending the resulting request to the MHI module. This notifies the MHI module to send more occurrences of DID k to the terminal. At this point, the Category Code is 11. The terminal could also have ignored the balance of DID k by echoing the DownloadMail Key field unaltered.
4. The MHI module sends three more occurrences of DID k in the response. DID k still does not fit into a single response, so the first byte of the Category Code remains as 1. This indicates to the terminal that more information for DID k exists. The second byte of the Category Code is changed back to 0 by the MHI module. The terminal can then change the second byte to 1 if the terminal is requesting more DID k information. However, at this point, the Category Code is 10.
5. The sequence described in steps 3 and 4 continues until DID k is complete or until the terminal does not request any more data.
6. The MHI module sends the last occurrence of DID k. The Category Code is changed to 00 indicating that DID k has completed. The response code in the standard message header is 880 (OK, no more data).

Appendix C

Track Data Processing

One of the functions of the MHI module is to translate the native message into the POS Standard Internal Message (PSTM) before sending the message to the Authorization module.

Native message track data is found in optional data fields. The FID for Track 2 customer or supervisor data is q or r, respectively. The FID for Track 1 customer or supervisor data is 2 or 3, respectively. These optional data fields are defined in section 4.

This appendix describes the actions the MHI module takes when translating track data to the PSTM under various scenarios.

For more information about the PSTM, refer to the ***BASE24-pos Transaction Processing Manual***. For more information about the Track 1 token, refer to the ***BASE24 Tokens Manual***.

PSTM Track Data

This appendix describes how the MHI module formats the PSTM based on the track data in the native message. The subsections that follow discuss these scenarios:

- The native message contains Track 2 data only.
- The native message contains Track 1 data only.
- The native message contains both Track 1 and electronically entered Track 2 data.
- The native message contains both Track 1 and Track 2 data, but the Track 2 data was manually entered.
- The native message contains no track data.

Note that the MHI module does not support manually entered Track 1 data.

Track 2 Data Only

When the native message contains only Track 2 data (the Track 2/Customer or Track 2/Supervisor data field contains a value; the Track 1/Customer or Track 1/Supervisor field is not present), the MHI module fills the PSTM.TRAN.TRACK2 field with the information from the appropriate Track 2 data field.

If the data is entered manually, the MHI module sets the first byte of the PSTM.TRAN.TRACK2 field to M and the PT-SRV-ENTRY-MDE field of the PSTM to 01 (manual). If an expiration date is not specified, the MHI module uses the appropriate default value. The MHI module also sets the COMPLETE-TRACK2-DATA field in the BASE24-pos Release 5.0 token to the corresponding value from the PTDF. The value in this field should not be used for further processing.

If the data is entered electronically, the MHI module sets the PT-SRV-ENTRY-MDE field in the PSTM to 02 (magnetic stripe). The MHI module also sets the COMPLETE-TRACK2-DATA field in the BASE24-pos Release 5.0 token to the corresponding value from the PTDF.

Track 1 Data Only

When the native message contains only Track 1 data (the Track 2/Customer or Track 2/Supervisor field is not present; the Track 1/Customer or Track 1/Supervisor field contains a value), the MHI module extracts the PAN and expiration date from the Track 1 field and places them in the PSTM.TRAN.TRACK2 field. The MHI module sets the first position of the field to M. However, the MHI module sets the PT-SRV-ENTRY-MDE field of the PSTM to 02 to indicate the data was read electronically.

The MHI module then moves the information from the Track 1 data field to the Track 1 token and adds the token to the PSTM. If an error occurs when adding the Track 1 token, the MHI module continues processing the transaction if possible. If the Track 1 token cannot be added, the MHI module sets the PT-SRV-ENTRY-MDE field to 01 to indicate that the track data is manually entered.

In addition, the MHI module sets the COMPLETE-TRACK2-DATA field in the BASE24-pos Release 5.0 token to the corresponding value from the PTDF.

Both Track 1 and Track 2 Data

If the native message contains both Track 1 and electronically entered Track 2 data (both the Track 2 and Track 1 fields contain values; the first byte of the Track 2 field is not M), the MHI module places the information from the Track 2 data field in the PSTM.TRAN.TRACK2 field. It sets the PT-SRV-ENTRY-MDE field of the PSTM to 02 to indicate the cardholder information was read electronically.

The MHI module then moves the Track 1 information from the Track 1 data field to the Track 1 token and adds the token to the PSTM. If an error occurs when adding the Track 1 token, the MHI module continues processing the transaction if possible.

In addition, the MHI module sets the COMPLETE-TRACK2-DATA field in the BASE24-pos Release 5.0 token to the corresponding value from the PTDF.

Track 1 Data and Manually Entered Track 2 Data

If the MHI module receives Track 1 data and manually entered Track 2 data (the first character of the Track 2/Customer or Track 2/Supervisor field is M; the Track 1/Customer or Track 1/Supervisor field contains a value), the MHI module ignores the manually entered data and processes the track data as described in the topic, “Track 1 Data Only.”

No Track Data

If the MHI module receives no track data (not all transaction types require track data), the MHI module determines whether an error condition exists and processes accordingly.

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